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CD4-SPECIFIC ANTIBODY CONSTRUCTS AND COMPOSITIONS AND USES THEREOF

Abstract

Disclosed herein are antibodies and antigen binding fragments thereof that specifically bind human CD4. Also disclosed are fusion proteins comprising a glycoprotein G of the Paramyxoviridae family and CD4 antibodies for targeting and transducing cells expressing CD4. Viral vectors and other compositions containing the fusion proteins, as well as methods of using the fusion proteins, are also disclosed.

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Background/Summary

FIELD
[0001] The present disclosure relates to antibodies or antigen binding fragments thereof that specifically bind human CD4. Also disclosed are fusion proteins comprising an envelope glycoprotein G, H, HN, and/or an F protein of the Paramyxoviridae family. Also disclosed are fusosomes comprising an envelope glycoprotein G, H, and/or an F protein of the Paramyxoviridae family. Fusosomes in one embodiment are gene therapy vectors pseudotyped with an envelope glycoprotein, including envelope glycoproteins G, H, HN and/or an F protein of the Paramyxoviridae family. Also disclosed are an envelope glycoprotein G, H, HN, and/or an F protein of the Paramyxoviridae family and a CD4 antibody, or an antigen binding fragment thereof, for targeting and transducing cells expressing CD4. Viral vectors and other compositions containing the fusion proteins, antibodies, or antigen binding fragments thereof, as well as methods of using the fusosomes, fusion proteins, antibodies, or antigen binding fragments thereof are also disclosed.
SUMMARY
[0002] CD4 (cluster of differentiation 4) is a transmembrane glycoprotein that serves as a co-receptor for the T cell receptor (TCR). CD4 serves multiple functions in immune responses against both external and internal challenges. In T cells, the CD4 co-receptor functions primarily to bind to a major histocompatibility complex (MHC) molecule to facilitate T cell signaling and aid with cytotoxic T cell antigen interactions. While CD4 is predominantly expressed on the surface of helper T cells, it can also be found on natural killer cells, cortical thymocytes, and dendritic cells. The CD4 molecule is also used as a marker for cytotoxic T cell populations.
[0003] T lymphocytes are common targets in gene therapy, even more so since chimeric antigen receptor (CAR) T cells have reached the clinic. Current approaches for T cell engineering mainly rely on ex vivo gene transfer methods. Following their isolation from either healthy donors or patients, lymphocytes are activated and subsequently transduced by lentiviral vectors. The modified lymphocytes are then expanded and either used in functional in vivo assays or used for in vivo applications.
[0004] Ex vivo modification of T lymphocytes, however, has its disadvantages. The complexity of the overall procedure, cost of the manufacturing process, and prolonged ex vivo culture negatively impact the quality of the final product. Methods that improve T lymphocyte engineering that use in vivo delivery platforms are needed.
[0005] In vivo delivery platforms using fusogenic glycoproteins of viral vectors have been shown

to be beneficial for targeting, binding, and transducing cells of interest. Certain fusogenic glycoproteins, however, may not be sufficiently stable or expressed on the surface of the viral vector. Thus, improved fusogenic glycoproteins, fusosomes and viral vectors containing those glycoproteins are needed. The provided disclosure addresses this need.

[0006] The present disclosure provides an antibody or antigen binding fragment thereof that specifically binds CD4, comprising certain heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and/or light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3). Another embodiment is an antibody or antigen binding fragment thereof specifically binding CD4, comprising certain heavy (VH) and/or light (VL) chain variable regions. The disclosure likewise provides for isolated nucleotides, vectors, and host cells comprising the anti-CD4 antibody or antigen binding fragment thereof.

[0007] The present disclosure also provides a fusion protein comprising a glycoprotein G (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), or a biologically active portion thereof of the Paramyxoviridae family and at least one disclosed CD4 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment is fused to the C-terminus of the G protein or the biologically active portion thereof.

[0008] The present disclosure also provides a fusosome comprising at least one antibody or antigen binding fragment thereof that specifically binds CD4, and at least one fusogen. The antibody or antigen binding fragment thereof that specifically binds CD4 may be any antibody or antigen binding fragment thereof that specifically binds CD4, including any antibody or antigen binding fragment thereof that specifically binds CD4 described herein. In some embodiments the fusogen and the antibody or antigen binding fragment thereof that specifically binds CD4 are linked within the fusosome, for example via a linker sequence. In some embodiments the fusogen and the antibody or antigen binding fragment thereof that specifically binds CD4 are not linked within the fusosome. In some embodiments the fusogen and the antibody or antigen binding fragment thereof that specifically binds CD4 are operably linked. The fusogen may be any fusogen, including any fusogen described herein. In some embodiments the fusogen is a G protein, including any G protein described herein. The present disclosure also provides a viral vector comprising a F protein molecule or biologically active portion thereof of the Paramyxoviridae family, an envelope glycoprotein G (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), or a biologically active portion thereof of the Paramyxoviridae family, and at least one disclosed CD4 antibody or antigen binding fragment thereof, wherein the antibody or antigen binding fragment thereof is attached to the C-terminus of the G protein or the biologically active portion thereof.

[0009] The present disclosure likewise relates to methods of selectively modulating and transducing CD4+ T cells using the disclosed fusosomes or viral vectors. Also disclosed are methods of delivering an exogenous agent to a subject, comprising administering to the subject the disclosed fusosomes or viral vectors, in which the fusosomes or viral vector further comprises an exogenous agent. The present disclosure also relates to methods of treating cancer in a subject, comprising administering to the subject the disclosed viral vectors, and corresponding first and second medical uses.

[0010] The present disclosure also provides compositions comprising the fusosomes or fusion proteins or viral vectors of the invention, comprising an antibody or antigen binding fragment thereof that specifically binds CD4, for use as a medicament.

[0011] The present disclosure also provides compositions comprising the fusosomes or fusion proteins or viral vectors of the invention, comprising an antibody or antigen binding fragment thereof that specifically binds CD4, for use in a method of treating cancer.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. **1** shows an exemplary system for administration of a lentiviral vector comprising a CD4 binding agent to a subject.

[0013] FIG. **2** shows off-target transduction of certain CD4 binders using CD4 knockout SupT1 cells and HEK-293T cells, as assessed by measuring percentage of GFP-expressing cells with flow cytometry.

[0014] FIG. **3** shows the flow cytometry data for CD4 retargeted fusogens on PBMCs.

[0015] FIG. **4** shows the percent of GFP+ cells of retargeted fusogens in donor PBMCs.

[0016] FIG. **5**A shows tumor burden at Day 21 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 integrating units (IU) of Binder 256, as assessed by bioluminescence imaging.

[0017] FIG. **5**B shows tumor burden at Day 21 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 IU of a CD8 Binder Control, as assessed by bioluminescence imaging.

[0018] FIG. **5**C shows tumor burden at Day 21 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 integrating units IU of Binder 75, as assessed by bioluminescence imaging.

[0019] FIG. **5**D shows the percentage of CD4+ T cells that express CAR at Day 15 in CD19+ tumor bearing mice treated with 2.5E6, 5E6, or 1E7 integrating units (IU) of CD4-targeted CD19 CAR fusosomes, as assessed by flow cytometry.

DETAILED DESCRIPTION

[0020] Unless defined otherwise, all terms of art, notations, and other technical and scientific terms or terminology used herein are intended to have the same meaning as is commonly understood by one of ordinary skill in the art to which the claimed subject matter pertains. In some cases, terms with commonly understood meanings are defined herein for clarity and/or for ready reference, and the inclusion of such definitions herein should not necessarily be construed to represent a substantial difference over what is generally understood in the art, unless such differences are expressly noted.

[0021] Unless defined otherwise, all technical and scientific terms, acronyms, and abbreviations used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure pertains. Unless indicated otherwise, abbreviations and symbols for chemical and biochemical names is per IUPAC-IUB nomenclature. Unless indicated otherwise, all numerical ranges are inclusive of the values defining the range as well as all integer values in-between.

[0022] As used herein, the articles “a” and “an” refer to one or to more than one (i.e. to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element.

[0023] As used herein, the term “about” will be understood by persons of ordinary skill in the art and will vary to some extent on the context in which it is used. In some embodiments, the term “about” when referring to a measurable value such as an amount, a temporal duration, and the like, is meant to encompass art-accepted variations based on standard errors in making such measurements. In some embodiments, the term “about” when referring to such values, is meant to encompass variations of ±20% or ±10%, more preferably ±5%, even more preferably ±1%, and still more preferably ±0.1% from the specified value, as such variations are appropriate to perform the disclosed methods.

[0024] As used herein, “CD4” or “cluster of differentiation 4” refers to a transmembrane glycoprotein which is a specific marker for a subclass of T cells (which includes helper T cells). The CD4 protein acts as a co-receptor together with the T cell receptor (TCR) to recognize antigen presentation by MHC class II cells. CD4 plays a role in the development of T cells and activation of mature T cells.

[0025] As used herein, “affinity” refers to the strength of the sum total of noncovalent interactions between a single binding site of a molecule (e.g., an antibody) and its binding partner (e.g., an antigen). The affinity of a molecule for its partner can generally be represented by the equilibrium dissociation constant (K.sub.D) (or its inverse equilibrium association constant, KA). Affinity can be measured by common methods known in the art, including those described herein. See, for example, surface plasmon resonance methods described in Pope M. E., Soste M. V., Eyford B. A., Anderson N. L., Pearson T. W., (2009) *J. Immunol. Methods.* 341(1-2):86-96, and methods described therein.

[0026] As used herein, “antibody” is meant in a broad sense and includes immunoglobulin molecules including monoclonal antibodies including murine, human, humanized, and chimeric antibodies, antibody fragments, bispecific or multispecific antibodies formed from at least two intact antibodies or antibody fragments, dimeric, tetrameric or multimeric antibodies, single chain antibodies, and any other modified configuration of the immunoglobulin molecule that comprises an antigen recognition site of the required specificity.

[0027] Immunoglobulins can be assigned to five major classes, namely IgA, IgD, IgE, IgG, and IgM, depending on the heavy chain constant domain amino acid sequence. IgA and IgG are further sub-classified to IgA1, IgA2, IgG1, IgG2, IgG3, and IgG4. Antibody light chains of any vertebrate species can be assigned to one of two types, namely kappa (κ) and lambda (A), based on the amino acid sequences of their constant domains.

[0028] As used herein, “antigen binding fragment” or “antibody fragment” refers to a portion of an immunoglobulin molecule that retains the heavy chain and/or the light chain antigen binding site, such as the heavy chain complementarity determining regions (HCDR) 1 (HCDR1), 2 (HCDR2), and 3 (HCDR3), the light chain complementarity determining regions (LCDR) 1 (LCDR1), 2 (LCDR2), and 3 (LCDR3), the heavy chain variable region (VH), or the light chain variable region (VL). Antibody fragments include a Fab fragment (a monovalent fragment comprising the VL or the VH); a F (ab) 2 fragment (a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region); a Fd fragment comprising the VH and CH1 domains; a Fv fragment comprising the VL and VH domains of a single arm of an antibody; a dAb fragment, which comprises a VH domain; and a variable domain (e.g., VNAR, VHH, etc.) from, e.g., human, shark, or camelid origin. VH and VL domains can be engineered and linked together via one or more synthetic linkers to form various types of single chain antibody designs in which the VH/VL domains pair intramolecularly, or intermolecularly in those cases in which the VH and VL domains are expressed by separate single chain antibody constructs, to form a monovalent antigen binding site, such as a single-chain Fv (scFv) or diabody. Such antibody fragments may be obtained using well known techniques and the fragments may be characterized in the same manner as are intact antibodies.

[0029] An antibody variable region comprises a “framework” region interrupted by three “antigen binding sites.” The antigen binding sites are defined using various terms, including, for example (i) “Complementarity Determining Regions” (CDRs), three in the VH (HCDR1, HCDR2, HCDR3) and three in the VL (LCDR1, LCDR2, LCDR3) (Wu and Kabat, *J Exp Med* 132:211-50, 1970; Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md., 1991), and (ii) “Hypervariable regions,” “HVR,” or “HV,” three in the VH (H1, H2, H3) and three in the VL (L1, L2, L3) (Chothia and Lesk *Mol Biol* 196:901-17, 1987). Other terms include “IMGT-CDRs” (Lefranc et al., *Dev Comparat Immunol* 27:55-77, 2003) and “Specificity Determining Residue Usage” (SDRU) (Almagro Mol Recognit, 17:132-43, 2004). The International ImMunoGeneTics (IMGT) database (http://www_imgt org) provides a standardized numbering and definition of antigen-binding sites. The correspondence between CDRs, HVs, and IMGT delineations is described in Lefranc et al., *Dev Comparat Immunol* 27:55-77, 2003.

[0030] The term “framework,” or “FR” or “framework sequence” refers to the remaining sequences of a variable region other than those sequences defined to be antigen binding sites. Because the antigen binding site can be defined by various terms as described above, the exact amino acid sequence of a framework depends on how the antigen-binding site was defined.

[0031] The term "CDR" denotes a complementarity determining region as defined by at least one manner of identification to one of skill in the art. The precise amino acid sequence boundaries of a given CDR or FR can be readily determined using any of a number of well-known schemes, including those described by Kabat et al. (1991), "Sequences of Proteins of Immunological Interest," 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD ("Kabat" numbering scheme); Al-Lazikani et al., (1997) JMB 273,927-948 ("Chothia" numbering scheme); MacCallum et al., J. Mol. Biol. 262:732-745 (1996), "Antibody-antigen interactions: Contact analysis and binding site topography," J. Mol. Biol. 262, 732-745." ("Contact" numbering scheme); Lefranc M P et al., "IMGT unique numbering for immunoglobulin and T cell receptor variable domains and Ig superfamily V-like domains," *Dev Comp Immunol,* 2003 January; 27(1):55-77 ("IMGT" numbering scheme); Honegger A and Plückthun A, "Yet another numbering scheme for immunoglobulin variable domains: an automatic modeling and analysis tool," J Mol Biol, 2001 Jun. 8; 309(3):657-70, ("Aho" numbering scheme); and Martin et al., "Modeling antibody hypervariable loops: a combined algorithm," PNAS, 1989, 86(23):9268-9272, ("AbM" numbering scheme).
[0032] The boundaries of a given CDR or FR may vary depending on the scheme used for identification. For example, the Kabat scheme is based on structural alignments, while the Chothia scheme is based on structural information. Numbering for both the Kabat and Chothia schemes is based upon the most common antibody region sequence lengths, with insertions accommodated by insertion letters, for example, "30a," and deletions appearing in some antibodies. The two schemes place certain insertions and deletions ("indels") at different positions, resulting in differential numbering. The Contact scheme is based on analysis of complex crystal structures and is similar in many respects to the Chothia numbering scheme. The AbM scheme is a compromise between the Kabat and Chothia definitions based on that used by Oxford Molecular's AbM antibody modeling software.
[0033] In some embodiments, CDRs are defined in accordance with any of the Chothia numbering schemes, the Kabat numbering scheme, the IMGT numbering scheme, a combination of Kabat, IMGT, and Chothia, the AbM definition, and/or the contact definition. A sdAb variable domain comprises three CDRs, designated CDR1, CDR2, and CDR3. Table 1, below, lists exemplary position boundaries of CDR-H1, CDR-H2, CDR-H3 as identified by Kabat, Chothia, AbM, and Contact schemes, respectively. For CDR-H1, residue numbering is listed using both the Kabat and Chothia numbering schemes. FRs are located between CDRs, for example, with FR-H1 located before CDR-H1, FR-H2 located between CDR-H1 and CDR-H2, FR-H3 located between CDR-H2 and CDR-H3 and so forth. It is noted that because the shown Kabat numbering scheme places insertions at H35A and H35B, the end of the Chothia CDR-H1 loop when numbered using the shown Kabat numbering convention varies between H32 and H34, depending on the length of the loop.
TABLE-US-00001 TABLE 1 Boundaries of CDRs according to various numbering schemes. CDR Kabat Chothia AbM Contact CDR-H1 H31--H35B H26--H32 . . . H26--H35B H30--H35B (Kabat 34 Numbering.sup.1) CDR-H1 H31--H35 H26--H32 H26--H35 H30--H35 (Chothia Numbering.sup.2) CDR-H2 H50--H65 H52--H56 H50--H58 H47--H58 CDR-H3 H95--H102 H95--H102 H95--H102 H93--H101 .sup.1Kabat et al. (1991), "Sequences of Proteins of Immunological Interest," 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD .sup.2Al-Lazikani et al., (1997) JMB 273,927-948.
[0034] Thus, unless otherwise specified, a "CDR" or "complementary determining region," or individual specified CDRs (e.g., CDR-H1, CDR-H2, CDR-H3), of a given antibody or region thereof, such as a variable region thereof, should be understood to encompass a (or the specific) complementary determining region as defined by any of the aforementioned schemes. For example, where it is stated that a particular CDR (e.g., a CDR-H3) contains the amino acid sequence of a corresponding CDR in a given sdAb amino acid sequence, it is understood that such a CDR has a

sequence of the corresponding CDR (e.g., CDR-H3) within the sdAb, as defined by any of the aforementioned schemes. It is understood that any antibody, such as a sdAb, includes CDRs and such can be identified according to any of the other aforementioned numbering schemes or other numbering schemes known to a skilled artisan.

[0035] As used herein, “Fv” refers to the minimum antibody fragment which contains a complete antigen-recognition and antigen-binding site. This region comprises a dimer of one heavy chain and one light chain variable domain in tight, non-covalent association. It is in this configuration that the three hypervariable regions of each variable domain interact to define an antigen-binding site on the surface of the VH-VL dimer. Collectively, the six hypervariable regions confer antigen-binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three hypervariable regions specific for an antigen) may have the ability to recognize and bind an antigen, although at a lower affinity than the entire binding site.

[0036] As used herein, “single-chain Fv” or “scFv” antibody fragments comprise the VH and VL domains of an antibody, wherein these domains are present in a single polypeptide chain. Preferably, the Fv polypeptide further comprises a linker (e.g., a polypeptide linker) between the VH and VL domains which enables the scFv to form the desired structure for antigen binding. For a review of scFv see Pluckthun in The Pharmacology of Monoclonal Antibodies, vol. 113, Rosenburg and Moore eds., Springer-Verlag, New York, pp. 269-315 (1994).

[0037] As used herein, “VHH” or “VHH antibodies” refer to single domain antibodies that comprise the variable (antigen binding) domain of the heavy chain antibody (HCAb or hcIgG) molecules produced by Camelidae family mammals (e.g., llamas, camels, and alpacas).

[0038] As used herein, “VNAR” or “VNAR antibodies” refer to single domain antibodies that comprise the variable (antibody binding) domain of the shark immunoglobulin new antigen receptors (IgNARs).

[0039] As used herein, the term “specifically binds” to a target molecule, such as an antigen, means that a binding molecule, such as a single domain antibody, reacts or associates more frequently, more rapidly, with greater duration, and/or with greater affinity with a particular target molecule than it does with alternative molecules. A binding molecule, such as a sdAb or scFv, “specifically binds” to a target molecule if it binds with greater affinity, avidity, more readily, and/or with greater duration than it binds to other molecules. It is understood that a binding molecule, such as a sdAb or scFv, that specifically binds to a first target may or may not specifically bind to a second target. As such, “specific binding” does not necessarily require (although it can include) exclusive binding.

[0040] As used herein, “percent (%) sequence identity” with respect to an amino acid or nucleic acid sequence is defined as the percentage of amino acid or nucleic acid residues in a candidate sequence that are identical with the amino acid or nucleic acid residues in another amino acid or nucleic acid sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Percent identity between nucleic acid sequences may be determined using a suite of commonly used and freely available sequence comparison algorithms provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S. F. et al. (1990) J. Mol. Biol. 215:403-410), which is available from several sources, including the NCBI, Bethesda, Md., and on the Internet at http://www.ncbi.nlm.nih.gov/BLAST/. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared.

[0041] An amino acid substitution may include but is not limited to the replacement of one amino acid in a polypeptide with another amino acid. Exemplary substitutions are shown in Table 2. Amino acid substitutions may be introduced into an antibody of interest and the products screened for a desired activity, for example, retained/improved binding.

TABLE-US-00002 TABLE 2 Original Residue Exemplary Substitutions Ala (A) Val; Leu; Ile Arg (R) Lys; Gln; Asn Asn (N) Gln; His; Asp, Lys; Arg Asp (D) Glu; Asn Cys (C) Ser; Ala Gln (Q) Asn; Glu Glu (E) Asp; Gln Gly (G) Ala His (H) Asn; Gln; Lys; Arg Ile (I) Leu; Val; Met; Ala; Phe; Norleucine Leu (L) Norleucine; Ile; Val; Met; Ala; Phe Lys (K) Arg; Gln; Asn Met (M) Leu; Phe; Ile Phe (F) Trp; Leu; Val; Ile; Ala; Tyr Pro (P) Ala Ser (S) Thr Thr (T) Val; Ser Trp (W) Tyr; Phe Tyr (Y) Trp; Phe; Thr; Ser Val (V) Ile; Leu; Met; Phe; Ala; Norleucine

[0042] Amino acids may be grouped according to common side-chain properties: [0043] (1) hydrophobic: Norleucine, Met, Ala, Val, Leu, Ile; [0044] (2) neutral hydrophilic: Cys, Ser, Thr, Asn, Gln; [0045] (3) acidic: Asp, Glu; [0046] (4) basic: His, Lys, Arg; [0047] (5) residues that influence chain orientation: Gly, Pro; [0048] (6) aromatic: Trp, Tyr, Phe.

[0049] Non-conservative substitutions will entail exchanging a member of one of these classes for another class. The term, “corresponding to” with reference to nucleotide or amino acid positions of a sequence, such as set forth in the Sequence Listing, refers to nucleotide or amino acid positions identified upon alignment with a target sequence based on structural sequence alignment or using a standard alignment algorithm, such as the GAP algorithm. For example, corresponding residues of a similar sequence (e.g. fragment or species variant) can be determined by alignment to a reference sequence by structural alignment methods. By aligning the sequences, one skilled in the art can identify corresponding residues, for example, using conserved and identical amino acid residues as guides.

[0050] The term “isolated” as used herein refers to a molecule that has been separated from at least some of the components with which it is typically found in nature or produced. For example, a polypeptide is referred to as “isolated” when it is separated from at least some of the components of the cell in which it was produced. When a polypeptide is secreted by a cell after expression, physically separating the supernatant containing the polypeptide from the cell that produced it is considered to be “isolating” the polypeptide. Similarly, a polynucleotide is referred to as “isolated” when it is not part of the larger polynucleotide (such as, for example, genomic DNA or mitochondrial DNA, in the case of a DNA polynucleotide) in which it is typically found in nature, or is separated from at least some of the components of the cell in which it was produced. Thus, a DNA polynucleotide that is contained in a vector inside a host cell may be referred to as “isolated.”

[0051] As used herein, “lipid particle” refers to any biological or synthetic particle that contains a bilayer of amphipathic lipids enclosing a lumen or cavity. Typically, a lipid particle does not contain a nucleus. Examples of lipid particles include nanoparticles, viral-derived particles, or cell-derived particles. Such lipid particles include, but are not limited to, viral particles (e.g. lentiviral particles), virus-like particles, viral vectors (e.g., lentiviral vectors), exosomes, enucleated cells, vesicles (e.g., microvesicles, membrane vesicles, extracellular membrane vesicles, plasma membrane vesicles, and giant plasma membrane vesicles), apoptotic bodies, mitoparticles, pyrenocytes, or lysosomes. In some embodiments, a lipid particle is a fusosome. In some embodiments, the lipid particle is not a platelet.

[0052] As used herein a “biologically active portion,” such as with reference to a protein such as a G protein or an F protein, refers to a portion of the protein that exhibits or retains an activity or property of the full-length of the protein. For example, a biologically active portion of an F protein retains fusogenic activity in conjunction with the G protein when each are embedded in a lipid bilayer. A biologically active portion of the G protein retains fusogenic activity in conjunction with an F protein when each is embedded in a lipid bilayer. The retained activity can include 10%-150% or more of the activity of a full-length or wild-type F protein or G protein. Examples of biologically active portions of F and G proteins include truncations of the cytoplasmic domain, e.g. truncations of up to 1, 2, 3, 4, 5, 6, 7, 8 9, 10, 11, 12, 13, 14, 15, 20, 22, 25, 30, 33, 34, 35, or more contiguous amino acids, see e.g. Khetawat and Broder 2010 Virology Journal 7:312; Witting et al. 2013 Gene Therapy 20:997-1005; published international; patent application No. WO/2013/148327.

[0053] As used herein, “G protein” refers to an envelope attachment glycoprotein G or biologically active portion thereof of the Paramyxoviridae family. “F protein” refers to a fusion protein F or biologically active portion thereof of the Paramyxoviridae family. “H protein” refers to an envelope attachment protein with haemagglutination activity. Morbilliviruses attachment proteins are designated H proteins. “HN protein” refers to an envelope attachment protein with haemagglutination-neuraminidase activity. Respiroviruses, rubulaviruses and avulaviruses attachment proteins are designated HN proteins. H, HN, and G proteins are cell attachment proteins that span the viral envelope and project from the surface as spikes. These proteins bind to proteins on the surface of target cells to facilitate cell entry.

[0054] The F and G proteins may be from a henipavirus, a Hendra (HeV) virus, or a Nipah (NiV) virus, and may be a wild-type protein or may be a variant thereof that exhibits reduced binding for the native binding partner. The F (fusion) and G (attachment) glycoproteins mediate cellular entry of Nipah virus. The G protein initiates infection by binding to the cellular surface receptor ephrin-B2 (EphB2) or EphB3. The subsequent release of the viral genome into the cytoplasm is mediated by the action of the F protein, which induces the fusion of the viral envelope with cellular membranes. The efficiency of transduction of targeted lipid particles can be improved by engineering hyperfusogenic mutations in one or both of the F protein (such as NiV-F) and G protein (such as NiV-G).

[0055] As used herein, “fusosome” refers to a particle containing a bilayer of amphipathic lipids enclosing a lumen or cavity and a fusogen that interacts with the amphipathic lipid bilayer. In some embodiments, the fusosome comprises a nucleic acid. In some embodiments, the fusosome is a membrane enclosed preparation. In some embodiments, the fusosome is derived from a source cell. In some embodiments the fusosme is a vector. In some embodiments the fusosome is an integrating vector. In some embodiments the fusosome is a viral vector. In some embodiments the fusosome is a lipid particle, including a targeted lipid particle, including any lipid particle or targeted lipid particle described herein. As used herein, “fusosome composition” refers to a composition comprising one or more fusosomes.

[0056] As used herein, “fusogen” refers to an agent or molecule that creates an interaction between two membrane enclosed lumens. In embodiments, the fusogen facilitates fusion of the membranes. In other embodiments, the fusogen creates a connection, e.g., a pore, between two lumens (e.g., a lumen of a retroviral vector and a cytoplasm of a target cell). In some embodiments, the fusogen comprises a complex of two or more proteins, e.g., wherein neither protein has fusogenic activity alone. In some embodiments, the fusogen comprises a targeting domain.

[0057] As used herein, a “re-targeted fusogen” refers to a fusogen that comprises a targeting moiety having a sequence that is not part of the naturally-occurring form of the fusogen. In embodiments, the fusogen comprises a different targeting moiety relative to the targeting moiety in the naturally-occurring form of the fusogen. In embodiments, the naturally-occurring form of the fusogen lacks a targeting domain, and the re-targeted fusogen comprises a targeting moiety that is absent from the naturally-occurring form of the fusogen. In embodiments, the fusogen is modified to comprise a targeting moiety. In embodiments, the fusogen comprises one or more sequence alterations outside of the targeting moiety relative to the naturally-occurring form of the fusogen, e.g., in a transmembrane domain, fusogenically active domain, or cytoplasmic domain.

[0058] As used herein, a “targeted envelope protein” refers to a polypeptide that contains a G protein (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), of the Paramyxoviridae family attached to a single domain antibody (sdAb) variable domain, such as a VL or VH sdAb, a scFv, a nanobody, a camelid VHH domain, a shark VNAR, or fragments thereof, that target a molecule on a desired cell type. In some such embodiments, the attachment may be direct or indirect via a linker, such as a polypeptide linker. The “targeted envelope protein” may also be referred to as a “fusion protein” comprising the G protein and antibodies or antigen binding fragments of the disclosure in which the antibody or antigen binding fragment is fused to the C-terminus of the G protein or a biologically active portion thereof.

[0059] As used herein, a "targeted lipid particle" refers to a lipid particle that contains a targeted envelope protein embedded in the lipid bilayer, e.g., targeting CD4. Such targeted lipid particles can be a viral particle, a virus-like particle, a nanoparticle, a vesicle, an exosome, a dendrimer, a lentivirus, a viral vector, an enucleated cell, a microvesicle, a membrane vesicle, an extracellular membrane vesicle, a plasma membrane vesicle, a giant plasma membrane vesicle, an apoptotic body, a mitoparticle, a pyrenocyte, a lysosome, another membrane enclosed vesicle, a lentiviral vector, a viral based particle, a virus like particle (VLP), or a cell derived particle.

[0060] As used herein, a "retroviral nucleic acid" refers to a nucleic acid containing at least the minimal sequence requirements for packaging into a retrovirus or retroviral vector, alone or in combination with a helper cell, helper virus, or helper plasmid. In some embodiments, the retroviral nucleic acid further comprises or encodes an exogenous agent, a positive target cell-specific regulatory element, a non-target cell-specific regulatory element (TCSRE), or a negative TCSRE. In some embodiments, the retroviral nucleic acid comprises one or more of (e.g., all of) a 5′ LTR (e.g., to promote integration), U3 (e.g., to activate viral genomic RNA transcription), R (e.g., a Tat-binding region), U5, a 3′ LTR (e.g., to promote integration), a packaging site (e.g., psi (′)), and RRE (e.g., to bind to Rev and promote nuclear export). The retroviral nucleic acid can comprise RNA (e.g., when part of a virion) or DNA (e.g., when being introduced into a source cell or after reverse transcription in a recipient cell). In some embodiments, the retroviral nucleic acid is packaged using a helper cell, helper virus, or helper plasmid which comprises one or more of (e.g., all of) gag, pol, and env.

[0061] As used herein, a "target cell" refers to a cell of a type to which it is desired that a targeted lipid particle delivers an exogenous agent. In embodiments, a target cell is a cell of a specific tissue type or class, e.g., an immune effector cell, e.g., a T cell. In some embodiments, a target cell is a diseased cell, e.g., a cancer cell. In some embodiments, the fusogen, e.g., a re-targeted fusogen, leads to preferential delivery of the exogenous agent to a target cell compared to a non-target cell.

[0062] As used herein a "non-target cell" refers to a cell of a type to which it is not desired that a targeted lipid particle delivers an exogenous agent. In some embodiments, a non-target cell is a cell of a specific tissue type or class. In some embodiments, a non-target cell is a non-diseased cell, e.g., a non-cancerous cell. In some embodiments, the fusogen, e.g., a re-targeted fusogen, leads to lower delivery of the exogenous agent to a non-target cell compared to a target cell.

[0063] The term "effective amount" as used herein means an amount of a pharmaceutical composition which is sufficient to significantly and positively modify the symptoms and/or conditions to be treated (e.g., provide a positive clinical response). The effective amount of the targeted lipid particles of the disclosure for use in a pharmaceutical composition will vary with the particular condition being treated, the severity of the condition, the duration of treatment, the nature of concurrent therapy, the particular lipid particle being employed, the particular pharmaceutically-acceptable excipient(s) and/or carrier(s) utilized, and like factors within the knowledge and expertise of the attending physician.

[0064] An "exogenous agent" as used herein with reference to a targeted lipid particle, refers to an agent that is neither comprised by nor encoded in the corresponding wild-type virus or fusogen made from a corresponding wild-type source cell. In some embodiments, the exogenous agent does not naturally exist, such as a protein or nucleic acid that has a sequence that is altered (e.g., by insertion, deletion, or substitution) relative to a naturally occurring protein. In some embodiments, the exogenous agent does not naturally exist in the source cell. In some embodiments, the exogenous agent exists naturally in the source cell but is exogenous to the virus. In some embodiments, the exogenous agent does not naturally exist in the recipient cell. In some embodiments, the exogenous agent exists naturally in the recipient cell, but is not present at a desired level or at a desired time. In some embodiments, the exogenous agent comprises DNA, RNA, or protein.

[0065] As used herein, a “promoter” refers to a cis-regulatory DNA sequence that, when operably linked to a gene coding sequence, drives transcription of the gene. The promoter may comprise one or more transcription factor binding sites. In some embodiments, a promoter works in concert with one or more enhancers which are distal to the gene.
[0066] As used herein, “operably linked” refers to a polynucleotide sequence that is joined to a regulatory region sequence in a manner that allows expression of the polynucleotide sequence. A regulatory region sequence directs transcription of a polynucleotide sequence, and can include enhancer sequences, response elements, protein recognition sites, inducible elements, promoter control elements, 5′ and 3′ untranslated regions protein binding sequences, transcriptional start sites, termination sequences, polyadenylation sequences, and introns.
[0067] As used herein, a composition refers to any mixture of two or more products, substances, or compounds, including cells. It may be a solution, a suspension, a liquid, a powder, a paste, aqueous, non-aqueous, or any combination thereof.
[0068] As used herein, the term “pharmaceutically acceptable” refers to a material, such as a carrier or diluent, which does not abrogate the biological activity or properties of a therapeutic compound, and is relatively nontoxic, i.e., the material may be administered to an individual without causing undesirable biological effects or interacting in a deleterious manner with any of the components of the composition in which it is contained.
[0069] As used herein, the term “pharmaceutical composition” refers to a mixture of at least one targeted lipid particle of the disclosure with other chemical components, such as carriers, stabilizers, diluents, dispersing agents, suspending agents, thickening agents, and/or excipients. The pharmaceutical composition facilitates administration of the targeted lipid particle to an organism. Multiple techniques of administering targeted lipid particles of the disclosure exist in the art including, but not limited to, intravenous, oral, aerosol, parenteral, ophthalmic, pulmonary, and topical administration.
[0070] A “disease” or “disorder” as used herein refers to a condition for which treatment is needed and/or desired.
[0071] As used herein, the terms “treat,” “treating,” or “treatment” refer to ameliorating a disease or disorder, e.g., slowing or arresting or reducing the development of the disease or disorder or reducing at least one of the clinical symptoms thereof. For purposes of this disclosure, ameliorating a disease or disorder can include obtaining a beneficial or desired clinical result that includes, but is not limited to, any one or more of: alleviation of one or more symptoms, diminishment of extent of disease, preventing or delaying spread (for example, metastasis, for example metastasis to the lung or to the lymph node) of disease, preventing or delaying recurrence of disease, delay or slowing of disease progression, amelioration of the disease state, inhibiting the disease or progression of the disease, inhibiting or slowing the disease or its progression, arresting its development, and remission (whether partial or total).
[0072] The terms “individual” and “subject” are used interchangeably herein to refer to an animal; for example a mammal. The terms include human and veterinary animals. In some embodiments, methods of treating animals, including, but not limited to, humans, rodents, simians, felines, canines, equines, bovines, porcines, ovines, caprines, mammalian laboratory animals, mammalian farm animals, mammalian sport animals, and mammalian pets, are provided. The animal can be male or female and can be any suitable age, including infant, juvenile, adolescent, adult, and geriatric. In some examples, an “individual” or “subject” refers to an animal in need of treatment for a disease or disorder. In some embodiments, the animal to receive the treatment is a “patient,” designating the fact that the animal has been identified as having a disorder of relevance to the treatment, or being at adequate risk of contracting the disorder. In particular embodiments, the animal is a human, such as a human patient.
CD4-Specific Antibodies
[0073] Described herein are novel antibodies and antigen binding fragments thereof that

specifically target and bind CD4. In some embodiments, the antibodies or antigen binding fragments thereof cross-react with cynomolgus (or "cyno") or *M. nemestrina* CD4. In some embodiments, the antibodies or antigen binding fragments thereof are single-chain variable fragments (scFvs) composed of the antigen-binding domains derived from the heavy (VH) and the light (VL) chains of an IgG molecule and connected via a linker domain. In some embodiments, the antibodies or antigen binding fragments thereof are VHHs or VNARs that correspond to the antigen binding domains of the camelid and shark IgG molecules, respectively. The present disclosure also provides polynucleotides encoding the antibodies and fragments thereof, vectors, and host cells, and methods of using the antibodies or antigen binding fragments thereof. In some embodiments, e.g., the antibodies or antigen binding fragments thereof fuse to a glycoprotein (G Protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein) of the Paramyxoviridae family for targeted binding and transduction to cells.

[0074] Sequences for exemplary antibodies and antigen binding fragments of the disclosure using the Kabat numbering scheme are shown in Tables 19-22 below. Sequences for exemplary HCDRs and LCDRs of the disclosure are shown in Table 22.

[0075] The sequences for the disclosed VH and VL domains are provided in Tables 20-21. Tables 23-24 provided herein show the CDR sequences of the disclosed antibodies and antigen binding fragments thereof using both Chothia and IMGT numbering schemes, respectively. The full CD4 binder sequences of the variant CD4 scFvs and VHHs of the disclosure are shown in Table 19.

[0076] In some embodiments, an antibody or antigen binding fragment thereof capable of binding CD4 is disclosed, comprising a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3), and the light chain variable region comprises three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3). In some embodiments, the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 comprise amino acid sequences of any one of the SEQ ID NOs recited in Table 22. In some embodiments, the heavy chain variable region (VH) comprises an amino acid sequence of any one of SEQ ID NOs: 256-511, 9447-9576, or 14000-14002 (Table 20) and the light chain variable region (VL) comprises an amino acid sequence of any one of SEQ ID NOs: 512-766 or 9577-9706 (Table 21).

[0077] In another embodiment, the antibody or antigen binding fragment thereof comprises a VH having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 256-511, 9447-9576, or 14000-14002.

[0078] In another embodiment, the antibody or antigen binding fragment thereof comprises a VL having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 512-766 or 9577-9706.

[0079] In another embodiment, the antibody or antigen binding fragment comprises a VH having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 256-511, 9447-9576, or 14000-14002 and a VL having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to a sequence selected from SEQ ID NOs: 512-766 or 9577-9706.

[0080] In another embodiment, the antibody or antigen binding fragment thereof comprises a VH having an amino acid sequence with at least 90%, 95%, 96%, 97%, 98%, 99%, or 100% identity to SEQ ID NO: 256.

[0081] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 304 and the VL of SEQ ID NO: 559.

[0082] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 331 and the VL of SEQ ID NO: 586.

[0083] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 9554 and the VL of SEQ ID NO: 9684.

[0084] In another embodiment, the antibody or antigen binding fragment thereof comprises the VH of SEQ ID NO: 256.

[0085] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of SEQ ID NOs: 1308, 1822, 2336, 5672, 6182, 6692, respectively.

[0086] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of SEQ ID NOs: 1376, 1890, 2404, 5740, 6250, 6760, respectively.

[0087] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of SEQ ID NOs: 10074, 10336, 10598, 12300, 12560, 12820, respectively.

[0088] In another embodiment, the antibody or antigen binding fragment thereof comprises the HCDR1, HCDR2, and HCDR3 of SEQ ID NOs: 1535, 2049, 2563, respectively.

[0089] In some embodiments, the single domain antibody is human or humanized. In some embodiments, the single domain antibody or portion thereof is naturally occurring. In some embodiments, the single domain antibody or portion thereof is synthetic.

[0090] In some embodiments, the single domain antibodies are antibodies whose complementary determining regions are part of a single domain polypeptide. In some embodiments, the single domain antibody is a heavy chain only antibody variable domain. In some embodiments, the single domain antibody does not include light chains.

[0091] In various embodiments, any of the antibodies or antigen binding fragments described herein comprise a heavy chain constant region and a light chain constant region. The heavy chain constant region may be an IgG, IgM, IgA, IgD, or IgE isotype, or a derivative or fragment thereof that retains at least one effector function of the intact heavy chain. The heavy chain constant region may be a human IgG isotype. The heavy chain constant region may be a human IgG1 or human IgG4 isotypes. The heavy chain constant region may be a human IgG1 isotype. The light chain constant region may be a human kappa light chain or lambda light chain or a derivative or fragment thereof that retains at least one effector function of the intact light chain. The light chain constant region may be a human kappa light chain.

[0092] In various embodiments, any of the disclosed antibodies or antigen binding fragments may be a rodent antibody or antigen binding fragment thereof, a chimeric antibody or an antigen binding fragment thereof, a CDR-grafted antibody or an antigen binding fragment thereof, or a humanized antibody or an antigen binding fragment thereof. In another embodiment, any of the disclosed antibodies or antigen binding fragments comprises human or human-derived heavy and light chain variable regions, including human frameworks or human frameworks with one or more backmutations. In various embodiments, any of the disclosed antibodies or antigen binding fragments may be a Fab, Fab′, F(ab′)2, Fd, scFv, (scFv)2, scFv-Fc, VHH, or Fv fragment.

[0093] Antibodies whose heavy chain CDR, light chain CDR, VH, or VL amino acid sequences differ insubstantially from those shown in Tables 19-24 are encompassed within the scope of the disclosure. Typically, this involves one or more conservative amino acid substitutions with an amino acid having similar charge, hydrophobic, or stereo chemical characteristics in the antigen-binding site or in the framework without adversely altering the properties of the antibody. Conservative substitutions may also be made to improve antibody properties, for example stability or affinity. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acid substitutions can be made to the VH or VL sequence. For example, a “conservative amino acid substitution” may involve a substitution of a native amino acid residue with a nonnative residue such that there is little or no effect on the polarity or charge of the amino acid residue at that position. Desired amino acid substitutions can be determined by those skilled in the art at the time such substitutions are desired. For example, amino acid substitutions can be used to identify important residues of the molecule sequence, or to increase or decrease the affinity of the molecules described herein. The following eight groups contain amino acids that are conservative amino acid substitutions for one another: 1)

Alanine (A), Glycine (G); 2) Aspartic acid (D), Glutamic acid (E); 3) Asparagine (N), Glutamine (Q); 4) Arginine (R), Lysine (K); 5) Isoleucine (I), Leucine (L), Methionine (M), Valine (V); 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W); 7) Serine(S), Threonine (T); and 8) Cysteine (C), Methionine (M).

[0094] In some embodiments, the antibody or antigen binding fragment binding CD4 is a single-chain variable fragment. In embodiments involving a single polypeptide containing both a heavy chain variable region and a light chain variable region, both orientations of these variable regions are contemplated. In some cases, the heavy chain variable region is on the N-terminal side of the light chain variable region, which means the heavy chain variable region is closer to the N-terminus of the polypeptide. In other cases, the light chain variable region is on the N-terminal side of the heavy chain variable region, which means the light chain variable region is closer to the N-terminus of the polypeptide than the heavy chain variable region.

[0095] In some embodiments, the scFv binding proteins comprise a linker. In some embodiments, the linker is between the heavy chain variable region (VH) and the light chain variable region (VL) (or vice versa). In some embodiments, the linker comprises the amino acid sequence of GS, GGS, GGGS (SEQ ID NO: 14125), GGGGS (SEQ ID NO: 9294), GGGGGS (SEQ ID NO: 9292), any one of SEQ ID NOs: 9312-9315, or combinations thereof. Substitutions to introduce new disulfide bonds are also within the scope of the disclosure, e.g., by making substitutions G44C in the VH FR 2 and G100C in the VL FR4.

[0096] In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to human CD4 with an affinity constant (K.sub.D) of about 1 nM to about 900 nM. In some embodiments, the K.sub.D to human CD4 is about 5 nM to about 500 nM, about 6 nM to about 10 nM, about 11 nM to about 20 nM, about 25 nM to about 40 nM, about 40 nM to about 60 nM, about 70 nM to about 90 nM, about 100 nM to about 120 nM, about 125 nM to about 140 nM, about 145 nM to about 160 nM, about 170 nM and to about 200 nM, about 210 nM to about 250 nM, about 260 nM to about 300 nM, about 310 nM to about 350 nM, about 360 nM to about 400 nM, about 410 nM to about 450 nM, and about 460 nM to about 500 nM. In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to human CD4 with an affinity constant (K.sub.D) of 500 nM, 400 nM, 300 nM, 200 nM, 100 nM, 50 nM, 20 nM, or 10 nM or lower. In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to human CD4 and CD4 of a non-human primate including cynomolgus, *M. mulatta* (rhesus monkey), or *M. nemestrina* CD4 with comparable binding affinity (K.sub.D).

[0097] In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to a non-human primate, cynomolgus, *M. mulatta* (rhesus monkey), or *N. nemestrina* CD4. In some embodiments, the anti-CD4 antibody or antigen binding binds to mouse, dog, pig, etc., CD4. In some embodiments, the K.sub.D to a non-human primate, cynomolgus or *M. nemestrina* CD4 is about 5 nM to about 500 nM, about 6 nM to about 10 nM, about 11 nM to about 20 nM, about 25 nM to about 40 nM, about 40 nM to about 60 nM, about 70 nM to about 90 nM, about 100 nM to about 120 nM, about 125 nM to about 140 nM, about 145 nM to about 160 nM, about 170 nM and to about 200 nM, about 210 nM to about 250 nM, about 260 nM to about 300 nM, about 310 nM to about 350 nM, about 360 nM to about 400 nM, about 410 nM to about 450 nM, and about 460 nM to about 500 nM. In some embodiments, the anti-CD4 antibody or antigen binding fragment binds to cynomolgus or *M. nemestrina* CD4 with an affinity constant (K.sub.D) of 500 nM, 400 nM, 300 nM, 200 nM, 100 nM, 50 nM, 20 nM, or 10 nM or lower.

[0098] An antibody or antigen binding fragment thereof that specifically binds CD4 refers to an antibody or binding fragment that preferentially binds to CD4, respectively, over other antigen targets. As used herein, references to an antibody that “specifically binds CD4” are interchangeable with an “anti-CD4” antibody or an “antibody that binds CD4.” In some embodiments, the antibody or binding fragment capable of binding to CD4 does so with higher affinity for that antigen than others. In some embodiments, the antibody or binding fragment capable of binding CD4 binds to that antigen with a K.sub.D of at least about 10-1, 10-2, 10-3, 10-4, 10-5, 10-6, 10-7, 10-8, 10-9, 10-10, 10-11, 10-12 or greater (or any value in between), e.g., as measured by surface plasmon resonance or other methods known to those skilled in the art.

[0099] Another embodiment of the disclosure is an isolated polynucleotide encoding any of the antibody heavy chain variable regions or the antibody light chain variable regions of the disclosure. Certain exemplary polynucleotides are disclosed herein, however, other polynucleotides which, given the degeneracy of the genetic code or codon preferences in a given expression system, encode the antibodies or antigen binding fragments thereof of the disclosure are also within the scope of the disclosure. The polynucleotide sequences encoding a VH or a VL or a fragment thereof of the antibody or antigen binding fragments thereof of the disclosure can be operably linked to one or more regulatory elements, such as a promoter and enhancer, that allow expression of the nucleotide sequence in the intended host cell. The polynucleotide may be a cDNA.

[0100] Another embodiment of the disclosure is a vector comprising the polynucleotide of the disclosure. Such vectors may be plasmid vectors, viral vectors, vectors for baculovirus expression, transposon-based vectors, or any other vector suitable for introduction of the polynucleotide of the disclosure into a given organism or genetic background by any means. For example, polynucleotides encoding light and heavy chain variable regions of the antibodies of the disclosure, optionally linked to constant regions, may be inserted into expression vectors. The light and heavy chains can be cloned in the same or different expression vectors. The DNA segments encoding immunoglobulin chains may be operably linked to control sequences in the expression vector(s) that ensure the expression of immunoglobulin polypeptides. Such control sequences include signal sequences, promoters (e.g., naturally associated or heterologous promoters), enhancer elements, and transcription termination sequences, and are chosen to be compatible with the host cell chosen to express the antibody. Once the vector has been incorporated into the appropriate host, the host is maintained under conditions suitable for high level expression of the proteins encoded by the incorporated polynucleotides.

[0101] Suitable expression vectors are typically replicable in the host organisms either as episomes or as an integral part of the host chromosomal DNA. Commonly, expression vectors contain selection markers such as ampicillin-resistance, hygromycin-resistance, tetracycline resistance, kanamycin resistance, or neomycin resistance to permit detection of those cells transformed with the desired DNA sequences. Suitable vectors, promoter, and enhancer elements are known in the art; many are commercially available for generating subject recombinant constructs.

[0102] Another embodiment of the disclosure is a host cell comprising the vector of the disclosure. The term “host cell” refers to a cell into which a vector has been introduced. It is understood that the term host cell is intended to refer not only to the particular subject cell but to the progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not be identical to the parent cell, but are still included within the scope of the term “host cell” as used herein. Such host cells may be eukaryotic cells, prokaryotic cells, plant cells, or archaeal cells. *Escherichia coli*, bacilli, such as *Bacillus subtilis*, and other Enterobacteriaceae, such as *Salmonella*, *Serratia*, and various *Pseudomonas* species are examples of prokaryotic host cells. Other microbes, such as yeast, are also useful for expression. *Saccharomyces* (e.g., *S. cerevisiae*) and *Pichia* are examples of suitable yeast host cells. Exemplary eukaryotic cells may be of mammalian, insect, avian, or other animal origins.

Fusion Proteins Targeting CD4

[0103] Also provided herein are fusion proteins targeting CD4 that may be exposed on the surface on a lipid particle or viral vector. In some embodiments the fusion protein comprises an envelope glycoprotein G, H, and/or an F protein of the Paramyxoviridae family. In some embodiments, the fusion protein contains a henipavirus envelope attachment glycoprotein G (G protein) or a biologically active portion thereof and a single domain antibody (sdAb) variable domain or a single chain variable fragment (scFv). The sdAb variable domain or scFv can be linked directly or indirectly to the G protein. In particular embodiments, the sdAb variable domain or scFv is linked to the C-terminus (C-terminal amino acid) of the G protein or the biologically active portion thereof. The linkage can be via a peptide linker, such as a flexible peptide linker. Table 25 provides a list of non-limiting examples of G proteins. Exemplary full length fusion protein sequences of the disclosure are disclosed in Table 19.

[0104] In some embodiments, the G protein is of the Paramyxoviridae family. In some embodiments the G protein is a Henipavirus G protein or a biologically active portion thereof. In some embodiments, the Henipavirus G protein is a Hendra (HeV) virus G protein, a Nipah (NiV) virus G-protein (NIV-G), a Cedar (CedPV) virus G-protein, a Mojiang virus G-protein, a bat Paramyxovirus G-protein, or a biologically active portion thereof. In some embodiments, the fusion protein is glycoprotein GP64 of baculovirus, or glycoprotein GP64 variant E45K/T259A. Non-limiting examples of G proteins include those disclosed in Table 25.

[0105] In some embodiments, the attachment G proteins are type II transmembrane glycoproteins containing an N-terminal cytoplasmic tail (e.g., corresponding to amino acids 1-49 of SEQ ID NO: 9266), a transmembrane domain (e.g., corresponding to amino acids 50-70 of SEQ ID NO: 9266), and an extracellular domain containing an extracellular stalk (e.g., corresponding to amino acids 71-187 of SEQ ID NO: 9266), and a globular head (corresponding to amino acids 188-602 of SEQ ID NO: 9266). In such embodiments, the N-terminal cytoplasmic domain is within the inner lumen of the lipid bilayer and the C-terminal portion is the extracellular domain that is exposed on the outside of the lipid bilayer. Regions of the stalk in the C-terminal region (e.g. corresponding to amino acids 159-167 of NiV-G) have been shown to be involved in interactions with F protein and triggering of F protein fusion (Liu et al. 2015 J of Virology 89:1838). In wild-type G protein, the globular head mediates receptor binding to henipavirus entry receptors eprhin B2 and ephrin B3, but is dispensable for membrane fusion (Brandel-Tretheway et al. Journal of Virology. 2019. 93(13)e00577-19). In particular embodiments herein, tropism of the G protein is altered by linkage of the G protein or biologically active fragment thereof (e.g. cytoplasmic truncation) to a sdAb variable domain. Binding of the G protein to a binding partner can trigger fusion mediated by a compatible F protein or a biologically active portion thereof. G protein sequences disclosed herein are predominantly disclosed as expressed sequences including an N-terminal methionine required for start of translation. As such N-terminal methionines are commonly cleaved co- or post-translationally, the mature protein sequences for all G protein sequences disclosed herein are also contemplated as lacking the N-terminal methionine.

[0106] G glycoproteins are highly conserved among henipavirus species. For example, the G proteins of NiV and HeV viruses share 79% amino acid identity. Studies have shown a high degree of compatibility among G proteins with F proteins of different species as demonstrated by heterotypic fusion activation (Brandel-Tretheway et al. *Journal of Virology.* 2019). As described further below, a targeted lipid particle can contain heterologous G and F proteins from different species.

[0107] In some embodiments, the G protein has a sequence set forth in any of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037, or is a functionally active variant or biologically active portion thereof that has a sequence that is at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% identical to any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037. In particular embodiments, the G protein or functionally active variant or biologically active portion is a protein that retains fusogenic activity in conjunction with a Henipavirus F protein, such as an F protein (e.g. NiV-F or HeV—F). Fusogenic activity includes the activity of the G protein in conjunction with a Henipavirus F protein to promote or facilitate fusion of two membrane lumens, such as the lumen of the targeted lipid particle having embedded in its lipid bilayer a henipavirus F and G protein, and a cytoplasm of a target cell, e.g. a cell that contains a surface receptor or molecule that is recognized or bound by the targeted envelope protein. In some embodiments, the F protein and G protein are from the same Henipavirus species (e.g. NiV-G and NiV-F). In some embodiments, the F protein and G protein are from different Henipavirus species (e.g. NiV-G and HeV-F).

[0108] In particular embodiments, the G protein has the sequence of amino acids set forth in SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037, or is a functionally active variant thereof or a biologically active portion thereof that retains fusogenic activity. In some embodiments, the functionally active variant comprises an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037 and retains fusogenic activity in conjunction with a Henipavirus F protein (e.g., NiV-F or HeV-F). In some embodiments, the biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037 and retains fusogenic activity in conjunction with a Henipavirus F protein (e.g., NiV-F or HeV-F).

[0109] Reference to retaining fusogenic activity includes activity (in conjunction with a Henipavirus F protein) that is at or about 10% to at or about 150% or more of the level or degree of binding of the corresponding wild-type G protein, such as set forth in any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037, such as at least or at least about 10% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 15% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 20% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 25% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 30% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 35% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 40% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 45% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 50% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 55% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 60% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 65% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 70% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 75% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 80% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 85% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 90% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 95% of the level or degree of fusogenic activity of the corresponding wild-type G protein, such as at least or at least about 100% of the level or degree of fusogenic activity of the corresponding wild-type G protein, or such as at least or at least about 120% of the level or

degree of fusogenic activity of the corresponding wild-type G protein.
[0110] In some embodiments, the G protein is a mutant G protein that is a functionally active variant or biologically active portion containing one or more amino acid mutations, such as one or more amino acid insertions, deletions, substitutions, or truncations. In some embodiments, the mutations described herein relate to amino acid insertions, deletions, substitutions, or truncations of amino acids compared to a reference G protein sequence. In some embodiments, the reference G protein sequence is the wild-type sequence of a G protein or a biologically active portion thereof. In some embodiments, the functionally active variant or the biologically active portion thereof is a mutant of a wild-type Hendra (HeV) virus G protein, a wild-type Nipah (NiV) virus G-protein (NiV-G), a wild-type Cedar (CedPV) virus G-protein, a wild-type Mojiang virus G-protein, a wild-type bat Paramyxovirus G-protein, or biologically active portions thereof. In some embodiments, the wild-type G protein has the sequence set forth in any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037.
[0111] In some embodiments, the G protein is a mutant G protein that is a biologically active portion that is an N-terminally and/or C-terminally truncated fragment of a wild-type Hendra (HeV) virus G protein, a wild-type Nipah (NiV) virus G-protein (NiV-G), a wild-type Cedar (CedPV) virus G-protein, a wild-type Mojiang virus G-protein, or a wild-type bat Paramyxovirus G-protein. In particular embodiments, the truncation is an N-terminal truncation of all or a portion of the cytoplasmic domain. In some embodiments, the mutant G protein is a biologically active portion that is truncated and lacks up to 49 contiguous amino acid residues at or near the N-terminus of the wild-type G protein, such as a wild-type G protein set forth in any one of SEQ ID NOs: 9266, 9274, 9285-9288, 9295, 9303, 9305-9037. In some embodiments, the mutant G protein is truncated and lacks up to 49 contiguous amino acids, such as up to 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 30, 38, 37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2 or 1 contiguous amino acid(s) at the N-terminus of the wild-type G protein.
[0112] In some embodiments, the G protein is a wild-type Nipah virus G (NiV-G) protein or a wild-type Hendra virus G protein, or is a functionally active variant or biologically active portion thereof. In some embodiments, the G protein is a NiV-G protein that has the sequence set forth in SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, or is a functional variant or a biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295.
[0113] In some embodiments, the G protein is a mutant NiV-G protein that is a biologically active portion of a wild-type NiV-G. In some embodiments, the biologically active portion is an N-terminally truncated fragment. In some embodiments, the mutant NiV-G protein is truncated and lacks up to 5 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 6 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 7 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 8 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 9 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 10 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 11

contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 12 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 13 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295), up to 14 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 15 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 16 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 17 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 18 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 19 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 20 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 21 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 22 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 23 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 24 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 25 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 26 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 27 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 28 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 29 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 30 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 31 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 32 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 33 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 34 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 35 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 36 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 37 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 38 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 39 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 40 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 41 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-

G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 42 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 43 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295), up to 44 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 618, or SEQ ID NO: 628), or up to 45 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295).

[0114] In some embodiments, the NiV-G protein is a biologically active portion that does not contain a cytoplasmic domain. In some embodiments, the NiV-G protein without the cytoplasmic domain is encoded by SEQ ID NO:9289.

[0115] In some embodiments, the mutant NiV-G protein comprises a sequence set forth in any of SEQ ID NOs: 601-606, 629-634, 612, 622, or 637, or is a functional variant thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, or at least at or about 87%, at least at or about 88%, or at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NOs: 9267-9269, 9296-9301, 9277, 9289, 9304.

[0116] In some embodiments, the mutant NiV-G protein has a 5 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9267 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9267, or as set forth in SEQ ID NO:9296 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9296 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9296.

[0117] In some embodiments, the mutant NiV-G protein has a 10 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9268 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9268, or such as set forth in SEQ ID NO:9297 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9297.

[0118] In some embodiments, the mutant NiV-G protein has a 15 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9269 or a functional variant thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9269, or such as set forth in SEQ ID NO:9298 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9298.

[0119] In some embodiments, the mutant NiV-G protein has a 20 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295) such as set forth in SEQ ID NO:9270, or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9270, or such as set forth in SEQ ID NO:9299 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9299.

[0120] In some embodiments, the mutant NiV-G protein has a 25 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9271 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9271, or such as set forth in SEQ ID NO:9300 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9300.

[0121] In some embodiments, the mutant NiV-G protein has a 30 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9273 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9273, or such as set forth in SEQ ID NO:9301 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9301.

[0122] In some embodiments, the mutant NiV-G protein has a 33 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295) or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9277, or such as set forth in SEQ ID NO:9302 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9302.

[0123] In some embodiments, the mutant NiV-G protein has a 34 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295), such as set forth in SEQ ID NO:9277 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9277, or such as set forth in SEQ ID NO:9302 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9302.

[0124] In a preferred embodiment, the NiV-G protein has a 34 amino acid truncation at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295) and one or more amino acid substitutions corresponding to amino acid substitutions selected from E501A, W504A, Q530A, and E533A with reference to the numbering set forth in

SEQ ID NO:9285.

[0125] In some embodiments, the mutant NiV-G protein lacks the N-terminal cytoplasmic domain of the wild-type NiV-G protein (SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO: 9295), such as set forth in SEQ ID NO:9289 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9289.

[0126] In some embodiments, the mutant G protein is a mutant HeV-G protein that has the sequence set forth in SEQ ID NO:9275 or 9303, or is a functional variant or biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9275 or 9303.

[0127] In some embodiments, the G protein is a mutant HeV-G protein that is a biologically active portion of a wild-type HeV-G. In some embodiments, the biologically active portion is an N-terminally truncated fragment. In some embodiments, the mutant HeV-G protein is truncated and lacks up to 5 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 6 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 7 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 8 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 9 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 10 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 11 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 12 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 13 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 14 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 15 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 16 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 17 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 18 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 19 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 20 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 21 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 22 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 23 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 24 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 25 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or

9303), up to 26 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 27 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 28 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 29 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 30 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 31 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 32 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 33 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 34 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 35 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 36 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 37 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 38 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 39 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 40 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 41 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 42 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 43 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), up to 44 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303), or up to 45 contiguous amino acid residues at or near the N-terminus of the wild-type HeV-G protein (SEQ ID NO:9275 or 9303).

[0128] In some embodiments, the HeV-G protein is a biologically active portion that does not contain a cytoplasmic domain. In some embodiments, the mutant HeV-G protein lacks the N-terminal cytoplasmic domain of the wild-type HeV-G protein (SEQ ID NO: 9275 or 9303), such as set forth in SEQ ID NO:9303 or a functional variant thereof having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9303.

[0129] In some embodiments, the G protein or the functionally active variant or biologically active portion thereof binds to Ephrin B2 or Ephrin B3. In some aspects, the G protein has the sequence of amino acids set forth in any one of SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or is a functionally active variant thereof or a biologically active portion thereof that is able to bind to Ephrin B2 or Ephrin B3. In some embodiments, the functionally active variant or biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO: 9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, and retains binding to Ephrhin B2 or B3.

[0130] Reference to retaining binding to Ephrin B2 or B3 includes binding that is at least or at least about 5% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 10% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 15% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 20% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 25% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion, 30% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO: 9288, or a functionally active variant or biologically active portion thereof, 35% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 40% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 45% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 50% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 55% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 60% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, 65% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO: 9295, SEQ ID NO: 9287, or SEQ ID NO: 9288, or a functionally active variant or biologically active portion thereof, 70% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 75% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 80% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 85% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, such as at least or at least about 90% of the level or degree of binding of the corresponding wild-type G protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO: 9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof, or such as at least or at least about 95% of the level or degree of binding of the corresponding wild-type protein, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9275, SEQ ID NO:9285, SEQ ID NO:9286, SEQ ID NO:9295, SEQ ID NO:9287, or SEQ ID NO:9288, or a functionally active variant or biologically active portion thereof. In some embodiments, the G protein is NiV-G or a functionally active variant or biologically active portion thereof and binds to Ephrin B2 or Ephrin B3.

[0131] In some aspects, the NiV-G has the sequence of amino acids set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:9295, or is a functionally active variant thereof or a biologically active portion thereof that is able to bind to Ephrin B2 or Ephrin B3. In some embodiments, the functionally active variant or biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO: 9295 and retains binding to Eprhin B2 or B3. Exemplary biologically active portions include N-terminally truncated variants lacking all or a portion of the cytoplasmic domain, e.g. 1 or more, such as 1 to 49 contiguous N-terminal amino acid residues, e.g. set forth in any one of SEQ ID NOS: 9267-9272, 9289, and 9296-9301.

[0132] Reference to retaining binding to Ephrin B2 or B3 includes binding that is at least or at least about 5% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 10% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 15% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 20% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:92954, 25% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, 30% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in S SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 35% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, 40% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, 45% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 50% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 55% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 60% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 65% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, 70% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, such as at least or at least about 75% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:9295, such as at least or at least about 80% of the level or degree of binding of the corresponding wild-type NIV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO:9295, such as at least or at least about 85% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO:9285, or SEQ ID NO: 9295, such as at least or at least about 90% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO:9266, SEQ ID NO: 9285, or SEQ ID NO:9295, or such as at least or at least about 95% of the level or degree of binding of the corresponding wild-type NiV-G, such as set forth in SEQ ID NO: 9266, SEQ ID NO:9285, or SEQ ID NO:9295.

[0133] In some embodiments, the G protein is HeV-G or a functionally active variant or biologically active portion thereof and binds to Ephrin B2 or Ephrin B3. In some aspects, the HeV-G has the sequence of amino acids set forth in SEQ ID NO:9275 or 9303, or is a functionally active variant thereof or a biologically active portion thereof that is able to bind to Ephrin B2 or Ephrin B3. In some embodiments, the functionally active variant or biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9275 or 9303 and retains binding to Eprhin B2 or B3. Exemplary biologically active portions include N-terminally truncated variants lacking all or a portion of the cytoplasmic domain, e.g. 1 or more, such as 1 to 49 contiguous N-terminal amino acid residues, e.g. set forth in any one of SEQ ID NO:9290.

[0134] Reference to retaining binding to Ephrin B2 or B3 includes binding that is at least or at least about 5% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 10% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 15% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 20% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 25% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 30% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 35% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 40% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 45% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 50% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 55% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 60% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 65% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, 70% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 75% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 80% of the level or degree of binding of the corresponding wild-type NIV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 85% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, such as at least or at least about 90% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or 9303, or such as at least or at least about 95% of the level or degree of binding of the corresponding wild-type HeV-G, such as set forth in SEQ ID NO:9275 or

9303.

[0135] In some embodiments, the G protein or the biologically thereof is a mutant G protein that exhibits reduced binding for the native binding partner of a wild-type G protein. In some embodiments, the mutant G protein or the biologically active portion thereof is a mutant of wild-type Niv-G and exhibits reduced binding to one or both of the native binding partners Ephrin B2 or Ephrin B3. In some embodiments, the mutant G-protein or the biologically active portion, such as a mutant NiV-G protein, exhibits reduced binding to the native binding partner. In some embodiments, the reduced binding to Ephrin B2 or Ephrin B3 is reduced by greater than at or about 5%, at or about 10%, at or about 15%, at or about 20%, at or about 25%, at or about 30%, at or about 40%, at or about 50%, at or about 60%, at or about 70%, at or about 80%, at or about 90%, or at or about 100%.

[0136] In some embodiments, the mutations described herein improve transduction efficiency. In some embodiments, the mutations described herein allow for specific targeting of other desired cell types that are not Ephrin B2 or Ephrin B3. In some embodiments, the mutations described herein result in at least the partial inability to bind at least one natural receptor, such as to reduce the binding to at least one of Ephrin B2 or Ephrin B3. In some embodiments, the mutations described herein interfere with natural receptor recognition.

[0137] In some embodiments, the mutant NiV-G protein or the biologically active portion thereof is truncated and lacks up to 5 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 6 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 7 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 8 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 9 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 10 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 11 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 12 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 13 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 14 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 15 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 16 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 17 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 18 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 19 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 20 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 21 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 22 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 23 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 24 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 25 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 26 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 27 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 28 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 29 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 30 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 31 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 32 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 33 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 34 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 35 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 36 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), 37 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 38 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285), 39 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO: 9285), or 40 contiguous amino acid residues at or near the N-terminus of the wild-type NiV-G protein (SEQ ID NO:9285).

[0138] In some embodiments, the G protein contains one or more amino acid substitutions in a residue that is involved in the interaction with one or both of Ephrin B2 and Ephrin B3. In some embodiments, the amino acid substitutions correspond to mutations E501A, W504A, Q530A, and E533A with reference to numbering set forth in SEQ ID NO:9285.

[0139] In some embodiments, the G protein is a mutant G protein containing one or more amino acid substitutions selected from the group consisting of E501A, W504A, Q530A, and E533A with reference to numbering set forth in SEQ ID NO:9285. In some embodiments, the G protein is a mutant G protein that contains one or more amino acid substitutions selected from the group consisting of E501A, W504A, Q530A, and E533A with reference to SEQ ID NO:9285 or a biologically active portion thereof containing an N-terminal truncation. In some embodiments, the G protein is a mutant G protein that contains one or more amino acid substitutions selected from the group consisting of E501A, W504A, Q530A, and E533A in combination with any one of the N-terminal truncations disclosed above with reference to SEQ ID NO:9285 or a biologically active portion thereof. In some embodiments, any of the mutant G proteins described above contains one, two, three, or all four amino acid selected from the group consisting of E501A, W504A, Q530A, and E533A with reference to numbering set forth in SEQ ID NO:9285, in all pairwise and triple combinations thereof.

[0140] In some embodiments, the mutant NiV-G protein has the amino acid sequence set forth in SEQ ID NO:9273 or 9302 or an amino acid sequence having at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9273 or 9302. In particular embodiments, the G protein has the sequence of amino acids set forth in SEQ ID NO:9273 or 9302.

[0141] In some embodiments, the targeted envelope protein contains a G protein or a functionally active variant or biologically active portion thereof and an sdAb variable domain, in which the targeted envelope protein exhibits increased binding for another molecule that is different from the native binding partner of a wild-type G protein. In some embodiments, the other molecule is a protein expressed on the surface of desired target cell. In some embodiments, the increased binding to the other molecule is increased by greater than at or about 25%, at or about 30%, at or about 40%, at or about 50%, at or about 60%, at or about 70%, at or about 80%, at or about 90%, or at or about 100%. In particular embodiments, the binding confers re-targeted binding compared to the binding of a wild-type G protein in which a new or different binding activity is conferred.

[0142] In some embodiments, the C-terminus of the single domain antibody is attached to the C-terminus of the G protein or biologically active portion thereof. In some embodiments, the N-terminus of the single domain antibody is exposed on the exterior surface of the lipid bilayer. In some embodiments, the N-terminus of the single domain antibody binds to a cell surface molecule of a target cell. In some embodiments, the single domain antibody specifically binds to a cell surface molecule present on a target cell. In some embodiments, the cell surface molecule is a protein, glycan, lipid, or low molecular weight molecule.

[0143] In some embodiments, the cell surface molecule of a target cell is an antigen or portion thereof. In some embodiments, the single domain antibody or portion thereof is an antibody having a single monomeric domain antigen binding/recognition domain that is able to bind selectively to a specific antigen. In some embodiments, the single domain antibody binds an antigen present on a target cell.
[0144] Exemplary cells include immune effector cells, peripheral blood mononuclear cells (PBMCs) such as lymphocytes (T cells, B cells, natural killer cells) and monocytes, granulocytes (neutrophils, basophils, eosinophils), macrophages, dendritic cells, cytotoxic T lymphocytes, polymorphonuclear cells (also known as PMNs, PMLs, or PMNLs), stem cells, embryonic stem cells (ESs or ECSs), neural stem cells, mesenchymal stem cells (MSCs), hematopoietic stem cells (HSCs), human myogenic stem cells, muscle-derived stem cells (MuStem), limbal epithelial stem cells, cardio-myogenic stem cells, cardiomyocytes, progenitor cells, allogenic cells, resident cardiac cells, induced pluripotent stem cells (iPSs or iPSCs), adipose-derived or phenotypic modified stem or progenitor cells, CD133+ cells, aldehyde dehydrogenase-positive cells (ALDH+), umbilical cord blood (UCB) cells, peripheral blood stem cells (PBSCs), neurons, neural progenitor cells, pancreatic beta cells, glial cells, or hepatocytes.
[0145] In some embodiments, the target cell is a cell of a target tissue. The target tissue can include liver, lungs, heart, spleen, pancreas, gastrointestinal tract, kidney, testes, ovaries, brain, reproductive organs, central nervous system, peripheral nervous system, skeletal muscle, endothelium, inner ear, or eye.
[0146] In some embodiments, the target cell is a muscle cell (e.g., skeletal muscle cell), kidney cell, liver cell (e.g. hepatocyte), or a cardiac cell (e.g. cardiomyocyte). In some embodiments, the target cell is a cardiac cell, e.g., a cardiomyocyte (e.g., a quiescent cardiomyocyte), a hepatoblast (e.g., a bile duct hepatoblast), an epithelial cell, a T cell (e.g. a naive T cell), a macrophage (e.g., a tumor infiltrating macrophage), or a fibroblast (e.g., a cardiac fibroblast).
[0147] In some embodiments, the target cell is a tumor-infiltrating lymphocyte, a T cell, a neoplastic or tumor cell, a virus-infected cell, a stem cell, a central nervous system (CNS) cell, a hematopoietic stem cell (HSC), a liver cell or a fully differentiated cell. In some embodiments, the target cell is a CD3+ T cell, a CD4+ T cell, a CD8+ T cell, a hepatocyte, a hematopoietic stem cell, a CD34+ hematopoietic stem cell, a CD105+ hematopoietic stem cell, a CD117+ hematopoietic stem cell, a CD105+ endothelial cell, a B cell, a CD20+ B cell, a CD19+ B cell, a cancer cell, a CD133+ cancer cell, an EpCAM+ cancer cell, a CD19+ cancer cell, a Her2/Neu+ cancer cell, a GluA2+ neuron, a GluA4+ neuron, a NKG2D+ natural killer cell, a SLC1A3+ astrocyte, a SLC7A10+ adipocyte, or a CD30+ lung epithelial cell.
[0148] In some embodiments, the target cell is an antigen presenting cell, an MHC class II+ cell, a professional antigen presenting cell, an atypical antigen presenting cell, a macrophage, a dendritic cell, a myeloid dendritic cell, a plasmacytoid dendritic cell, a CD11c+ cell, a CD11b+ cell, a splenocyte, a B cell, a hepatocyte, a endothelial cell, or a non-cancerous cell. In some embodiments, the cell surface molecule is any one of CD4.
[0149] In some embodiments, the G protein or functionally active variant or biologically active portion thereof is linked directly to the sdAb variable domain (e.g., a VHH) or scFv. In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N′-single domain antibody-C′)-(C′-G protein-N′). In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N′-scFv-C′)-(C′-G protein-N′).
[0150] In some embodiments, the G protein or functionally active variant or biologically active portion thereof is linked indirectly via a linker to the sdAb variable domain or scFv. In some embodiments, the linker is a peptide linker, such as a polypeptide linker. In some embodiments, the linker is a chemical linker.
[0151] In some embodiments, the linker is a peptide linker and the targeted envelope protein is a fusion protein containing the G protein or functionally active variant or biologically active portion

thereof linked via a peptide linker to the sdAb variable domain or svFv. In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N′-single domain antibody-C′)-Linker-(C′-G protein-N′). In some embodiments, the targeted envelope protein is a fusion protein that has the following structure: (N′-scFv-C′)-Linker-(C′-G protein-N′). In some embodiments, the peptide linker is a polypeptide linker up to 65 amino acids in length. In some embodiments, the peptide linker comprises from or from about 2 to 65 amino acids, 2 to 60 amino acids, 2 to 56 amino acids, 2 to 52 amino acids, 2 to 48 amino acids, 2 to 44 amino acids, 2 to 40 amino acids, 2 to 36 amino acids, 2 to 32 amino acids, 2 to 28 amino acids, 2 to 24 amino acids, 2 to 20 amino acids, 2 to 18 amino acids, 2 to 14 amino acids, 2 to 12 amino acids, 2 to 10 amino acids, 2 to 8 amino acids, 2 to 6 amino acids, 6 to 65 amino acids, 6 to 60 amino acids, 6 to 56 amino acids, 6 to 52 amino acids, 6 to 48 amino acids, 6 to 44 amino acids, 6 to 40 amino acids, 6 to 36 amino acids, 6 to 32 amino acids, 6 to 28 amino acids, 6 to 24 amino acids, 6 to 20 amino acids, 6 to 18 amino acids, 6 to 14 amino acids, 6 to 12 amino acids, 6 to 10 amino acids, 6 to 8 amino acids, 8 to 65 amino acids, 8 to 60 amino acids, 8 to 56 amino acids, 8 to 52 amino acids, 8 to 48 amino acids, 8 to 44 amino acids, 8 to 40 amino acids, 8 to 36 amino acids, 8 to 32 amino acids, 8 to 28 amino acids, 8 to 24 amino acids, 8 to 20 amino acids, 8 to 18 amino acids, 8 to 14 amino acids, 8 to 12 amino acids, 8 to 10 amino acids, 10 to 65 amino acids, 10 to 60 amino acids, 10 to 56 amino acids, 10 to 52 amino acids, 10 to 48 amino acids, 10 to 44 amino acids, 10 to 40 amino acids, 10 to 36 amino acids, 10 to 32 amino acids, 10 to 28 amino acids, 10 to 24 amino acids, 10 to 20 amino acids, 10 to 18 amino acids, 10 to 14 amino acids, 10 to 12 amino acids, 12 to 65 amino acids, 12 to 60 amino acids, 12 to 56 amino acids, 12 to 52 amino acids, 12 to 48 amino acids, 12 to 44 amino acids, 12 to 40 amino acids, 12 to 36 amino acids, 12 to 32 amino acids, 12 to 28 amino acids, 12 to 24 amino acids, 12 to 20 amino acids, 12 to 18 amino acids, 12 to 14 amino acids, 14 to 65 amino acids, 14 to 60 amino acids, 14 to 56 amino acids, 14 to 52 amino acids, 14 to 48 amino acids, 14 to 44 amino acids, 14 to 40 amino acids, 14 to 36 amino acids, 14 to 32 amino acids, 14 to 28 amino acids, 14 to 24 amino acids, 14 to 20 amino acids, 14 to 18 amino acids, 18 to 65 amino acids, 18 to 60 amino acids, 18 to 56 amino acids, 18 to 52 amino acids, 18 to 48 amino acids, 18 to 44 amino acids, 18 to 40 amino acids, 18 to 36 amino acids, 18 to 32 amino acids, 18 to 28 amino acids, 18 to 24 amino acids, 18 to 20 amino acids, 20 to 65 amino acids, 20 to 60 amino acids, 20 to 56 amino acids, 20 to 52 amino acids, 20 to 48 amino acids, 20 to 44 amino acids, 20 to 40 amino acids, 20 to 36 amino acids, 20 to 32 amino acids, 20 to 28 amino acids, 20 to 26 amino acids, 20 to 24 amino acids, 24 to 65 amino acids, 24 to 60 amino acids, 24 to 56 amino acids, 24 to 52 amino acids, 24 to 48 amino acids, 24 to 44 amino acids, 24 to 40 amino acids, 24 to 36 amino acids, 24 to 32 amino acids, 24 to 30 amino acids, 24 to 28 amino acids, 28 to 65 amino acids, 28 to 60 amino acids, 28 to 56 amino acids, 28 to 52 amino acids, 28 to 48 amino acids, 28 to 44 amino acids, 28 to 40 amino acids, 28 to 36 amino acids, 28 to 34 amino acids, 28 to 32 amino acids, 32 to 65 amino acids, 32 to 60 amino acids, 32 to 56 amino acids, 32 to 52 amino acids, 32 to 48 amino acids, 32 to 44 amino acids, 32 to 40 amino acids, 32 to 38 amino acids, 32 to 36 amino acids, 36 to 65 amino acids, 36 to 60 amino acids, 36 to 56 amino acids, 36 to 52 amino acids, 36 to 48 amino acids, 36 to 44 amino acids, 36 to 40 amino acids, 40 to 65 amino acids, 40 to 60 amino acids, 40 to 56 amino acids, 40 to 52 amino acids, 40 to 48 amino acids, 40 to 44 amino acids, 44 to 65 amino acids, 44 to 60 amino acids, 44 to 56 amino acids, 44 to 52 amino acids, 44 to 48 amino acids, 48 to 65 amino acids, 48 to 60 amino acids, 48 to 56 amino acids, 48 to 52 amino acids, 50 to 65 amino acids, 50 to 60 amino acids, 50 to 56 amino acids, 50 to 52 amino acids, 54 to 65 amino acids, 54 to 60 amino acids, 54 to 56 amino acids, 58 to 65 amino acids, 58 to 60 amino acids, or 60 to 65 amino acids. In some embodiments, the peptide linker is a peptide that is 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, or 65 amino acids in length.

[0152] In particular embodiments, the linker is a flexible peptide linker. In some such embodiments, the linker is 1-20 amino acids, such as 1-20 amino acids predominantly composed of glycine. In some embodiments, the linker is 1-20 amino acids, such as 1-20 amino acids predominantly composed of glycine and serine. In some embodiments, the linker is a flexible peptide linker containing amino acids Glycine and Serine, referred to as GS-linkers. In some embodiments, the peptide linker includes the sequences GS, GGS, GGGGS (SEQ ID NO:9294), GGGGGS (SEQ ID NO:9292) or combinations thereof. In some embodiments, the peptide linker is a polypeptide linker that has the sequence (GGS) n (SEQ ID NO: 14126), wherein n is 1 to 10. In some embodiments, the peptide linker is a polypeptide linker that has the sequence (GGGGS) n, (SEQ ID NO:9293) wherein n is 1 to 10. In some embodiments, the peptide linker is a polypeptide linker that has the sequence (GGGGGS) n (SEQ ID NO:9284), wherein n is 1 to 6.

[0153] Also provided herein are polynucleotides comprising a nucleic acid sequence encoding a targeted envelope protein. In some embodiments, the polynucleotides comprise a nucleic acid sequence encoding a G protein or biologically active portion thereof. In some embodiments, the polynucleotides further comprise a nucleic acid sequence encoding a single domain antibody (sdAb) variable domain or scFv or biologically active portion thereof. The polynucleotides may include a sequence of nucleotides encoding any of the targeted envelope proteins described above. The polynucleotide can be a synthetic nucleic acid. Also provided are expression vectors containing any of the provided polynucleotides.

[0154] In some of any embodiments, expression of natural or synthetic nucleic acids is typically achieved by operably linking a nucleic acid encoding the gene of interest to a promoter and incorporating the construct into an expression vector. In some embodiments, vectors are suitable for replication and integration in eukaryotes. In some embodiments, cloning vectors contain transcription and translation terminators, initiation sequences, and promoters useful for expression of the desired nucleic acid sequence. In some of any embodiments described herein, a plasmid comprises a promoter suitable for expression in a cell.

[0155] In some embodiments, the polynucleotides contain at least one promoter that is operatively linked to control expression of the targeted envelope protein containing the G protein and the single domain antibody (sdAb) variable domain or scFv. For expression of the targeted envelope protein, at least one element in each promoter functions to position the start site for RNA synthesis. The best-known example of this is the TATA box, but in some promoters lacking a TATA box, such as the promoter for the mammalian terminal deoxynucleotidyl transferase gene and the promoter for the SV40 genes, a discrete element overlying the start site itself helps to fix the place of initiation.

[0156] In some embodiments, additional promoter elements, e.g., enhancers, regulate the frequency of transcriptional initiation. In some embodiments, additional promoter elements are located in the region 30-110 bp upstream of the start site, although a number of promoters have been shown to contain functional elements downstream of the start site as well. In some embodiments, spacing between promoter elements frequently is flexible, so that promoter function is preserved when elements are inverted or moved relative to one another. In some embodiments, such as with the thymidine kinase (tk) promoter, the spacing between promoter elements is increased to 50 bp apart before activity begins to decline. In some embodiments, depending on the promoter, individual elements function either cooperatively or independently to activate transcription.

[0157] A promoter may be one naturally associated with a gene or polynucleotide sequence, as may be obtained by isolating the 5′ non-coding sequences located upstream of the coding segment and/or exon. Such a promoter can be referred to as “endogenous.” Similarly, an enhancer may be one naturally associated with a polynucleotide sequence, located either downstream or upstream of that sequence. Alternatively, certain advantages will be gained by positioning the coding polynucleotide segment under the control of a recombinant or heterologous promoter, which refers to a promoter that is not normally associated with a polynucleotide sequence in its natural environment. A recombinant or heterologous enhancer refers also to an enhancer not normally associated with a polynucleotide sequence in its natural environment. Such promoters or enhancers

may include promoters or enhancers of other genes, and promoters or enhancers isolated from any other prokaryotic, viral, or eukaryotic cell, and promoters or enhancers not “naturally occurring,” i.e., containing different elements of different transcriptional regulatory regions, and/or mutations that alter expression. In addition to producing nucleic acid sequences of promoters and enhancers synthetically, sequences may be produced using recombinant cloning and/or nucleic acid amplification technology, including PCR, in connection with the compositions disclosed herein.
[0158] In some embodiments, a suitable promoter is the immediate early cytomegalovirus (CMV) promoter sequence. In some embodiments, the promoter sequence is a strong constitutive promoter sequence capable of driving high levels of expression of any polynucleotide sequence operatively linked thereto. In some embodiments, a suitable promoter is Elongation Growth Factor-la (EF-I a). In some embodiments, other constitutive promoter sequences are also used, including, but not limited to the simian virus 40 (SV40) early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, MoMuLV promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the myosin promoter, the hemoglobin promoter, and the creatine kinase promoter.
[0159] In some embodiments, the promoter is an inducible promoter. In some embodiments, the inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence to which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. In some embodiments, inducible promoters comprise a metallothionine promoter, a glucocorticoid promoter, a progesterone promoter, and a tetracycline promoter.
[0160] In some embodiments, exogenously controlled inducible promoters are used to regulate expression of the G protein and single domain antibody (sdAb) variable domain or scFv. For example, radiation-inducible promoters, heat-inducible promoters, and/or drug-inducible promoters can be used to selectively drive transgene expression in, for example, targeted regions. In such embodiments, the location, duration, and level of transgene expression is regulated by the administration of the exogenous source of induction.
[0161] In some embodiments, expression of the targeted envelope protein containing a G protein and single domain antibody (sdAb) variable domain or scFv is regulated using a drug-inducible promoter. For example, in some cases, the promoter, enhancer, or transactivator comprises a Lac operator sequence, a tetracycline operator sequence, a galactose operator sequence, a doxycycline operator sequence, a rapamycin operator sequence, a tamoxifen operator sequence, or a hormone-responsive operator sequence, or an analog thereof. In some instances, the inducible promoter comprises a tetracycline response element (TRE). In some embodiments, the inducible promoter comprises an estrogen response element (ERE), which can activate gene expression in the presence of tamoxifen. In some instances, a drug-inducible element, such as a TRE, is combined with a selected promoter to enhance transcription in the presence of drug, such as doxycycline. In some embodiments, the drug-inducible promoter is a small molecule-inducible promoter.
[0162] Any of the provided polynucleotides can be modified to remove CpG motifs and/or to optimize codons for translation in a particular species, such as human, canine, feline, equine, ovine, bovine, etc. species. In some embodiments, the polynucleotides are optimized for human codon usage (i.e., human codon-optimized). In some embodiments, the polynucleotides are modified to remove CpG motifs. In other embodiments, the provided polynucleotides are modified to remove CpG motifs and are codon-optimized, such as human codon-optimized. Methods of codon optimization and CpG motif detection and modification are well-known. Typically, polynucleotide optimization enhances transgene expression, increases transgene stability and preserves the amino acid sequence of the encoded polypeptide.
[0163] In order to assess the expression of the targeted envelope protein, the expression vector to be introduced into a cell can also contain either a selectable marker gene or a reporter gene or both

to facilitate identification and selection of expressing particles, e.g. viral particles. In other embodiments, the selectable marker is carried on a separate piece of DNA and used in a co-transfection procedure. Both selectable markers and reporter genes may be flanked with appropriate regulatory sequences to enable expression in the host cells. Useful selectable markers are known in the art and include, for example, antibiotic-resistance genes, such as neo and the like.

[0164] Reporter genes are used for identifying potentially transfected cells and for evaluating the functionality of regulatory sequences. Reporter genes that encode for easily assayable proteins are well known in the art. In general, a reporter gene is a gene that is not present in or expressed by the recipient organism or tissue and that encodes a protein whose expression is manifested by some easily detectable property, e.g., enzymatic activity. Expression of the reporter gene is assayed at a suitable time after the DNA has been introduced into the recipient cells.

[0165] Suitable reporter genes may include genes encoding luciferase, beta-galactosidase, chloramphenicol acetyl transferase, secreted alkaline phosphatase, or the green fluorescent protein gene (see, e.g., Ui-Tei et al., 2000, FEBS Lett. 479:79-82). Suitable expression systems are well known and may be prepared using well known techniques or obtained commercially. Internal deletion constructs may be generated using unique internal restriction sites or by partial digestion of non-unique restriction sites. Constructs may then be transfected into cells that display high levels of the desired polynucleotide and/or polypeptide expression. In general, the construct with the minimal 5′ flanking region showing the highest level of expression of reporter gene is identified as the promoter. Such promoter regions may be linked to a reporter gene and used to evaluate agents for the ability to modulate promoter-driven transcription.

Lipid Particles Targeting CD4

[0166] Also provided herein are targeted lipid particles (e.g. targeting CD4), such as targeted viral vectors, that comprise a F protein molecule or biologically active portion thereof of the Paramyxoviridae family, and a fusion protein comprising (i) an envelope attachment glycoprotein G (G protein), hemagglutinin (H Protein), or hemagglutinin-neuraminidase (HN Protein), or a biologically active portion thereof of the Paramyxoviridae family, and (ii) a single domain antibody (sdAb) variable domain or scFv, wherein the single domain antibody variable domain or scFv is attached to the C-terminus of the G protein or the biologically active portion, wherein each is exposed on the outer surface of the targeted lipid particle. In particular embodiments, the provided targeted lipid particles exhibit fusogenic activity, which is mediated by the targeted envelope protein that facilitates binding to a target cell and contains the G protein or biologically active portion thereof, and the F protein or biologically active portion thereof that is involved in facilitating the merger or fusion of the two lumens of the lipid particle and the target cell membranes. Table 25 provides non-limiting examples of G and F proteins for use in the targeted lipid particles of the disclosure.

[0167] In some embodiments, the targeted lipid particle provided herein (e.g. targeted lentiviral vector) has increased or greater expression of the targeted envelope protein compared to a reference lipid particle (e.g. reference lentiviral vector) that incorporates a similar envelope protein but that is fused to an alternative targeting moiety other than a sdAb variable domain, such as a single chain variable fragment (scFv). In some embodiments, the targeted lipid particles are produced by pseudotyping of viral vectors (e.g lentiviral particles) following co-transfection of the packaging cells with the transfer, envelope, and gag-pol plasmids.

[0168] In some embodiments, the expression is increased by at or greater than 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 125%, 150%, 200%, 300%, 400%, 500% or more, compared to a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference lipid particle containing a similar envelope protein but that is fused to an scFv. In some examples, the expression is increased by at or greater than 1.5-fold, 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, 15-fold, 20-fold, 30-fold or more, compared to a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference lipid particle containing a similar envelope protein but that is fused to an scFv. In some embodiments, expression is assayed in vitro using flow cytometry, e.g. FACs. In some embodiments, expression can be depicted as the number or density of targeted envelope protein on the surface of a targeted lipid particle (e.g. targeted lentiviral vector). In some embodiments, expression is depicted as the mean fluorescent intensity (MFI) of surface expression of the targeted envelope protein on the surface of a targeted lipid particle (e.g. targeted lentiviral vector). In some embodiments, expression is depicted as the percent of lipid particle (e.g. lentiviral vectors) in a population that are surface positive for the targeted envelope protein.

[0169] In some embodiments, in a population of targeted lipid particles (e.g. targeted lentiviral vectors) greater than at or about 50% of the lipid particles are surface positive for the targeted envelope protein. For example, in a population of provided targeted lipid particle (e.g. targeted lentiviral vectors) greater than at or about 55%, greater than at or about 60%, greater than at or about 65%, greater than at or about 70%, or greater than at or about 75% of the viral vectors in the population are surface positive for the targeted envelope protein.

[0170] In some embodiments, titer of the targeted lipid particles following introduction into target cells, such as by transduction (e.g. transduced cells), is increased compared to titer into the same target cells of reference lipid particles (e.g. reference lentiviral vector) that incorporate a similar envelope protein but fused to an alternative targeting moiety other than a sdAb variable domain, such as a single chain variable fragment (scFv). Typically, the alternative targeting moiety recognizes or binds the same target molecule as the sdAb variable domain of the targeted envelope protein of the targeted lipid particles. In some embodiments, the titer is increased by at or greater than 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 125%, 150%, 200%, 300%, 400%, 500% or more, compared to titer of a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference lipid particle containing a similar envelope protein but that is fused to an scFv. In some examples, the titer is increased by at or greater than 1.5-fold, 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, 15-fold, 20-fold, 30-fold or more, compared to the titer of a reference lipid particle (e.g. reference lentiviral vector), e.g. a reference viral vector containing a similar envelope protein but that is fused to an scFv. In some embodiments, the titer of the targeted lipid particles in target cells (e.g. transduced cells) is greater than at or about 1×10.sup.6 transduction units (TU)/mL. For example, the titer of the targeted lipid particles in target cells (e.g. transduced cells) is greater than at or about 2×10.sup.6 TU/mL, greater than at or about 3×10.sup.6 TU/mL, greater than at or about 4×10.sup.6 TU/mL, greater than at or about 5×10.sup.6 TU/mL, greater than at or about 6×10.sup.6 TU/mL, greater than at or about 7×10.sup.6 TU/mL, greater than at or about 8×10.sup.6 TU/mL, greater than at or about 9×10.sup.6 TU/mL, or greater than at or about 1×10.sup.7 TU/mL.

A. F Proteins

[0171] In some embodiments, the targeted lipid particle comprises one or more fusogens, e.g. F proteins of the Paramyxoviridae family. In some embodiments, the targeted lipid particle contains an exogenous or overexpressed fusogen. In some embodiments, the fusogen is disposed in the lipid bilayer. In some embodiments, the fusogen facilitates the fusion of the targeted particle's lipid bilayer to a membrane. In some embodiments, the membrane is a plasma cell membrane.

[0172] In some embodiments, fusogens comprise protein based, lipid based, and chemical based fusogens. In some embodiments, the targeted lipid particle comprises a first fusogen comprising a protein fusogen and a second fusogen comprising a lipid fusogen or chemical fusogen. In some embodiments, the fusogen binds a fusogen binding partner on a target cell surface.

[0173] In some embodiments, the fusogen comprises a protein with a hydrophobic fusion polypeptide domain. In some embodiments the fusogen comprises an F protein of the Paramyxoviridae family. In some embodiments the fusogen contains a Nipah virus protein F, a measles virus F protein, a tupaia paramyxovirus F protein, a paramyxovirus F protein, a Hendra virus F protein, a Henipavirus F protein, a Morbilivirus F protein, a respirovirus F protein, a Sendai virus F protein, a rubulavirus F protein, or an avulavirus F protein, or a biologically active portion

thereof.
[0174] In some embodiments, the fusion protein is a hemagglutinin-neuraminidase (HN) of the Paramyxoviridae family and/or F protein of the Paramyxoviridae family. In some embodiments, the respiratory paramyxovirus is a Sendai virus. The HN and F glycoproteins of Sendai viruses function to attach to sialic acids via the HN protein, and to mediate cell fusion for entry to cells via the F protein. In some embodiments, the fusion protein is a F and/or HN protein from the murine parainfluenza virus type 1 (See e.g., U.S. Pat. No. 10,704,061).
[0175] In some embodiments, the N-terminal hydrophobic fusion polypeptide domain of the F protein molecule or biologically active portion thereof is exposed on the outside of a lipid bilayer.
[0176] F proteins of henipaviruses are encoded as F.sub.0 precursors containing a signal polypeptide (e.g. corresponding to amino acid residues 1-26 of SEQ ID NO: 592). Following cleavage of the signal polypeptide, the mature F.sub.0 (e.g. SEQ ID NO: 593) is transported to the cell surface, then endocytosed and cleaved by cathepsin L (e.g. between amino acids 109-110 of SEQ ID NO: 592) into the mature fusogenic subunits F1 (e.g. corresponding to amino acids 110-546 of SEQ ID NO:9258; set forth in SEQ ID NO:9261) and F2 (e.g. corresponding to amino acid residues 27-109 of SEQ ID NO:1; set forth in SEQ ID NO:9260). The F1 and F2 subunits are associated by a disulfide bond and recycled back to the cell surface. The F1 subunit contains the fusion polypeptide domain located at the N terminus of the F1 subunit (e.g., corresponding to amino acids 110-129 of SEQ ID NO:9258) where it is able to insert into a cell membrane to drive fusion. In some cases, fusion activity is blocked by association of the F protein with G protein, until G engages with a target molecule resulting in its disassociation from F and exposure of the fusion polypeptide to mediate membrane fusion.
[0177] Among different henipavirus species, the sequence and activity of the F protein is highly conserved. For example, the F protein of NiV and HeV viruses share 89% amino acid sequence identity. Further, in some cases, the henipavirus F proteins exhibit compatibility with G proteins from other species to trigger fusion (Brandel-Tretheway et al. Journal of Virology. 2019. 93(13):e00577-19). In some aspects of the provided targeted lipid particle, the F protein is heterologous to the G protein, i.e., the F and G protein or biologically active portions thereof are from different henipavirus species. For example, the F protein is from Hendra virus and the G protein is from Nipah virus. In other aspects, the F protein can be a chimeric F protein containing regions of F proteins from different species of Henipavirus. In some embodiments, switching a region of amino acid residues of the F protein from one species of Henipavirus to another can result in fusion to the G protein of the species comprising the amino acid insertion. (Brandel-Tretheway et al. 2019). In some cases, the chimeric F protein contains an extracellular domain from one henipavirus species and a transmembrane and/or cytoplasmic domain from a different henipavirus species. For example, the F protein may contain an extracellular domain of Hendra virus and a transmembrane/cytoplasmic domain of Nipah virus. F protein sequences disclosed herein are predominantly disclosed as expressed sequences including an N-terminal signal sequence. Such N-terminal signal sequences are commonly cleaved co- or post-translationally, thus the mature protein sequences for all F protein sequences disclosed herein are also contemplated as lacking the N-terminal signal sequence.
[0178] In some embodiments, the F protein is encoded by a polynucleotide sequence that encodes the sequence set forth by any one of SEQ ID NOs: 592, 593, 608, 614-616, or 641-644, or is a functionally active variant or a biologically active portion thereof that has a sequence that is at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% identical to any one of SEQ ID NOS: 592, 593, 608, 614-616, or 641-644. In particular embodiments, the F protein or the functionally active variant or biologically active portion thereof retains fusogenic activity in conjunction with a Henipavirus G protein, such as a G protein set forth

herein. Fusogenic activity includes the activity of the F protein in conjunction with a Henipavirus G protein to promote or facilitate fusion of two membrane lumens, such as the lumen of the targeted lipid particle having embedded in its lipid bilayer a henipavirus F and G protein, and a cytoplasm of a target cell, e.g., a cell that contains a surface receptor or molecule that is recognized or bound by the targeted envelope protein. In some embodiments, the F protein and G protein are from the same Henipavirus species (e.g. NiV-G and NiV-F). In some embodiments, the F protein and G protein are from different Henipavirus species (e.g., NiV-G and HeV-F). In particular embodiments, the F protein of the functionally active variant or biologically active portion retains the cleavage site cleaved by cathepsin L (e.g., corresponding to the cleavage site between amino acids 109-110 of SEQ ID NO:9258).

[0179] In particular embodiments, the F protein has the sequence of amino acids set forth in SEQ ID NO:9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO: 9282, SEQ ID NO:9283, SEQ ID NO:9308, SEQ ID NO:9309, SEQ ID NO:9310, or SEQ ID NO:9311 or is a functionally active variant thereof or a biologically active portion thereof that retains fusogenic activity. In some embodiments, the functionally active variant comprises an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO:9282, SEQ ID NO:9283, SEQ ID NO: 9308, SEQ ID NO: 9309, SEQ ID NO:9310, or SEQ ID NO:9311 and retains fusogenic activity in conjunction with a Henipavirus G protein (e.g., NiV-G or HeV-G). In some embodiments, the biologically active portion has an amino acid sequence having at least at or about 80%, at least at or about 85%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO:9282, SEQ ID NO:9283, SEQ ID NO:9308, SEQ ID NO:9309, SEQ ID NO:9310, or SEQ ID NO: 9311 and retains fusogenic activity in conjunction with a Henipavirus G protein (e.g., NiV-G or HeV-G).

[0180] Reference to retaining fusogenic activity includes activity (in conjunction with a Henipavirus G protein) that is at or about 10% to at or about 150% or more of the level or degree of binding of the corresponding wild-type F protein, such as set forth in SEQ ID NO:9258, SEQ ID NO:9259, SEQ ID NO:9274, SEQ ID NO:9281, SEQ ID NO: 9282, SEQ ID NO: 9283, SEQ ID NO:9308, SEQ ID NO:9309, SEQ ID NO:9310, or SEQ ID NO:9311, such as at least or at least about 10% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 15% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 20% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 25% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 30% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 35% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 40% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 45% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 50% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 55% of the level or degree of fusogenic activity of the corresponding wild-type f protein, such as at least or at least about 60% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 65% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 70% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 75% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 80% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 85% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 90% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 95% of the level or degree of fusogenic activity of the corresponding wild-type F protein, such as at least or at least about 100% of the level or degree of fusogenic activity of the corresponding wild-type F protein, or such as at least or at least about 120% of the level or degree of fusogenic activity of the corresponding wild-type F protein.

[0181] In some embodiments, the F protein is a mutant F protein that is a functionally active fragment or a biologically active portion containing one or more amino acid mutations, such as one or more amino acid insertions, deletions, substitutions, or truncations. In some embodiments, the mutations described herein relate to amino acid insertions, deletions, substitutions, or truncations of amino acids compared to a reference F protein sequence. In some embodiments, the reference F protein sequence is the wild-type sequence of an F protein or a biologically active portion thereof. In some embodiments, the mutant F protein or the biologically active portion thereof is a mutant of a wild-type Hendra (Hev) virus F protein, a Nipah (NiV) virus F-protein, a Cedar (CedPV) virus F protein, a Mojiang virus F protein, or a bat Paramyxovirus F protein. In some embodiments, the wild-type F protein is encoded by a sequence of nucleotides that encodes any one of SEQ ID NO: 592, 593, 608, 614-616, or 641-644.

[0182] In some embodiments, the mutant F protein is a biologically active portion of a wild-type F protein that is an N-terminally and/or C-terminally truncated fragment. In some embodiments, the mutant F protein or the biologically active portion of a wild-type F protein thereof comprises one or more amino acid substitutions. In some embodiments, the mutations described herein improve transduction efficiency. In some embodiments, the mutations described herein increase fusogenic capacity. Exemplary mutations include any as described, see e.g. Khetawat and Broder 2010 Virology Journal 7:312; Witting et al. 2013 Gene Therapy 20:997-1005; published international; patent application No. WO/2013/148327.

[0183] In some embodiments, the mutant F protein is a biologically active portion that is truncated and lacks up to 20 contiguous amino acid residues at or near the C-terminus of the wild-type F protein, such as a wild-type F protein encoded by a sequence of nucleotides encoding the F protein set forth in any one of SEQ ID NOS: 592, 593, 608, or 614-616. In some embodiments, the mutant F protein is truncated and lacks up to 19 contiguous amino acids, such as up to 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 contiguous amino acid(s) at the C-terminus of the wild-type F protein.

[0184] In some embodiments, the F protein or the functionally active variant or biologically active portion thereof comprises an F1 subunit or a fusogenic portion thereof. In some embodiments, the F1 subunit is a proteolytically cleaved portion of the F0 precursor. In some embodiments, the F0 precursor is inactive. In some embodiments, the cleavage of the F0 precursor forms a disulfide-linked F1+F2 heterodimer. In some embodiments, the cleavage exposes the fusion polypeptide and produces a mature F protein. In some embodiments, the cleavage occurs at or around a single basic residue. In some embodiments, the cleavage occurs at Arginine 109 of NiV-F protein. In some embodiments, cleavage occurs at Lysine 109 of the Hendra virus F protein.

[0185] In some embodiments, the F protein is a wild-type Nipah virus F (NiV-F) protein or is a functionally active variant or biologically active portion thereof. In some embodiments, the F.sub.0 precursor is encoded by a sequence of nucleotides encoding the sequence set forth in SEQ ID NO:9258. The encoding nucleic acid can encode a signal polypeptide sequence that has the sequence MVVILDKRCY CNLLILILMI SECSVG (SEQ ID NO:9291) or another signal polypeptide sequence. In some embodiments, the F protein has the sequence set forth in SEQ ID NO:9259. In some examples, the F protein is cleaved into an F1 subunit comprising the sequence set forth in SEQ ID NO:9261 and an F2 subunit comprising the sequence set forth in SEQ ID NO:9260.

[0186] In some embodiments, the F protein is a NiV-F protein that is encoded by a sequence of nucleotides encoding the sequence set forth in SEQ ID NO:9258, or is a functionally active variant or biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9258. In some embodiments, the NiV-F-protein has the sequence of set forth in SEQ ID NO:9259, or is a functionally active variant or a biologically active portion thereof that has an amino acid sequence having at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89%, at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9259. In particular embodiments, the F protein or the functionally active variant or biologically active portion thereof retains the cleavage site cleaved by cathepsin L (e.g., corresponding to the cleavage site between amino acids 109-110 of SEQ ID NO:9258).

[0187] In some embodiments, the F protein or the functionally active variant or the biologically active portion thereof includes an F1 subunit that has the sequence set forth in SEQ ID NO:9261, or an amino acid sequence having, at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89% at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9261.

[0188] In some embodiments, the F protein or the functionally active variant or biologically active portion thereof includes an F2 subunit that has the sequence set forth in SEQ ID NO: 9260, or an amino acid sequence having, at least at or about 80%, at least at or about 81%, at least at or about 82%, at least at or about 83%, at least at or about 84%, at least at or about 85%, at least at or about 86%, at least at or about 87%, at least at or about 88%, at least at or about 89% at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO: 9260.

[0189] In some embodiments, the F protein is a mutant NiV-F protein that is a biologically active portion thereof that is truncated and lacks up to 20 contiguous amino acid residues at or near the C-terminus of the wild-type NiV-F protein (e.g., set forth SEQ ID NO: 9259). In some embodiments, the mutant NiV-F protein comprises an amino acid sequence set forth in SEQ ID NO:9262. In some embodiments, the mutant NiV-F protein has a sequence that has at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9262. In some embodiments, the mutant F protein contains an F1 protein that has the sequence set forth in SEQ ID NO:9263. In some embodiments, the mutant F protein has a sequence that has at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9263.

[0190] In some embodiments, the F protein is a mutant NiV-F protein that is a biologically active portion thereof that comprises a 20 amino acid truncation at or near the C-terminus of the wild-type NiV-F protein (SEQ ID NO:9259); and a point mutation on an N-linked glycosylation site. In some embodiments, the mutant NiV-F protein comprises an amino acid sequence set forth in SEQ ID NO:9264. In some embodiments, the mutant NiV-F protein has a sequence that has at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9264.
[0191] In some embodiments, the F protein is a mutant NiV-F protein that is a biologically active portion thereof that comprises a 22 amino acid truncation at or near the C-terminus of the wild-type NiV-F protein (SEQ ID NO:9259). In some embodiments, the NiV-F protein comprises an amino acid sequence set forth in SEQ ID NO:9265. In some embodiments, the NiV-F protein has a sequence with at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9265. In particular embodiments, the variant F protein is a mutant Niv-F protein that has the sequence of amino acids set forth in SEQ ID NO:9280. In some embodiments, the NiV-F protein has a sequence with at least at or about 90%, at least at or about 91%, at least at or about 92%, at least at or about 93%, at least at or about 94%, at least at or about 95%, at least at or about 96%, at least at or about 97%, at least at or about 98%, or at least at or about 99% sequence identity to SEQ ID NO:9280.
[0192] It has been reported that the henipavirus F proteins from various species exhibit compatibility with G proteins from other species to trigger fusion (Brandel-Tretheway et al. Journal of Virology. 2019. 93(13):e00577-19). In some aspects of the provided lentiviral vector, the F protein is heterologous to the G protein, i.e. the F and G protein or biologically active portions are from different henipavirus species. For example, the G protein is from Hendra virus and the F protein is a NiV-F as described. In other aspects, the F and/or G protein can be a chimeric F and/or G protein containing regions of F and/or G proteins from different species of Henipavirus. In some embodiments, replacing a portion of the F protein with amino acids from a heterologous sequence of Henipavirus results in fusion to the G protein with the heteroglous sequence. (Brandel-Tretheway et al. 2019). In some cases, the chimeric F and/or G protein contains an extracellular domain from one henipavirus species and a transmembrane and/or cytoplasmic domain from a different henipavirus species. For example, the F protein contains an extracellular domain of Hendra virus and a transmembrane/cytoplasmic domain of Nipah virus.
B. Lipid Bilayer
[0193] In some embodiments, the targeted lipid particle includes a naturally derived bilayer of amphipathic lipids that encloses a lumen or cavity. In some embodiments, the targeted lipid particle comprises a lipid bilayer as the outermost surface. In some embodiments, the lipid bilayer encloses a lumen. In some embodiments, the lumen is aqueous. In some embodiments, the lumen is in contact with the hydrophilic head groups on the interior of the lipid bilayer. In some embodiments, the lumen is a cytosol. In some embodiments, the cytosol contains cellular components present in a source cell. In some embodiments, the cytosol does not contain cellular components present in a source cell. In some embodiments, the lumen is a cavity. In some embodiments, the cavity contains an aqueous environment. In some embodiments, the cavity does not contain an aqueous environment.
[0194] In some aspects, the lipid bilayer is derived from a source cell during a process to produce a lipid-containing particle. In some embodiments, the lipid bilayer includes membrane components of the cell from which the lipid bilayer is produced, e.g., phospholipids, membrane proteins, etc. In some embodiments, the lipid bilayer includes a cytosol that includes components found in the cell from which the lipid bilayer is produced, e.g., solutes, proteins, nucleic acids, etc., but not all of the

components of a cell, e.g., it lacks a nucleus. In some embodiments, the lipid bilayer is considered to be exosome-like. The lipid particle may vary in size, and in some instances have a diameter ranging from 30 and 300 nm, such as from 30 and 150 nm, and including from 40 to 100 nm.

[0195] In some embodiments, the lipid bilayer is a viral envelope. In some embodiments, the viral envelope is obtained from a source cell. In some embodiments, the viral envelope is obtained from the source cell plasma membrane. In some embodiments, the lipid bilayer is obtained from a membrane other than the plasma membrane of a host cell. In some embodiments, the viral envelope lipid bilayer is embedded with viral proteins, including viral glycoproteins.

[0196] In other aspects, the lipid bilayer includes synthetic lipid complex. In some embodiments, the lipid bilayer is a liposome that includes a synthetic lipid complex. In some embodiments, the lipid particle is a vesicular structure characterized by a phospholipid bilayer membrane and an inner aqueous medium. In some embodiments, the lipid bilayer has multiple lipid layers separated by aqueous medium. In some embodiments, the lipid bilayer forms spontaneously when phospholipids are suspended in an excess of aqueous solution. In some examples, the lipid components undergo self-rearrangement before the formation of closed structures and entrap water and dissolved solutes between the lipid bilayers. In some embodiments the lipid bilayer is a fusosome.

[0197] In some embodiments, a targeted envelope protein and fusogen, such as any described above including any that are exogenous or overexpressed relative to the source cell, is disposed in the lipid bilayer.

[0198] In some embodiments, the targeted lipid particle comprises several different types of lipids. In some embodiments, the lipids are amphipathic lipids. In some embodiments, the amphipathic lipids are phospholipids. In some embodiments, the phospholipids comprise phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol, and phosphatidylserine. In some embodiments, the lipids comprise DMPC, DOPC, and DSPC.

[0199] In some embodiments, the bilayer is comprised of one or more lipids of the same or different type. In some embodiments, the source cell comprises a cell selected from CHO cells, BHK cells, MDCK cells, C3H 10T1/2 cells, FLY cells, Psi-2 cells, BOSC 23 cells, PA317 cells, WEHI cells, COS cells, BSC 1 cells, BSC 40 cells, BMT 10 cells, VERO cells, W138 cells, MRC5 cells, A549 cells, HT1080 cells, 293 cells, 293T cells, B-50 cells, 3T3 cells, NIH3T3 cells, HepG2 cells, Saos-2 cells, Huh7 cells, Hela cells, W163 cells, 211 cells, and 211A cells.

C. Exogenous Agent

[0200] In some embodiments, the targeted lipid particle further comprises an agent that is exogenous relative to the source cell (also referred to herein as a “cargo” or “payload”). In some embodiments, the exogenous agent is a small molecule, a protein, or a nucleic acid (e.g., a DNA, a chromosome (e.g. a human artificial chromosome), an RNA, e.g., an mRNA or miRNA). In some embodiments, the exogenous agent or cargo encodes a cytosolic protein. In some embodiments the exogenous agent or cargo comprises or encodes a membrane protein. In some embodiments, the exogenous agent or cargo comprises a therapeutic agent. In some embodiments, the therapeutic agent is chosen from one or more of a protein, e.g., an enzyme, a transmembrane protein, a receptor, an antibody; a nucleic acid, e.g., DNA, a chromosome (e.g. a human artificial chromosome), RNA, mRNA, siRNA, miRNA; or a small molecule.

[0201] In some embodiments, the exogenous agent is present in at least, or no more than, 10, 20, 50, 100, 200, 500, 1,000, 2,000, 5,000, 10,000, 20,000, 50,000, 100,000, 200,000, 500,000, 1,000,000, 5,000,000, 10,000,000, 50,000,000, 100,000,000, 500,000,000, or 1,000,000,000 copies. In some embodiments, the targeted lipid particle has an altered, e.g., increased or decreased level of one or more endogenous molecules, e.g., protein or nucleic acid (e.g., in some embodiments, endogenous relative to the source cell, and in some embodiments, endogenous relative to the target cell), e.g., due to treatment of the source cell, e.g., mammalian source cell with a siRNA or gene editing enzyme. In some embodiments, the endogenous molecule is present in at least, or no more than, 10, 20, 50, 100, 200, 500, 1,000, 2,000, 5,000, 10,000, 20,000, 50,000, 100,000, 200,000, 500,000, 1,000,000, 5,000,000, 10,000,000, 50,000,000, 100,000,000, 500,000,000, or 1,000,000,000 copies. In some embodiments, the endogenous molecule (e.g., an RNA or protein) is present at a concentration of at least 1, 2, 3, 4, 5, 10, 20, 50, 100, 500, 10.sup.3, 5.0×10.sup.3, 10.sup.4, 5.0×10.sup.4, 10.sup.5, 5.0×10.sup.5, 10.sup.6, 5.0×10.sup.6, 1.0×10.sup.7, 5.0×10.sup.7, or 1.0×10.sup.8, greater than its concentration in the source cell. In some embodiments, the endogenous molecule (e.g., an RNA or protein) is present at a concentration of at least 1, 2, 3, 4, 5, 10, 20, 50, 100, 500, 10.sup.3, 5.0×10.sup.3, 10.sup.4, 5.0×10.sup.4, 10.sup.5, 5.0×10.sup.5, 10.sup.6, 5.0×10.sup.6, 1.0×10.sup.7, 5.0×10.sup.7, or 1.0×10.sup.8 less than its concentration in the source cell.

[0202] In some embodiments, the targeted lipid particle (e.g., targeted viral vector) delivers to a target cell at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, or 99% of the cargo (e.g., a therapeutic agent, e.g., an exogenous therapeutic agent) comprised by the targeted lipid particle. In some embodiments, the targeted lipid particle that fuses with the target cell(s) delivers to the target cell an average of at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, or 99% of the cargo (e.g., a therapeutic agent, e.g., an exogenous therapeutic agent) comprised by the targeted lipid particle that fuses with the target cell(s). In some embodiments, the targeted lipid particle composition delivers to a target tissue at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, or 99% of the cargo (e.g., a therapeutic agent, e.g., an exogenous therapeutic agent) comprised by the targeted lipid particle composition.

[0203] In some embodiments, the exogenous agent or cargo is not expressed naturally in the cell from which the targeted lipid particle is derived. In some embodiments, the exogenous agent or cargo is expressed naturally in the cell from which the viral vector is derived. In some embodiments, the exogenous agent or cargo is loaded into the targeted lipid particle via expression in the cell from which the viral vector is derived (e.g. expression from DNA or mRNA introduced via transfection, transduction, or electroporation). In some embodiments, the exogenous agent or cargo is expressed from DNA integrated into the genome or maintained episomally. In some embodiments, expression of the exogenous agent or cargo is constitutive. In some embodiments, expression of the exogenous agent or cargo is induced. In some embodiments, expression of the exogenous agent or cargo is induced immediately prior to generating the targeted lipid particle. In some embodiments, expression of the exogenous agent or cargo is induced at the same time as expression of the fusogen.

[0204] In some embodiments, the exogenous agent or cargo is loaded into the viral vector via electroporation into the viral vector itself or into the cell from which the viral vector is derived. In some embodiments, the exogenous agent or cargo is loaded into the viral vector via transfection (e.g., of a DNA or mRNA encoding the cargo) into the viral vector itself or into the cell from which the viral vector is derived.

[0205] In some embodiments, the exogenous agent or cargo includes one or more nucleic acid sequences, one or more amino acid sequences, a combination of nucleic acid sequences and/or amino acid sequences, one or more organelles, and any combination thereof. In some embodiments, the exogenous agent or cargo includes one or more cellular components. In some embodiments, the exogenous agent or cargo includes one or more cytosolic and/or nuclear components.

[0206] In some embodiments, the exogenous agent or cargo includes a nucleic acid, e.g., DNA, nDNA (nuclear DNA), mtDNA (mitochondrial DNA), protein coding DNA, gene, operon, chromosome, genome, transposon, retrotransposon, viral genome, intron, exon, modified DNA, mRNA (messenger RNA), tRNA (transfer RNA), modified RNA, microRNA, siRNA (small interfering RNA), tmRNA (transfer messenger RNA), rRNA (ribosomal RNA), mtRNA (mitochondrial RNA), snRNA (small nuclear RNA), small nucleolar RNA (snoRNA), SmY RNA (mRNA trans-splicing RNA), gRNA (guide RNA), TERC (telomerase RNA component), aRNA (antisense RNA), cis-NAT (Cis-natural antisense transcript), CRISPR RNA (crRNA), lncRNA (long noncoding RNA), piRNA (piwi-interacting RNA), shRNA (short hairpin RNA), tasiRNA (trans-acting siRNA), eRNA (enhancer RNA), satellite RNA, pcRNA (protein coding RNA), dsRNA (double stranded RNA), RNAi (interfering RNA), circRNA (circular RNA), reprogramming RNAs, aptamers, and any combination thereof. In some embodiments, the nucleic acid is a wild-type nucleic acid. In some embodiments, the nucleic acid is a mutant nucleic acid. In some embodiments the nucleic acid is a fusion or chimera of multiple nucleic acid sequences.

[0207] In some embodiments, the exogenous agent or cargo includes a nucleic acid. For example, the exogenous agent or cargo may comprise RNA to enhance expression of an endogenous protein, or a siRNA or miRNA that inhibits protein expression of an endogenous protein. For example, the endogenous protein may modulate structure or function in the target cells. In some embodiments, the cargo includes a nucleic acid encoding an engineered protein that modulates structure or function in the target cells. In some embodiments, the exogenous agent or cargo is a nucleic acid that targets a transcriptional activator that modulate structure or function in the target cells.

[0208] In some embodiments, the exogenous agent or cargo includes a polypeptide, e.g., enzymes, structural proteins, signaling proteins, regulatory proteins, transport proteins, sensory proteins, motor proteins, defense proteins, storage proteins, transcription factors, antibodies, cytokines, hormones, catabolic proteins, anabolic proteins, proteolytic proteins, metabolic proteins, kinases, transferases, hydrolases, lyases, isomerases, ligases, enzyme modulator proteins, protein binding polypeptides, lipid binding proteins, membrane fusion proteins, cell differentiation proteins, epigenetic proteins, cell death proteins, nuclear transport proteins, nucleic acid binding proteins, reprogramming proteins, DNA editing proteins, DNA repair proteins, DNA recombination proteins, transposase proteins, DNA integration proteins, targeted endonucleases (e.g. Zinc-finger nucleases, transcription-activator-like nucleases (TALENs), cas9 and homologs thereof), recombinases, and any combination thereof. In some embodiments the protein targets a protein in the cell for degradation. In some embodiments the protein targets a protein in the cell for degradation by localizing the protein to the proteasome. In some embodiments, the protein is a wild-type protein. In some embodiments, the protein is a mutant protein. In some embodiments the protein is a fusion or chimeric protein.

[0209] In some embodiments, the exogenous agent or cargo includes a small molecule, e.g., ions (e.g. Ca.sup.2+, C1−, Fe.sup.2+), carbohydrates, lipids, reactive oxygen species, reactive nitrogen species, isoprenoids, signaling molecules, heme, peptide cofactors, electron accepting compounds, electron donating compounds, metabolites, ligands, and any combination thereof. In some embodiments the small molecule is a pharmaceutical that interacts with a target in the cell. In some embodiments the small molecule targets a protein in the cell for degradation. In some embodiments the small molecule targets a protein in the cell for degradation by localizing the protein to the proteasome. In some embodiments that small molecule is a proteolysis targeting chimera molecule (PROTAC).

[0210] In some embodiments, the exogenous agent or cargo includes a mixture of proteins, nucleic acids, or metabolites, e.g., multiple amino acids, multiple nucleic acids, multiple small molecules; combinations of nucleic acids, amino acids, and small molecules; ribonucleoprotein complexes (e.g. Cas9-gRNA complex); multiple transcription factors, multiple epigenetic factors, reprogramming factors (e.g. Oct4, Sox2, cMyc, and Klf4); multiple regulatory RNAs; and any combination thereof.

[0211] In some embodiments, the exogenous agent or cargo includes one or more organelles, e.g., chondriosomes, mitochondria, lysosomes, nucleus, cell membrane, cytoplasm, endoplasmic reticulum, ribosomes, vacuoles, endosomes, spliceosomes, polymerases, capsids, acrosome, autophagosome, centriole, glycosome, glyoxysome, hydrogenosome, melanosome, mitosome, myofibril, cnidocyte, peroxisome, proteasome, vesicle, stress granule, networks of organelles, and any combination thereof.

[0212] In some embodiments, the exogenous agent encodes a therapeutic agent or a diagnostic agent. In some embodiments, the therapeutic agent is a chimeric antigen receptor (CAR) or T-cell receptor (TCR). In some embodiments, the CAR targets a tumor antigen selected from CD19, CD20, CD22, or BCMA. In another embodiment, the CAR is engineered to comprise an intracellular signaling domain of the T cell antigen receptor complex zeta chain (e.g., CD3 zeta). In a preferred embodiment, the intracellular domain is selected from a CD137 (4-1BB) signaling domain, a CD28 signaling domain, and a CD3zeta signaling domain.

D. Methods of Generating Targeted Lipid Particles Derived from Virus

[0213] Provided herein are targeted lipid particles that are derived from virus, such as viral particles or virus-like particles, including those derived from retroviruses or lentiviruses. In some embodiments, the targeted lipid particle's bilayer of amphipathic lipids is or comprises the viral envelope. In some embodiments, the targeted lipid particle's bilayer of amphipathic lipids is or comprises lipids derived from a producer cell. In some embodiments, the viral envelope comprises a fusogen, e.g., a fusogen that is endogenous to the virus or a pseudotyped fusogen. In some embodiments, the targeted lipid particle's lumen or cavity comprises a viral nucleic acid, e.g., a retroviral nucleic acid, e.g., a lentiviral nucleic acid. In some embodiments, the viral nucleic acid is a viral genome. In some embodiments, the targeted lipid particle further comprises one or more viral non-structural proteins, e.g., in its cavity or lumen. In some embodiments, the targeted lipid particle is or comprises a virus-like particle (VLP). In some embodiments, the VLP does not comprise an envelope. In some embodiments, the VLP comprises an envelope.

[0214] In some embodiments, the viral particle or virus-like particle, such as a retrovirus or retrovirus-like particle, comprises one or more of a Gag polyprotein, polymerase (e.g., Pol), integrase (IN, e.g., a functional or non-functional variant), protease (PR), and a fusogen. In some embodiments, the targeted lipid particle further comprises Rev. In some embodiments, one or more of the aforesaid proteins are encoded in the retroviral genome, and in some embodiments, one or more of the aforesaid proteins are provided in trans, e.g., by a helper cell, helper virus, or helper plasmid. In some embodiments, the targeted lipid particle nucleic acid (e.g., retroviral nucleic acid) comprises one or more of the following nucleic acid sequences: 5′ LTR (e.g., comprising U5 and lacking a functional U3 domain), Psi packaging element (Psi), Central polypurine tract (cPPT) Promoter operatively linked to the payload gene, payload gene (optionally comprising an intron before the open reading frame), Poly A tail sequence, WPRE, and 3′ LTR (e.g., comprising U5 and lacking a functional U3). In some embodiments the targeted lipid particle nucleic acid further comprises one or more insulator elements. In some embodiments, the recognition sites are situated between the poly A tail sequence and the WPRE.

[0215] In some embodiments, the targeted lipid particle comprises supramolecular complexes formed by viral proteins that self-assemble into capsids. In some embodiments, the targeted lipid particle is a viral particle or virus-like particle derived from viral capsids. In some embodiments, the targeted lipid particle is a viral particle or virus-like particle derived from viral nucleocapsids. In some embodiments, the targeted lipid particle comprises nucleocapsid-derived proteins that retain the property of packaging nucleic acids. In some embodiments, the viral particles or virus-like particles comprise only viral structural glycoproteins. In some embodiments, the targeted lipid particle does not contain a viral genome.

[0216] In some embodiments, the targeted lipid particle packages nucleic acids from host cells during the expression process. In some embodiments, the nucleic acids do not encode any genes involved in virus replication. In particular embodiments, the targeted lipid particle is a virus-like particle, e.g. retrovirus-like particle such as a lentivirus-like particle, that is replication defective.

[0217] In some cases, the targeted lipid particle is a viral particle that is morphologically indistinguishable from the wild type infectious virus. In some embodiments, the viral particle presents the entire viral proteome as an antigen. In some embodiments, the viral particle presents

only a portion of the proteome as an antigen.
[0218] In some embodiments, the viral particle or virus-like particle is produced utilizing proteins (e.g., envelope proteins) from a virus within the Paramyxoviridae family. In some embodiments, the Paramyxoviridae family comprises members within the Henipavirus genus. In some embodiments, the Henipavirus is or comprises a Hendra (HeV) or a Nipah (NiV) virus. In particular embodiments, the viral particles or virus-like particles incorporate a targeted envelope protein and fusogen.
[0219] In some embodiments, viral particles or virus-like particles is produced in multiple cell culture systems including bacteria, mammalian cell lines, insect cell lines, yeast, and plant cells.
[0220] Suitable cell lines which can be used include, for example, CHO cells, BHK cells, MDCK cells, C3H 10T1/2 cells, FLY cells, Psi-2 cells, BOSC 23 cells, PA317 cells, WEHI cells, COS cells, BSC 1 cells, BSC 40 cells, BMT 10 cells, VERO cells, W138 cells, MRC5 cells, A549 cells, HT1080 cells, 293 cells, 293T cells, B-50 cells, 3T3 cells, NIH3T3 cells, HepG2 cells, Saos-2 cells, Huh7 cells, Hela cells, W163 cells, 211 cells, 211A cells, and cyno and *Macaca nemestrina* cell lines. In embodiments, the packaging cells are 293 cells, 293T cells, or A549 cells.
[0221] In some embodiments, a source cell line includes a cell line which is capable of producing recombinant retroviral particles, comprising a producer cell line and a transfer vector construct comprising a packaging signal. Methods of preparing viral stock solutions are illustrated by, e.g., Y. Soneoka et al. (1995) *Nucl. Acids Res.* 23:628-633, and N. R. Landau et al. (1992) *J. Virol.* 66:5110-5113, which are incorporated herein by reference.
[0222] In some embodiments, the assembly of a viral particle or virus-like particle is initiated by binding of the core protein to a unique encapsidation sequence within the viral genome (e.g. UTR with stem-loop structure). In some embodiments, the interaction of the core with the encapsidation sequence facilitates oligomerization.
[0223] In some embodiments, the targeted lipid particle is a virus-like particle which comprises a sequence that is devoid of or lacking viral RNA. In some embodiments, such particles are the result of removing or eliminating the viral RNA from the sequence. In some embodiments, this is achieved by using an endogenous packaging signal binding site on Gag. In some embodiments, the endogenous packaging signal binding site is on Pol. In some embodiments, the RNA which is to be delivered will contain a cognate packaging signal. In some embodiments, a heterologous binding domain (which is heterologous to Gag) located on the RNA to be delivered, and a cognate binding site located on Gag or Pol, is used to ensure packaging of the RNA to be delivered. In some embodiments, the heterologous sequence is non-viral or it could be viral, in which case it is derived from the same virus or a different virus. In some embodiments, the vector particles could be used to deliver therapeutic RNA, in which case functional integrase and/or reverse transcriptase is not required. In some embodiments, the vector particles could also be used to deliver a therapeutic gene of interest, in which case Pol is typically included. In some embodiments, the retroviral nucleic acid comprises one or more of (e.g., all of): a 5′ promoter (e.g., to control expression of the entire packaged RNA), a 5′ LTR (e.g., that includes R (polyadenylation tail signal) and/or U5 which includes a primer activation signal), a primer binding site, a Psi packaging signal, a RRE element for nuclear export, a promoter directly upstream of the transgene to control transgene expression, a transgene (or other exogenous agent element), a polypurine tract, and a 3′ LTR (e.g., that includes a mutated U3, a R, and U5). In some embodiments, the retroviral nucleic acid further comprises one or more of a cPPT, a WPRE, and/or an insulator element.
[0224] A retrovirus typically replicates by reverse transcription of its genomic RNA into a linear double-stranded DNA copy and subsequently covalently integrates its genomic DNA into a host genome. Illustrative retroviruses suitable for use in particular embodiments, include, but are not limited to: Moloney murine leukemia virus (M-MuLV), Moloney murine sarcoma virus (MOMSV), Harvey murine sarcoma virus (HaMuSV), murine mammary tumor virus (MuMTV), gibbon ape leukemia virus (GaLV), feline leukemia virus (FLV), spumavirus, Friend murine

leukemia virus, Murine Stem Cell Virus (MSCV), Rous Sarcoma Virus (RSV), and other lentiviruses.

[0225] In some embodiments the retrovirus is a Gammaretrovirus. In some embodiments the retrovirus is an Epsilonretrovirus. In some embodiments the retrovirus is an Alpharetrovirus. In some embodiments the retrovirus is a Betaretrovirus. In some embodiments the retrovirus is a Deltaretrovirus. In some embodiments the retrovirus is a Lentivirus. In some embodiments the retrovirus is a Spumaretrovirus. In some embodiments the retrovirus is an endogenous retrovirus.

[0226] Illustrative lentiviruses include, but are not limited to: HIV (human immunodeficiency virus; including HIV type 1, and HIV type 2); visna-maedi virus (VMV) virus; the caprine arthritis-encephalitis virus (CAEV); equine infectious anemia virus (EIAV); feline immunodeficiency virus (FIV); bovine immune deficiency virus (BIV); and simian immunodeficiency virus (SIV). In some embodiments, HIV based vector backbones (i.e., HIV cis-acting sequence elements) are used.

[0227] In some embodiments, a vector herein is a nucleic acid molecule capable transferring or transporting another nucleic acid molecule. The transferred nucleic acid is generally linked to, e.g., inserted into, the vector nucleic acid molecule. A vector may include sequences that direct autonomous replication in a cell, or may include sequences sufficient to allow integration into host cell DNA. Useful vectors include, for example, plasmids (e.g., DNA plasmids or RNA plasmids), transposons, cosmids, bacterial artificial chromosomes, and viral vectors. Useful viral vectors include, e.g., replication defective retroviruses and lentiviruses.

[0228] In some embodiments, a viral vector comprises a nucleic acid molecule (e.g., a transfer plasmid) that includes virus-derived nucleic acid elements that typically facilitate transfer of the nucleic acid molecule or integration into the genome of a cell or to a viral particle that mediates nucleic acid transfer. Viral particles will typically include various viral components and sometimes also host cell components in addition to nucleic acid(s). In some embodiments, a viral vector comprises e.g., a virus or viral particle capable of transferring a nucleic acid into a cell, or the transferred nucleic acid (e.g., as naked DNA). In some embodiments, a viral vectors and transfer plasmids comprise structural and/or functional genetic elements that are primarily derived from a virus. A retroviral vector can comprise a viral vector or plasmid containing structural and functional genetic elements, or portions thereof, that are primarily derived from a retrovirus. A lentiviral vector can comprise a viral vector or plasmid containing structural and functional genetic elements, or portions thereof, including LTRs that are primarily derived from a lentivirus.

[0229] In embodiments, a lentiviral vector (e.g., lentiviral expression vector) comprises a lentiviral transfer plasmid (e.g., as naked DNA) or an infectious lentiviral particle. With respect to elements such as cloning sites, promoters, regulatory elements, heterologous nucleic acids, etc., it is to be understood that the sequences of these elements can be present in RNA form in lentiviral particles and can be present in DNA form in DNA plasmids.

[0230] In some embodiments, the viral vector further comprises a vector-surface targeting moiety which specifically binds to a target ligand. In some embodiments, the vector-surface targeting moiety is a polypeptide. In some embodiments, a nucleic acid encoding the Paramyxovirus envelope protein (e.g. G protein) is modified with a targeting moiety to specifically bind to a target molecule on a target cells. In some embodiments, the targeting moiety is any targeting protein, including but not necessarily limited to antibodies and antigen binding fragments thereof.

[0231] In some embodiments, in the vectors described herein at least part of one or more protein coding regions that contribute to or are essential for replication are absent compared to the corresponding wild-type virus. In some embodiments, the viral vector is replication-defective. In some embodiments, the vector is capable of transducing a target non-dividing host cell and/or integrating its genome into a host genome.

[0232] In some embodiments, different cells differ in their usage of particular codons. In some embodiments, this codon bias corresponds to a bias in the relative abundance of particular tRNAs in the cell type. In some embodiments, by altering the codons in the sequence so that they are tailored to match with the relative abundance of corresponding tRNAs, it is possible to increase expression. In some embodiments, it is possible to decrease expression by deliberately choosing codons for which the corresponding tRNAs are known to be rare in the particular cell type. In some embodiments, an additional degree of translational control is available. An additional description of codon optimization is found, e.g., in WO 99/41397, which is herein incorporated by reference in its entirety.

[0233] Conventional techniques for generating retrovirus vectors (and, in particular, lentivirus vectors) with or without the use of packaging/helper vectors are known to those skilled in the art and may be used to generate targeted lipid particles according to the present disclosure. (See, e.g., Derse and Newbold 1993 Virology 194:530-6; Maury et al. 1994 Virology 200:632-42; Wanisch et al. 2009. Mol Ther. 1798:1316-1332; Martarano et al. 1994 J. Virol. 68:3102-11; Naldini et al., (1996a, 1996b, and 1998); Zufferey et al., 1999, J. Virol., 73:2886; Huang et al., Mol. Cell. Biol., 5:3864; Liu et al., 1995, Genes Dev., 9:1766; Cullen et al., 1991. J. Virol. 65:1053; and Cullen et al., 1991. Cell 58:423; Dull et al., 1998, U.S. Pat. Nos. 6,013,516; and 5,994,136; PCT patent applications WO 99/15683, WO 98/17815, WO 99/32646, and WO 01/79518). Conventional techniques relating to packaging vectors and producer cells known in the art may also be used according to the present disclosure. (See, e.g., Yao et al, 1998; Jones et al, 2005.)

[0234] Provided herein are targeted lipid particles that comprise a naturally derived membrane. In some embodiments, the naturally derived membrane comprises membrane vesicles prepared from cells or tissues. In some embodiments, the targeted lipid particle comprises a vesicle that is obtainable from a cell. In some embodiments, the targeted lipid particle comprises a microvesicle, an exosome, a membrane enclosed body, an apoptotic body (from apoptotic cells), a particle (which may be derived from e.g. platelets), an ectosome (derivable from, e.g., neutrophiles and monocytes in serum), a prostatosome (obtainable from prostate cancer cells), or a cardiosome (derivable from cardiac cells).

[0235] In some embodiments, the source cell is an endothelial cell, a fibroblast, a blood cell (e.g., a macrophage, a neutrophil, a granulocyte, a leukocyte), a stem cell (e.g., a mesenchymal stem cell, an umbilical cord stem cell, bone marrow stem cell, a hematopoietic stem cell, an induced pluripotent stem cell e.g., an induced pluripotent stem cell derived from a subject's cells), an embryonic stem cell (e.g., a stem cell from embryonic yolk sac, placenta, umbilical cord, fetal skin, adolescent skin, blood, bone marrow, adipose tissue, erythropoietic tissue, hematopoietic tissue), a myoblast, a parenchymal cell (e.g., hepatocyte), an alveolar cell, a neuron (e.g., a retinal neuronal cell), a precursor cell (e.g., a retinal precursor cell, a myeloblast, myeloid precursor cells, a thymocyte, a meiocyte, a megakaryoblast, a promegakaryoblast, a melanoblast, a lymphoblast, a bone marrow precursor cell, a normoblast, or an angioblast), a progenitor cell (e.g., a cardiac progenitor cell, a satellite cell, a radial glial cell, a bone marrow stromal cell, a pancreatic progenitor cell, an endothelial progenitor cell, a blast cell), or an immortalized cell (e.g., HeEa, HEK293, HFF-I, MRC-5, WI-38, IMR 90, IMR 91, PER.C6, HT-1080, or BJ cell). In some embodiments, the source cell is other than a 293 cell, HEK cell, human endothelial cell, or a human epithelial cell, monocyte, macrophage, dendritic cell, or stem cell.

[0236] In some embodiments, the targeted lipid particle has a density of <1, 1-1.1, 1.05-1.15, 1.1-1.2, 1.15-1.25, 1.2-1.3, 1.25-1.35, or >1.35 g/ml. In some embodiments, the targeted lipid particle composition comprises less than 0.01%, 0.05%, 0.1%, 0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 4%, 5%, or 10% source cells by protein mass, or less than 0.01%, 0.05%, 0.1%, 0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 4%, 5%, or 10% of cells having a functional nucleus.

[0237] In embodiments, the targeted lipid particle has a size, or the population of targeted lipid particles have an average size, that is less than about 0.01%, 0.05%, 0.1%, 0.5%, 1%, 2%, 3%, 4%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, of that of the source cell.

[0238] In some embodiments the targeted lipid particle comprises an extracellular vesicle, e.g., a cell-derived vesicle comprising a membrane that encloses an internal space and has a smaller diameter than the cell from which it is derived. In embodiments the extracellular vesicle has a diameter from 20 nm to 1000 nm. In embodiments the targeted lipid particle comprises an apoptotic body, a fragment of a cell, a vesicle derived from a cell by direct or indirect manipulation, a vesiculated organelle, and a vesicle produced by a living cell (e.g., by direct plasma membrane budding or fusion of the late endosome with the plasma membrane). In embodiments the extracellular vesicle is derived from a living or dead organism, explanted tissues or organs, or cultured cells.

[0239] In embodiments, the targeted lipid particle comprises a nanovesicle, e.g., a cell-derived small (e.g., from 20 to 250 nm in diameter, or from 30 to 150 nm in diameter) vesicle comprising a membrane that encloses an internal space, and which is generated from said cell by direct or indirect manipulation. The production of nanovesicles can, in some instances, result in the destruction of the source cell. The nanovesicle may comprise a lipid or fatty acid and a polypeptide.

[0240] In embodiments, the targeted lipid particle comprises an exosome. In embodiments, the exosome is a cell-derived small (e.g., from 20 to 300 nm in diameter, or from 40 to 200 nm in diameter) vesicle comprising a membrane that encloses an internal space, and which is generated from said cell by direct plasma membrane budding or by fusion of the late endosome with the plasma membrane. In embodiments, production of exosomes does not result in the destruction of the source cell. In embodiments, the exosome comprises a lipid or fatty acid and a polypeptide.

[0241] In some embodiments, the targeted lipid particle is derived from a source cell with a genetic modification which results in increased expression of an immunomodulatory agent. In some embodiments, the immunosuppressive agent is on an exterior surface of the cell. In some embodiments, the immunosuppressive agent is incorporated into the exterior surface of the targeted lipid particle. In some embodiments, the targeted lipid particle comprises an immunomodulatory agent attached to the surface of the solid particle by a covalent or non-covalent bond.

A. Generation of Cell-Derived Particles

[0242] In some embodiments, targeted lipid particles are generated by inducing budding of an exosome, microvesicle, membrane vesicle, extracellular membrane vesicle, plasma membrane vesicle, giant plasma membrane vesicle, apoptotic body, mitoparticle, pyrenocyte, lysosome, or other membrane enclosed vesicle.

[0243] In some embodiments, targeted lipid particles are generated by inducing cell enucleation. Enucleation may be performed using assays such as genetic, chemical (e.g., using Actinomycin D, see Bayona-Bafaluy et al., “A chemical enucleation method for the transfer of mitochondrial DNA to p° cells” Nucleic Acids Res. 2003 Aug. 15; 31(16):e98), or mechanical methods (e.g., squeezing or aspiration, see Lee et al., “A comparative study on the efficiency of two enucleation methods in pig somatic cell nuclear transfer: effects of the squeezing and the aspiration methods.” Anim Biotechnol. 2008; 19(2):71-9), or combinations thereof.

[0244] In some embodiments, the targeted lipid particles are generated by inducing cell fragmentation. In some embodiments, cell fragmentation is performed using the following methods, including, but not limited to: chemical methods, mechanical methods (e.g., centrifugation (e.g., ultracentrifugation, or density centrifugation), freeze-thaw, or sonication), or combinations thereof.

[0245] In some embodiments, the targeted lipid particle is a microvesicle. In some embodiments the microvesicle has a diameter of about 100 nm to about 2000 nm. In some embodiments, a targeted lipid particle comprises a cell ghost. In some embodiments, a vesicle is a plasma membrane vesicle, e.g., a giant plasma membrane vesicle.

[0246] In some embodiments, a characteristic of a targeted lipid particle is described by comparison to a reference cell. In embodiments, the reference cell is the source cell. In embodiments, the reference cell is a HeLa, HEK293, HFF-1, MRC-5, WI-38, IMR 90, IMR 91, PER.C6, HT-1080, or BJ cell. In some embodiments, for example when the source cell used to make the targeted lipid particle is not available for testing after the targeted lipid particle is made, a characteristic of a population of targeted lipid particle is described by comparison to a population of reference cells, e.g., a population of source cells, or a population of HeLa, HEK293, HFF-1, MRC-5, WI-38, IMR 90, IMR 91, PER.C6, HT-1080, or BJ cells.

Pharmaceutical Compositions

[0247] The present disclosure also provides, in some aspects, a pharmaceutical composition comprising the targeted lipid particle (e.g., targeted viral vectors) composition described herein and a pharmaceutically acceptable carrier. The pharmaceutical compositions can include any of the described targeted lipid particles.

[0248] In some embodiments, the targeted lipid particle meets a pharmaceutical or good manufacturing practices (GMP) standard. In some embodiments, the targeted lipid particle is made according to good manufacturing practices (GMP). In some embodiments, the targeted lipid particle has a pathogen level below a predetermined reference value, e.g., is substantially free of pathogens. In some embodiments, the targeted lipid particle has a contaminant level below a predetermined reference value, e.g., is substantially free of contaminants. In some embodiments, the targeted lipid particle has low immunogenicity.

[0249] In some embodiments, provided herein are the use of pharmaceutical compositions to practice the methods of the disclosure. Such a pharmaceutical composition may comprise at least one targeted lipid particle of the disclosure in a form suitable for administration to a subject, or the pharmaceutical composition may comprise at least one targeted lipid particle of the disclosure and one or more pharmaceutically acceptable carriers, one or more additional ingredients, or some combination of these.

[0250] In some embodiments, the relative amounts of the targeted lipid particle, the pharmaceutically acceptable carrier, and any additional ingredients in a pharmaceutical composition of the disclosure will vary, depending upon the identity, size, and condition of the subject treated and further depending upon the route by which the composition is to be administered. In some embodiments, the composition comprises from 0.1% to 100% (w/w) of the targeted lipid particles of the disclosure.

[0251] In some embodiments, pharmaceutical compositions that are useful in the methods of the disclosure are suitably developed for intravenous, intratumoral, oral, rectal, vaginal, parenteral, topical, pulmonary, intranasal, buccal, ophthalmic, or another route of administration. In some embodiments, a composition useful within the methods of the disclosure are directly administered to the skin, vagina or any other tissue of a mammal. In some embodiments, formulations include liposomal preparations, resealed erythrocytes containing the targeted lipid particles of the disclosure, and immunologically based formulations. In some embodiments, the route(s) of administration will be readily apparent to the skilled artisan and will depend upon any number of factors including the type and severity of the disease being treated, the type and age of the veterinary or human subject being treated, and the like.

[0252] In some embodiments, formulations of the pharmaceutical compositions described herein are prepared by any method known or hereafter developed in the art of pharmacology. In some embodiments, preparatory methods include the step of bringing the targeted lipid particles of the disclosure into association with a carrier or one or more other accessory ingredients, and then, if necessary or desirable, shaping or packaging the product into a desired single- or multi-dose unit.

[0253] In some embodiments, a "unit dose" is a discrete amount of the pharmaceutical composition comprising a predetermined amount of the targeted lipid particles of the disclosure. In some embodiments, the amount is generally equal to the dosage that would be administered to a subject or a convenient fraction of such a dosage such as, for example, one-half or one-third of such a dosage. In some embodiments, the unit dosage form is for a single daily dose or one of multiple daily doses (e.g., about 1 to 4 or more times per day). In some embodiments, when multiple daily doses are used, the unit dosage form is the same or different for each dose.

[0254] In some embodiments, although the descriptions of pharmaceutical compositions provided herein are principally directed to pharmaceutical compositions that are suitable for ethical administration to humans, it will be understood by the skilled artisan that such compositions are generally suitable for administration to animals of all sorts. In some embodiments, modification of pharmaceutical compositions suitable for administration to humans in order to render the compositions suitable for administration to various animals is well understood, and the ordinarily skilled veterinary pharmacologist may design and perform such modification with merely ordinary, if any, experimentation. In some embodiments, subjects to which administration of the pharmaceutical compositions of the disclosure is contemplated include humans and other primates, mammals including commercially relevant mammals such as cattle, pigs, horses, sheep, cats, and dogs.

[0255] In some of any embodiments, the compositions of the disclosure are formulated using one or more pharmaceutically acceptable excipients or carriers. In some embodiments, the pharmaceutical compositions of the disclosure comprise a therapeutically effective amount of a targeted lipid particle of the disclosure and a pharmaceutically acceptable carrier. In some embodiments, pharmaceutically acceptable carriers that are useful, include, but are not limited to, glycerol, water, saline, ethanol, and other pharmaceutically acceptable salt solutions such as phosphates and salts of organic acids. Examples of these and other pharmaceutically acceptable carriers are described in Remington's Pharmaceutical Sciences (1991, Mack Publication Co., New Jersey).

[0256] In some embodiments, the carrier is a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. In some embodiments, the proper fluidity is maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. In some embodiments, prevention of the action of microorganisms is achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In some embodiments, it is preferable to include isotonic agents, for example, sugars, sodium chloride, or polyalcohols such as mannitol and sorbitol, in the composition. In some embodiments, prolonged absorption of the injectable compositions is brought about by including in the composition an agent that delays absorption, for example, aluminum monostearate or gelatin. In some embodiments, the pharmaceutically acceptable carrier is not DMSO alone.

[0257] In some embodiments, formulations are employed in admixtures with conventional excipients, i.e., pharmaceutically acceptable organic or inorganic carrier substances suitable for oral, vaginal, parenteral, nasal, intravenous, subcutaneous, enteral, or any other suitable mode of administration, known to the art. In some embodiments, the pharmaceutical preparations are sterilized and, if desired, mixed with auxiliary agents, e.g., lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure buffers, coloring, flavoring, and/or aromatic substances and the like. In some embodiments, pharmaceutical preparations are also combined with other active agents, e.g., other analgesic agents.

[0258] In some embodiments, “additional ingredients” include, but are not limited to, one or more of the following: excipients; surface active agents; dispersing agents; inert diluents; granulating and disintegrating agents; binding agents; lubricating agents; sweetening agents; flavoring agents; coloring agents; preservatives; physiologically degradable compositions such as gelatin; aqueous vehicles and solvents; oily vehicles and solvents; suspending agents; dispersing or wetting agents; emulsifying agents, demulcents; buffers; salts; thickening agents; fillers; emulsifying agents; antioxidants; antibiotics; antifungal agents; stabilizing agents; and pharmaceutically acceptable polymeric or hydrophobic materials. In some embodiments, “additional ingredients” that are included in the pharmaceutical compositions of the disclosure are known in the art and described, for example in Genaro, ed. (1985, Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pa.), which is incorporated herein by reference.

[0259] In some embodiments, the composition of the disclosure comprises a preservative from about 0.005% to 2.0% by total weight of the composition. In some embodiments, the preservative is used to prevent spoilage in the case of exposure to contaminants in the environment. In some embodiments, examples of preservatives useful in accordance with the disclosure included but are not limited to those selected from the group consisting of benzyl alcohol, sorbic acid, parabens, imidurea and combinations thereof. In some embodiments, a particularly preferred preservative is a combination of about 0.5% to 2.0% benzyl alcohol and 0.05% to 0.5% sorbic acid.
[0260] In some embodiments, liquid suspensions are prepared using conventional methods to achieve suspension of the targeted lipid particles of the disclosure in an aqueous or oily vehicle. In some embodiments, aqueous vehicles include, for example, water, and isotonic saline. In some embodiments, oily vehicles include, for example, almond oil, oily esters, ethyl alcohol, vegetable oils such as *arachis*, olive, sesame, or coconut oil, fractionated vegetable oils, and mineral oils such as liquid paraffin. In some embodiments, liquid suspensions further comprise one or more additional ingredients including, but not limited to, suspending agents, dispersing or wetting agents, emulsifying agents, demulcents, preservatives, buffers, salts, flavorings, coloring agents, and sweetening agents. In some embodiments, oily suspensions further comprise a thickening agent. In some embodiments, suspending agents include, but are not limited to, sorbitol syrup, hydrogenated edible fats, sodium alginate, polyvinylpyrrolidone, gum tragacanth, gum acacia, and cellulose derivatives such as sodium carboxymethylcellulose, methylcellulose, hydroxypropylmethylcellulose. In some embodiments, dispersing or wetting agents include, but are not limited to, naturally-occurring phosphatides such as lecithin, condensation products of an alkylene oxide with a fatty acid, with a long chain aliphatic alcohol, with a partial ester derived from a fatty acid and a hexitol, or with a partial ester derived from a fatty acid and a hexitol anhydride (e.g., polyoxyethylene stearate, heptadecaethyleneoxycetanol, polyoxyethylene sorbitol monooleate, and polyoxyethylene sorbitan monooleate, respectively). Known emulsifying agents include, but are not limited to, lecithin, and acacia. Known preservatives include, but are not limited to, methyl, ethyl, or n-propyl-para-hydroxybenzoates, ascorbic acid, and sorbic acid. Known sweetening agents include, for example, glycerol, propylene glycol, sorbitol, sucrose, and saccharin. Known thickening agents for oily suspensions include, for example, beeswax, hard paraffin, and cetyl alcohol.
[0261] In some embodiments, liquid solutions of the targeted lipid particles of the disclosure in aqueous or oily solvents are prepared in substantially the same manner as liquid suspensions, the primary difference being that the targeted lipid particles of the disclosure is dissolved, rather than suspended in the solvent. As used herein, an “oily” liquid is one which comprises a carbon-containing liquid molecule and which exhibits a less polar character than water. In some embodiments, liquid solutions of the pharmaceutical composition of the disclosure comprise each of the components described with regard to liquid suspensions, it being understood that suspending agents will not necessarily aid dissolution of the targeted lipid particles of the disclosure in the solvent. In some embodiments, aqueous solvents include, for example, water, and isotonic saline. In some embodiments, oily solvents include, for example, almond oil, oily esters, ethyl alcohol, vegetable oils such as *arachis*, olive, sesame, or coconut oil, fractionated vegetable oils, and mineral oils such as liquid paraffin.
[0262] In some embodiments, powdered and granular formulations of a pharmaceutical preparation of the disclosure are prepared using known methods. In some embodiments, formulations are administered directly to a subject, used, for example, to form tablets, to fill capsules, or to prepare an aqueous or oily suspension or solution by addition of an aqueous or oily vehicle thereto. In some of any embodiments, formulations further comprise one or more of dispersing or wetting agent, a suspending agent, and a preservative. Additional excipients, such as fillers and sweetening, flavoring, or coloring agents, are also included in these formulations.
[0263] In some embodiments, a pharmaceutical composition of the disclosure is also prepared,

packaged, or sold in the form of oil-in-water emulsion or a water-in-oil emulsion. In some embodiments, the oily phase is a vegetable oil such as olive or arachis oil, a mineral oil such as liquid paraffin, or a combination of these. In some embodiments, compositions further comprise one or more emulsifying agents such as naturally occurring gums such as gum acacia or gum tragacanth, naturally-occurring phosphatides such as soybean or lecithin phosphatide, esters or partial esters derived from combinations of fatty acids and hexitol anhydrides such as sorbitan monooleate, and condensation products of such partial esters with ethylene oxide such as polyoxyethylene sorbitan monooleate. In some embodiments, emulsions also contain additional ingredients including, for example, sweetening or flavoring agents.

Methods of Treatment

[0264] In some embodiments, the targeted lipid particles (e.g. targeted viral vectors) provided herein, or pharmaceutical compositions thereof as described herein are administered to a subject, e.g. a mammal, e.g. a human. In such embodiments, the subject is at risk of, has a symptom of, or is diagnosed with or identified as having, a particular disease or condition. In some embodiments, the subject has cancer. In some embodiments, the subject has an infectious disease. In some embodiments, the targeted lipid particle contains nucleic acid sequences encoding an exogenous agent for treating the disease or condition in the subject. For example, the exogenous agent is one that targets or is specific for a protein of a neoplastic cells and the targeted lipid particle is administered to a subject for treating a tumor or cancer in the subject. In another example, the exogenous agent is an inflammatory mediator or immune molecule, such as a cytokine, and targeted lipid particle is administered to a subject for treating any condition in which it is desired to modulate (e.g., increase) the immune response, such as a cancer or infectious disease. In some embodiments, the targeted lipid particle is administered in an effective amount or dose to effect treatment of the disease, condition, or disorder. Provided herein are uses of any of the provided targeted lipid particles in such methods and treatments, and in the preparation of a medicament in order to carry out such therapeutic methods. In some embodiments, the methods are carried out by administering the targeted lipid particle or compositions comprising the same, to the subject having, having had, or suspected of having the disease or condition or disorder. In some embodiments, the methods thereby treat the disease or condition or disorder in the subject. Also provided herein are uses of any of the compositions, such as pharmaceutical compositions provided herein, for the treatment of a disease, condition or disorder associated with a particular gene or protein targeted by or provided by the exogenous agent.

[0265] In some embodiments, the provided methods or uses involve administration of a pharmaceutical composition comprising oral, inhaled, transdermal or parenteral (including intravenous, intratumoral, intraperitoneal, intramuscular, intracavity, and subcutaneous) administration. In some embodiments, the targeted lipid particle is administered alone or formulated as a pharmaceutical composition. In some embodiments, the targeted lipid particle or compositions described herein are administered to a subject, e.g., a mammal, e.g., a human. In some of any embodiments, the subject is at risk of, has a symptom of, or is diagnosed with or identified as having, a particular disease or condition (e.g., a disease or condition described herein). In some embodiments, the disease is a disease or disorder. In some embodiments, the disease is a B cell malignancy.

[0266] In some embodiments, the targeted lipid particles are administered in the form of a unit-dose composition, such as a unit dose oral, parenteral, transdermal, or inhaled composition. In some embodiments, the compositions are prepared by admixture and are adapted for oral, inhaled, transdermal, or parenteral administration, and as such may be in the form of tablets, capsules, oral liquid preparations, powders, granules, lozenges, reconstitutable powders, injectable, and infusible solutions or suspensions, or suppositories or aerosols.

[0267] In some embodiments, the regimen of administration affects what constitutes an effective amount. In some embodiments, the therapeutic formulations are administered to the subject either prior to or after a diagnosis of disease. In some embodiments, several divided dosages, as well as staggered dosages are administered daily or sequentially, or the dose is continuously infused, or is a bolus injection. In some embodiments, the dosages of the therapeutic formulations are proportionally increased or decreased as indicated by the exigencies of the therapeutic or prophylactic situation.

[0268] In some embodiments, the administration of the compositions of the present disclosure to a subject, preferably a mammal, more preferably a human, are carried out using known procedures, at dosages and for periods of time effective to prevent or treat disease. In some embodiments, an effective amount of the targeted lipid particle of the disclosure necessary to achieve a therapeutic effect varies according to factors such as the activity of the particular lipid particle employed; the time of administration; the rate of excretion; the duration of the treatment; other drugs, compounds or materials used in combination with the targeted lipid particle of the disclosure; the state of the disease or disorder, age, sex, weight, condition, general health and prior medical history of the subject being treated, and like factors well-known in the medical arts. In some embodiments, the dosage regimens are adjusted to provide the optimum therapeutic response. In some embodiments, several divided doses are administered daily or the dose is proportionally reduced as indicated by the exigencies of the therapeutic situation. One of ordinary skill in the art would be able to study the relevant factors and make the determination regarding the effective amount of the therapeutic targeted lipid particle of the disclosure without undue experimentation.

[0269] In some embodiments, dosage levels of the targeted lipid particles in the pharmaceutical compositions of this disclosure are varied so as to obtain an amount that is effective to achieve the desired therapeutic response for a particular subject, composition, and mode of administration, without being toxic to the subject.

[0270] A medical doctor, e.g., physician or veterinarian, having ordinary skill in the art may readily determine and prescribe the effective amount of the pharmaceutical composition required. In some embodiments, the physician or veterinarian could start doses of the targeted lipid particles of the disclosure employed in the pharmaceutical composition at levels lower than that required in order to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved.

[0271] In some embodiments, the term “container” includes any receptacle for holding the pharmaceutical composition. In some embodiments, the container is the packaging that contains the pharmaceutical composition. In other embodiments, the container is not the packaging that contains the pharmaceutical composition, i.e., the container is a receptacle, such as a box or vial that contains the packaged pharmaceutical composition or unpackaged pharmaceutical composition and the instructions for use of the pharmaceutical composition. It should be understood that the instructions for use of the pharmaceutical composition may be contained on the packaging containing the pharmaceutical composition, and as such the instructions form an increased functional relationship to the packaged product. In some embodiments, instructions contain information pertaining to the pharmaceutical composition's ability to perform its intended function, e.g., treating or preventing a disease in a subject, or delivering an imaging or diagnostic agent to a subject.

[0272] In some embodiments, routes of administration of any of the compositions disclosed herein include oral, nasal, rectal, parenteral, sublingual, transdermal, transmucosal (e.g., sublingual, lingual, (trans) buccal, (trans) urethral, vaginal (e.g., trans- and perivaginally), (intra) nasal, and (trans) rectal), intravesical, intrapulmonary, intraduodenal, intragastrical, intrathecal, subcutaneous, intramuscular, intradermal, intra-arterial, intravenous, intrabronchial, inhalation, and topical administration.

[0273] In some of any embodiments, suitable compositions and dosage forms include, for example, tablets, capsules, caplets, pills, gel caps, troches, dispersions, suspensions, solutions, syrups, granules, beads, transdermal patches, gels, powders, pellets, magmas, lozenges, creams, pastes, plasters, lotions, discs, suppositories, liquid sprays for nasal or oral administration, dry powder or aerosolized formulations for inhalation, compositions and formulations for intravesical administration, and the like.

[0274] In some embodiments, the targeted lipid particle composition comprising an exogenous agent or cargo, are used to deliver such exogenous agent or cargo to a cell tissue or subject. In some embodiments, delivery of a cargo by administration of a targeted lipid particle composition described herein modify cellular protein expression levels. In certain embodiments, the administered composition directs upregulation (via expression in the cell, delivery in the cell, or induction within the cell) of one or more cargo (e.g., a polypeptide or mRNA) that provide a functional activity which is substantially absent or reduced in the cell in which the polypeptide is delivered. In some embodiments, the missing functional activity is enzymatic, structural, or regulatory in nature. In some embodiments, the administered composition directs up-regulation of one or more proteins that increases (e.g., synergistically) a functional activity which is present but substantially deficient in the cell in which the protein is upregulated. In some of any embodiments disclosed herein, the administered composition directs downregulation of (via expression in the cell, delivery in the cell, or induction within the cell) one or more cargo (e.g., a protein, siRNA, or miRNA) that repress a functional activity which is present or upregulated in the cell in which the protein, siRNA, or miRNA is delivered. In some embodiments, the upregulated functional activity is enzymatic, structural, or regulatory in nature. In some embodiments, the administered composition directs down-regulation of one or more proteins that decreases (e.g., synergistically) a functional activity which is present or upregulated in the cell in which the protein is downregulated. In some embodiments, the administered composition directs upregulation of certain functional activities and downregulation of other functional activities.

[0275] In some of any embodiments, the targeted lipid particle composition (e.g., one comprising mitochondria or DNA) mediates an effect on a target cell, and the effect lasts for at least 1, 2, 3, 4, 5, 6, or 7 days, 2, 3, or 4 weeks, or 1, 2, 3, 6, or 12 months. In some embodiments (e.g., wherein the targeted viral vector composition comprises an exogenous protein), the effect lasts for less than 1, 2, 3, 4, 5, 6, or 7 days, 2, 3, or 4 weeks, or 1, 2, 3, 6, or 12 months.

[0276] In some of any embodiments, the targeted lipid particle composition described herein is delivered ex-vivo to a cell or tissue, e.g., a human cell or tissue. In embodiments, the composition improves function of a cell or tissue ex-vivo, e.g., improves cell viability, respiration, or other function (e.g., another function described herein).

[0277] In some embodiments, the composition is delivered to an ex vivo tissue that is in an injured state (e.g., from trauma, disease, hypoxia, ischemia or other damage).

[0278] In some embodiments, the composition is delivered to an ex-vivo transplant (e.g., a tissue explant or tissue for transplantation, e.g., a human vein, a musculoskeletal graft such as bone or tendon, cornea, skin, heart valves, nerves; or an isolated or cultured organ, e.g., an organ to be transplanted into a human, e.g., a human heart, liver, lung, kidney, pancreas, intestine, thymus, eye). In some embodiments, the composition is delivered to the tissue or organ before, during and/or after transplantation.

[0279] In some embodiments, the composition is delivered, administered, or contacted with a cell, e.g., a cell preparation. In some embodiments, the cell preparation is a cell therapy preparation (a cell preparation intended for administration to a human subject). In embodiments, the cell preparation comprises cells expressing a T-cell receptor (TCR) or chimeric antigen receptor (CAR), e.g., expressing a recombinant CAR. The cells expressing the CAR may be, e.g., T cells, Natural Killer (NK) cells, cytotoxic T lymphocytes (CTL), regulatory T cells. In embodiments, the cell preparation is a neural stem cell preparation. In embodiments, the cell preparation is a mesenchymal stem cell (MSC) preparation. In embodiments, the cell preparation is a hematopoietic stem cell (HSC) preparation. In embodiments, the cell preparation is an islet cell preparation.

[0280] In some embodiments, the viral vector comprising an anti-CD4 sdAb or scFv composition described herein is used to deliver a CAR or TCR. In some embodiments, the viral vector transduces a cell expressing CD4 (e.g., a CD4+ T cell) and expresses and amplifies the CAR or TCR. The amplified CAR or TCR T cells then mediate targeted cell killing. Thus, the disclosure includes the use of viral vector comprising an anti-CD4 scFv fusogen construct to elicit an immune response specific to the antigen binding moiety of the CAR or TCR. In some embodiments, the CAR is used to target a tumor antigen selected from CD19, CD20, CD22, or BCMA. In another embodiment, the CAR is engineered to comprise an intracellular signaling domain of the T cell antigen receptor complex zeta chain (e.g., CD3 zeta). In a preferred embodiment, the intracellular domain is selected from a CD137 (4-1BB) signaling domain, a CD28 signaling domain, and a CD3zeta signaling domain.

Engineered Receptor Payloads

[0281] In some embodiments, the targeted lipid particles (e.g. targeted viral vectors) disclosed herein encode an engineered receptor. In some embodiments, the cells for use in or administered in connection with the provided methods contain or are engineered to contain an engineered receptor, e.g., an engineered antigen receptor, such as a chimeric antigen receptor (CAR). Also provided are populations of such cells, compositions containing such cells and/or enriched for such cells, such as in which cells of a certain type such as T cells or CD4+ cells are enriched or selected. Among the compositions are pharmaceutical compositions and formulations for administration, such as for adoptive cell therapy. Also provided are therapeutic methods for administering the cells and compositions to subjects, e.g., patients, in accord with the provided methods, and/or with the provided articles of manufacture or compositions.

[0282] In some embodiments, gene transfer is accomplished without first stimulating the cells, such as by combining it with a stimulus that induces a response such as proliferation, survival, and/or activation, e.g., as measured by expression of a cytokine or activation marker, followed by introduction of the nucleic acids, e.g., by transduction, into the stimulated cells, and optionally incubation or expansion in culture to numbers sufficient for clinical applications.

[0283] The viral vectors may express recombinant receptors, such as antigen receptors including chimeric antigen receptors (CARs), and other antigen-binding receptors such as transgenic T cell receptors (TCRs). Also among the receptors are other chimeric receptors.

a. Chimeric Antigen Receptors (CARs)

[0284] In some embodiments of the provided methods and uses, chimeric receptors, such as a CARs, contain one or more domains that combine an antigen- or ligand-binding domain (e.g. antibody or antibody fragment) that provides specificity for a desired antigen (e.g., tumor antigen) with intracellular signaling domains. In some embodiments, the intracellular signaling domain is a stimulating or an activating intracellular domain portion, such as a T cell stimulating or activating domain, providing a primary activation signal or a primary signal. In some embodiments, the intracellular signaling domain contains or additionally contains a costimulatory signaling domain to facilitate effector functions. In some embodiments, chimeric receptors when genetically engineered into immune cells modulate T cell activity, and, in some cases, modulate T cell differentiation or homeostasis, thereby resulting in genetically engineered cells with improved longevity, survival and/or persistence in vivo, such as for use in adoptive cell therapy methods.

[0285] Exemplary antigen receptors, including CARs, and methods for engineering and introducing such receptors into cells, include those described, for example, in W0200014257, WO2013126726, WO2012/129514, WO2014031687, WO2013/166321, WO2013/071154, WO2013/123061, U.S. patent app. Pub. Nos. US2002131960, US2013287748, US20130149337, U.S. Pat. Nos. 6,451,995, 7,446,190, 8,252,592, 8,339,645, 8,398,282, 7,446,179, 6,410,319, 7,070,995, 7,265,209, 7,354,762, 7,446,191, 8,324,353, and 8,479,118, and European patent app. No. EP2537416, and/or those described by Sadelain et al., Cancer Discov. 2013 April; 3(4): 388-398; Davila et al. (2013) PLOS ONE 8(4):e61338; Turtle et al., Curr. Opin. Immunol., 2012 October; 24(5): 633-39; Wu et al., Cancer, 2012 Mar. 18(2): 160-75. In some aspects, the antigen receptors include a CAR as described in U.S. Pat. No. 7,446,190, and those described in WO/2014055668. Examples of the CARs include CARs as disclosed in any of the aforementioned publications, such as WO2014031687, U.S. Pat. Nos. 8,339,645, 7,446,179, US 2013/0149337, U.S. Pat. Nos. 7,446,190, 8,389,282, Kochenderfer et al., (2013) Nature Reviews Clinical Oncology, 10, 267-276; Wang et al. (2012) J. Immunother. 35(9): 689-701; and Brentjens et al., Sci Transl Med. 2013 5(177). See also WO2014031687, U.S. Pat. Nos. 8,339,645, 7,446,179, US 2013/0149337, U.S. Pat. Nos. 7,446,190, and 8,389,282. The recombinant receptors, such as CARs, generally include an extracellular antigen binding domain, such as a portion of an antibody molecule, generally a variable heavy (VH) chain region and/or variable light (VL) chain region of the antibody, e.g., an scFv antibody fragment. In some embodiments, the antigen binding domain of the CAR molecule comprises an antibody, an antibody fragment, an scFv, a Fv, a Fab, a (Fab′)2, a single domain antibody (SdAb), a VH or VL domain, or a camelid VHH domain.

[0286] In some embodiments, the antigen targeted by the receptor is a polypeptide. In some embodiments, it is a carbohydrate or other molecule. In some embodiments, the antigen is selectively expressed or overexpressed on cells of the disease or condition, e.g., the tumor or pathogenic cells, as compared to normal or non-targeted cells or tissues. In other embodiments, the antigen is expressed on normal cells and/or is expressed on the engineered cells.

[0287] In some embodiments, the antigen targeted by the receptor includes antigens associated with a B cell malignancy, such as any of a number of known B cell marker. In some embodiments, the antigen targeted by the receptor is CD20, CD19, CD22, ROR1, CD45, CD47, CD21, CD5, CD33, Igkappa, Iglambda, CD79a, CD79b or CD30.

[0288] In some embodiments, the chimeric antigen receptor includes an extracellular portion containing an antibody or antibody fragment. In some aspects, the chimeric antigen receptor includes an extracellular portion containing the antibody or fragment and an intracellular signaling domain. In some embodiments, the antibody or fragment includes an scFv.

[0289] In some embodiments, the antigen targeted by the antigen-binding domain is CD19. In some aspects, the antigen-binding domain of the recombinant receptor, e.g., CAR, and the antigen-binding domain binds, such as specifically binds or specifically recognizes, a CD19, such as a human CD19. In some embodiments, the scFv contains a VH and a VL derived from an antibody or an antibody fragment specific to CD19. In some embodiments, the antibody or antibody fragment that binds CD19 is a mouse derived antibody such as FMC63 and SJ25C1. In some embodiments, the antibody or antibody fragment is a human antibody, e.g., as described in U.S. Patent Publication No. US 2016/0152723.

[0290] In some embodiments, the antigen is CD19. In some embodiments, the scFv contains a VH and a VL derived from an antibody or an antibody fragment specific to CD19. In some embodiments, the antibody or antibody fragment that binds CD19 is a mouse derived antibody such as FMC63 and SJ25C1. In some embodiments, the antibody or antibody fragment is a human antibody, e.g., as described in U.S. Patent Publication No. US 2016/0152723.

[0291] In some embodiments, the scFv is derived from FMC63. FMC63 generally refers to a mouse monoclonal IgGI antibody raised against Nalm-1 and -16 cells expressing CD19 of human origin (Fing, N. R., et al. (1987). Leucocyte typing III. 302).

[0292] In some embodiments, the antigen targeted by the antigen-binding domain is BCMA. In some aspects, the antigen-binding domain of the recombinant receptor, e.g., CAR, and the antigen-binding domain binds, such as specifically binds or specifically recognizes, a BCMA, such as a human BCMA. In some embodiments, the antigen-binding domain is a fully human VH sdAb disclosed in US2020/0138865 (disclosed herein by reference in its entirety), e.g., FHVH74, FHVH32, FHVH33, or FHVH93.

Antigen Binding Domain (ABD) Targets an Antigen Characteristic of a Neoplastic or Cancer Cell

[0293] In some embodiments, the antigen binding domain (ABD) targets an antigen characteristic of a neoplastic cell. In other words, the antigen binding domain targets an antigen expressed by a neoplastic or cancer cell. In some embodiments, the ABD binds a tumor associated antigen. In some embodiments, the antigen characteristic of a neoplastic cell (e.g., antigen associated with a neoplastic or cancer cell) or a tumor associated antigen is selected from a cell surface receptor, an ion channel-linked receptor, an enzyme-linked receptor, a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, histidine kinase associated receptor, epidermal growth factor receptors (EGFR) (including ErbB1/EGFR, ErbB2/HER2, ErbB3/HER3, and ErbB4/HER4), fibroblast growth factor receptors (FGFR) (including FGF1, FGF2, FGF3, FGF4, FGF5, FGF6, FGF7, FGF18, and FGF21), vascular endothelial growth factor receptors (VEGFR) (including VEGF-A, VEGF-B, VEGF-C, VEGF-D, and PIGF), RET Receptor and the Eph Receptor Family (including EphA1, EphA2, EphA3, EphA4, EphA5, EphA6, EphA7, EphA8, EphA9, EphA10, EphB1, EphB2. EphB3, EphB4, and EphB6), CXCR1, CXCR2, CXCR3, CXCR4, CXCR6, CCR1, CCR2, CCR3, CCR4, CCR5, CCR6, CCR8, CFTR, CIC-1, CIC-2, CIC-4, CIC-5, CIC-7, CIC-Ka, CIC-Kb, Bestrophins, TMEM16A, GABA receptor, glycin receptor, ABC transporters, NAV1.1, NAV1.2, NAV1.3, NAV1.4, NAV1.5, NAV1.6, NAV1.7, NAV1.8, NAV1.9, sphingosin-1-phosphate receptor (S1P1R), NMDA channel, transmembrane protein, multispan transmembrane protein, T-cell receptor motifs, T-cell alpha chains, T-cell β chains, T-cell γ chains, T-cell δ chains, CCR7, CD3, CD4, CD5, CD7, CD8, CD11b, CD11c, CD16, CD19, CD20, CD21, CD22, CD25, CD28, CD34, CD35, CD40, CD45RA, CD45RO, CD52, CD56, CD62L, CD68, CD80, CD95, CD117, CD127, CD133, CD137 (4-1BB), CD163, F4/80, IL-4Ra, Sca-1, CTLA-4, GITR, GARP, LAP, granzyme B, LFA-1, transferrin receptor, NKp46, perforin, CD4+, Th1, Th2, Th17, Th40, Th22, Th9, Tfh, canonical Treg. FoxP3+, Tr1, Th3, Treg17, TREG; CDCP, NT5E, EpCAM, CEA, gpA33, mucins, TAG-72, carbonic anhydrase IX, PSMA, folate binding protein, gangliosides (e.g., CD2, CD3, GM2), Lewis-γ.sup.2, VEGF, VEGFR 1/2/3, αVβ3, α5β1, ErbB1/EGFR, ErbB1/HER2, ErB3, c-MET, IGF1R, EphA3, TRAIL-R1, TRAIL-R2, RANKL, FAP, Tenascin, PDL-1, BAFF, HDAC, ABL, FLT3, KIT, MET, RET, IL-1β, ALK, RANKL, mTOR, CTLA-4, IL-6, IL-6R, JAK3, BRAF, PTCH, Smoothened, PIGF, ANPEP, TIMP1, PLAUR, PTPRJ, LTBR, ANTXR1, folate receptor alpha (FRa), ERBB2 (Her2/neu), EphA2, IL-13Ra2, epidermal growth factor receptor (EGFR), mesothelin, TSHR, CD19, CD123, CD22, CD30, CD171, CS-1, CLL-1, CD33, EGFRvIII, GD2, GD3, BCMA, MUC16 (CA125), L1CAM, LeY, MSLN, IL13Rα1, L1-CAM, Tn Ag, prostate specific membrane antigen (PSMA), ROR1, FLT3, FAP, TAG72, CD38, CD44v6, CEA, EPCAM, B7H3, KIT, interleukin-11 receptor a (IL-11Ra), PSCA, PRSS21, VEGFR2, LewisY, CD24, platelet-derived growth factor receptor-beta (PDGFR-beta), SSEA-4, CD20, MUC1, NCAM, Prostase, PAP, ELF2M, Ephrin B2, IGF-1 receptor, CAIX, LMP2, gpl00, bcr-abl, tyrosinase, Fucosyl GM1, sLe, GM3, TGS5, HMWMAA, o-acetyl-GD2, folate receptor beta, TEM1/CD248, TEM7R, CLDN6, GPRC5D, CXORF61, CD97, CD179a, ALK, Polysialic acid, PLACI, GloboH, NY-BR-1, UPK2, HAVCR1, ADRB3, PANX3, GPR20, LY6K, OR51E2, TARP, WT1, NY-ESO-1, LAGE-la, MAGE-A1, legumain, HPV E6, E7, ETV6-AML, sperm protein 17, XAGE1, Tie 2, MAD-CT-1, MAD-CT-2, major histocompatibility complex class I-related gene protein (MR1), urokinase-type plasminogen activator receptor (uPAR), Fos-related antigen 1, p53, p53 mutant, prostein, survivin, telomerase, PCTA-1/Galectin 8, MelanA/MART1, Ras mutant, hTERT, sarcoma translocation breakpoints, ML-IAP, ERG (TMPRSS2 ETS fusion gene), NA17, PAX3, androgen receptor, cyclin B1, MYCN, RhoC, TRP-2, CYPIB I, BORIS, SART3, PAX5, OY-TES1, LCK, AKAP-4, SSX2, RAGE-1, human telomerase reverse transcriptase, RU1, RU2, intestinal carboxyl esterase, mut hsp70-2, CD79a, CD79b, CD72, LAIR1, FCAR, LILRA2, CD300LF, CLEC12A, BST2, EMR2, LY75, GPC3, FCRL5, IGLL1, a neoantigen, CD133, CD15, CD184, CD24, CD56, CD26, CD29, CD44, HLA-A, HLA-B, HLA-C, (HLA-A,B,C) CD49f, CD151 CD340, CD200, tkrA, trkB, or trkC, or an antigenic fragment or antigenic portion thereof.

ABD Targets an Antigen Characteristic of a T Cell

[0294] In some embodiments, the antigen binding domain targets an antigen characteristic of a T cell. In some embodiments, the ABD binds an antigen associated with a T cell. In some instances, such an antigen is expressed by a T cell or is located on the surface of a T cell. In some embodiments, the antigen characteristic of a T cell or the T cell associated antigen is selected from a cell surface receptor, a membrane transport protein (e.g., an active or passive transport protein such as, for example, an ion channel protein, a pore-forming protein, etc.), a transmembrane receptor, a membrane enzyme, and/or a cell adhesion protein characteristic of a T cell. In some embodiments, an antigen characteristic of a T cell is a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, histidine kinase associated receptor, AKT1; AKT2; AKT3; ATF2; BCL10; CALM1; CD3D (CD30); CD3E (CD38); CD3G (CD3γ); CD4; CD8; CD28; CD45; CD80 (B7-1); CD86 (B7-2); CD247 (CD32); CTLA-4 (CD152); ELK1; ERK1 (MAPK3); ERK2; FOS; FYN; GRAP2 (GADS); GRB2; HLA-DRA; HLA-DRB1; HLA-DRB3; HLA-DRB4; HLA-DRB5; HRAS; IKBKA (CHUK); IKBKB; IKBKE; IKBKG (NEMO); IL2; ITPR1; ITK; JUN; KRAS2; LAT; LCK; MAP2K1 (MEK1); MAP2K2 (MEK2); MAP2K3 (MKK3); MAP2K4 (MKK4); MAP2K6 (MKK6); MAP2K7 (MKK7); MAP3K1 (MEKK1); MAP3K3; MAP3K4; MAP3K5; MAP3K8; MAP3K14 (NIK); MAPK8 (JNK1); MAPK9 (JNK2); MAPK10 (JNK3); MAPK11 (p383); MAPK12 (p38γ); MAPK13 (p380); MAPK14 (p38a); NCK; NFAT1; NFAT2; NFKB1; NFKB2; NFKBIA; NRAS; PAK1; PAK2; PAK3; PAK4; PIK3C2B; PIK3C3 (VPS34); PIK3CA; PIK3CB; PIK3CD; PIK3R1; PKCA; PKCB; PKCM; PKCQ; PLCY1; PRF1 (Perforin); PTEN; RAC1; RAF1; RELA; SDF1; SHP2; SLP76; SOS; SRC; TBK1; TCRA; TEC; TRAF6; VAV1; VAV2; or ZAP70.

ABD Targets an Antigen Characteristic of an Autoimmune or Inflammatory Disorder

[0295] In some embodiments, the antigen binding domain targets an antigen characteristic of an autoimmune or inflammatory disorder. In some embodiments, the ABD binds an antigen associated with an autoimmune or inflammatory disorder. In some instances, the antigen is expressed by a cell associated with an autoimmune or inflammatory disorder. In some embodiments, the autoimmune or inflammatory disorder is selected from chronic graft-vs-host disease (GVHD), lupus, arthritis, immune complex glomerulonephritis, goodpasture syndrome, uveitis, hepatitis, systemic sclerosis or scleroderma, type I diabetes, multiple sclerosis, cold agglutinin disease, Pemphigus vulgaris, Grave's disease, autoimmune hemolytic anemia, Hemophilia A, Primary Sjogren's Syndrome, thrombotic thrombocytopenia purrpura, neuromyelits optica, Evan's syndrome, IgM mediated neuropathy, cryoglobulinemia, dermatomyositis, idiopathic thrombocytopenia, ankylosing spondylitis, bullous pemphigoid, acquired angioedema, chronic urticarial, antiphospholipid demyelinating polyneuropathy, and autoimmune thrombocytopenia or neutropenia or pure red cell aplasias, while exemplary non-limiting examples of alloimmune diseases include allosensitization (see, for example, Blazar et al., 2015, Am. J. Transplant, 15(4):931-41) or xenosensitization from hematopoietic or solid organ transplantation, blood transfusions, pregnancy with fetal allosensitization, neonatal alloimmune thrombocytopenia, hemolytic disease of the newborn, sensitization to foreign antigens such as can occur with replacement of inherited or acquired deficiency disorders treated with enzyme or protein replacement therapy, blood products, and gene therapy. In some embodiments, the antigen characteristic of an autoimmune or inflammatory disorder is selected from a cell surface receptor, an ion channel-linked receptor, an enzyme-linked receptor, a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, or histidine kinase associated receptor.

[0296] In some embodiments, an antigen binding domain of a CAR binds to a ligand expressed on B cells, plasma cells, or plasmablasts. In some embodiments, an antigen binding domain of a CAR binds to CD10, CD19, CD20, CD22, CD24, CD27, CD38, CD45R, CD138, CD319, BCMA, CD28, TNF, interferon receptors, GM-CSF, ZAP-70, LFA-1, CD3 gamma, CD5 or CD2. See, e.g., US 2003/0077249; WO 2017/058753; WO 2017/058850, the contents of which are herein incorporated by reference.

ABD Targets an Antigen Characteristic of Senescent Cells

[0297] In some embodiments, the antigen binding domain targets an antigen characteristic of senescent cells, e.g., urokinase-type plasminogen activator receptor (uPAR). In some embodiments, the ABD binds an antigen associated with a senescent cell. In some instances, the antigen is expressed by a senescent cell. In some embodiments, the CAR is used for treatment or prophylaxis of disorders characterized by the aberrant accumulation of senescent cells, e.g., liver and lung fibrosis, atherosclerosis, diabetes and osteoarthritis.

ABD Targets an Antigen Characteristic of an Infectious Disease

[0298] In some embodiments, the antigen binding domain targets an antigen characteristic of an infectious disease. In some embodiments, the ABD binds an antigen associated with an infectious disease. In some instances, the antigen is expressed by a cell affected by an infectious disease. In some embodiments, wherein the infectious disease is selected from HIV, hepatitis B virus, hepatitis C virus, Human herpes virus, Human herpes virus 8 (HHV-8, Kaposi sarcoma-associated herpes virus (KSHV)), Human T-lymphotrophic virus-1 (HTLV-1), Merkel cell polyomavirus (MCV), Simian virus 40 (SV40), Epstein-Barr virus, CMV, human papillomavirus. In some embodiments, the antigen characteristic of an infectious disease is selected from a cell surface receptor, an ion channel-linked receptor, an enzyme-linked receptor, a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, histidine kinase associated receptor, HIV Env, gpl20, or CD4-induced epitope on HIV-1 Env.

ABD Binds to a Cell Surface Antigen of a Cell

[0299] In some embodiments, an antigen binding domain binds to a cell surface antigen of a cell. In some embodiments, a cell surface antigen is characteristic of (e.g., expressed by) a particular or specific cell type. In some embodiments, a cell surface antigen is characteristic of more than one type of cell.

[0300] In some embodiments, a CAR antigen binding domain binds a cell surface antigen characteristic of a T cell, such as a cell surface antigen on a T cell. In some embodiments, an antigen characteristic of a T cell is a cell surface receptor, a membrane transport protein (e.g., an active or passive transport protein such as, for example, an ion channel protein, a pore-forming protein, etc.), a transmembrane receptor, a membrane enzyme, and/or a cell adhesion protein characteristic of a T cell. In some embodiments, an antigen characteristic of a T cell is a G protein-coupled receptor, receptor tyrosine kinase, tyrosine kinase associated receptor, receptor-like tyrosine phosphatase, receptor serine/threonine kinase, receptor guanylyl cyclase, or histidine kinase associated receptor.

[0301] In some embodiments, an antigen binding domain of a CAR binds a T cell receptor. In some embodiments, a T cell receptor is AKT1; AKT2; AKT3; ATF2; BCL10; CALM1; CD3D (CD30); CD3E (CD38); CD3G (CD3γ); CD4; CD8; CD28; CD45; CD80 (B7-1); CD86 (B7-2); CD247 (CD3ζ); CTLA-4 (CD152); ELK1; ERK1 (MAPK3); ERK2; FOS; FYN; GRAP2 (GADS); GRB2; HLA-DRA; HLA-DRB1; HLA-DRB3; HLA-DRB4; HLA-DRB5; HRAS; IKBKA (CHUK); IKBKB; IKBKE; IKBKG (NEMO); IL2; ITPR1; ITK; JUN; KRAS2; LAT; LCK; MAP2K1 (MEK1); MAP2K2 (MEK2); MAP2K3 (MKK3); MAP2K4 (MKK4); MAP2K6 (MKK6); MAP2K7 (MKK7); MAP3K1 (MEKK1); MAP3K3; MAP3K4; MAP3K5; MAP3K8; MAP3K14 (NIK); MAPK8 (JNK1); MAPK9 (JNK2); MAPK10 (JNK3); MAPK11 (p38B); MAPK12 (p38γ); MAPK13 (p380); MAPK14 (p38a); NCK; NFAT1; NFAT2; NFKB1; NFKB2; NFKBIA; NRAS; PAK1; PAK2; PAK3; PAK4; PIK3C2B; PIK3C3 (VPS34); PIK3CA; PIK3CB; PIK3CD; PIK3R1; PKCA; PKCB; PKCM; PKCQ; PLCY1; PRF1 (Perforin); PTEN; RAC1; RAF1; RELA; SDF1; SHP2; SLP76; SOS; SRC; TBK1; TCRA; TEC; TRAF6; VAV1; VAV2; or ZAP70.

Transmembrane Domain

[0302] In some embodiments, the CAR transmembrane domain comprises at least a transmembrane region of the alpha, beta or zeta chain of a T cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, or functional variant thereof. In some embodiments, the transmembrane domain comprises at least a transmembrane region(s) of CD4, 4-1BB/CD137, CD28, CD34, CD4, FcεRIγ, CD16, OX40/CD134, CD3ζ, CD3ε, CD3γ, CD3δ, TCRα, TCRβ, TCRζ, CD32, CD64, CD64, CD45, CD5, CD9, CD22, CD37, CD80, CD86, CD40, CD40L/CD154, VEGFR2, FAS, and FGFR2B, or functional variant thereof. antigen binding domain binds

Signaling Domain or Plurality of Signaling Domains

[0303] In some embodiments, a CAR described herein comprises one or at least one signaling domain selected from one or more of B7-1/CD80; B7-2/CD86; B7-H1/PD-L1; B7-H2; B7-H3; B7-H4; B7-H6; B7-H7; BTLA/CD272; CD28; CTLA-4; Gi24/VISTA/B7-H5; ICOS/CD278; PD-1; PD-L2/B7-DC; PDCD6); 4-1BB/TNFSF9/CD137; 4-1BB Ligand/TNFSF9; BAFF/BLyS/TNFSF13B; BAFF R/TNFRSF13C; CD27/TNFRSF7; CD27 Ligand/TNFSF7; CD30/TNFRSF8; CD30 Ligand/TNFSF8; CD40/TNFRSF5; CD40/TNFSF5; CD40 Ligand/TNFSF5; DR3/TNFRSF25; GITR/TNFRSF18; GITR Ligand/TNFSF18; HVEM/TNFRSF14; LIGHT/TNFSF14; Lymphotoxin-alpha/TNF-beta; OX40/TNFRSF4; OX40 Ligand/TNFSF4; RELT/TNFRSF19L; TACI/TNFRSF13B; TL1A/TNFSF15; TNF-alpha; TNF RII/TNFRSF1B); 2B4/CD244/SLAMF4; BLAME/SLAMF8; CD2; CD2F-10/SLAMF9; CD48/SLAMF2; CD58/LFA-3; CD84/SLAMF5; CD229/SLAMF3; CRACC/SLAMF7; NTB-A/SLAMF6; SLAM/CD150); CD2; CD7; CD53; CD82/Kai-1; CD90/Thyl; CD96; CD160; CD200; CD300a/LMIR1; HLA Class I; HLA-DR; Ikaros; Integrin alpha 4/CD49d; Integrin alpha 4 beta 1; Integrin alpha 4 beta 7/LPAM-1; LAG-3; TCL1A; TCL1B; CRTAM; DAP12; Dectin-1/CLEC7A; DPPIV/CD26; EphB6; TIM-1/KIM-1/HAVCR; TIM-4; TSLP; TSLP R; lymphocyte function associated antigen-1 (LFA-1); NKG2C, a CD3 zeta domain, an immunoreceptor tyrosine-based activation motif (ITAM), CD27, CD28, 4-1BB, CD134/OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83, or functional fragment thereof.

[0304] In some embodiments, the at least one signaling domain comprises a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In other embodiments, the at least one signaling domain comprises (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof. In yet other embodiments, the at least one signaling domain comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof. In some embodiments, the at least one signaling domain comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

[0305] In some embodiments, the at least two signaling domains comprise a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In other embodiments, the at least two signaling domains comprise (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof. In yet other embodiments, the at least one signaling domain comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof. In some embodiments, the at least two signaling domains comprise a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

[0306] In some embodiments, the at least three signaling domains comprise a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In other embodiments, the at least three signaling domains comprise (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof. In yet other embodiments, the least three signaling domains comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof. In some embodiments, the at least three signaling domains comprise a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

[0307] In some embodiments, the CAR comprises a CD3 zeta domain or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof. In some embodiments, the CAR comprises (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; and (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof.

[0308] In some embodiments, the CAR comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; and (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof.

[0309] In some embodiments, the CAR comprises (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain, or a 4-1BB domain, or functional variant thereof, and/or (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof.

[0310] In some embodiments, the CAR comprises a (i) a CD3 zeta domain, or an immunoreceptor tyrosine-based activation motif (ITAM), or functional variant thereof; (ii) a CD28 domain or functional variant thereof; (iii) a 4-1BB domain, or a CD134 domain, or functional variant thereof; and (iv) a cytokine or costimulatory ligand transgene.

Domain which Upon Successful Signaling of the CAR Induces Expression of a Cytokine Gene

[0311] In some embodiments, a first, second, third, or fourth generation CAR further comprises a domain which upon successful signaling of the CAR induces expression of a cytokine gene. In some embodiments, a cytokine gene is endogenous or exogenous to a target cell comprising a CAR which comprises a domain which upon successful signaling of the CAR induces expression of a cytokine gene. In some embodiments, a cytokine gene encodes a pro-inflammatory cytokine. In some embodiments, a cytokine gene encodes IL-1, IL-2, IL-9, IL-12, IL-18, TNF, or IFN-gamma, or functional fragment thereof. In some embodiments, a domain which upon successful signaling of the CAR induces expression of a cytokine gene is or comprises a transcription factor or functional domain or fragment thereof. In some embodiments, a domain which upon successful signaling of the CAR induces expression of a cytokine gene is or comprises a transcription factor or functional domain or fragment thereof. In some embodiments, a transcription factor or functional domain or fragment thereof is or comprises a nuclear factor of activated T cells (NFAT), an NF-kB, or functional domain or fragment thereof. See, e.g., Zhang. C. et al., Engineering CAR-T cells. Biomarker Research. 5:22 (2017); WO 2016126608; Sha, H. et al. Chimaeric antigen receptor T-cell therapy for tumour immunotherapy. Bioscience Reports Jan. 27, 2017, 37 (1).

[0312] In some embodiments, the CAR further comprises one or more spacers, e.g., wherein the spacer is a first spacer between the antigen binding domain and the transmembrane domain. In some embodiments, the first spacer includes at least a portion of an immunoglobulin constant region or variant or modified version thereof. In some embodiments, the spacer is a second spacer between the transmembrane domain and a signaling domain. In some embodiments, the second spacer is an oligopeptide, e.g., wherein the oligopeptide comprises glycine and serine residues such as but not limited to glycine-serine doublets. In some embodiments, the CAR comprises two or more spacers, e.g., a spacer between the antigen binding domain and the transmembrane domain and a spacer between the transmembrane domain and a signaling domain.

[0313] In some embodiments, any one of the cells described herein comprises a nucleic acid encoding a CAR or a first generation CAR. In some embodiments, a first generation CAR comprises an antigen binding domain, a transmembrane domain, and signaling domain. In some embodiments, a signaling domain mediates downstream signaling during T cell activation.

[0314] In some embodiments, the methods and compositions disclosed herein comprise a nucleic acid encoding a CAR or a second generation CAR. In some embodiments, a second generation CAR comprises an antigen binding domain, a transmembrane domain, and two signaling domains. In some embodiments, a signaling domain mediates downstream signaling during T cell activation. In some embodiments, a signaling domain is a costimulatory domain. In some embodiments, a costimulatory domain enhances cytokine production, CAR-T cell proliferation, and/or CAR-T cell persistence during T cell activation.

[0315] In some embodiments, any one of the compositions and methods described herein comprises a nucleic acid encoding a CAR or a third generation CAR. In some embodiments, a third generation CAR comprises an antigen binding domain, a transmembrane domain, and at least three signaling domains. In some embodiments, a signaling domain mediates downstream signaling during T cell activation. In some embodiments, a signaling domain is a costimulatory domain. In some embodiments, a costimulatory domain enhances cytokine production, CAR-T cell proliferation, and or CAR-T cell persistence during T cell activation. In some embodiments, a third generation CAR comprises at least two costimulatory domains. In some embodiments, the at least two costimulatory domains are not the same.

[0316] In some embodiments, any one of the compositions and methods described herein comprises a nucleic acid encoding a CAR or a fourth generation CAR. In some embodiments, a fourth generation CAR comprises an antigen binding domain, a transmembrane domain, and at least two, three, or four signaling domains. In some embodiments, a signaling domain mediates downstream signaling during T cell activation. In some embodiments, a signaling domain is a costimulatory domain. In some embodiments, a costimulatory domain enhances cytokine production, CAR-T cell proliferation, and or CAR-T cell persistence during T cell activation.

Abd Comprising an Antibody or Antigen-Binding Portion Thereof

[0317] In some embodiments, a CAR antigen binding domain is or comprises an antibody or antigen-binding portion thereof. In some embodiments, a CAR antigen binding domain is or comprises an scFv or Fab. In some embodiments, a CAR antigen binding domain comprises an scFv or Fab fragment of a CD19 antibody; CD22 antibody; T-cell alpha chain antibody; T-cell β chain antibody; T-cell γ chain antibody; T-cell δ chain antibody; CCR7 antibody; CD3 antibody; CD4 antibody; CD5 antibody; CD7 antibody; CD8 antibody; CD11b antibody; CD11c antibody; CD16 antibody; CD20 antibody; CD21 antibody; CD25 antibody; CD28 antibody; CD34 antibody; CD35 antibody; CD40 antibody; CD45RA antibody; CD45RO antibody; CD52 antibody; CD56 antibody; CD62L antibody; CD68 antibody; CD80 antibody; CD95 antibody; CD117 antibody; CD127 antibody; CD133 antibody; CD137 (4-1 BB) antibody; CD163 antibody; F4/80 antibody; IL-4Ra antibody; Sca-1 antibody; CTLA-4 antibody; GITR antibody GARP antibody; LAP antibody; granzyme B antibody; LFA-1 antibody; MR1 antibody; uPAR antibody; or transferrin receptor antibody.

[0318] In some embodiments, a CAR comprises a signaling domain which is a costimulatory domain. In some embodiments, a CAR comprises a second costimulatory domain. In some embodiments, a CAR comprises at least two costimulatory domains. In some embodiments, a CAR comprises at least three costimulatory domains. In some embodiments, a CAR comprises a costimulatory domain selected from one or more of CD27, CD28, 4-1BB, CD134/OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83. In some embodiments, if a CAR comprises two or more costimulatory domains, two costimulatory domains are different. In some embodiments, if a CAR comprises two or more costimulatory domains, two costimulatory domains are the same.

[0319] In addition to the CARs described herein, various chimeric antigen receptors and nucleotide sequences encoding the same are known in the art and would be suitable for fusosomal delivery and reprogramming of target cells in vivo and in vitro as described herein. See, e.g., WO2013040557; WO2012079000; WO2016030414; Smith T, et al., Nature Nanotechnology. 2017. DOI: 10.1038/NNANO.2017.57, the disclosures of which are herein incorporated by reference.

Additional Descriptions of CARS

[0320] In certain embodiments, the compositions and methods comprise a polynucleotide encoding a CAR. CARs (also known as chimeric immunoreceptors, chimeric T cell receptors, or artificial T cell receptors) are receptor proteins that have been engineered to give host cells (e.g., T cells) the new ability to target a specific protein. The receptors are chimeric because they combine both antigen-binding and T cell activating functions into a single receptor. The polycistronic vector of the present disclosure may be used to express one or more CARs in a host cell (e.g., a T cell) for use in therapies against various target antigens. The CARs expressed by the one or more expression cassettes may be the same or different. In these embodiments, the CAR comprises an extracellular binding domain (also referred to as a “binder”) that specifically binds a target antigen, a transmembrane domain, and an intracellular signaling domain. In certain embodiments, the CAR further comprises one or more additional elements, including one or more signal peptides, one or more extracellular hinge domains, and/or one or more intracellular costimulatory domains. Domains may be directly adjacent to one another, or there may be one or more amino acids linking the domains. The nucleotide sequence encoding a CAR may be derived from a mammalian sequence, for example, a mouse sequence, a primate sequence, a human sequence, or combinations thereof. In the cases where the nucleotide sequence encoding a CAR is non-human, the sequence of the CAR may be humanized. The nucleotide sequence encoding a CAR may also be codon-optimized for expression in a mammalian cell, for example, a human cell. In any of these embodiments, the nucleotide sequence encoding a CAR may be at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to any of the nucleotide sequences disclosed herein. The sequence variations may be due to codon-optimalization, humanization, restriction enzyme-based cloning scars, and/or additional amino acid residues linking the functional domains, etc.

[0321] In certain embodiments, the CAR comprises a signal peptide at the N-terminus. Non-limiting examples of signal peptides include CD4 signal peptide, IgK signal peptide, and granulocyte-macrophage colony-stimulating factor receptor subunit alpha (GMCSFR-α, also known as colony stimulating factor 2 receptor subunit alpha (CSF2RA)) signal peptide, and variants thereof, the amino acid sequences of which are provided in Table 3 below.

TABLE-US-00003 TABLE 3 Exemplary sequences of signal peptides SEQ ID NO: Sequence Description 14003 MALPVTALLLPLALLLHA CD8α signal ARP peptide 14004 METDTLLLWVLLLWVPGS IgK signal TG peptide 14005 MLLLVTSLLLCELPHPAF GMCSFR-α (CSF2RA) LLIP signal peptide

[0322] In certain embodiments, the extracellular binding domain of the CAR comprises one or more antibodies specific to one target antigen or multiple target antigens. The antibody may be an antibody fragment, for example, an scFv, or a single-domain antibody fragment, for example, a VHH. In certain embodiments, the scFv may comprise a heavy chain variable region (V.sub.H) and a light chain variable region (V.sub.L) of an antibody connected by a linker. The V.sub.H and the

V.sub.L may be connected in either order, i.e., V.sub.H-linker-V.sub.L or V.sub.L-linker-V.sub.H. Non-limiting examples of linkers include Whitlow linker, (G4S) n (n can be a positive integer, e.g., 1, 2, 3, 4, 5, 6, etc. (SEQ ID NO: 14127)) linker, and variants thereof. In certain embodiments, the antigen is an antigen that is exclusively or preferentially expressed on tumor cells, or an antigen that is characteristic of an autoimmune or inflammatory disease. Exemplary target antigens include, but are not limited to, CD5, CD19, CD20, CD22, CD23, CD30, CD70, Kappa, Lambda, and B cell maturation agent (BCMA), G-protein coupled receptor family C group 5 member D (GPRC5D) (associated with leukemias); CS1/SLAMF7, CD38, CD138, GPRC5D, TACI, and BCMA (associated with myelomas); GD2, HER2, EGFR, EGFRvIII, B7H3, PSMA, PSCA, CAIX, CD171, CEA, CSPG4, EPHA2, FAP, FRa, IL-13Ra, Mesothelin, MUC1, MUC16, and ROR1 (associated with solid tumors). In any of these embodiments, the extracellular binding domain of the CAR is codon-optimized for expression in a host cell or have variant sequences to increase functions of the extracellular binding domain.

[0323] In certain embodiments, the CAR comprises a hinge domain, also referred to as a spacer. The terms "hinge" and "spacer" may be used interchangeably in the present disclosure. Non-limiting examples of hinge domains include CD4 hinge domain, CD28 hinge domain, IgG4 hinge domain, IgG4 hinge-CH2-CH3 domain, and variants thereof, the amino acid sequences of which are provided in Table 4 below.

TABLE-US-00004 TABLE 4 Exemplary sequences of hinge domains SEQ ID NO: Sequence Description 14006 TTTPAPRPPTPAPTI-ASQPLSL CD8α hinge RPEACRPAAGGAVHTRGLDFACD domain 14007 IEVMYPPPYLDNEKSNGTIIHVK CD28 hinge GKHLCP-SPLFPGPSKP domain 14013 AAAIEVMYPPPYLDNEKSNGTII CD28 hinge HVKGKHLCP-SPLFPGPSKP domain 14008 ESKYGPPCPPCP IgG4 hinge domain 14009 ESKYGPPCPSCP IgG4 hinge domain 14010 ESKYGPPCPPCPAPEFLGGPSVF IgG4 hinge-LFPPKPKD-TLMISRTPEVTCVV CH2-CH3 VDVSQEDPEVQFNWY-VDGVEVH domain NAKTKPREEQFNSTYRVVSVLTV LHQDWLNGKEYKCKVSNKGLPSS IEK-TISKAKGQPREPQVYTLPP -SQEEMTKNQVSLTCLVKGFYPS DIAVEWESNGQPENNYKTTPPVL DSDGSFFLYSRL-TVDKSRWQEG NVFSCSVM-HEALHNHYTQKSLS LSLGK

[0324] In certain embodiments, the transmembrane domain of the CAR comprises a transmembrane region of the alpha, beta, or zeta chain of a T cell receptor, CD28, CD38, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, or a functional variant thereof, including the human versions of each of these sequences. In other embodiments, the transmembrane domain comprises a transmembrane region of CD4, 4-1BB/CD137, CD28, CD34, CD8α, CD8β, FcεRIγ, CD16, OX40/CD134, CD3ζ, CD3ε, CD3γ, CD3δ, TCRα, TCRβ, TCRζ, CD32, CD64, CD64, CD45, CD5, CD9, CD22, CD37, CD80, CD86, CD40, CD40L/CD154, VEGFR2, FAS, and FGFR2B, or a functional variant thereof, including the human versions of each of these sequences. Table 5 provides the amino acid sequences of a few exemplary transmembrane domains.

TABLE-US-00005 TABLE 5 Exemplary sequences of transmembrane domains SEQ ID NO: Sequence Description 14011 IYIWAPLAGTCGVL CD8α LLSLVITLYC transmembrane domain 14012 FWVLVVVGGVLACY CD28 SLLVTVAFIIFWV transmembrane domain 14014 MFWVLVVVGGVLAC CD28 YSLLVTVAFIIFWV transmembrane domain

[0325] In certain embodiments, the intracellular signaling domain and/or intracellular costimulatory domain of the CAR comprises one or more signaling domains selected from B7-1/CD80, B7-2/CD86, B7-H1/PD-L1, B7-H2, B7-H3, B7-H4, B7-H6, B7-H7, BTLA/CD272, CD28, CTLA-4, Gi24/VISTA/B7-H5, ICOS/CD278, PD-1, PD-L2/B7-DC, PDCD6, 4-1BB/TNFSF9/CD137, 4-1BB Ligand/TNFSF9, BAFF/BLyS/TNFSF13B, BAFF R/TNFRSF13C, CD27/TNFRSF7, CD27 Ligand/TNFSF7, CD30/TNFRSF8, CD30 Ligand/TNFSF8, CD40/TNFRSF5, CD40/TNFSF5, CD40 Ligand/TNFSF5, DR3/TNFRSF25, GITR/TNFRSF18, GITR Ligand/TNFSF18, HVEM/TNFRSF14, LIGHT/TNFSF14, Lymphotoxin-alpha/TNFβ, OX40/TNFRSF4, OX40 Ligand/TNFSF4, RELT/TNFRSF19L, TACI/TNFRSF13B, TL1A/TNFSF15, TNFα, TNF RII/TNFRSF1B, 2B4/CD244/SLAMF4, BLAME/SLAMF8, CD2, CD2F-10/SLAMF9, CD48/SLAMF2, CD58/LFA-3, CD84/SLAMF5, CD229/SLAMF3, CRACC/SLAMF7, NTB-A/SLAMF6, SLAM/CD150, CD2, CD7, CD53, CD82/Kai-1, CD90/Thyl, CD96, CD160, CD200, CD300a/LMIR1, HLA Class I, HLA-DR, Ikaros, Integrin alpha 4/CD49d, Integrin alpha 4 beta 1, Integrin alpha 4 beta 7/LPAM-1, LAG-3, TCL1A, TCL1B, CRTAM, DAP12, Dectin-1/CLEC7A, DPPIV/CD26, EphB6, TIM-1/KIM-1/HAVCR, TIM-4, TSLP, TSLP R, lymphocyte function associated antigen-1 (LFA-1), NKG2C, CD3ζ, an immunoreceptor tyrosine-based activation motif (ITAM), CD27, CD28, 4-1BB, CD134/OX40, CD30, CD40, PD-1, ICOS, lymphocyte function-associated antigen-1 (LFA-1), CD2, CD7, LIGHT, NKG2C, B7-H3, a ligand that specifically binds with CD83, and a functional variant thereof including the human versions of each of these sequences. In some embodiments, the intracellular signaling domain and/or intracellular costimulatory domain comprises one or more signaling domains selected from a CD37 domain, an ITAM, a CD28 domain, 4-1BB domain, or a functional variant thereof. Table 6 provides the amino acid sequences of a few exemplary intracellular costimulatory and/or signaling domains. In certain embodiments, as in the case of tisagenlecleucel as described below, the CD34 signaling domain of SEQ ID NO:14017 has a mutation, e.g., a glutamine (Q) to lysine (K) mutation, at amino acid position 14 (see SEQ ID NO:14018).

TABLE-US-00006 TABLE 6 Exemplary sequences of intracellular costimulatory and/or signaling domains SEQ ID NO: Sequence Description 14015 KRGRKKLLY-IFKQPFMRPVQ 4-1BB costimulatory TTQEEDGCSCRF-PEEEEGGC domain EL 14016 RSKRS-RLLHSDYMNMTPRR CD28 costimulatory PGPTRKHYQPYAPPRDFAAY domain RS 14017 RVKFSRSADAPA-YQQGQNQ CD3ζ signaling LYNELNLGR-REEYDVLDKR domain RGRDPEMGGKPRR-KNPQEG LYNEL-QKDKMAEAYSEIGM KGERRRGKGHDGLYQGLSTA TKDTYDALHMQALPPR 14018 RVKFSRSADAPA-YKQGQNQ CD3ζ signaling LYNELNLGR-REEYDVLDKR domain RGRDPEMGGKPRR-KNPQEG (with Q to K LYNEL-QKDKMAEAYSEIGM mutation at KGERRRGKGHDGLYQGLSTA position 14) TKDTYDALHMQALPPR

[0326] In certain embodiments where the polycistronic vector encodes two or more CARs, the two or more CARs comprise the same functional domains, or one or more different functional domains, as described. For example, the two or more CARs comprise different signal peptides, extracellular binding domains, hinge domains, transmembrane domains, costimulatory domains, and/or intracellular signaling domains, in order to minimize the risk of recombination due to sequence similarities. Or, alternatively, the two or more CARs comprise the same domains. In the cases where the same domain(s) and/or backbone are used, it is optional to introduce codon divergence at the nucleotide sequence level to minimize the risk of recombination.

CD19 CAR

[0327] In some embodiments, the CAR is a CD19 CAR (“CD19-CAR”), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR. In some embodiments, the CD19 CAR comprises a signal peptide, an extracellular binding domain that specifically binds CD19, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.

[0328] In some embodiments, the signal peptide of the CD19 CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO:14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO:14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.

[0329] In some embodiments, the extracellular binding domain of the CD19 CAR is specific to CD19, for example, human CD19. The extracellular binding domain of the CD19 CAR can be codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain. In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv.

[0330] In some embodiments, the extracellular binding domain of the CD19 CAR comprises an scFv derived from the FMC63 monoclonal antibody (FMC63), which comprises the heavy chain variable region (V.sub.H) and the light chain variable region (V.sub.L) of FMC63 connected by a linker. FMC63 and the derived scFv have been described in Nicholson et al., Mol. Immun. 34(16-17): 1157-1165 (1997) and PCT Application Publication No. WO2018/213337. In some embodiments, the amino acid sequences of the entire FMC63-derived scFv (also referred to as FMC63 scFv) and its different portions are provided in Table 7 below. In some embodiments, the CD19-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14019, 14020, or 14025, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO:14019, 14020, or 14025. In some embodiments, the CD19-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14021-14023 and 14026-14028. In some embodiments, the CD19-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14021-14023. In some embodiments, the CD19-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14026-14028. In any of these embodiments, the CD19-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD19 CAR comprises or consists of the one or more CDRs as described herein.

[0331] In some embodiments, the linker linking the V.sub.H and the V.sub.L portions of the scFv is a Whitlow linker having an amino acid sequence set forth in SEQ ID NO:14024. In some embodiments, the Whitlow linker is replaced by a different linker, for example, a 3×G4S linker (SEQ ID NO: 9313) having an amino acid sequence set forth in SEQ ID NO: 14030, which gives rise to a different FMC63-derived scFv having an amino acid sequence set forth in SEQ ID NO:14029. In certain of these embodiments, the CD19-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO:14029 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO:14029.

TABLE-US-00007 TABLE 7 Exemplary sequences of anti-CD19 scFv and components SEQ ID NO: Amino Acid Sequence Description 14019

DIQMTQTTSSLSASLGDRVTIS- Anti-CD19 FMC63 scFv CRASQDISKY-LNWYQQKPDGT entire sequence, with VKLLI-YHTSRLHSGVPSRFSG Whitlow linker SGSGTDYSLTISNLEQEDIATY FCQQGN-TLPYTFGGGTKLEIT-GSTSGSGKPGSGEGSTKGEV-K LQESGPGLVAPSQSLSVTCTVS GVSLPDYGVSWIRQP-PRKGLE LGVIWGSET-TYYNSALKSRLT IIKDNSKSQVFLK-MNSLQTDD TAIYYCAKHYYYGGSYAMDYWG QGTSVTVSS 14020 DIQMTQTTSSLSASLGDRVTIS-Anti-CD19 FMC63 scFv CRASQDISKY-LNWYQQKPDGTV light chain variable KLLI-YHTSRLHSGVPSRFSGSG region SGTDYSLTISNLEQEDIATYFCQ QGN-TLPYTFGGGTKLEIT 14021 QDISKY Anti-CD19 FMC63 scFv light chain CDR1 HTS Anti-CD19 FMC63 scFv light chain CDR2 14023 QQGNTLPYT Anti-CD19 FMC63 scFv light chain CDR3 14024 GSTSGSGKPGSGEGSTKG Whitlow linker 14025 EVKLQESGPGLVAP-SQSLSV Anti-CD19 FMC63 scFv TCTVSGVSLPDY-GVSWIRQP-heavy chain variable PRKGLEWLGVIWGSET-TYYN region SALKSRLTIIKDNSKSQVFLK MNSLQTDD-TAIYYCAKHYYY GGSYAMDYWGQGTSVTVSS 14026 GVSLPDYG Anti-CD19 FMC63 scFv heavy chain CDR1 14027 IWGSETT Anti-CD19 FMC63 scFv heavy chain CDR2 14028 AKHYYYGGSYAMDY Anti-CD19 FMC63 scFv heavy chain CDR3 14029 DIQMTQTTSSLSASLGDRVTIS- Anti-CD19 FMC63 scFv CRASQDISKY-LNWYQQKPDGT entire sequence, with VKLLI-YHTSRLHSGVPSRFSG 3xG.sub.4S linker (SEQ ID SGSGTDYSLTISNLEQEDIATY NO: 9313) FCQQGN-TLPYTFGGGTKLEIT- GGGGSGGGGSGGGGSEV-KLQE SGPGLVAPSQSLSVTCTVSGVS LPDYGVSWIRQP-PRKGLEWLG VIWGSET-TYYNSALKSRLTII KDNSKSQVFLK-MNSLQTDDTA IYYCAKHYYYGGSYAMDYWGQG TSVTVSS 14030 GGGGSGGGGSGGGGS 3xG.sub.4S linker

[0332] In some embodiments, the extracellular binding domain of the CD19 CAR is derived from an antibody specific to CD19, including, for example, SJ25C1 (Bejcek et al., Cancer Res. 55:2346-2351 (1995)), HD37 (Pezutto et al., J. Immunol. 138(9):2793-2799 (1987)), 4G7 (Meeker et al., Hybridoma 3:305-320 (1984)), B43 (Bejcek (1995)), BLY3 (Bejcek (1995)), B4 (Freedman et al., 70:418-427 (1987)), B4 HB12b (Kansas & Tedder, J. Immunol. 147:4094-4102 (1991); Yazawa et al., Proc. Natl. Acad. Sci. USA 102:15178-15183 (2005); Herbst et al., J. Pharmacol. Exp. Ther. 335:213-222 (2010)), BU12 (Callard et al., J. Immunology, 148(10):2983-2987 (1992)), and CLB-CD19 (De Rie Cell. Immunol. 118:368-381 (1989)). In any of these 10 embodiments, the extracellular binding domain of the CD19 CAR comprises or consists of the V.sub.H, the V.sub.L, and/or one or more CDRs of any of the antibodies.

[0333] In some embodiments, the hinge domain of the CD19 CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO:14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14008 or SEQ ID NO:14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0334] In some embodiments, the transmembrane domain of the CD19 CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0335] In some embodiments, the intracellular costimulatory domain of the CD19 CAR comprises a 4-1BB costimulatory domain. 4-1BB, also known as CD137, transmits a potent costimulatory signal to T cells, promoting differentiation and enhancing long-term survival of T lymphocytes. In some embodiments, the 4-1BB costimulatory domain is human. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain. CD28 is another co-stimulatory molecule on T cells. In some embodiments, the CD28 costimulatory domain is human. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016. In some embodiments, the intracellular costimulatory domain of the CD19 CAR comprises a 4-1BB costimulatory domain and a CD28 costimulatory domain as described.

[0336] In some embodiments, the intracellular signaling domain of the CD19 CAR comprises a CD3 zeta (2) signaling domain. CD37 associates with T cell receptors (TCRs) to produce a signal and contains immunoreceptor tyrosine-based activation motifs (ITAMs). The CD37 signaling domain refers to amino acid residues from the cytoplasmic domain of the zeta chain that are sufficient to functionally transmit an initial signal necessary for T cell activation. In some embodiments, the CD3ζ signaling domain is human. In some embodiments, the CD37 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0337] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR, including, for example, a CD19 CAR comprising the CD19-specific scFv having sequences set forth in SEQ ID NO: 14019 or SEQ ID NO: 14029, the CD4 hinge domain of SEQ ID NO:14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least

97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the CD19 CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0338] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR, including, for example, a CD19 CAR comprising the CD19-specific scFv having sequences set forth in SEQ ID NO: 14019 or SEQ ID NO: 14029, the IgG4 hinge domain of SEQ ID NO:14008 or SEQ ID NO: 14009, the CD28 transmembrane domain of SEQ ID NO:14012, the 4-1 BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the CD19 CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0339] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR, including, for example, a CD19 CAR comprising the CD19-specific scFv having sequences set forth in SEQ ID NO: 14019 or SEQ ID NO: 14029, the CD28 hinge domain of SEQ ID NO: 14007, the CD28 transmembrane domain of SEQ ID NO: 14012, the CD28 costimulatory domain of SEQ ID NO: 14016, the CD33 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the CD19 CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0340] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR as set forth in SEQ ID NO: 14031 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14031 (see Table 8). The encoded CD19 CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14032 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14032, with the following components: CD4 signal peptide, FMC63 scFv (V.sub.L-Whitlow linker-V.sub.H), CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD3ζ signaling domain.

[0341] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a commercially available embodiment of CD19 CAR. Non-limiting examples of commercially available embodiments of CD19 CARs expressed and/or encoded by T cells include tisagenlecleucel, lisocabtagene maraleucel, axicabtagene ciloleucel, and brexucabtagene autoleucel.

[0342] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding tisagenlecleucel or portions thereof. Tisagenlecleucel comprises a CD19 CAR with the following components: CD4 signal peptide, FMC63 scFv (V.sub.L-3×G4S linker-V.sub.H), CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD37 signaling domain. The nucleotide and amino acid sequence of the CD19 CAR in tisagenlecleucel are provided in Table 8, with annotations of the sequences provided in Table 9.

[0343] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding lisocabtagene maraleucel or portions thereof. Lisocabtagene maraleucel comprises a CD19 CAR with the following components: GMCSFR-α or CSF2RA signal peptide, FMC63 scFv (V.sub.L-Whitlow linker-V.sub.H), IgG4 hinge domain, CD28 transmembrane domain, 4-1BB costimulatory domain, and CD37 signaling domain. The nucleotide and amino acid sequence of the CD19 CAR in lisocabtagene maraleucel are provided in Table 8, with annotations of the sequences provided in Table 10.

[0344] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding axicabtagene ciloleucel or portions thereof. Axicabtagene ciloleucel comprises a CD19 CAR with the following components: GMCSFR-α or CSF2RA signal peptide, FMC63 scFv (V.sub.L-Whitlow linker-V.sub.H), CD28 hinge domain, CD28 transmembrane domain, CD28 costimulatory domain, and CD33 signaling domain. The nucleotide and amino acid sequence of the CD19 CAR in axicabtagene ciloleucel are provided in Table 8, with annotations of the sequences provided in Table 11.

[0345] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding brexucabtagene autoleucel or portions thereof. Brexucabtagene autoleucel comprises a CD19 CAR with the following components: GMCSFR-α signal peptide, FMC63 scFv, CD28 hinge domain, CD28 transmembrane domain, CD28 costimulatory domain, and CD3ζ signaling domain.

[0346] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD19 CAR as set forth in SEQ ID NO: 14033, 14035, or 14037, or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14033, 14035, or 14037. The encoded CD19 CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14034, 14036, or 14038, respectively, or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14034, 14036, or 14038, respectively.

TABLE-US-00008 TABLE 8 Exemplary sequences of CD19 CARs SEQ ID NO: Sequence Description 14031 atggccttaccagtgaccgccttgctcctgccgct Exemplary CD19 ggcctt-gctgctccac-gccgccaggccggacat CAR nucleotide ccagatgacacagactacatcctccctgtctgcct sequence ctctgggagacagagtcaccatcagtt-gcagggc aagtcaggacatt-agtaaatatttaaattggtat cagcagaaaccagatggaactgttaaactcctgat ctaccatacatcaagattacactcag-gagtccca tcaaggttcag-tggcagtgggtctggaacagatt attctctcaccattagcaacctggagcaagaagat attgccacttacttttt-gccaacagggtaatacgc ttccgtacac-gttcggaggggggaccaagctgga gatcacaggctccacctctggatccggcaagcccg -gatctggcgagggatccaccaagggcgaggtgaa actg-cag-gagtcaggacctggcctggtggcgcc ctcacagagcctgtccgtcacatgcactgtctcag gggtctcattacccgac-tatggtgtaagctggat -tcgccagcctccacgaaagggtctggagtggctg ggagtaatatggggtagtgaaaccacatacta-ta attcagctctcaaatccagactgac-catcatcaa ggacaactccaagagccaagttttcttaaaaatga acagtctg-caaactgatgacacagccatttacta ctgtgccaaacattattactacggtggtagctatg ctatggac-tactggggccaaggaacctcagtcac -cgtctcctcaaccacgacgccagcgccgcgacca ccaacaccggcgcccaccatcgcgtcg-cagcccc tgtccctgcgcccagaggcgtgccggccagcggcg gggggcgcagtgcacacgagggggctg-gacttcg cctgtga-tatctacatctgggcgcccttggccgg gacttgtggggtccttctcctgtcactggttatca ccctttactgcaaacgggg-cagaaagaaactcct gtata-tattcaaacaaccatttatgagaccagta caaactactcaagaggaagatggctgtagctgccg atttccagaagaa-gaagaaggaggatgtgaactg agag-tgaagttcagcaggagcgcagacgcccccg cgtaccagcagggccagaaccagctctataacgag ctcaatctag-gacgaagagaggagtacgatgttt t-ggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaaccctcaggaaggcc tg-tacaatgaactgcagaaagataa-gatggcgg aggcctacagtgagattgggatgaaaggcgagcgc cggaggggcaaggggcacgatggcctttac-cagg gtctcagtacagccac-caaggacacctacgacgc ccttcacatgcaggccctgccccctcgc 14032 MAL-PVTALLLPLALLLHAARPDIQMTQTTSSLSA Exemplary CD19 SLGDRVTISCRASQDISKY-LNWYQQKPDGTVKLL CAR amino acid I-YHTSRLHSGVPSRFSGSGSGTDYS-LTISNLEQ sequence EDIATYFCQQGNTLPYTFGGGTKLEITGSTSGSGK PGSGEGSTKGEV-KLQESGPGLVAP-SQSLSVTCT VSGVSLPDY-GVSWIRQPPRKGLEWLGVIWGSETT YYNSALKSRLTIIKDNSKSQVFLKMNSLQTDD-TA

IYYCAKHYYYGGSYAMDYWGQGTSVTVSSTTTPAP RPPTPAPTI-ASQPLSLRPEACRPAAGGAVHTRGL D-FACDIYIWAPLAG-TCGVLLLSLVITLYCKRGR KKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEE-G GCELRVKFSRSADAPA-YQQGQNQLYNELNLGR-R EEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNEL-Q KDKMAEAYSEIGMKGERRRGKGH-DGLYQGLSTAT KDTYDALHMQALPPR 14033 atggccttaccagtgaccgccttgctcctgccgct Tisagenlecleucel ggcctt-gctgctccac-gccgccaggccggacat CD19 CAR ccagatgacacagactacatcctccctgtctgcct nucleotide ctctgggagacagagtcaccatcagtt-gcagggc sequence aagtcaggacatt-agtaaatatttaaattggtat cagcagaaaccagatggaactgttaaactcctgat ctaccatacatcaagattacactcag-gagtccca tcaaggttcag-tggcagtgggtctggaacagatt attctctcaccattagcaacctggagcaagaagat attgccacttacttt-gccaacagggtaatacgc ttccgtacac-gttcggaggggggaccaagctgga gatcacaggtggcggtggctcgggcggtggtgggt cgggtggcggcg-gatctgaggtgaaactgcagga gtcag-gacctggcctggtggcgccctcacagagc ctgtccgtcacatgcactgtctcaggggtctcatt acccgactatggtgtaa-gctggattcgccagcct ccac-gaaagggtctggagtggctgggagtaatat ggggtagtgaaaccacatactataattcagctctc aaatccagactgac-catcatcaaggacaactcca a-gagccaagttttcttaaaaatgaacagtctgca aactgatgacacagccatttactactgtgccaaac attattactac-ggtggtagctatgctatggac-t actggggccaaggaacctcagtcaccgtctcctca accacgacgccagcgccgcgaccaccaacac-cgg cgcccaccatcgcgtcg-cagcccctgtccctgcg cccagaggcgtgccggccagcggcggggggcgcag tgcacacgagggggctg-gacttcgcctgtga-ta tctacatctgggcgcccttggccgggacttgtggg gtccttctcctgtcactggttatcaccctttactg caaacgggg-cagaaagaaactcctgtata-tatt caaacaaccatttatgagaccagtacaaactactc aagaggaagatggctgtagctgccgatttccagaa gaa-gaagaaggaggatgtgaactgagag-tgaag ttcagcaggagcgcagacgcccccgcgtacaagca gggccagaaccagctctataacgagctcaatctag -gacgaagagaggagtacgatgtttt-ggacaaga gacgtggccgggaccctgagatggggggaaagccg agaaggaagaaccctcaggaaggcctg-tacaatg aactgcagaaagataa-gatggcggaggcctacag tgagattgggatgaaaggcgagcgccggaggggca aggggcacgatggcctttac-cagggtctcagtac agccac-caaggacacctacgacgcccttcacatg caggccctgccccctcgc 14034 MAL-PVTALLLPLALLLHAARPDIQMTQTTSSLSA Tisagenlecleucel SLGDRVTISCRASQDISKY-LNWYQQKPDGTVKLL CD19 CAR I-YHTSRLHSGVPSRFSGSGSGTDYS-LTISNLEQ amino acid EDIATYFCQQGNTLPYTFGGGTKLEITGGGGSGGG sequence GSGGGGSEV-KLQESGPGLVAP-SQSLSVTCTVSG VSLPDYGVSWIRQP-PRKGLEWLGVIWGSETTYYN SALKSRLTIIKDNSKSQVFLKMNSLQTDD-TAIYY CAKHYYYGGSYAMDYWGQGTSVTVSSTTTPAPRPP TPAPTI-ASQPLSLRPEACRPAAGGAVHTRGLD-F ACDIYIWAPLAG-TCGVLLLSLVITLYCKRGRKKL LYIFKQPFMRPVQTTQEEDGCSCRFPEEEE-GGCE LRVKFSRSADAPA-YKQGQNQLYNELNLGR-REEY DVLDKRRGRDPEMGGKPRRKNPQEGLYNEL-QKDK MAEAYSEIGMKGERRRGKGH-DGLYQGLSTATKDT YDALHMQALPPR 14035 atgctgctgctggtgac-cagcctgctgctgtgcg Lisocabtagene agctgccccaccccgcctttctgctgatccccgac maraleucel atccagatgacccagaccac-ctccagcctgagcg CD19 CAR nucleotide ccagcctgggcgac-cgggtgaccatcagctgccg sequence ggccagccaggacatcagcaagtacctgaactggt atcagcagaagcccgacgg-caccgtcaagctgct gatctac-cacaccagccggctgcacagcggcgtg cccagccggtttagcggcagcggctccggcaccga ctacagcctgac-catctccaacctggaacaggaa ga-tatcgccacctacttttgccagcagggcaaca cactgccctacacctttggcggcggaacaaagctg gaaatcaccgg-cagcacctccggcagcggcaa-g cctggcagcggcgagggcagcaccaagggcgaggt gaagctgcaggaaa-gcggccctggcctggtggcc cccagccagagcctgagcgtgacctgcaccgtgag cggcgtgagcctgcccgactac-ggcgtgagctgg atccgg-cagcccccaggaagggcctggaatggc tgggcgtgatctggggcagcgagaccacctactac aacagcgccctgaa-gagccggctgac-catcatc aaggacaacagcaagagccaggtgttcctgaagat gaacagcctgcagaccgacgacac-cgccatctac tactgcgccaagcactactactac-ggcggcagct acgccatggactactggggccagggcaccagcgtg accgtgagcagcgaatctaagtacggac-cgccct gcccccctt-gccctatgttctgggtgctggtggt ggtcggaggcgtgctggcctgctacagcctgctgg tcaccgtggccttcatcatcttttt-gggtgaaacg gggcagaaa-gaaactcctgtatatattcaaacaa ccatttatgagaccagtacaaactactcaagagga agatggctgtagctgccgat-ttccagaagaagaa gaaggag-gatgtgaactgcgggtgaagttcagca gaagcgccgacgcccctgcctaccagcagggccag aatcagctgtacaac-gagctgaacctgggcagaa gggaa-gagtacgacgtcctggataagcggagagg ccgggaccctgagatgggcggcaagcctcggcgga agaacccccag-gaaggcctgtataacgaactg-c agaaagacaagatggccgaggcctacagcgagatc ggcatgaagggcgagcggaggcggggcaagggcca c-gacggcctgtatcagggcctgtccac-cgccac caaggatacctacgacgccctgcacatgcaggccc tgcccccaagg 14036 MLLLVTSLLLCELPHPAFL-LIPDIQMTQTTSSLS Lisocabtagene ASLGDRVTIS-CRASQDISKY-LNWYQQKPDGTVK maraleucel LLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQ CD19 CAR EDIATY-FCQQGNTLPYTFGGGTKLEIT-GSTSGS amino acid GKPGSGEGSTKGEVKLQESGPGLVAPSQSLSVTCT sequence VSGVSLPDY-GVSWIRQPPRKGLEWLGVIWGSET- TYYNSALKSRLTIIK-DNSKSQVFLKMNSLQTDDT AIYYCAKHYYYGGSYAMDYWGQGTSVTVSSESKYG PPCPPCPMFWVLVVVGGVLACYSLLVTVAFI-IFW VKRGRKKLLY-IFKQPFMRPVQTTQEEDGCSCRFP EEEEGGCELRVKFSRSADAPA-YQQGQNQLYNELN LGR-REEYDVLDKRRGRDPEMGGKPRR-KNPQEGL YNELQKDKMAEAYSEIGMKGERRRGKGHDGLYQGL STAT-KDTYDALHMQALPPR 14037 atgcttctcctggtgacaagccttctgctctgtga Axicabtagene ci- gttac-cacacccag-cattcctcctgatcccaga loleucel CD19 catccagatgacacagactacatcctccctgtctg CAR nucleotide cctctctgggagacagagtcac-catcagttgcag sequence ggcaagtcag-gacattagtaaatatttaaattgg tatcagcagaaaccagatggaactgttaaactcct gatctaccatacatcaagat-tacactcaggag-t cccatcaaggttcagtggcagtgggtctggaacag attattctctcaccattagcaacctggagcaagaa gatattgccac-ttacttttgccaacagggtaa-t acgcttccgtacacgttcggaggggggactaagtt ggaaataacaggctccacctctggatccggcaagc ccg-gatctggcgagggatccac-caagggcgagg tgaaactgcaggagtcaggacctggcctggtggcg ccctcacagagcctgtccgtcacatgcac-tgtct caggggtctcattacccgac-tatggtgtaagctg gattcgccagcctccacgaaagggtctggagtggc tgggagtaatatggggtagtgaaac-cacatacta -taattcagctctcaaatccagactgaccatcatc aaggacaactccaagagccaagttttcttaaaaat gaacagtctg-caaactgatgacacagccatttac tactgtgccaaacattattactacggtggtagcta tgctatggac-tactggggtcaaggaacctcagtc ac-cgtctcctcagcggccgcaattgaagttatgt atcctcctccttacctagacaatgagaagagcaat ggaac-cattatccatgtgaaagggaaacaccttt -gtccaagtcccctatttcccggacccttctaagcc cttttgggtgctggtggtggttgggggagtcctgg cttgctatagctt-gctagtaacag-tggccttta ttattttctgggtgaggagtaagaggagcaggctc ctgcacagtgac-tacatgaacatgactccccgcc gccccgggcccacccg-caagcattac-cagccct atgccccaccacgcgacttcgcagcctatcgctcc agagtgaagttcagcaggagcgcagac-gcccccg cgtaccagcagggccagaaccagctc-tataacga gctcaatctaggacgaagagaggagtacgatgttt tggacaagagacgtggccgggaccctga-gatggg gggaaagccgagaaggaa-gaaccctcaggaaggc ctgtacaatgaactgcagaaagataagatggcgga ggcctacagtgagattgg-gatgaaaggcgagcgc cggaggggcaaggggg-cacgatggcctttaccagg gtctcagtacagccaccaaggacacctacgac-gc ccttcacatgcaggccctgccccctcgc 14038 MLLLVTSLLLCELPHPAFL-LIPDIQMTQTTSSLS Axicabtagene ci- ASLGDRVTIS-CRASQDISKY-LNWYQQKPDGTVK loleucel CD19 LLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQ CAR amino acid EDIATY-FCQQGNTLPYTFGGGTKLEIT-GSTSGS sequence GKPGSGEGSTKGEVKLQESGPGLVAPSQSLSVTCT VSGVSLPDY-GVSWIRQPPRKGLEWLGVIWGSET- TYYNSALKSRLTIIK-DNSKSQVFLKMNSLQTDDT AIYYCAKHYYYGGSYAMDYWGQGTSVTVSSAAAIE VMYPPPYLDNEKSNGTIIHVKGKHLCPSPLF-PGP SKPFWVLVVVGGVLACYSLLVTVAFII-FWVRSKR SRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAAY RSRVKFSRSADAPA-YQQGQNQLYNELNLGR-REE YDVLDKRRGRD-PEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRGKGHDGLYQGLSTAT-KD TYDALHMQALPPR

TABLE-US-00009 TABLE 9 Annotation of tisagenlecleucel CD19 CAR sequences Nucleotide Amino Acid Feature Sequence Position Sequence Position CD8a signal peptide 1-63 1-21 FMC63 scFv 64-789 22-263 (V.sub.L-3xG.sub.4S linker-V.sub.H) CD8α hinge domain 790-924 264-308 CD8α transmembrane domain 925-996 309-332 4-1BB costimulatory domain 997-1122

333-374 CD3ζ signaling domain 1123-1458 375-486
TABLE-US-00010 TABLE 10 Annotation of lisocabtagene maraleucel CD19 CAR sequences Nucleotide Amino Acid Feature Sequence Position Sequence Position GMCSFR-a signal peptide 1-66 1-22 FMC63 scFv 67-801 23-267 (V.sub.L-Whitlow linker-V.sub.H) IgG4 hinge domain 802-837 268-279 CD28 transmembrane domain 838-921 280-307 4-1BB costimulatory domain 922-1047 308-349 CD3ζ signaling domain 1048-1383 350-461
TABLE-US-00011 TABLE 11 Annotation of axicabtagene ciloleucel CD19 CAR sequences Nucleotide Amino Acid Feature Sequence Position Sequence Position CSF2RA signal peptide 1-66 1-22 FMC63 scFv 67-801 23-267 (V.sub.L-Whitlow linker-V.sub.H) CD28 hinge domain 802-927 268-309 CD28 transmembrane domain 928-1008 310-336 CD28 costimulatory domain 1009-1131 337-377 CD3ζ signaling domain 1132-1467 378-489
[0347] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding CD19 CAR as set forth in SEQ ID NO: 14033, 14035, or 14037, or at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14033, 14035, or 14037. The encoded CD19 CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14034, 14036, or 14038, respectively, or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14034, 14036, or 14038, respectively.
CD20 CAR
[0348] In some embodiments, the CAR is a CD20 CAR ("CD20-CAR"), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR. CD20 is an antigen found on the surface of B cells as early at the pro-B phase and progressively at increasing levels until B cell maturity, as well as on the cells of most B-cell neoplasms. CD20 positive cells are also sometimes found in cases of Hodgkins disease, myeloma, and thymoma. In some embodiments, the CD20 CAR comprises a signal peptide, an extracellular binding domain that specifically binds CD20, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.
[0349] In some embodiments, the signal peptide of the CD20 CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO:14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO:14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.
[0350] In some embodiments, the extracellular binding domain of the CD20 CAR is specific to CD20, for example, human CD20. The extracellular binding domain of the CD20 CAR is codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain. In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv.

[0351] In some embodiments, the extracellular binding domain of the CD20 CAR is derived from an antibody specific to CD20, including, for example, Leu16, IF5, 1.5.3, rituximab, obinutuzumab, ibritumomab, ofatumumab, tositumumab, odronextamab, veltuzumab, ublituximab, and ocrelizumab. In any of these embodiments, the extracellular binding domain of the CD20 CAR comprises or consists of the V.sub.H, the V.sub.L, and/or one or more CDRs of any of the antibodies.

[0352] In some embodiments, the extracellular binding domain of the CD20 CAR comprises an scFv derived from the Leu16 monoclonal antibody, which comprises the heavy chain variable region (V.sub.H) and the light chain variable region (V.sub.L) of Leu16 connected by a linker. See Wu et al., Protein Engineering. 14(12):1025-1033 (2001). In some embodiments, the linker is a 3×G4S linker (SEQ ID NO: 9313). In other embodiments, the linker is a Whitlow linker as described herein. In some embodiments, the amino acid sequences of different portions of the entire Leu16-derived scFv (also referred to as Leu16 scFv) and its different portions are provided in Table 12 below. In some embodiments, the CD20-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14039, 14040, or 14044, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14039, 14040, or 14044. In some embodiments, the CD20-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14041-14043, 14045 and 14046. In some embodiments, the CD20-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14041-14043. In some embodiments, the CD20-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14045-14046. In any of these embodiments, the CD20-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD20 CAR comprises or consists of the one or more CDRs as described herein.

TABLE-US-00012 TABLE 12 Exemplary sequences of anti-CD20 scFv and components

SEQ ID NO:	Amino Acid Sequence	Description
14039	DIVLTQSPAILSASPGEKVTMT- CRASSSVNYMDWYQKKPGSSPKP- WIYATSNLAS- GVPARFSGSGSGTSYSLTISRVEAE DAATYYCQQWS- FNPPTFGGGTKLEIKGSTSGSGKP GSGEGSTKGEVQLQQSGAELVKP- GASVKMSCKASGYTFTSYN- MHWVKQTPGQGLEWIGAI- YPGNGDTSYNQKFKGKATLTADKS SSTAYMQLSSLTSED- SADYYCARSNYYGSSYWFFDVW- GAGTTVTVSS	Anti-CD20 Leu16 scFv entire sequence, with Whitlow linker
14040	DIVLTQSPAILSASPGEKVTMT- CRASSSVNYMDWYQKKPGSSPKP- WIYATSNLAS- GVPARFSGSGSGTSYSLTISRVEAE DAATYYCQQWS- FNPPTFGGGTKLEIK	Anti-CD20 Leu16 scFv light chain variable region
14041	RASSSVNYMD	Anti-CD20 Leu16 scFv light chain CDR1
14042	ATSNLAS	Anti-CD20 Leu16 scFv light chain CDR2
14043	QQWSFNPPT	Anti-CD20 Leu16 scFv light chain CDR3
14044	EVQLQQSGAELVKPGASVKMSCK- ASGYTFTSYN- MHWVKQTPGQGLEWIGAIYPGNG- DTSYNQKFKGKATLTADKSSSTAY MQLSSLTSED- SADYYCARSNYYGSSYWFFDVW- GAGTTVTVSS	Anti-CD20 Leu16 scFv heavy chain
14045	SYNMH	Anti-CD20 Leu16 scFv heavy chain CDR1
14046	AIYPGNGDTSYNQKFKG	Anti-CD20 Leu16 scFv heavy chain CDR2

[0353] In some embodiments, the hinge domain of the CD20 CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least

96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14008 or SEQ ID NO: 14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0354] In some embodiments, the transmembrane domain of the CD20 CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0355] In some embodiments, the intracellular costimulatory domain of the CD20 CAR comprises a 4-1BB costimulatory domain, for example, a human 4-1BB costimulatory domain. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain, for example, a human CD28 costimulatory domain. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016.

[0356] In some embodiments, the intracellular signaling domain of the CD20 CAR comprises a CD3 zeta (2) signaling domain, for example, a human CD3ζ signaling domain. In some embodiments, the CD32 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0357] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD4 hinge domain of SEQ ID NO: 14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD3ζ signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0358] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD28 hinge domain of SEQ ID NO: 14007, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0359] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0360] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD4 hinge domain of SEQ ID NO: 14006, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0361] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the CD28 hinge domain of SEQ ID NO: 14007, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0362] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD20 CAR, including, for example, a CD20 CAR comprising the CD20-specific scFv having sequences set forth in SEQ ID NO: 14039, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

CD22 CAR

[0363] In some embodiments, the CAR is a CD22 CAR (“CD22-CAR”), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR. CD22, which is a transmembrane protein found mostly on the surface of mature B cells that functions as an inhibitory receptor for B cell receptor (BCR) signaling. CD22 is expressed in 60-70% of B cell lymphomas and leukemias (e.g., B-chronic lymphocytic leukemia, hairy cell leukemia, acute lymphocytic leukemia (ALL), and Burkitt's lymphoma) and is not present on the cell surface in early stages of B cell development or on stem cells. In some embodiments, the CD22 CAR comprises a signal peptide, an extracellular binding domain that specifically binds CD22, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.

[0364] In some embodiments, the signal peptide of the CD22 CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.

[0365] In some embodiments, the extracellular binding domain of the CD22 CAR is specific to CD22, for example, human CD22. The extracellular binding domain of the CD22 CAR is codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain. In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv.

[0366] In some embodiments, the extracellular binding domain of the CD22 CAR is derived from an antibody specific to CD22, including, for example, SM03, inotuzumab, epratuzumab, moxetumomab, and pinatuzumab. In any of these embodiments, the extracellular binding domain of the CD22 CAR comprises or consists of the V.sub.H, the V.sub.L, and/or one or more CDRs of any of the antibodies.

[0367] In some embodiments, the extracellular binding domain of the CD22 CAR comprises an scFv derived from the m971 monoclonal antibody (m971), which comprises the heavy chain variable region (V.sub.H) and the light chain variable region (V.sub.L) of m971 connected by a linker. In some embodiments, the linker is a 3×G4S linker (SEQ ID NO: 9313). In other embodiments, the Whitlow linker is used instead. In some embodiments, the amino acid sequences of the entire m971-derived scFv (also referred to as m971 scFv) and its different portions are provided in Table 13 below. In some embodiments, the CD22-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14047, 14048, or 14052, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14047, 14048, or 14052. In some embodiments, the CD22-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14049-14051 and 14053-14055. In some embodiments, the CD22-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14049-14051. In some embodiments, the CD22-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14053-14055. In any of these embodiments, the CD22-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD22 CAR comprises or consists of the one or more CDRs as described herein.

[0368] In some embodiments, the extracellular binding domain of the CD22 CAR comprises an scFv derived from m971-L7, which is an affinity matured variant of m971 with significantly improved CD22 binding affinity compared to the parental antibody m971 (improved from about 2 nM to less than 50 pM). In some embodiments, the scFv derived from m971-L7 comprises the V.sub.H and the V.sub.L of m971-L7 connected by a 3×G4S linker (SEQ ID NO: 9313). In other embodiments, the Whitlow linker is used instead. In some embodiments, the amino acid sequences of the entire m971-L7-derived scFv (also referred to as m971-L7 scFv) and its different portions are provided in Table K below. In some embodiments, the CD22-specific scFv comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14056, 14057, or 14061, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14056, 14057, or 14061. In some embodiments, the CD22-specific scFv comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14058-14060 and 14062-14064. In some embodiments, the CD22-specific scFv comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14058-14060. In some embodiments, the CD22-specific scFv comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14062-14064. In any of these embodiments, the CD22-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the CD22 CAR comprises or consists of the one or more CDRs as described herein.

TABLE-US-00013 TABLE 13 Exemplary sequences of anti-CD22 scFv and components SEQ ID NO: Amino Acid Sequence Description 14047 QVQLQQSGPGLVKP- Anti-CD22 m971 scFv SQTLSLTCAISGDSVSS- entire sequence, with NSAAWNWIRQSPSR- 3xG.sub.4S linker (SEQ ID GLEWLGRTYYRSKWYNDYAVSVK NO: 9313) SRITINPDTSKNQFSLQLNSVTPED- TAVYYCAREVTGDLEDAFD- IWGQGTMVTVSSGGGGSGGGGSG GGGSDIQMTQSPSSLSASVG- DRVTITCRASQTI- WSYLNWYQQRPGKAPNLLIYAAS- SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA- TYYCQQSYSIPQTFGQGTKLEIK 14048 QVQLQQSGPGLVKP- Anti-CD22 m971 scFv SQTLSLTCAISGDSVSS- heavy chain variable NSAAWNWIRQSPSR- region GLEWLGRTYYRSKWYNDYAVSVK SRITINPDTSKNQFSLQLNSVTPED- TAVYYCAREVTGDLEDAFD- IWGQGTMVTVSS 14049 GDSVSSNSAA Anti-CD22 m971 scFv heavy chain CDR1 14050 TYYRSKWYN Anti-CD22 m971 scFv heavy chain CDR2 14051 AREVTGDLEDAFDI Anti-CD22 m971 scFv heavy chain CDR3 14052 DIQMTQSPSSLSASVG- Anti-CD22 m971 scFv DRVTITCRASQTI- light chain WSYLNWYQQRPGKAPNLLIYAAS- SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA- TYYCQQSYSIPQTFGQGTKLEIK 14053 QTIWSY Anti-CD22 m971 scFv light chain CDR1 AAS Anti-CD22 m971 scFv light chain CDR2 14055 QQSYSIPQT Anti-CD22 m971 scFv light chain CDR3 14056 QVQLQQSGPGMVKP- Anti-CD22 m971-L7 SQTLSLTCAISGDSVSS- scFv entire sequence, NSVAWNWIRQSPSR- with 3xG.sub.4S linker GLEWLGRTYYRST- (SEQ ID NO: 9313) WYNDYAVSMKSRITINPDTNKNQFS LQLNSVTPEDTAVYYCAREV- TGDLEDAFD- IWGQGTMVTVSSGGGGSGGGGSG GGGSDIQMIQSPSSLSASVG- DRVTITCRASQTI- WSYLNWYRQRPGEAPNLLIYAAS- SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA- TYYCQQSYSIPQTFGQGTKLEIK 14057 QVQLQQSGPGMVKP- Anti-CD22 m971-L7 SQTLSLTCAISGDSVSS- scFv heavy chain vari- NSVAWNWIRQSPSR- able region GLEWLGRTYYRST- WYNDYAVSMKSRITINPDTNKNQFS LQLNSVTPEDTAVYYCAREV- TGDLEDAFDIWGQGTMVTVSS 14058 GDSVSSNSVA Anti-CD22 m971-L7 scFv heavy chain CDR1 14059 TYYRSTWYN Anti-CD22 m971-L7 scFv heavy chain CDR2 14060 AREVTGDLEDAFDI Anti-CD22 m971-L7 scFv heavy chain CDR3 14061 DIQMIQSPSSLSASVG- Anti-CD22 m971-L7 DRVTITCRASQTI- scFv light chain varia- WSYLNWYRQRPGEAPNLLIYAAS- ble region SLQSGVPSRFSGRGSGTDFTLTISS LQAEDFA- TYYCQQSYSIPQTFGQGTKLEIK 14062 QTIWSY Anti-CD22 m971-L7 scFv light chain CDR1 AAS Anti-CD22 m971-L7 scFv light chain CDR2 14064 QQSYSIPQT Anti-CD22 m971-L7 scFv light chain CDR3

[0369] In some embodiments, the extracellular binding domain of the CD22 CAR comprises immunotoxins HA22 or BL22. Immunotoxins BL22 and HA22 are therapeutic agents that comprise an scFv specific for CD22 fused to a bacterial toxin, and thus can bind to the surface of the cancer cells that express CD22 and kill the cancer cells. BL22 comprises a dsFv of an anti-CD22 antibody, RFB4, fused to a 38-kDa truncated form of Pseudomonas exotoxin A (Bang et al., Clin. Cancer Res., 11:1545-50 (2005)). HA22 (CAT8015, moxetumomab pasudotox) is a mutated, higher affinity version of BL22 (Ho et al., J. Biol. Chem., 280(1): 607-17 (2005)). Suitable sequences of antigen binding domains of HA22 and BL22 specific to CD22 are disclosed in, for example, U.S. Pat. Nos. 7,541,034; 7,355,012; and 7,982,011.

[0370] In some embodiments, the hinge domain of the CD22 CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14008 or SEQ ID NO: 14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0371] In some embodiments, the transmembrane domain of the CD22 CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO:14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0372] In some embodiments, the intracellular costimulatory domain of the CD22 CAR comprises a 4-1BB costimulatory domain, for example, a human 4-1BB costimulatory domain. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain, for example, a human CD28 costimulatory domain. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016.

[0373] In some embodiments, the intracellular signaling domain of the CD22 CAR comprises a CD3 zeta (2) signaling domain, for example, a human CD37 signaling domain. In some embodiments, the CD34 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0374] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD4 hinge domain of SEQ ID NO:9, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0375] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD28 hinge domain of SEQ ID NO: 14007, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0376] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD33 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0377] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD4 hinge domain of SEQ ID NO:9, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0378] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the CD28 hinge domain of SEQ ID NO: 14007, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD37 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

[0379] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a CD22 CAR, including, for example, a CD22 CAR comprising the CD22-specific scFv having sequences set forth in SEQ ID NO: 14047 or SEQ ID NO: 14056, the IgG4 hinge domain of SEQ ID NO: 14008 or SEQ ID NO: 14009, the CD28 transmembrane domain of SEQ ID NO: 14012, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof.

BCMA CAR

[0380] In some embodiments, the CAR is a BCMA CAR ("BCMA-CAR"), and in these embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR. BCMA is a tumor necrosis family receptor (TNFR) member expressed on cells of the B cell lineage, with the highest expression on terminally differentiated B cells or mature B lymphocytes. BCMA is involved in mediating the survival of plasma cells for maintaining long-term humoral immunity. The expression of BCMA has been recently linked to a number of cancers, such as multiple myeloma, Hodgkin's and non-Hodgkin's lymphoma, various leukemias, and glioblastoma. In some embodiments, the BCMA CAR comprises a signal peptide, an extracellular binding domain that specifically binds BCMA, a hinge domain, a transmembrane domain, an intracellular costimulatory domain, and/or an intracellular signaling domain in tandem.

[0381] In some embodiments, the signal peptide of the BCMA CAR comprises a CD4 signal peptide. In some embodiments, the CD4 signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14003 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14003. In some embodiments, the signal peptide comprises an IgK signal peptide. In some embodiments, the IgK signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14004 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14004. In some embodiments, the signal peptide comprises a GMCSFR-α or CSF2RA signal peptide. In some embodiments, the GMCSFR-α or CSF2RA signal peptide comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14005 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14005.

[0382] In some embodiments, the extracellular binding domain of the BCMA CAR is specific to BCMA, for example, human BCMA. The extracellular binding domain of the BCMA CAR can be codon-optimized for expression in a host cell or to have variant sequences to increase functions of the extracellular binding domain.

[0383] In some embodiments, the extracellular binding domain comprises an immunogenically active portion of an immunoglobulin molecule, for example, an scFv. In some embodiments, the

extracellular binding domain of the BCMA CAR is derived from an antibody specific to BCMA, including, for example, belantamab, erlanatamab, teclistamab, LCAR-B38M, and ciltacabtagene. In any of these embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the V.sub.H, the V.sub.L, and/or one or more CDRs of any of the antibodies.
[0384] In some embodiments, the extracellular binding domain of the BCMA CAR comprises an scFv derived from C11D5.3, a murine monoclonal antibody as described in Carpenter et al., Clin. Cancer Res. 19(8):2048-2060 (2013). See also PCT Application Publication No. WO2010/104949. The C11D5.3-derived scFv may comprise the heavy chain variable region (V.sub.H) and the light chain variable region (V.sub.L) of C11D5.3 connected by the Whitlow linker, the amino acid sequences of which is provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14065, 14066, or 14067, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14065, 14066, or 14067. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14067-14069 and 14071-14073. In some embodiments, the BCMA-specific extracellular binding domain comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14067-14069. In some embodiments, the BCMA-specific extracellular binding domain comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14071-14073. In any of these embodiments, the BCMA-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.
[0385] In some embodiments, the extracellular binding domain of the BCMA CAR comprises an scFv derived from another murine monoclonal antibody, C12A3.2, as described in Carpenter et al., Clin. Cancer Res. 19(8):2048-2060 (2013) and PCT Application Publication No. WO2010/104949, the amino acid sequence of which is also provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14074, 14075, or 14079, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14074, 14075, or 14079. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14076-14078 and 14080-14082. In some embodiments, the BCMA-specific extracellular binding domain comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14076-14078. In some embodiments, the BCMA-specific extracellular binding domain comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14080-14082. In any of these embodiments, the BCMA-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.
[0386] In some embodiments, the extracellular binding domain of the BCMA CAR comprises a murine monoclonal antibody with high specificity to human BCMA, referred to as BB2121 in Friedman et al., Hum. Gene Ther. 29(5):585-601 (2018)). See also, PCT Application Publication No. WO2012163805.

[0387] In some embodiments, the extracellular binding domain of the BCMA CAR comprises single variable fragments of two heavy chains (VHH) that bind to two epitopes of BCMA as described in Zhao et al., J. Hematol. Oncol. 11(1):141 (2018), also referred to as LCAR-B38M. See also, PCT Application Publication No. WO2018/028647.

[0388] In some embodiments, the extracellular binding domain of the BCMA CAR comprises a fully human heavy-chain variable domain (FHVH) as described in Lam et al., Nat. Commun. 11(1):283 (2020), also referred to as FHVH33. See also, PCT Application Publication No. WO2019/006072. The amino acid sequences of FHVH33 and its CDRs are provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14083 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14083. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14084-14086. In any of these embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.

[0389] In some embodiments, the extracellular binding domain of the BCMA CAR comprises an scFv derived from CT103A (or CAR0085) as described in U.S. Pat. No. 11,026,975 B2, the amino acid sequence of which is provided in Table 14 below. In some embodiments, the BCMA-specific extracellular binding domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14087, 14088, or 14092, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14087, 14088, or 14092. In some embodiments, the BCMA-specific extracellular binding domain comprises one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14089-14091 and 14093-14095. In some embodiments, the BCMA-specific extracellular binding domain comprises a light chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14089-14091. In some embodiments, the BCMA-specific extracellular binding domain comprises a heavy chain with one or more CDRs having amino acid sequences set forth in SEQ ID NOs: 14093-14095. In any of these embodiments, the BCMA-specific scFv comprises one or more CDRs comprising one or more amino acid substitutions, or comprising a sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical), to any of the sequences identified. In some embodiments, the extracellular binding domain of the BCMA CAR comprises or consists of the one or more CDRs as described herein.

[0390] Additionally, CARs and binders directed to BCMA have been described in U.S. Application Publication Nos. 2020/0246381 A1 and 2020/0339699 A1.

TABLE-US-00014 TABLE 14 Exemplary sequences of anti-BCMA binder and components SEQ ID NO: Amino Acid Sequence Description 14065 DIVLTQSPASLAMSLGKRATIS- Anti-BCMA C11D5.3 CRASESVSVIGAHLIHWYQQKPGQ scFv entire sequence, PPKLLIYLASN- with Whitlow linker LETGVPARFSGSGSGTDFT-LTIDPVEEDDVAIYSCLQSRIFPRT- FGGGTKLEIKGSTSGSGKPGSGEG STKGQIQLVQSGPELKKPGETVKIS CKASGYTFTDYSINWVKRAPGKGL KWMGWIN-TETREPAYAYDFRGRFAFSLETSAS TAYLQINNLKYEDTATYFCAL-DYSYAMDYWGQGTSVTVSS 14066 DIVLTQSPASLAMSLGKRATIS- Anti-BCMA C11D5.3 CRASESVSVIGAHLIHWYQQKPGQ scFv light chain PPKLLIYLASN- variable region

LETGVPARFSGSGSGTDFT- LTIDPVEEDDVAIYSCLQSRIFPRT- FGGGTKLEIK 14067 RASESVSVIGAHLIH Anti-BCMA C11D5.3 scFv light chain CDR1 14068 LASNLET Anti-BCMA C11D5.3 scFv light chain CDR2 14069 LQSRIFPRT Anti-BCMA C11D5.3 scFv light chain CDR3 14070 QIQLVQSGPELKKPGETVKISCK- Anti-BCMA C11D5.3 ASGYTFTDYSINWVKRAPGKGLK- scFv heavy chain vari- WMGWIN- able region TETREPAYAYDFRGRFAFSLETSAS TAYLQINNLKYEDTATYFCAL- DYSYAMDYWGQGTSVTVSS 14071 DYSIN Anti-BCMA C11D5.3 scFv heavy chain CDR1 14072 WINTETREPAYAYDFRG Anti-BCMA C11D5.3 scFv heavy chain CDR2 14073 DYSYAMDY Anti-BCMA C11D5.3 scFv heavy chain CDR3 14074 DIVLTQSPPSLAMSLGKRATIS- Anti-BCMA C12A3.2 CRASESVTILGSHLI- scFv entire sequence, YWYQQKPGQPPTLLIQ- with Whitlow linker LASNVQTGVPARFSGSGSRTDFTL TIDPVEEDDVAVYYCLQSRTIPRT- FGGGTKLEIKGSTSGSGKPGSGEG STKGQIQLVQSGPELKKPGETVKIS CKASGYTFRHYSMNWVKQAPGKG LKWMGRINTESGVPIYADD- FKGRFAFSVETSASTAYL- VINNLKDEDTASYFCSNDYLYSLD- FWGQGTALTVSS 14075 DIVLTQSPPSLAMSLGKRATIS- Anti-BCMA C12A3.2 CRASESVTILGSHLI- scFv light chain YWYQQKPGQPPTLLIQ- variable region LASNVQTGVPARFSGSGSRTDFTL TIDPVEEDDVAVYYCLQSRTIPRT- FGGGTKLEIK 14076 RASESVTILGSHLIY Anti-BCMA C12A3.2 scFv light chain CDR1 14077 LASNVQT Anti-BCMA C12A3.2 scFv light chain CDR2 14078 LQSRTIPRT Anti-BCMA C12A3.2 scFv light chain CDR3 14079 QIQLVQSGPELKKPGETVKISCK- Anti-BCMA C12A3.2 ASGYTFRHYSMNWVKQAPGKGLK- scFv heavy chain WMGRINTESGVPIYADDFKGRFAF variable region SVETSASTAYL- VINNLKDEDTASYFCSNDYLYSLD- FWGQGTALTVSS 14080 HYSMN Anti-BCMA C12A3.2 scFv heavy chain CDR1 14081 RINTESGVPIYADDFKG Anti-BCMA C12A3.2 scFv heavy chain CDR2 14082 DYLYSLDF Anti-BCMA C12A3.2 scFv heavy chain CDR3 14083 EVQLLESGGGLVQPGGSLRLS- Anti-BCMA FHVH33 CAASGFTFSSYAMSWVR- entire sequence QAPGKGLEWVSSISGSGDYIY- YADSVKGRFTISRDISKNTLYLQMN SLRAEDTAVYYCAKEGTGANSSLA DYRGQGTLVTVSS 14084 GFTFSSYA Anti-BCMA FHVH33 CDR1 14085 ISGSGDYI Anti-BCMA FHVH33 CDR2 14086 AKEGTGANSSLADY Anti-BCMA FHVH33 CDR3 14087 DIQMTQSPSSLSASVG- Anti-BCMA CT103A DRVTITCRASQSIS- scFv entire sequence, SYLNWYQQKPGKAPKLLIYAAS- with Whitlow linker SLQSGVPSRFSGSGSGTDFTLTISS LQPEDFA- TYYCQQKYDLLTFGGGTKVEIKGST SGSGKPGSGEGSTKGQLQLQESG PGLVKPSETLSLTCTVSGGSIS- SSSYYWGWIRQPPGKGLEWIG- SISYSGSTYYNPSLKSRVTISVDTSK NQFSLKLSSVTAADTAVYYCARDR GDTILDVWGQGTMVTVSS 14088 DIQMTQSPSSLSASVG- Anti-BCMA CT103A DRVTITCRASQSIS- scFv light chain SYLNWYQQKPGKAPKLLIYAAS- variable region SLQSGVPSRFSGSGSGTDFTLTISS LQPEDFA- TYYCQQKYDLLTFGGGTKVEIK 14089 QSISSY Anti-BCMA CT103A scFv light chain CDR1 AAS Anti-BCMA CT103A scFv light chain CDR2 14091 QQKYDLLT Anti-BCMA CT103A scFv light chain CDR3 14092 QLQLQESGPGLVKP- Anti-BCMA CT103A SETLSLTCTVSGGSIS- scFv heavy chain SSSYYWGWIRQPPGKGLEWIG- variable region SISYSGSTYYNPSLKSRVTISVDTSK NQFSLKLSSVTAADTAVYYCARDR GDTILDVWGQGTMVTVSS 14093 GGSISSSSYY Anti-BCMA CT103A scFv heavy chain CDR1 14094 ISYSGST Anti-BCMA CT103A scFv heavy chain CDR2 14095 ARDRGDTILDV Anti-BCMA CT103A scFv heavy chain CDR3

[0391] In some embodiments, the hinge domain of the BCMA CAR comprises a CD4 hinge domain, for example, a human CD4 hinge domain. In some embodiments, the CD4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14006 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14006. In some embodiments, the hinge domain comprises a CD28 hinge domain, for example, a human CD28 hinge domain. In some embodiments, the CD28 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14007 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14007. In some embodiments, the hinge domain comprises an IgG4 hinge domain, for example, a human IgG4 hinge domain. In some embodiments, the IgG4 hinge domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14008 or SEQ ID NO: 14009, or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14008 or SEQ ID NO: 14009. In some embodiments, the hinge domain comprises a IgG4 hinge-Ch2-Ch3 domain, for example, a human IgG4 hinge-Ch2-Ch3 domain. In some embodiments, the IgG4 hinge-Ch2-Ch3 domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14010 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14010.

[0392] In some embodiments, the transmembrane domain of the BCMA CAR comprises a CD4 transmembrane domain, for example, a human CD4 transmembrane domain. In some embodiments, the CD4 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14011 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14011. In some embodiments, the transmembrane domain comprises a CD28 transmembrane domain, for example, a human CD28 transmembrane domain. In some embodiments, the CD28 transmembrane domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14012 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14012.

[0393] In some embodiments, the intracellular costimulatory domain of the BCMA CAR comprises a 4-1BB costimulatory domain, for example, a human 4-1BB costimulatory domain. In some embodiments, the 4-1BB costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14015 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14015. In some embodiments, the intracellular costimulatory domain comprises a CD28 costimulatory domain, for example, a human CD28 costimulatory domain. In some embodiments, the CD28 costimulatory domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14016 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14016.

[0394] In some embodiments, the intracellular signaling domain of the BCMA CAR comprises a CD3 zeta (2) signaling domain, for example, a human CD3ζ signaling domain. In some embodiments, the CD37 signaling domain comprises or consists of an amino acid sequence set forth in SEQ ID NO: 14017 or an amino acid sequence that is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in SEQ ID NO: 14017.

[0395] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR, including, for example, a BCMA CAR comprising any of the BCMA-specific extracellular binding domains as described, the CD4 hinge domain of SEQ ID NO: 14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the 4-1BB costimulatory domain of SEQ ID NO: 14015, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the BCMA CAR additionally comprises a signal peptide (e.g., a CD4 signal peptide) as described.

[0396] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR, including, for example, a BCMA CAR comprising any of the BCMA-specific extracellular binding domains as described, the CD4 hinge domain of SEQ ID NO: 14006, the CD4 transmembrane domain of SEQ ID NO: 14011, the CD28 costimulatory domain of SEQ ID NO: 14016, the CD34 signaling domain of SEQ ID NO: 14017, and/or variants (i.e., having a sequence that is at least 80% identical, for example, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99 identical to the disclosed sequence) thereof. In any of these embodiments, the BCMA CAR additionally comprises a signal peptide as described.

[0397] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a BCMA CAR as set forth in SEQ ID NO: 14096 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the nucleotide sequence set forth in SEQ ID NO: 14096 (see Table 15). The encoded BCMA CAR has a corresponding amino acid sequence set forth in SEQ ID NO: 14097 or is at least 80% identical (e.g., at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to the amino acid sequence set forth in of SEQ ID NO: 14097, with the following components: CD4 signal peptide, CT103A scFv (V.sub.L-Whitlow linker-V.sub.H), CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD3ζ signaling domain.

[0398] In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding a commercially available embodiment of BCMA CAR, including, for example, idecabtagene vicleucel (ide-cel, also called bb2121). In some embodiments, the polycistronic vector comprises an expression cassette that contains a nucleotide sequence encoding idecabtagene vicleucel or portions thereof. Idecabtagene vicleucel comprises a BCMA CAR with the following components: the BB2121 binder, CD4 hinge domain, CD4 transmembrane domain, 4-1BB costimulatory domain, and CD37 signaling domain.

TABLE-US-00015 TABLE 15 Exemplary sequences of BCMA CARs SEQ ID NO: Sequence Description 14096 atggccttaccagtgaccgccttgctcctgccgctggcctt- Exemplary BCMA gctgctccac- CAR nucleotide gccgccaggccggacatccagatgacccagtctccatc sequence ctccctgtctgcatctgtaggagacagagtcaccatcac- ttgccgggcaagtcagagcatt- agcagctatttaaattggtatcagcagaaaccagggaaa gccctaagctcctgatctatgctgcatccagttt- gcaaagtggggtcccatcaaggttcag- tggcagtggatctgggacagatttcactctcaccatcagc agtctgcaacctgaagattttgcaacttactactgtcag- caaaaatacgacctcctcac- ttttggcggagggaccaaggttgagatcaaaggcagca ccagcggctccggcaagcctggctctggcgagggcag- cacaaagggacagctgcagctgcag- gagtcgggcccaggactggtgaagccttcggagaccct gtccctcacctgcactgtctctggtggctccatcagcag- tagtagttactactggggctg- gatccgccagcccccagggaaggggctggagtggattg ggagtatctcctatagtgggagcacctacta- caacccgtccctcaagagtcgagtcac- catatccgtagacacgtccaagaaccagttctccctgaag ctgagttctgtgaccgccgcagacacggcggtg- tactactgcgccagagatcgtggagacac- catactagacgtatggggtcagggtacaatggtcaccgtc agctcattcgtgcccgtgttcctgcccgccaaacctac- caccacccctgcccctagac- ctcccaccccagccccaacaatcgccagccagcctctgt ctctgcggcccgaagcctgtagacctgctgccggcg- gagccgtgcacaccagaggcctggacttcgcctgcgacatctacatctgggcccctctggccg gcac- ctgtggcgtgctgctgctgagcctggtgatcaccctg-tactgcaaccaccg- gaacaaacggggcagaaagaaactcctgtatatattca aacaaccatttatgagaccagtacaaactactcaagag-gaagatggctgtagctgccgat- ttccagaagaagaagaaggaggatgtgaactgagagtg aagttcagcagatccgccgacgcccctgcctaccag- cagggacagaaccagctgtacaac-gagctgaacctgggcagacgggaagagtacgacgtgct ggacaagcggagaggccgggaccccgagatgggcg-gaaagcccagacggaagaacccccag- gaaggcctgtataacgaactgcagaaagacaagatgg ccgaggcctacagcgagatcgg-catgaagggcgagcggaggcgcggcaagggccac- gatggcctgtaccagggcctgagcaccgccaccaagga cacctacgacgccctg- cacatgcaggccctgccccccaga 14097 MAL- Exemplary BCMA PVTALLLPLALLLHAARPDIQMTQSPSSLS CAR amino acid ASVGDRVTITCRASQSIS-sequence SYLNWYQQKPGKAPKLLIYAAS- SLQSGVPSRFSGSGSGTDFT-LTISSLQPEDFATYYCQQKYDLLTFGGGT KVEIKGSTSGSGKPGSGEGSTKGQLQLQ ESGPGLVKPSETLSLTCTVSGGSIS- SSSYYWGWIRQPPGKGLEWIG-SISYSGSTYYNPSLKSRVTISVDTSKNQF SLKLSSVTAADTAVYYCARDRGDTIL-DVWGQGTMVTVSS- FVPVFLPAKPTTTPAPRPPTPAP-TIASQPLSLRPEACRPAAGGAVHTRGLDF ACDIYIWAPLAGTCGVLLLSLVITLYC-NHRNKRGRKKLLY- IFKQPFMRPVQTTQEEDGCSCRFPEEEE GGCELRVKFSRSADAPA-YQQGQNQLYNELNLGR- REEYDVLDKRRGRDPEMGGKPRR-KNPQEGLYNELQKDKMAEAYSEIGMKGE RRRGKGHDGLYQGLSTAT-KDTYDALHMQALPPR

[0399] In some embodiments, the antibody portion of the recombinant receptor, e.g., CAR, further includes a spacer between the transmembrane domain and extracellular antigen binding domain. In some embodiments, the spacer includes at least a portion of an immunoglobulin constant region, such as a hinge region, e.g., an IgG4 hinge region, and/or a CH1/CL and/or Fc region. In some embodiments, the constant region or portion is of a human IgG, such as IgG4 or IgGI. In some aspects, the portion of the constant region serves as a spacer region between the antigen-recognition component, e.g., scFv, and transmembrane domain. The spacer can be of a length that provides for increased responsiveness of the cell following antigen binding, as compared to in the absence of the spacer. Exemplary spacers include, but are not limited to, those described in Hudecek et al. (2013) *Clin. Cancer Res.*, 19:3153, WO2014031687, U.S. Pat. No. 8,822,647 or published app. No. US 2014/0271635. In some embodiments, the constant region or portion is of a human IgG, such as IgG4 or IgGI.

[0400] In some embodiments, the antigen receptor comprises an intracellular domain linked directly or indirectly to the extracellular domain. In some embodiments, the chimeric antigen receptor includes a transmembrane domain linking the extracellular domain and the intracellular signaling domain. In some embodiments, the intracellular signaling domain comprises an ITAM. For example, in some aspects, the antigen recognition domain (e.g. extracellular domain) generally is linked to one or more intracellular signaling components, such as signaling components that mimic activation through an antigen receptor complex, such as a TCR complex, in the case of a CAR, and/or signal via another cell surface receptor. In some embodiments, the chimeric receptor comprises a transmembrane domain linked or fused between the extracellular domain (e.g. scFv) and intracellular signaling domain. Thus, in some embodiments, the antigen-binding component (e.g., antibody) is linked to one or more transmembrane and intracellular signaling domains.

[0401] In some embodiments, a transmembrane domain that naturally is associated with one of the domains in the receptor, e.g., CAR, is used. In some instances, the transmembrane domain is selected or modified by amino acid substitution to avoid binding of such domains to the transmembrane domains of the same or different surface membrane proteins to minimize interactions with other members of the receptor complex.

[0402] The transmembrane domain in some embodiments is derived either from a natural or from a synthetic source. Where the source is natural, the domain in some aspects is derived from any membrane-bound or transmembrane protein. Transmembrane regions include those derived from (i.e. comprise at least the transmembrane region(s) of) the alpha, beta or zeta chain of the T-cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154. Alternatively, the transmembrane domain in some embodiments is synthetic. In some aspects, the synthetic transmembrane domain comprises predominantly hydrophobic residues such as leucine and valine. In some aspects, a triplet of phenylalanine, tryptophan and valine will be found at each end of a synthetic transmembrane domain. In some embodiments, the linkage is by linkers, spacers, and/or transmembrane domain(s). In some aspects, the transmembrane domain contains a transmembrane portion of CD28.

[0403] In some embodiments, the extracellular domain and transmembrane domain is linked directly or indirectly. In some embodiments, the extracellular domain and transmembrane are linked by a spacer, such as any described herein. In some embodiments, the receptor contains extracellular portion of the molecule from which the transmembrane domain is derived, such as a CD28 extracellular portion.

[0404] Among the intracellular signaling domains are those that mimic or approximate a signal through a natural antigen receptor, a signal through such a receptor in combination with a costimulatory receptor, and/or a signal through a costimulatory receptor alone. In some embodiments, a short oligo- or polypeptide linker, for example, a linker of 2 to 10 amino acids in length, such as one containing glycines and serines, e.g., glycine-serine doublet, is present and forms a linkage between the transmembrane domain and the cytoplasmic signaling domain of the CAR.

[0405] T cell activation is in some aspects described as being mediated by two classes of cytoplasmic signaling sequences: those that initiate antigen-dependent primary activation through the TCR (primary cytoplasmic signaling sequences), and those that act in an antigen-independent manner to provide a secondary or co-stimulatory signal (secondary cytoplasmic signaling sequences). In some aspects, the CAR includes one or both of such signaling components.

[0406] The receptor, e.g., the CAR, generally includes at least one intracellular signaling component or components. In some aspects, the CAR includes a primary cytoplasmic signaling sequence that regulates primary activation of the TCR complex. Primary cytoplasmic signaling sequences that act in a stimulatory manner may contain signaling motifs which are known as immunoreceptor tyrosine-based activation motifs or ITAMs. Examples of ITAM containing primary cytoplasmic signaling sequences include those derived from CD3 zeta chain, FcR gamma, CD3 gamma, CD3 delta and CD3 epsilon. In some embodiments, cytoplasmic signaling molecule(s) in the CAR contain(s) a cytoplasmic signaling domain, portion thereof, or sequence derived from CD3 zeta.

[0407] In some embodiments, the receptor includes an intracellular component of a TCR complex, such as a TCR CD3 chain that mediates T-cell activation and cytotoxicity, e.g., CD3 zeta chain. Thus, in some aspects, the antigen-binding portion is linked to one or more cell signaling modules. In some embodiments, cell signaling modules include a CD3 transmembrane domain, CD3 intracellular signaling domains, and/or other CD transmembrane domains. In some embodiments, the intracellular component is or includes a CD3-zeta intracellular signaling domain. In some embodiments, the intracellular component is or includes a signaling domain from a Fc receptor gamma chain. In some embodiments, the receptor, e.g., CAR, includes the intracellular signaling domain and further includes a portion, such as a transmembrane domain and/or hinge portion, of one or more additional molecules such as CD8, CD4, CD25, or CD16. For example, in some aspects, the CAR or other chimeric receptor is a chimeric molecule of CD3-zeta (CD3-z) or Fc receptor gamma and a portion of one of CD8, CD4, CD25 or CD16.

[0408] In some embodiments, upon ligation of the CAR or other chimeric receptor, the cytoplasmic domain or intracellular signaling domain of the receptor activates at least one of the normal effector functions or responses of the immune cell, e.g., T cell engineered to express the CAR. For example, in some contexts, the CAR induces a function of a T cell such as cytolytic activity or T-helper activity, such as secretion of cytokines or other factors. In some embodiments, a truncated portion of an intracellular signaling domain of an antigen receptor component or costimulatory molecule is used in place of an intact immunostimulatory chain, for example, if it transduces the effector function signal. In some embodiments, the intracellular signaling domain or domains include the cytoplasmic sequences of the T cell receptor (TCR), and in some aspects also those of co-receptors that in the natural context act in concert with such receptors to initiate signal transduction following antigen receptor engagement.

[0409] In the context of a natural TCR, full activation generally requires not only signaling through the TCR, but also a costimulatory signal. Thus, in some embodiments, to promote full activation, a component for generating secondary or co-stimulatory signal is also included in the CAR. In other embodiments, the CAR does not include a component for generating a costimulatory signal. In some aspects, an additional CAR is expressed in the same cell and provides the component for generating the secondary or costimulatory signal.

[0410] In some embodiments, the chimeric antigen receptor contains an intracellular domain of a T cell costimulatory molecule. In some embodiments, the CAR includes a signaling domain and/or transmembrane portion of a costimulatory receptor, such as CD28, 4-1BB, 0X40, DAP10, and ICOS. In some aspects, the same CAR includes both the activating and costimulatory components. In some embodiments, the chimeric antigen receptor contains an intracellular domain derived from a T cell costimulatory molecule or a functional variant thereof, such as between the transmembrane domain and intracellular signaling domain. In some aspects, the T cell costimulatory molecule is CD28 or 41BB.

[0411] In some embodiments, the activating domain is included within one CAR, whereas the costimulatory component is provided by another CAR recognizing another antigen. In some embodiments, the CARs include activating or stimulatory CARs, costimulatory CARs, both expressed on the same cell (see WO2014/055668). In some aspects, the cells include one or more stimulatory or activating CARs and/or a costimulatory CAR. In some embodiments, the cells further include inhibitory CARs (iCARs, see Fedorov et al., Sci. Transl. Medicine, 5(215) (December 2013), such as a CAR recognizing an antigen other than the one associated with and/or specific for the disease or condition whereby an activating signal delivered through the disease-targeting CAR is diminished or inhibited by binding of the inhibitory CAR to its ligand, e.g., to reduce off-target effects.

[0412] In certain embodiments, the intracellular signaling domain comprises a CD28 transmembrane and signaling domain linked to a CD3 (e.g., CD3-zeta) intracellular domain. In some embodiments, the intracellular signaling domain comprises a chimeric CD28 and CD137 (4-1BB, TNFRSF9) co-stimulatory domains, linked to a CD3 zeta intracellular domain.

[0413] In some embodiments, the CAR encompasses one or more, e.g., two or more, costimulatory domains and an activation domain, e.g., primary activation domain, in the cytoplasmic portion. Exemplary CARs include intracellular components of CD3-zeta, CD28, and 4-1BB.

[0414] In some embodiments the intracellular signaling domain includes intracellular components of a 4-1BB signaling domain and a CD3-zeta signaling domain. In some embodiments, the intracellular signaling domain includes intracellular components of a CD28 signaling domain and a CD3zeta signaling domain.

[0415] In some embodiments, the CAR comprises an extracellular antigen binding domain (e.g., antibody or antibody fragment, such as an scFv) that binds to an antigen (e.g. tumor antigen), a spacer (e.g. containing a hinge domain, such as any as described herein), a transmembrane domain (e.g. any as described herein), and an intracellular signaling domain (e.g. any intracellular signaling domain, such as a primary signaling domain or costimulatory signaling domain as described herein). In some embodiments, the intracellular signaling domain is or includes a primary cytoplasmic signaling domain. In some embodiments, the intracellular signaling domain additionally includes an intracellular signaling domain of a costimulatory molecule (e.g., a costimulatory domain). Examples of exemplary components of a CAR are described in Table 16. In provided aspects, the sequences of each component in a CAR can include any combination listed in Table 16.

TABLE-US-00016 TABLE 16 SEQ ID Component Sequence NO: Extracellular binding domain Anti-CD19 scFv DIQMTQTTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDG 14098 (FMC63) TVKLLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATY FCQQGNTLPYTFGGGTKLEITGSTSGSGKPGSGEGSTKGEV KLQESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKG LEWLGVIWGSETTYYNSALKSRLTIIKDNSKSQVFLKMNSLQT DDTAIYYCAKHYYYGGSYAMDYWGQGTSVTVSS Anti-CD19 scFv DIQMTQTTSSLSASLGDRVTISCRASQDISKYLNWYQQKPDG 14099 (FMC63) TVKLLIYHTSRLHSGVPSRFSGSGSGTDYSLTISNLEQEDIATY FCQQGNTLPYTFGGGTKLEITGGGGSGGGGSGGGGSEVKL QESGPGLVAPSQSLSVTCTVSGVSLPDYGVSWIRQPPRKGLE WLGVIWGSETTYYNSALKSRLTIIKDNSKSQVFLKMNSLQTDD TAIYYCAKHYYYGGSYAMDYWGQGTSVTVSS Anti-BCMA sdAb QVQLVESGGGLVQPGGSLRLSCAASGFTFTNHAMSWVRQA 14100 (FHVH74) PGKGLELVSSISGNGRTTYYADSVKGRFTISRDISKNTLDLQM NSLRAEDTAVYYCAKDGGETLVDSRGQGTLVTVSS Anti-BCMA sdAb QVQLVESGGGLVQPGGSLRLSCAASGFTFSSHAMTWVRQAP 14101 (FHVH32) GKGLEWVAAISGSGDFTHYADSVKGRFTISRDNSKNTVSLQM NNLRAEDTAVYYCAKDEDGGSLLGYRGQGTLVTVSS Anti-BCMA sdAb EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAP 14102 (FHVH33) GKGLEWVSSISGSGDYIYYADSVKGRFTISRDISKNTLYLQMN SLRAEDTAVYYCAKEGTGANSSLADYRGQGTLVTVSS Anti-BCMA sdAb EVQLLESGGGLIQPGGSLRLSCAASGFTFSSHAMTWVRQAP 14103 (FHVH93) GKGLEWVSAISGSGDYTHYADSVKGRFTISRDNSKNTVYLQM NSLRAEDSAVYYCAKDEDGGSLLGHRGQGTLVTVSS Spacer (e.g. hinge) IgG4 Hinge ESKYGPPCPPCP 14104 CD8 Hinge TTTPAPRPPTPAPTIASQPLSLRPE 14105 CD28 IEVMYPPPYLDNEKSNGTIIHVKGKHLCPSPLFPGPSKP 14106 Transmembrane CD8 ACRPAAGGAVHTRGLDFACDIYIWAPLAGTCGVLLLSLVITLY 14107 C CD28 FWVLVVVGGVLACYSLLVTVAFIIFWV 14108 CD28 FWVLVVVGGVLACYSLLVTVAFIIFWV 14109 Costimulatory domain CD28 RSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAAYRS 14110 4-1BB KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL 14111 Primary Signaling Domain RVKFSRSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR 14112 DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERR RGKGHDGLYQGLSTATKDTYDALHMQALPPR CD3zeta RVKFSRSADAPAYKQGQNQLYNELNLGRREEYDVLDKRRGR 14113 DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERR RGKGHDGLYQGLSTATKDTYDALHMQALPPR

[0416] In some embodiments, the antigen receptor further includes a marker and/or cells expressing the CAR or other antigen receptor further include a surrogate marker, such as a cell surface marker, which is used to confirm transduction or engineering of the cell to express the receptor. In some aspects, the marker includes all or part (e.g., truncated form) of CD34, a NGFR, or epidermal growth factor receptor, such as truncated version of such a cell surface receptor (e.g., tEGFR). In some embodiments, the nucleic acid encoding the marker is operably linked to a polynucleotide encoding a linker sequence, such as a cleavable linker sequence, e.g., T2A. For example, a marker, and optionally a linker sequence, can be any as disclosed in published patent application No. WO2014031687. For example, the marker can be a truncated EGFR (tEGFR) that is, optionally,

linked to a linker sequence, such as a T2A cleavable linker sequence.
[0417] In some embodiments, the marker is a molecule, e.g., cell surface protein, not naturally found on T cells or not naturally found on the surface of T cells, or a portion thereof. In some embodiments, the molecule is a non-self molecule, e.g., non-self protein, i.e., one that is not recognized as “self” by the immune system of the host into which the cells will be adoptively transferred.
[0418] In some embodiments, the marker serves no therapeutic function and/or produces no effect other than to be used as a marker for genetic engineering, e.g., for selecting cells successfully engineered. In other embodiments, the marker is a therapeutic molecule or molecule otherwise exerting some desired effect, such as a ligand for a cell to be encountered in vivo, such as a costimulatory or immune checkpoint molecule to enhance and/or dampen responses of the cells upon adoptive transfer and encounter with ligand.
[0419] In some cases, CARs are referred to as first, second, and/or third generation CARs. In some aspects, a first generation CAR is one that solely provides a CD3-chain induced signal upon antigen binding; in some aspects, a second-generation CAR is one that provides such a signal and costimulatory signal, such as one including an intracellular signaling domain from a costimulatory receptor such as CD28 or CD 137; in some aspects, a third generation CAR is one that includes multiple costimulatory domains of different costimulatory receptors.
[0420] For example, in some embodiments, the CAR contains an antibody, e.g., an antibody fragment, a transmembrane domain that is or contains a transmembrane portion of CD28 or a functional variant thereof, and an intracellular signaling domain containing a signaling portion of CD28 or functional variant thereof and a signaling portion of CD3 zeta or functional variant thereof. In some embodiments, the CAR contains an antibody, e.g., antibody fragment, a transmembrane domain that is or contains a transmembrane portion of CD28 or a functional variant thereof, and an intracellular signaling domain containing a signaling portion of a 4-IBB or functional variant thereof and a signaling portion of CD3 zeta or functional variant thereof. In some such embodiments, the receptor further includes a spacer containing a portion of an Ig molecule, such as a human Ig molecule, such as an Ig hinge, e.g. an IgG4 hinge, such as a hinge-only spacer.
[0421] In some aspects, the spacer contains only a hinge region of an IgG, such as only a hinge of IgG4 or IgG. In other embodiments, the spacer is or contains an Ig hinge, e.g., an IgG4-derived hinge, optionally linked to a CH2 and/or CH3 domains. In some embodiments, the spacer is an Ig hinge, e.g., an IgG4 hinge, linked to CH2 and CH3 domains. In some embodiments, the spacer is an Ig hinge, e.g., an IgG4 hinge, linked to a CH3 domain only. In some embodiments, the spacer is or comprises a glycine-serine rich sequence or other flexible linker such as known flexible linkers.
[0422] For example, in some embodiments, the CAR includes an antibody such as an antibody fragment, including scFvs, a spacer, such as a spacer containing a portion of an immunoglobulin molecule, such as a hinge region and/or one or more constant regions of a heavy chain molecule, such as an Ig-hinge containing spacer, a transmembrane domain containing all or a portion of a CD28-derived transmembrane domain, a CD28-derived intracellular signaling domain, and a CD3 zeta signaling domain. In some embodiments, the CAR includes an antibody or fragment, such as scFv, a spacer such as any of the Ig-hinge containing spacers, a CD28-derived transmembrane domain, a 4-IBB-derived intracellular signaling domain, and a CD3 zeta-derived signaling domain.
[0423] The recombinant receptors, such as CARs, expressed by the cells administered to the subject generally recognize or specifically bind to a molecule that is expressed in, associated with, and/or specific for the disease or condition or cells thereof being treated. Upon specific binding to the molecule, e.g., antigen, the receptor generally delivers an immunostimulatory signal, such as an ITAM-transduced signal, into the cell, thereby promoting an immune response targeted to the disease or condition. For example, in some embodiments, the cells express a CAR that specifically binds to an antigen expressed by a cell or tissue of the disease or condition or associated with the disease or condition.

b. T Cell Receptors Antigen Receptors (TCRs)

[0424] In some embodiments, engineered cells, such as T cells, used in connection with the provided methods, uses, articles of manufacture or compositions are cells that express a T cell receptor (TCR) or antigen-binding portion thereof that recognizes a protein epitope or T cell epitope of a target protein, such as an antigen of a tumor, viral or autoimmune protein.

[0425] In some embodiments, a “T cell receptor” or “TCR” is a molecule that contains variable α and β chains (also known as TCRalpha and TCRbeta, respectively) or a variable γ and δ chains (also known as TCRgamma and TCRdelta, respectively), or antigen-binding portions thereof, and which is capable of specifically binding to a polypeptide bound to an MHC molecule. In some embodiments, the TCR is in the αβ form. Typically, TCRs that exist in αβ and γδ forms are generally structurally similar, but T cells expressing them may have distinct anatomical locations or functions. A TCR can be found on the surface of a cell or in soluble form. Generally, a TCR is found on the surface of T cells (or T lymphocytes) where it is generally responsible for recognizing antigens bound to major histocompatibility complex (MHC) molecules.

[0426] Unless otherwise stated, the term “TCR” should be understood to encompass full TCRs as well as antigen-binding portions or antigen-binding fragments thereof. In some embodiments, the TCR is an intact or full-length TCR, including TCRs in the ab form or gd form. In some embodiments, the TCR is an antigen-binding portion that is less than a full-length TCR but that binds to a specific peptide bound in an MHC molecule, such as binds to an MHC-peptide complex. In some cases, an antigen-binding portion or fragment of a TCR can contain only a portion of the structural domains of a full-length or intact TCR, but yet is able to bind the peptide epitope, such as MHC-peptide complex, to which the full TCR binds. In some cases, an antigen-binding portion contains the variable domains of a TCR, such as variable a chain and variable b chain of a TCR, sufficient to form a binding site for binding to a specific MHC-peptide complex. Generally, the variable chains of a TCR contain complementarity determining regions involved in recognition of the peptide, MHC and/or MHC-peptide complex.

c. Multi-Targeting

[0427] In some embodiments, the cells used in connection with the provided methods, uses, articles of manufacture and compositions include cells employing multi-targeting strategies, such as expression of two or more genetically engineered receptors on the cell, each recognizing the same of a different antigen and typically each including a different intracellular signaling component. Such multi-targeting strategies are described, for example, in WO 2014055668 (describing combinations of activating and costimulatory CARs, e.g., targeting two different antigens present individually on off-target, e.g., normal cells, but present together only on cells of the disease or condition to be treated) and Fedorov et al., Sci. Transl. Medicine, 5(215) (2013) (describing cells expressing an activating and an inhibitory CAR, such as those in which the activating CAR binds to one antigen expressed on both normal or non-diseased cells and cells of the disease or condition to be treated, and the inhibitory CAR binds to another antigen expressed only on the normal cells or cells which it is not desired to treat).

[0428] For example, in some embodiments, the cells include a receptor expressing a first genetically engineered antigen receptor (e.g., CAR) which is capable of inducing an activating or stimulatory signal to the cell, generally upon specific binding to the antigen recognized by the first receptor, e.g., the first antigen. In some embodiments, the cell further includes a second genetically engineered antigen receptor (e.g., CAR), e.g., a chimeric costimulatory receptor, which is capable of inducing a costimulatory signal to the immune cell, generally upon specific binding to a second antigen recognized by the second receptor. In some embodiments, the first antigen and second antigen are the same. In some embodiments, the first antigen and second antigen are different.

[0429] In some embodiments, the first and/or second genetically engineered antigen receptor (e.g. CAR) is capable of inducing an activating signal to the cell. In some embodiments, the receptor includes an intracellular signaling component containing ITAM or ITAM-like motifs. In some embodiments, the activation induced by the first receptor involves a signal transduction or change in protein expression in the cell resulting in initiation of an immune response, such as ITAM phosphorylation and/or initiation of ITAM-mediated signal transduction cascade, formation of an immunological synapse and/or clustering of molecules near the bound receptor (e.g. CD4 or CD8, etc.), activation of one or more transcription factors, such as NF-KB and/or AP-1, and/or induction of gene expression of factors such as cytokines, proliferation, and/or survival.

[0430] In some embodiments, the first and/or second receptor includes intracellular signaling domains or regions of costimulatory receptors such as CD28, CD137 (4-1BB), OX40, and/or ICOS. In some embodiments, the first and second receptor include an intracellular signaling domain of a costimulatory receptor that are different. In some embodiments, the first receptor contains a CD28 costimulatory signaling region and the second receptor contain a 4-IBB co-stimulatory signaling region or vice versa.

[0431] In some embodiments, the first and/or second receptor includes both an intracellular signaling domain containing ITAM or ITAM-like motifs and an intracellular signaling domain of a costimulatory receptor.

[0432] In some embodiments, the first receptor contains an intracellular signaling domain containing ITAM or ITAM-like motifs and the second receptor contains an intracellular signaling domain of a costimulatory receptor. The costimulatory signal in combination with the activating signal induced in the same cell is one that results in an immune response, such as a robust and sustained immune response, such as increased gene expression, secretion of cytokines and other factors, and T cell mediated effector functions such as cell killing.

[0433] In some embodiments, neither ligation of the first receptor alone nor ligation of the second receptor alone induces a robust immune response. In some aspects, if only one receptor is ligated, the cell becomes tolerized or unresponsive to antigen, or inhibited, and/or is not induced to proliferate or secrete factors or carry out effector functions. In some such embodiments, however, when the plurality of receptors are ligated, such as upon encounter of a cell expressing the first and second antigens, a desired response is achieved, such as full immune activation or stimulation, e.g., as indicated by secretion of one or more cytokine, proliferation, persistence, and/or carrying out an immune effector function such as cytotoxic killing of a target cell.

[0434] In some embodiments, the two receptors induce, respectively, an activating and an inhibitory signal to the cell, such that binding by one of the receptors to its antigen activates the cell or induces a response, but binding by the second inhibitory receptor to its antigen induces a signal that suppresses or dampens that response. Examples are combinations of activating CARs and inhibitory CARs or iCARs. Such a strategy may be used, for example, in which the activating CAR binds an antigen expressed in a disease or condition but which is also expressed on normal cells, and the inhibitory receptor binds to a separate antigen which is expressed on the normal cells but not cells of the disease or condition.

[0435] In some embodiments, the multi-targeting strategy is employed in a case where an antigen associated with a particular disease or condition is expressed on a non-diseased cell and/or is expressed on the engineered cell itself, either transiently (e.g., upon stimulation in association with genetic engineering) or permanently. In such cases, by requiring ligation of two separate and individually specific antigen receptors, specificity, selectivity, and/or efficacy may be improved.

[0436] In some embodiments, the plurality of antigens, e.g., the first and second antigens, are expressed on the cell, tissue, or disease or condition being targeted, such as on the cancer cell. In some aspects, the cell, tissue, disease or condition is multiple myeloma or a multiple myeloma cell. In some embodiments, one or more of the plurality of antigens generally also is expressed on a cell which it is not desired to target with the cell therapy, such as a normal or non-diseased cell or tissue, and/or the engineered cells themselves. In such embodiments, by requiring ligation of multiple receptors to achieve a response of the cell, specificity and/or efficacy is achieved.

d. Chimeric Auto-Antibody Receptor (CAAR)

[0437] In some embodiments, the recombinant receptor is a chimeric autoantibody receptor (CAAR). In some embodiments, the CAAR binds, e.g., specifically binds, or recognizes, an autoantibody. In some embodiments, a cell expressing the CAAR, such as a T cell engineered to express a CAAR, is used to bind to and kill autoantibody-expressing cells, but not normal antibody expressing cells. In some embodiments, CAAR-expressing cells are used to treat an autoimmune disease associated with expression of self-antigens, such as autoimmune diseases. In some embodiments, CAAR-expressing cells target B cells that ultimately produce the autoantibodies and display the autoantibodies on their cell surfaces, marking these B cells as disease-specific targets for therapeutic intervention. In some embodiments, CAAR-expressing cells are used to efficiently target and kill the pathogenic B cells in autoimmune diseases by targeting the disease-causing B cells using an antigen-specific chimeric autoantibody receptor. In some embodiments, the recombinant receptor is a CAAR, such as any described in U.S. Patent Application Pub. No. US 2017/0051035.

[0438] In some embodiments, the CAAR comprises an autoantibody binding domain, a transmembrane domain, and one or more intracellular signaling region or domain (also interchangeably called a cytoplasmic signaling domain or region). In some embodiments, the intracellular signaling region comprises an intracellular signaling domain. In some embodiments, the intracellular signaling domain is or comprises a primary signaling domain, a signaling domain that is capable of stimulating and/or inducing a primary activation signal in a T cell, a signaling domain of a T cell receptor (TCR) component (e.g. an intracellular signaling domain or region of a CD3-zeta) chain or a functional variant or signaling portion thereof), and/or a signaling domain comprising an immunoreceptor tyrosine-based activation motif (ITAM).

[0439] In some embodiments, the autoantibody binding domain comprises an autoantigen or a fragment thereof. The choice of autoantigen can depend upon the type of autoantibody being targeted. For example, the autoantigen may be chosen because it recognizes an autoantibody on a target cell, such as a B cell, associated with a particular disease state, e.g. an autoimmune disease, such as an autoantibody-mediated autoimmune disease. In some embodiments, the autoimmune disease includes pemphigus vulgaris (PV). Exemplary autoantigens include desmoglein 1 (Dsgl) and Dsg3.

[0440] In some embodiments, the encoded nucleic acid is operatively linked to a “positive target cell-specific regulatory element” (or positive TCSRE). In some embodiments, the positive TCSRE is a functional nucleic acid sequence. In some embodiments, the positive TCSRE comprises a promoter or enhancer. In some embodiments, the TCSRE is a nucleic acid sequence that increases the level of an exogenous agent in a target cell. In some embodiments, the positive target cell-specific regulatory element comprises a T cell-specific promoter, a T cell-specific enhancer, a T cell-specific splice site, a T cell-specific site extending half-life of an RNA or protein, a T cell-specific mRNA nuclear export promoting site, a T cell-specific translational enhancing site, or a T cell-specific post-translational modification site. In some embodiments, the T cell-specific promoter is a promoter described in Immgen consortium, herein incorporated by reference in its entirety, e.g., the T cell-specific promoter is an IL2RA (CD25), LRRC32, FOXP3, or IKZF2 promoter. In some embodiments, the T cell-specific promoter or enhancer is a promoter or enhancer described in Schmidl et al., Blood. 2014 Apr. 24; 123(17):e68-78., herein incorporated by reference in its entirety. In some embodiments, the T cell-specific promoter is a transcriptionally active fragment of any of the foregoing. In some embodiments, the T-cell specific promoter is a variant having at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity to any of the foregoing.

[0441] In some embodiments, the encoded nucleic acid is operatively linked to a “negative target cell-specific regulatory element” (or negative TCSRE). In some embodiments, the negative TCSRE is a functional nucleic acid sequence. In some embodiments, the negative TCSRE is a miRNA recognition site that causes degradation of inhibition of the viral vector in a non-target cell. In some embodiments, the exogenous agent is operatively linked to a “non-target cell-specific regulatory element” (or NTCSRE). In some embodiments, the NTCSRE comprises a nucleic acid sequence that decreases the level of an exogenous agent in a non-target cell compared to in a target cell. In some embodiments, the NTCSRE comprises a non-target cell-specific miRNA recognition sequence, non-target cell-specific protease recognition site, non-target cell-specific ubiquitin ligase site, non-target cell-specific transcriptional repression site, or non-target cell-specific epigenetic repression site. In some embodiments, the NTCSRE comprises a tissue-specific miRNA recognition sequence, tissue-specific protease recognition site, tissue-specific ubiquitin ligase site, tissue-specific transcriptional repression site, or tissue-specific epigenetic repression site. In some embodiments, the NTCSRE comprises a non-target cell-specific miRNA recognition sequence, non-target cell-specific protease recognition site, non-target cell-specific ubiquitin ligase site, non-target cell-specific transcriptional repression site, or non-target cell-specific epigenetic repression site.

[0442] In some embodiments, the NTCSRE comprises a non-target cell-specific miRNA recognition sequence and the miRNA recognition sequence is able to be bound by one or more of miR3 1, miR363, or miR29c. In some embodiments, the NTCSRE is situated or encoded within a transcribed region encoding the exogenous agent, optionally wherein an RNA produced by the transcribed region comprises the miRNA recognition sequence within a UTR or coding region.

[0443] In some embodiments, the viral vector comprising an anti-CD4 scFv or sdAb composition described herein are administered to a subject, e.g., a mammal, e.g., a human. In such embodiments, the subject is at risk of, has a symptom of, or is diagnosed with or identified as having, a particular disease or condition (e.g., a disease or condition described herein).

[0444] In some aspects, resting or non-activated T cells are contacted with a viral vector of the disclosure (e.g., a retroviral vector or lentiviral vector) that includes a CD4 binding agent. The contacting may be performed in vitro (e.g., with T cells derived from a healthy donor or a donor in need of cellular therapy) or in vivo by administration of the viral vector to a subject. In some embodiments the process comprises a) obtaining whole blood from the subject; b) collecting the fraction of blood containing leukocyte components including CD4+ T cells; c) contacting the leukocyte components including CD4+ T cells with a composition comprising the lentiviral vector to create a transfection mixture; and d) reinfusing the contacted leukocyte components including CD4+ T cells and/or the transfection mixture to the subject, thereby administering the lipid particle and/or payload gene to the subject. In some embodiments, the T cells (e.g. CD4+ T cells) are not activated during the method. In some embodiments, step (c) of the method is carried out for no more than 24 hours, e.g., no more than 20, 16, 12, 8, 6, 5, 4, 3, 2, or 1 hour.

[0445] In some embodiments, the method according to the present disclosure is capable of delivering a lentiviral particle to an ex vivo system. The method includes the use of a combination of various apheresis machine hardware components, a software control module, and a sensor module to measure citrate or other solute levels in-line to ensure the maximum accuracy and safety of treatment prescriptions, and the use of replacement fluids designed to fully exploit the design of the system according to the present methods. It is understood that components described for one system according to the present invention can be implemented within other systems according to the present invention as well.

[0446] In some embodiments, the method for administration of the lentiviral vector to the subject comprises the use of a blood processing set for obtaining whole blood from the subject, a separation chamber for collecting the fraction of blood containing leukocyte components including CD4+ T cells, a contacting container for contacting the CD4+ T cells with the composition comprising the lentiviral vector, and a further fluid circuit for reinfusion of CD4+ T cells to the patient. In some embodiments, the method further comprises any of i) a washing component for concentrating T cells, and ii) a sensor and/or module for monitoring cell density and/or concentration. In some embodiments, the methods allow processing of blood directly from the patient, transduction with the lentiviral vector, and reinfusion directly to the patient without any steps of selection for T cells or for CD4+ T cells. Further the methods also can be carried out without cryopreserving or freezing any cells before or between any one or more of the steps, such that there is no step of formulating cells with a cryoprotectant, e.g. DMSO. In some embodiments, the provided methods do not include a lymphodepletion regimen. In some embodiments, the method including steps (a)-(d) are carried out for a time of no more than 24 hours, such as between 2 hours and 12 hours, for example 3 hours to 6 hours.

[0447] In some embodiments, the method for administration of the lentiviral vector to the subject comprises the use of a blood processing set for obtaining whole blood from the subject, a separation chamber for collecting the fraction of blood containing leukocyte components including CD4+ T cells, a contacting container for contacting the CD4+ T cells with the composition comprising the lentiviral vector, and a further fluid circuit for reinfusion of CD4+ T cells to the patient. In some embodiments, the method further comprises any of i) a washing component for concentrating T cells, and ii) a sensor and/or module for monitoring cell density and/or concentration. In some embodiments, the methods allow processing of blood directly from the patient, transduction with the lentiviral vector, and reinfusion directly to the patient without any steps of selection for T cells or for CD4+ T cells. Further the methods also can be carried out without cryopreserving or freezing any cells before or between any one or more of the steps, such that there is no step of formulating cells with a cryoprotectant, e.g. DMSO. In some embodiments, the provided methods do not include a lymphodepletion regimen. In some embodiments, the method including steps (a)-(d) are carried out for a time of no more than 24 hours, such as between 2 hours and 12 hours, for example 3 hours to 6 hours.

[0448] Also provided herein are systems for administration of a lentiviral vector comprising a CD4 binding agent to a subject. An exemplary system for administration is shown in FIG. **1**.

[0449] In some embodiments, the resting or non-activated T cells are not treated with one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines. In some embodiments, the resting or non-activated T cells are not treated with any of one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines.

[0450] In additional aspects, the application includes methods of administration to a subject of a viral vector that includes an anti-CD4 binding agent, wherein the subject is not administered or has not been administered a T cell activating treatment. In some embodiments, the T cell activating treatment includes one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines. In some embodiments, the subject is not administered or has not been administered any of one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines. In some embodiments, the T cell activating treatment is lymphodepletion. In certain embodiments, the subject is not administered or has not been administered the T cell activating treatment within 1 month before or after administration of the viral vector. In some embodiments, the subject is not administered or has not been administered the T cell activating treatment within 1 month before administration of the viral vector, such as within or at or about 4 weeks, 3 weeks, 2 weeks or 1 weeks, such as at or about 1 day, 2 days, 3 days, 4 days, 5 days, 6 days or 7 days before administration of the viral vector. In some embodiments, the subject is not administered the T cell activating treatment within 1 month after administration of the viral vector, such as within or at or about 4 weeks, 3 weeks, 2 weeks or 1 weeks, such as at or about 1 day, 2 days, 3 days, 4 days, 5 days, 6 days or 7 days after administration of the viral vector.

[0451] In some aspects, the viral vectors of the disclosure do not include one or more T cell stimulatory molecules (e.g., an anti-CD-3 antibody), one or more T cell costimulatory molecules, and/or one or more T cell activating cytokines.

[0452] The use of anti-CD3 antibodies is well-known for activation of T cells. The anti-CD3 antibodies can be of any species, e.g., mouse, rabbit, human, humanized, or camelid. Exemplary antibodies include OKT3, CRIS-7, I2C the anti-CD3 antibody included in DYNABEADS Human T-Activator CD3/CD28 (Thermo Fisher), and the anti-CD3 domains of approved and clinically studied molecules such as blinatumomab, catumaxomab, fotetuzumab, teclistamab, ertumaxomab, epcoritamab, talquetamab, odronextamab, cibistamab, obrindatamab, tidutamab, duvortuxizumab, solitomab, eluvixtamab, pavurutamab, tepoditamab, vibecotamab, plamotamab, glofitamab, etevritamab, and tarlatamab.

[0453] In some embodiments, the one or more T cell costimulatory molecules include CD28 ligands (e.g., CD80 and CD86); antibodies that bind to CD28 such as CD28.2, the anti-CD28 antibody included in DYNABEADS Human T-Activator CD3/CD28 (Thermo Fisher) and anti-CD28 domains disclosed in US2020/0199234, US2020/0223925, US2020/0181260, US2020/0239576, US2020/0199233, US2019/0389951, US2020/0299388, US2020/0399369, and US2020/0140552; CD137 ligand (CD137L); anti-CD137 antibodies such as urelumab and utomilumab; ICOS ligand (ICOS-L); and anti-ICOS antibodies such as feladilimab, vopratelimab, and the anti-ICOS domain of izuralimab.

[0454] In some embodiments, the one or more T cell activating cytokines include IL-2, IL-7, IL-15, IL-21, interferons (e.g., interferon-gamma), and functional variants and modified versions thereof.

[0455] Lymhpodepletion may be induced by various treatments that destroy lymphocytes and T cells in the subject. For example, the lymphodepletion may include myeloablative chemotherapies, such as fludarabine, cyclophosphamide, bendamustine, and combinations thereof. Lymphodepletion may also be induced by irradiation (e.g., full-body irradiation) of the subject.

[0456] In some embodiments, the source of targeted lipid particles is the same subject that is administered a targeted lipid particle composition. In other embodiments, they are different. In some embodiments, the source of targeted lipid particles and recipient tissue is autologous (from the same subject) or heterologous (from different subjects). In some embodiments, the donor tissue for targeted lipid particle compositions described herein is a different tissue type than the recipient tissue. In some embodiments, the donor tissue is muscular tissue and the recipient tissue is connective tissue (e.g., adipose tissue). In other embodiments, the donor tissue and recipient tissue are of the same or different type, but from different organ systems.

[0457] In some embodiments, the targeted lipid particles (e.g, viral vector) composition described herein are administered to a subject having a cancer, an autoimmune disease, an infectious disease, a metabolic disease, a neurodegenerative disease, or a genetic disease (e.g., enzyme deficiency). In some embodiments, the subject is in need of regeneration.

[0458] In some embodiments, the cancer is a T cell-mediated cancer. In another embodiment, the antigen binding moiety portion of a CAR is designed to treat a particular cancer. In some embodiments, the targeted lipid particle is used to treat cancers and disorders including but not limited to non-Hodgkin lymphoma (NHL), acute lymphocytic leukemia (ALL), chronic lymphocytic leukemia (CLL), multiple myeloma, and the like. In some embodiments, the targeted lipid particle is used to treat B cell malignancies, e.g., refractory B cell malignancies.

[0459] In some embodiments, the targeted lipid particle is co-administered with an inhibitor of a protein that inhibits membrane fusion. For example, Suppressyn is a human protein that inhibits cell-cell fusion (Sugimoto et al., “A novel human endogenous retroviral protein inhibits cell-cell fusion” *Scientific Reports* 3:1462 DOI: 10.1038/srep01462). In some embodiments, the targeted lipid particle is co-administered with an inhibitor of sypressyn, e.g., a siRNA or inhibitory antibody.

EXAMPLES

[0460] The present disclosure may be further described by the following non-limiting examples, in which standard techniques known to the skilled artisan and techniques analogous to those described in these examples may be used where appropriate. It is understood that the skilled artisan will envision additional embodiments consistent with the disclosure provided herein.

Example 1 Characterization of CD4 Binders

[0461] Binders were selected that demonstrated detectable CD4 binding in solution. To assay the ability of the binders to direct cell-specific transduction, the binders were used to generate binder (e.g., as scFv)-Nipah G glycoprotein fusions as described in WO2017182585 for pseudotyping of lentiviral vectors. The Nipah G-linker-binder construct was codon optimized for expression in human cells and sub-cloned into an expression vector for lentivirus generation.

[0462] Crude lentiviral production was performed as follows: HEK-293LX cells were plated 24 hours in advance of transfection. On the day of transfection, HEK-293LX cells were transfected with a lentiviral packaging plasmid, the lentiviral transfer plasmid encoding GFP (pSFFV-GFP), and the plasmids encoding for Nipah G protein retargeted for CD4 receptor targeting (NIV-G (CD4)) and Nipah F fusion protein (NiV-Fd22). To harvest the lentivirus, supernatant was removed from the HEK293LX cells and spun at 1000×g for 5 minutes. The supernatant was removed and immediately added to CD4-positive target cells or T cells, or frozen at −80° C. for later use. Lentiviral vectors were produced in both adherent cells and in suspension. In certain experiments, the vectors were filtered using a 0.45 μm filter and concentrated by ultracentrifugation.

[0463] To estimate activity of the CD4-retargeted vectors, the supernatants were diluted 1:5 and used to transduce SupT1 cells. After 5 days, the transduced cells were assayed for GFP expression by flow cytometry. The percentage of live cells expressing GFP is shown in Table 17, column “Single point SupT1”. Titer of certain CD4-retargeted vectors was determined by multi-point dilution of vector for transduction of SupT1 cells (adherent and suspension production) (Table 17, columns “Multiple point SupT1 Adh.” and “Multiple point SupT1 Susp.”). Titer was similarly determined using HEK-293T cells overexpressing *Macaca nemestrina* CD4 to estimate cross-reactivity of the vectors (Table 17, column “Multiple point 293oeNemestrinaCD4”).

TABLE-US-00017 TABLE 17 Multiple point Multiple point Single point Multiple point SupT1 SupT1 Susp. 293oeNemestri- SupT1 1:5 dilution Adh. titer titer naCD4 Adh. titer Binder GFP (% GFP) [Log(TU/mL)] [Log(TU/mL)] [Log(TU/mL)] 265 24.1 ± 10.8 5.8 ± 0.25 266 13.1 ± 1.6 5.51 ± 0.05 267 10.1 ± 5.8 5.27 ± 0.25 4.7 263 4.5 ± 7 268 34.7 ± 7.8 6.1 ± 0.09 5.65 ± 0.57 270 56.5 ± 12.1 6.53 ± 0.3 5.32 ± 0.23 271 52.9 ± 11.8 6.51 ± 0.21 5.38 ± 0.28 273 44 ± 12.2 6.29 ± 0.26 5.7 ± 0.51 5.31 ± 0.76 274 23 ± 18 5.8 ± 0.36 5.61 ± 0.13 275 53.9 ± 3.2 6.46 ± 0.24 5.45 ± 0.14 276 9.4 ± 0.1 5.35 ± 0.05 277 4.2 ± 1.5 5.10 278 36.6 ± 5 6.12 ± 0.04 5.3 279 37.5 ± 34.8 6.13 ± 0.67 5.71 ± 0.19 264 11.3 ± 0 257 47.3 ± 1.1 6.1 ± 0.15 280 25.9 ± 2.1 6 ± 0.04 5.34 ± 0.12 282 12.8 ± 1.1 5.51 ± 0.05 5.19 ± 0.12 283 41.1 ± 8.2 6.26 ± 0.11 5.24 ± 0.09 284 17.9 ± 2.3 5.65 ± 0.07 285 5.9 ± 1.8 5.24 286 59.3 ± 25.7 6.23 ± 0.39 5.46 ± 0.03 6.81 ± 0.06 258 49.6 ± 0.8 6.1 ± 0.17 287 38.8 ± 22.8 5.96 ± 0.49 5.28 ± 0.2 288 5.3 ± 0.5 5.18 289 9.8 ± 2.8 5.39 ± 0.12 290 27.7 ± 14.5 5.91 ± 0.4 5.36 ± 0.19 291 0.6 ± 0.5 260 0.2 ± 0 292 21.8 ± 8.3 5.9 ± 0.3 5.63 ± 0.29 6.59 ± 0.03 293 3.9 ± 0.3 294 11.6 ± 2.3 5.45 ± 0.08 295 11.3 ± 8.9 5.74 ± 0.28 5.2 ± 0.19 296 5.8 ± 2.4 5.10 261 3.1 ± 0 297 31.8 ± 4.4 5.89 ± 0.26 5.34 ± 0.14 298 19.9 ± 0.3 5.79 ± 0.12 300 16.3 ± 4.1 5.59 ± 0.12 301 28.2 ± 8.7 5.93 ± 0.21 5.34 ± 0.2 302 0.2 ± 0.2 304 20.7 ± 9.9 5.74 ± 0.28 5.15 ± 0.11 305 30.7 ± 9.9 6.01 ± 0.15 5.26 ± 0.26 306 0.3 ± 0 307 4.7 ± 2 5.24 314 25 ± 4.5 5.89 ± 0.12 5.43 ± 0.21 315 21.3 ± 5.9 5.79 ± 0.14 5.61 ± 0.37 316 0.2 ± 0.1 317 12.5 ± 4.4 5.41 ± 0.15 4.85 319 35.4 ± 17.2 6.16 ± 0.11 5.31 ± 0.18 320 14 ± 7.4 5.44 ± 0.3 5.08 ± 0.21 322 59.7 ± 10.3 6.44 ± 0.2 5.57 ± 0.26 323 20.5 ± 4.5 5.77 ± 0.13 324 14.2 ± 6.9 5.41 ± 0.18 5.02 ± 0.42 4.62 325 21.4 ± 5.1 5.82 ± 0.17 5.44 ± 0.14 326 43.7 ± 7.7 6.25 ± 0.15 5.54 ± 0.25 327 0.3 ± 0 262 0 ± 0 328 23.5 ± 9.1 5.76 ± 0.16 5.49 ± 0.28 329 10.5 ± 2.4 5.34 ± 0.15 4.92 ± 0.11 330 14.1 ± 0.8 5.56 ± 0.02 5.17 ± 0.1 331 24.6± 1.2 5.92 ± 0.04 5.21 ± 0.06 333 22.9 ± 13.6 5.89 ± 0.26 4.85 5.78 ± 1 334 5.5 ± 2.1 5.24 5.25 ± 0.1 335 17.6 ± 2.3 5.64 ± 0.05 5.21 ± 0.12 336 0.1 ± 0.1 337 27.4 ± 6.5 6.01 ± 0.11 5.25 ± 0.07 339 23.1 ± 5.2 5.8 ± 0.13 5.09 ± 0.09 340 13 ± 2.2 5.43 ± 0.15 5.15 ± 0.08 109 17.7 ± 3.5 5.66 ± 0.12

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3.1 5.74 ± 0.19 356 2.4 ± 0.4 357 0.9 ± 0.1 5.41 358 39.6 ± 15 6.19 ± 0.17 5.1 359 0.9 ± 0.1 360 0.2 ± 0.1 361 12.7 ± 2 5.49 ± 0.07 362 8.6 ± 1.3 5.35 ± 0.07 363 13.3 ± 3 5.51 ± 0.1 364 60.5 ± 12.3 6.44 ± 0.36 5.8 5.75 ± 0.69 259 54.9 ± 6.8 6.24 ± 0.13 5.97 ± 0.12 365 2.4 ± 0.3 366 20.5 ± 2 5.83 ± 0.14 367 51.8 ± 16.3 6.54 ± 0.19 5.49 ± 0.07 368 7 ± 0.4 5.24 ± 0 369 37.7 ± 18.8 6.13 ± 0.25 5.33 ± 0.04 6.59 ± 0.24 370 11 ± 1 5.44 ± 0.06 371 5.9 ± 0.8 5.14 ± 0.06 5.4 ± 0.07 372 2.5 ± 0.7 373 0.1 ± 0.1 374 24 ± 4.2 6.02 ± 0.11 375 24.6 ± 0.5 6.03 ± 0.04 376 8.8 ± 1.1 5.35 ± 0.07 377 0.7 ± 0.1 378 21 ± 0.2 5.88 ± 0 5.18 379 45.9 ± 12.1 6.23 ± 0.24 5.64 ± 0.02 380 6.7 ± 0.7 5.21 ± 0.05 381 1.4 ± 0.7 382 49 ± 0.2 6.2 ± 0.16 5.1 6.67 ± 0.07 383 11.3 ± 0.9 5.46 ± 0.03 384 51.2 ± 1.1 6.23 ± 0.18 5.24 385 17.6 ± 8.9 5.62 ± 0.26 6.57 386 45.1 ± 13.3 6.25 ± 0.15 5.24 ± 0.09 6.24 ± 0.09 223 2.4 ± 0.4 224 6.45 ± 0.04 106 0.9 ± 0.1 6.45 ± 0.04 225 1.5 ± 0.1 226 0.1 ± 0 228 15.5 ± 1.1 5.39 ± 0.16 6.21 ± 0.1 229 0.6 ± 0.4 107 0.1 ± 0.1 238 0±0 239 0 ± 0 240 0.1 ± 0 241 41.1 ± 2.1 5.77 ± 0.24 5.99 ± 0.11 243 1.7 ± 0.6 256 60.5 ± 11.3 6.31 ± 0.4 6.37 ± 1.18 5.91 ± 1.33

[0464] Titer was also determined on SupT1 cells following concentration of the viral vector compositions (Table 26).

TABLE-US-00018 TABLE 26 SupT1 titer SupT1 titer SupT1 titer (concentrated, CD4 Nemestrina (crude, (crude, suspension, domain Cross- adherent, suspension, PE) TU/mL Binder Modality specificity Reactivity PE) TU/mL PE) TU/mL [conc factor]* 387 DARPin Domain 1 Cross- 6.50E6 ± 2.33E6* 1.2E8 reactive 1.47E6 [200X] 256 VHH Domain 2 Cross- 4.73E6 ± 2.34E6* 5.22E7 (VHH4)* reactive 2.79E6 [200X] 388* VHH Domain 3/4* Cross- 1.18E ± Too low* 5.45E6 reactive 1.26E6 [200] 279 scFv Domain 3/4 No cross- 2.85E6 ± 5.50E5 ± 1.43E8 reactivity 3.35E6 2.46E5 [200X] 286 scFv Domain 3/4 Cross- 2.27E6 ± 2.88E5 ± 5.47E7 reactive 1.78E6 1.77E4 [200X] 287 scFv Domain 2 No cross- 1.73E6 ± 1.94E5 ± 3.83E7 reactivity 1.29E6 8.40E4 [200X] 322 scFv Domain 3/4 No cross- 3.03E6 ± 4.40E5 ± 4.66E7 reactivity 1.42E6 3.25E5 [200X] 47 scFv Domain 1/2 Cross- 4.39E6 ± 6.47E5 ± 6.86E7 reactive 1.98E6 3.21E5 [200X] 48 scFv Domain 1/2 Cross- 5.91E6 ± 6.93E5 ± 8.75E7 reactive 4.47E6 3.22E5 [200X] 75 scFv Domain 2 Cross- 3.80E6 ± 1.44E6 1.36E8 reactive 1.75E6 [200X] 364 scFv Domain 2 Cross- 3.41E6 ± 7.42E5 5.67E7 reactive 1.70E6 [200X]

[0465] Off-target transduction directed by certain binders was determined using CD4 knockout SupT1 cells and HEK-293T cells, which were determined to be negative for CD4 expression. CD4-retargeted vectors expressing GFP were produced in either adherent or suspension culture, as described above, and used to transduce CD4 knockout SupT1 cells and HEK-293T cells at a single dilution. The percentage of GFP-expressing cells was determined by flow cytometry (FIG. **2**).

[0466] Transduction efficiency on human PBMCs and Pan T cells was also determined for CD4-targeted lentiviral vectors. Concentrated vector was produced as described above and used to transduce human PBMCs from 3 donors, and the transduced cells were assayed for GFP expression by flow cytometry five days after transduction. The flow cytometry results are presented in FIG. **3**. The percent of live cells that were GFP+ are shown in FIG. **4** The vectors showed a strong specificity for CD4+ cells vs. CD4− cells.

[0467] Binding kinetics of certain CD4 binders were assayed using biolayer interferometry (BLI). The CD4 binders were expressed as homodimers with mouse Fc. Human CD4-Fc was used as the capture reagent. Kinetic parameters are shown in Table 18 below.

TABLE-US-00019 TABLE 18 Binder Conc [nM] K.sub.D [nM] K.sub.on [1/Ms] K.sub.off [1/s] 387 250 7 2.47E+04 1.75E−05 256 500 5.14 8.95E+04 4.60E−04 279 125 5 4.17E+04 2.10E−04 286 125 13.8 7.19E+02 2.70E−04 322 62.5 1.7 9.94E+04 1.75E−04 47 62.5 0.3 4.79E+05 5.70E−05 75 62.5 0.7 3.12E+05 2.05E−04 364 62.5 7.54 3.04E+04 2.30E−06

Example 2. Transduction of Resting T Helper Cells Using a CD4 Targeted Fusogen to Generate CAR T Cells

[0468] A CD19-specific CAR encoding 4-1BB and the CD3zeta endodomains (CD 19 CAR) was generated to examine CD4+ CAR T transduction efficiency and functionality. PBMCs were thawed and activated with anti-CD3/anti-CD28 beads and exposed to GFP-expressing CD4 fusosomes (Binder 256), and specificity of targeting CD4+ T cells was measured by flow cytometry.

[0469] CD19 CAR fusosomes targeting CD4 were used to test transduction efficiency against activated (CD3/CD28 or IL-7 treated) or resting CD4+ T cells, and to measure T cell function against CD19+ and CD19 CRISPR/Cas9-knockout lymphoma cells (Nalm-6) (e.g., tumor co-culture and rechallenge assays and cytokine production) in vitro. Vector copy number (VCN) was determined by a multiplex digital droplet polymerase chain reaction (ddPCR) assay and reported as copies per diploid genome (c/dg). CD4-targeted CD 19 CAR fusosomes could efficiently transduce both activated (34%+1.5% CD4+ CAR+; 0.54±0.18 c/dg), and resting CD4-selected T cells, albeit at a lower expression and integration level (20%±0.5% CD4+ CAR+; 0.28±0.14 c/dg). Resting CD4-transduced CART cells demonstrated specific cytotoxicity and cytokine production (GM-CSF, IFN-γ, TNF-α, IL-2, IL-6, and IL-10) against CD19+ Nalm-6 cells, but did not recognize CD19 knockout tumor cells. In long-term co-culture assays (9-day) with repetitive stimulation with fresh tumor cells (3×), CD4+CD 19 CAR T cells transduced without prior activation continued to show potent tumor cell killing.

[0470] CD4-specific fusosomes encoding LVV were observed to efficiently deliver an integrating CAR payload to resting and activated CD4+ T cells. Modified CD4+ CAR T cells demonstrated potent anti-tumor activity against CD19+ tumor cells. These data are consistent with a finding that targeting the CD4 co-receptor through in vivo delivery using a novel pseudotyped LVV can produce functional CAR T cells.

Example 3. CD4-Targeted Fusosomes Reduce CD19+ Tumor Burden In Vivo

[0471] CD4-targeted CD19 CAR fusosomes (lentiviral vector) were generated substantially as described above and assessed for their ability to reduce tumor burden in vivo. The fusosomes were pseudotyped with Nipah virus fusogen retargeted with CD4 Binder 256, CD4 Binder 75, and a CD8 Binder Control. NSG mice (n=5/group) were injected with 1E6 Nalm6-Luc leukemia B cells via intravenous (IV) tail injection, followed three days later by an IV tail injection of 1E7 human peripheral blood mononuclear cells (hPBMC). A day after hPBMC injection, 2.5E6, 5E6, or 1E7 integrating units (IU) of CD4-targeted CD19 CAR fusosomes were injected into separate groups of mice via the same route and volume. Beginning 1 day following fusosome injection, Nalm6 tumor progression was tracked via bioluminescent imaging (BLI) weekly throughout the duration of the study. Tumor growth data is represented as total flux (photons/sec). In addition, peripheral blood was collected from all animals on study day 15 to assess the presence of CAR+ T− cells in peripheral blood using flow cytometry. The CD19 CAR contained an anti-scFv directed against CD19 and an intracellular signaling domain containing intracellular components of 4-1BB and CD3-zeta.

[0472] As shown in FIGS. **5**A and **5**B, dose dependent tumor control was observed with Binder 256 and CD8 Control Binder at Day 21. Binder 75 showed a reduced ability to control Nalm6 growth, as shown in FIG. **5**C. As shown in FIG. **5**D, expression of the CAR in CD4+ T cells was dose-dependent. These data indicate that in vivo delivery of a CD19 CAR transgene payload with CD4-targeted fusosomes in CD19+ tumor bearing mice demonstrates robust production of CAR T cells and CD19+ tumor eradication.

TABLE-US-00020 TABLE 19 Full Binder Sequences Table 19-Full length sequences Binder SEQ ID Sequence Name NO:
QVQLQQSGAELMKPGASVKISCKATGYTFSSYWIEWVKQRPGHGLEWIGEIFPGS 1 1
GHTSFNEKFKGKATFTADTSSNTAYIQLSSLTSEDSAVYYCARRGYGYDEGFDYWG
QGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINSY
LSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYCL
QYDEFPPTFGAGTKLELK
QVQLQQSGAELMKPGASVKISCKATGYTLSSYWIEWVKQRPGHGLEWIGEILPGS 2 2
GSTSYNEKFKGKATFTADTSSSTAYMQLSSLTSEDSAVYYCARRGYGYDEGFDYW

GQGTTLTVSSGGGSGGGGSGGGGSDIKMTQSPSSMYASLGERVTITCKASQDINS
YLSWFQQKPGKSPKTLIYRANRLVDGVPSRFSGSGSGQDYSLTISSLEYEDMGIYYC
LQYDEFPPTFGAGTKLELK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATSYNQKFK 3 3
GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGEYFDYWGQGTTLTVSSGG
GSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGK
SPQLLVFAATYLADGVPSRFSGSGSGTQYSLKINSLQSEDFGNYYCQHFWGTPWT
FGGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK 4 4
GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGG
GSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGK
SPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWT
FGGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTSYNQKFK 5 5
GKATFTVDTSSSTAYMHFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGG
GSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGK
SPQVLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGSPWT
FGGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK 6 6
GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGG
GSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGK
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GGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGANGYNQKFK 7 7
GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGG
GSGGGGSGGGGSDIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGK
SPQLLVYAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWT
FGGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK 8 8
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SPQVLVYAATNVADGVPSRFSGSGSGTQYSLKINSLQSEDFGSYYCQHFWGTPWT
FGGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK 9 9
GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGG
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FGGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK 10 10
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LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGVTGYNQKFK 11 11
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TFGGGTKLEIK
LVKTGASVKISCKASGYSFTGFYMHWVKQSHGKGLEWIGYISSYNGATGYNQKFK 112 12
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GGGTKLEIK
LVKTGASVKISCKASGYSFTGYYMHWVKQSHGKSLEWIGYISSYNGATGYNQKFK 13 13
GKATFTVDTSSSTAYMQFNSLTSEDSAVYYCARGRYGDYFDYWGQGTTLTVSSGG
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SGSSYYNEKFKGKATLTADKSSNTAYMQLSSLTSEDSAVYFCARRGERGPWFAYW
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NWYQQKPGKPPSFLIYYATELAEGVPSRFSGSGSGSDYSLTISNLESEDFADYYCLQ
FYEFPYTFGGGTKLEIK
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DVWGTGTTVTVSSGGGSGGGGSGGGGSDIQMTQSPASLSASVGETVTITCRASG
NIHNYLAWYQQKQGKSPQLLVYNAKTLADGVPSRFSGSGSGTQYSLKINSLQPED
FGSYYCQHFWSTPWTFGGGTKLEIK
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TABLE-US-00022 TABLE 21 VL Sequences Table 21-VL sequences Binder SEQ ID Sequence Name NO:
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EIVMTQSPATLSVSPGERATLSCRASRSVSTYLAWYQQKPGQAPRLLIYGASTRATGI 315
9635 PARFSGSGSGTEFTLTISSLQSEDFAVYYCQQYDGSPYTFGQGTKLEIK
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9636 NLETGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQTPPAFGPGTKVDIK
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9637 STRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSAPPTFGQGTKVEIK

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9638 STRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSDPITFGQGTKLEIK
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9639 SARESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSIPIAFGQGTRLEIK
DIVMTQSPDSLAVSLGERATINCKSSQSVLSSSYNKNYLAWYQQKPGQPPKLLIYWA 320
9640 STRDSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSIPYTFGQGTKLEIK
DIVMTQSPDSLAVSLGERATINCKSSQSVLSTSYNKNYLAWYQQKPGQPPKLLIYW 321
9641 ASTRASGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYTTPPTFGQGTKVEIK
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9642 ASTRQSGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPYTFGQGTKLEIK
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9644 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPTFGQGTKVEIK
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9645 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQEADSFPLTFGGGTKVEIK
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9646 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPWTFGQGTKLEIK
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9647 PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPFTFGQGTRLEIK
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9648 GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPLTFGGGTKVEIK
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9649 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPFTFGPGTKVDIK
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9650 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLNSYPLTFGGGTKVEIK
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9651 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSAPLTFGGGTKLEIK
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9653 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIK
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9654 LQSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQARQTPLTFGQGTRLEIK
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9655 LQSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQTLQTPFTFGPGTKVDIK
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9656 RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQTPLTFGPGTKVDIK
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9657 ASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPLTFGGGTKVEIK
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9659 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPFTFGPGTKVDIK
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9660 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSFPTFGGGTKVEIK
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9661 PARFSGSGSGTEFTLTISSLQSEDFAVYYCQQYYDTPYTFGQGTKLEIKR*
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9662 STREPGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSTPPTFGGGTKLEIKR*

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9664 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTYSTPWTFGQGTRLEIKR*
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9665 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSSSFPLTFGQGTKVEIKR*
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9666 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLYNFPYTFGGGTKVEIKR*
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9667 PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNTPLTFGQGTKLEIKR
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9671 GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTNSFPLTFGQGTKLEIKR*
DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPGKAPKLLIYDASNLETGV 352
9672 PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGGGTKVEIKR*
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9673 RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALQSPWTFGQGTKLEIKR*
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9674 ASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYSSPLTFGGGTKVEIKR *
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9676 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYKTPYTFGQGTKLEIKR*
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9681 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCLQDYSYPLTFGQGTKVEIKR*
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9682 ASTRAAGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYYYTSTFGQGTKVEIKR*
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9683 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPPTFGGGTKVEIKR*
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9688 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSDSFPYTFGQGTKVEIKR*
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9689 IPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQHDSYPLTFGGGTKVEIKR*
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9690 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPPTFGQGTRLEIKR*
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9691 PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQTDSTFITFGQGTKVEIKR*
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9694 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPYTFGQGTKVEIKR*
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9695 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQANRFPLTFGQGTKLEIKR*
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9696 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYNFPATFGQGTRLEIKR*
DIVMTQSPLSLPVTPGEPASISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSN 377
9697 RASGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQGTHWPETFGQGTKVEIKR*
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9698 PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSTPLTFGQGTKLEIKR*
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9699 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYILPLTFGGGTKVEIKR*
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9700 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSAPYTFGQGTKLEIKR*
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9701 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSYSSPWTFGQGTRLEIKR*
DIQMTQSPSSLSASVGDRVTITCRASQGIDSWLAWYQQKPGKAPKLLIYAASTLESG 382
9702 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPLTFGGGTKVEIKR*
DIQMTQSPSSLSASVGDRVTITCRASQNIGTWLAWYQQKPGKAPKLLIYRASSLESG 383
9703 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQAYSFPWTFGQGTKLEIKR*
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9704 GVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQADSFPPTFGQGTKVEIKR*
DIQMTQSPSSLSASVGDRVTITCRASQDISSYLAWYQQKPGKAPKLLIYAASTLQSG 385
9705 VPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQLNRYPITFGQGTKVEIKR*
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9706 PSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYDSSFITFGQGTRLEIKR*
TABLE-US-00023 TABLE 22 CDRs using the Kabat Numbering Scheme Table 22- Kabat CDR Sequence HCDR1 HCDR2 HCDR3 Binder SEQ ID SEQ ID SEQ ID Name Sequence NO: Sequence NO: Sequence NO: 109 SYWIE 1280 EILPGSGSTNYNEKFKG 1794 RAYGYDGGFDY 2308 110 SYWIE 1281 EILPGSGSTNYIEKFKG 1795 RAYGYDEGFDY 2309 111 SYWIE 1282 EILPGSDSTSYNEKFKG 1796 RAYGYDEGFDY 2310 112 IYWIE 1283 EILPGSGSTNYNEKVKG 1797 RAYGYDGGFDY 2311 1 SYWIE 1284 EIFPGSGHTSFNEKFKG 1798 RGYGYDEGFDY 2312 113 SYWIE 1285 EILPGSGSTNYNEKFKG 1799 RGYGYDEGFDY 2313 114 SYWIE 1286 EILPGSGSTNYNEKFKG 1800 RGYGYDEGFDY 2314 115 NYWIE 1287 EILPGSGSTSYNEKFKG 1801 RGYGYDEGFDY 2315 23 SYWIE 1288 EILPGSGSTSYNEKFKG 1802 RGYGYDEGFDY 2316 116 SYWIE 1289 EILPGSGYTSYIEQFKG 1803

RGYGYDEGFDY 2317 117 SYWIE 1290 EILPGSGSTSYNEKFKD 1804
RAYGYDEGFDY 2318 2 SYWIE 1291 EILPGSGSTSYNEKFKG 1805
RGYGYDEGFDY 2319 118 SYWIE 1292 EVLPGSGSTSYNEKFKG 1806
RAYGYDEGFDY 2320 119 SYWIE 1293 EISPGSGSTNYNEKFKG 1807
RGYGYDEGFDY 2321 120 TYWIE 1294 EILPGSGTPNYNEKFKG 1808
RAYGYDAGFDY 2322 121 SYWIE 1295 EILPGSGSTSCNEKFKG 1809
RGYGYDEGFDY 2323 122 SYWIE 1296 EILPGSGRTSYIEKFKG 1810
RGYGYDEGFDY 2324 123 SYWIE 1297 EILPGSGRTSYIEKFKG 1811
RGYGYDEGFDY 2325 38 SYWIE 1298 EILPGSGSTSYNEKFKD 1812
RAYGYDEGFDY 2326 39 SYWIE 1299 EVLPGSGSTSYNEKFKG 1813
RAYGYDEGFDY 2327 40 SYWIE 1300 EILPGSGRTSYIEKFKG 1814
RGYGYDEGFDY 2328 41 SYWIE 1301 EVLPGSGSTSYNEKFKG 1815
RAYGYDEGFDY 2329 42 SYWIE 1302 EILPGSGRTSYIEKFKG 1816
RGYGYDEGFDY 2330 43 SYWIE 1303 EILPGSGSTNYNEKFKG 1817
RAYGYDGGFDY 2331 44 SYWIE 1304 EILPGSDSTSYNEKFKG 1818
RAYGYDEGFDY 2332 45 IYWIE 1305 EILPGSGSTNYNEKVKG 1819
RAYGYDGGFDY 2333 46 SYWIE 1306 EIFPGSGHTSFNEKFKG 1820
RGYGYDEGFDY 2334 47 NYWIE 1307 EILPGSGSTSYNEKFKG 1821
RGYGYDEGFDY 2335 47 NYWIE 1307 EILPGSGSTSYNEKFKG 1821
RGYGYDEGFDY 2335 48 SYWIE 1308 EILPGSGSTSYNEKFKG 1822
RGYGYDEGFDY 2336 48 SYWIE 1308 EILPGSGSTSYNEKFKG 1822
RGYGYDEGFDY 2336 49 SYWIE 1309 EILPGSGYTSYIEQFKG 1823
RGYGYDEGFDY 2337 50 SYWIE 1310 EILPGSGSTSYNEKFKG 1824
RGYGYDEGFDY 2338 51 SYWIE 1311 EISPGSGSTNYNEKFKG 1825
RGYGYDEGFDY 2339 52 TYWIE 1312 EILPGSGTPNYNEKFKG 1826
RAYGYDAGFDY 2340 53 SYWIE 1313 EILPGSGSTSCNEKFKG 1827
RGYGYDEGFDY 2341 54 SYWIE 1314 EILPGSGRTSYIEKFKG 1828
RGYGYDEGFDY 2342 55 NYWIE 1315 EILPGSGSTSYNEKFKG 1829
RGYGYDEGFDY 2343 56 SYWIE 1316 EILPGSGSTSYNEKFKG 1830
RGYGYDEGFDY 2344 3 GYYMH 1317 YISSYNGATSYNQKFKG 1831
GRYGEYFDY 2345 4 GYYMH 1318 YISSYNGVTGYNQKFKG 1832
GRYGDYFDY 2346 5 GYYMH 1319 YISSYNGVTSYNQKFKG 1833
GRYGDYFDY 2347 6 GYYMH 1320 YISSYNGVTGYNQKFKG 1834
GRYGDYFDY 2348 7 GYYMH 1321 YISSYNGANGYNQKFKG 1835
GRYGDYFDY 2349 8 GYYMH 1322 YISSYNGVTGYNQKFKG 1836
GRYGDYFDY 2350 9 GYYMH 1323 YISSYNGVTGYNQKFKG 1837
GRYGDYFDY 2351 10 GYYMH 1324 YISSYNGVTGYNQKFKG 1838
GRYGDYFDY 2352 11 GYYMH 1325 YISSYNGVTGYNQKFKG 1839
GRYGDYFDY 2353 12 GFYMH 1326 YISSYNGATGYNQKFKG 1840 GRYGDYFDY
2354 13 GYYMH 1327 YISSYNGATGYNQKFKG 1841 GRYGDYFDY 2355 57
GYYMH 1328 YISSYNGVTGYNQKFKG 1842 GRYGDYFDY 2356 58 GYYMH
1329 YISSYNGATSYNQKFKG 1843 GRYGEYFDY 2357 124 SYGVH 1330
VIWRGGSTDYNAAFMS 1844 NLYGHYVMDY 2358 125 SYGVH 1331
VIWRGGSTDYNAAFMS 1845 NLYGHYVMDY 2359 126 SYGVH 1332
VIWRGGSTDNNAAFMS 1846 NLYGHYVMDY 2360 127 RYGVH 1333
VIWRGGSTDHNAAFMS 1847 NLYGHYVMDY 2361 128 TYGVH 1334
VIWRGGSTDYNAAFMS 1848 NLYGHYVMDY 2362 129 SYGVH 1335
VIWRGGSTDYNAAFMS 1849 NLYGHYVMDY 2363 130 RYGVH 1336
VIWRGGSTDHNAAFMS 1850 NLYGHYVMDY 2364 59 SYGVH 1337
VIWRGGSTDYNAAFMS 1851 NLYGHYVMDY 2365 60 SYGVH 1338

VIWRGGSTDYNAAFMS	1852	NLYGHYVMDY	2366	61	SYGVH	1339
VIWRGGSTDNNAAFMS	1853	NLYGHYVMDY	2367	62	RYGVH	1340
VIWRGGSTDHNAAFMS	1854	NLYGHYVMDY	2368	63	TYGVH	1341
VIWRGGSTDYNAAFMS	1855	NLYGHYVMDY	2369	131	SYWMH	1342
MIHPNSGSTNYNEKFKS	1856	WGDGYSFAY	2370	132	SYWMH	1343
MIHPNSGSTNYNEKFKS	1857	WGDGYSFAY	2371	133	TYWMH	1344
MIHPNSDNTNYNEKFKS	1858	WGDGYSFAY	2372	14	SYWMH	1345
MIHPNSGTTNYNEKFKS	1859	WGDGYSFAY	2373	134	SYWMH	1346
MIHPNSGNTNYNEKFKS	1860	WGDGYSFAY	2374	64	SYWMH	1347
MIHPNSGSTNYNEKFKS	1861	WGDGYSFAY	2375	65	SYWMH	1348
MIHPNSGTTNYNEKFKS	1862	WGDGYSFAY	2376	135	DYVIN	1349
EIYPGSGSTYYNEKFKG	1863	RGERGPWFAY	2377	136	DYVIN	1350
EIYPGSGSSYYNEKFKG	1864	RGERGPWFAY	2378	137	DYVIN	1351
EIYPGSGSSYYNEKFRG	1865	RGERGPWFAY	2379	138	DYVIN	1352
EIYPGSGSSYYNEKFKG	1866	RGERGPWFAY	2380	15	DYVIN	1353
EIYPGSGSSYYNEKFKG	1867	RGERGPWFAY	2381	66	DYVIN	1354
EIYPGSGSTYYNEKFKG	1868	RGERGPWFAY	2382	67	DYVIN	1355
EIYPGSGSSYYNEKFKG	1869	RGERGPWFAY	2383	68	DYVIN	1356
EIYPGSGSSYYNEKFRG	1870	RGERGPWFAY	2384	69	DYVIN	1357
EIYPGSGSSYYNEKFKG	1871	RGERGPWFAY	2385	24	NYWMN	1358
LIDPSDSETHYNQVFKD	1872	YDVYYRFAY	2386	139	NYWMN	1359
MIDPSDSETHYNQMFKD	1873	YDGYYRFAY	2387	140	NYWMN	1360
MIDPSDSETHFNQMFKD	1874	YDIYYRFAY	2388	16	SYWMH	1361
MIHPNSGSTNYNEKFKS	1875	PGGYGFVY	2389	141	SYWMH	1362
MIHPNSDSTNYNEKFKS	1876	PGGYGFAD	2390	142	TYWMH	1363
MIHPNSGSTNYNEKFKS	1877	PGGYGFTY	2391	143	SYWMH	1364
MIHPNSGSPNYNEKFKS	1878	PGGYGFAY	2392	70	SYWMH	1365
MIHPNSGSPNYNEKFKS	1879	PGGYGFAY	2393	25	SYWIN	1366
NIYPSDSYTNYNQKFKD	1880	GNYIDY	2394	144	SYWIN	1367
NIYPSDSYTNYNQKFKD	1881	GNYIDY	2395	145	DYWIN	1368
NIYPSDSYTNYNQKFKD	1882	GNYIDY	2396	146	DYVIS	1369
EIYPGSGSSYYNEKFKG	1883	PGDLGFAY	2397	147	DYVIS	1370
EIYPGSGSNYYNEKFKG	1884	PGDLGFAY	2398	148	DYVIS	1371
EIYPGSGSSYYNEKFKG	1885	PGDLGFAY	2399	71	DYVIS	1372
EIYPGSGSSYYNEKFKG	1886	PGDLGFAY	2400	72	DYVIS	1373
EIYPGSGSSYYNEKFKG	1887	PGDLGFAY	2401	73	DYVIS	1374
EIYPGSGSNYYNEKFKG	1888	PGDLGFAY	2402	74	DYVIS	1375
EIYPGSGSSYYNEKFKG	1889	PGDLGFAY	2403	74	DYVIS	1375
EIYPGSGSSYYNEKFKG	1889	PGDLGFAY	2403	75	DYVIS	1376
EIYPGSGSSYYNEKFKG	1890	PGDLGFAY	2404	75	DYVIS	1376
EIYPGSGSSYYNEKFKG	1890	PGDLGFAY	2404	76	DYVIS	1377
EIYPGSGSSYYNEKFKG	1891	PGDLGFAY	2405	149	NYGVH	1378
VVWAGGITNYNWALMS	1892	GDGYDDGYAMDY	2406	150	SYGVH	1379
VLWAGGITNYNSALMS	1893	GDGYDDGYAMDY	2407	26	SYGVH	1380
VIWAGGTTNYNSALMS	1894	GDGYDDGYAMDY	2408	77	NYGVH	1381
VVWAGGITNYNWALMS	1895	GDGYDDGYAMDY	2409	78	SYGVH	1382
VLWAGGITNYNSALMS	1896	GDGYDDGYAMDY	2410	79	SYGVH	1383
VIWAGGTTNYNSALMS	1897	GDGYDDGYAMDY	2411	151	SYWMY	1384
MIDPSDSETRLNQKFKD	1898	TRNY	2412	152	SYWMY	1385
MIDPSDSETRLNQKFKD	1899	TRNY	2413	153	SYWMY	1386

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WIDPENGDTEYAPKFQG 1902 PLLRYSSSMDY 2416 156 DYYMH 1389
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EINPSNGGTNFNEKFKS 1909 YYTHETYYYAMDY 2423 158 DYYMH 1396
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5702 GASNRYT 6212 GQSYSYPFT 6722 VVTYVS 61 ASEN- 5703 GASNRYT 6213 GQSYSYPFT 6723 VVTYVS 62 ASEN- 5704 GASNRYT 6214 GQSYSYPFT 6724 VVTYVS 63 ASEN- 5705 GASNRYT 6215 GQSYSYPFT 6725 VVTYVS 131 SASSSVSY 5706 DTSKLAS 6216 QQWSSNPLY 6726 MH 132 SASSSVSY 5707 DTSKLAS 6217 QQWSSNPHV 6727 MH 133 SASSSVSY 5708 DTSKLAS 6218 QQWSSNPLY 6728 MH 14 SASSSVSY 5709 DTSKLAS 6219 QQWSSNPLY 6729 MH 134 SASSSVSY 5710 DTSKLAS 6220 QQWSSNPLY 6730 MH 64 ASSSVSYM 5711 DTSKLAS 6221 QQWSSNPLY 6731 H 65 ASSSVSYM 5712 DTSKLAS 6222 QQWSSNPLY 6732 H 135 KASQSVD 5713 AASNLES 6223 QQSNEDPLT 6733 YD- GDSYMN 136 KASQSVD 5714 AASNLES 6224 QQSNEDPLT 6734 YD- GDSYMN 137 KASQSVD 5715 AASNLQS 6225 QQSNEDPLT 6735 YD- GDSYMN 138 KASQSVD 5716 AASNLES 6226 QQSNEDPLT 6736 YD- GDSYMN 15 KASQSVD 5717 AASNLES 6227 QQSNEDPLT 6737 YD- GDSYMN 66 ASQSVDY 5718 AASNLES 6228 QQSNEDPLT 6738 DGDSYMN 67 ASQSVDY 5719 AASNLES 6229 QQSNEDPLT 6739 DGDSYMN 68 ASQSVDY 5720 AASNLQS 6230 QQSNEDPLT 6740 DGDSYMN 69 ASQSVDY 5721 AASNLES 6231 QQSNEDPLT 6741 DGDSYMN 24 KASQDINK 5722 YTSTLQP 6232 LQYDNLMYT 6742 YIA 139 KASQDINK 5723 YTSTLQP 6233 LQYDILMYT 6743 YIA 140 KASQDINK 5724 YTSTLQP 6234 LQYDILMYT 6744 YIA 16 QATQDIV- 5725 YATELAE 6235 LQFYEFPLT 6745 KNLN 141 QATQDIV- 5726 YATELAE 6236 LQFYEFPLT 6746 KNLN 142 QATQDIV- 5727 YATELAE 6237 LQFYEFPLT 6747 KNLN 143 QATQDIV- 5728 YATELAE 6238 LQFYEFPLT 6748 KNLN 70 ATQDIV- 5729 YATELAE 6239 LQFYEFPLT 6749 KNLN 25 RAS- 5730 YTSNLAP 6240 QQFTSSHTF 6750 SSVNYMY 144 RAS- 5731 YTSNLAP 6241 QQFTSSHTF 6751 SSVNYMY 145 RAS- 5732 YTSNLAP 6242 QQFTSSHTF 6752 SSVNYMY 146 KASQSVD 5733 AASNLES 6243 QQSNKDPLT 6753 YD- GDSYMN 147 KASQSVD 5734 AASNLES 6244 QQSNEDPLT 6754 YD- GDSYMN 148 KASQSVD 5735 AASNLES 6245 QQSNKDPFT 6755 YD- GDSYMN 71 ASQSVDY 5736 AASNLES 6246 QQSNKDPLT 6756 DGDSYMN 72 ASQSVDY 5737 AASNLES 6247 QQSNKDPLT 6757 DGDSYMN 73 ASQSVDY 5738 AASNLES 6248 QQSNEDPLT 6758 DGDSYMN 74 ASQSVDY 5739 AASNLES 6249 QQSNKDPFT 6759 DGDSYMN 75 ASQSVDY 5740 AASNLES 6250 QQSNKDPFT 6760 DGDSYMN 76 ASQSVDY 5741 AASNLES 6251 QQSNKDPFT 6761 DGDSYMN 149 RASQSVST 5742 YASNLES 6252 QHSWEIPLT 6762 SSYSYMH 150 RASQSVST 5743 YASNLES 6253 QHSWEIPLT 6763 SSYSYMH 26 RASQSVST 5744 YASNLES 6254 QHSWEIPLT 6764 SSYSYMH 77 ASQSVSTS 5745 YASNLES 6255 QHSWEIPLT 6765 SYSYMH 78 ASQSVSTS 5746 YASNLES 6256 QHSWEIPLT 6766 SYSYMH 79 ASQSVSTS 5747 YASNLES 6257 QHSWEIPLT 6767 SYSYMH 151 RASQDISN 5748 STSRLHS 6258 QQGNTLPWT 6768 YLN 152 RASQDISN 5749 STSRLHS 6259 QQGNALPWT 6769 YLN 153 RASQDISN 5750 STSRLHS 6260 QQGNTLPWT 6770 YLN 154 RSST- 5751 GTSNRAP 6261 ALWYSTHYV 6771 GAVTTSN YAN 155 RSST- 5752 GTSNRAP 6262 ALWYSTHYV 6772 GAVTTSN YAN 156 RSST- 5753 GTSNRAP 6263 ALWYSTHYV 6773 GAVTTSN YAN 80 SST- 5754 GTSNRAP 6264 ALWYSTHYV 6774 GAVTTSN YAN 81 SST- 5755 GTSNRAP 6265 ALWYSTHYV 6775 GAVTTSN YAN 82 SST- 5756 GTSNRAP 6266 ALWYSTHYV 6776 GAVTTSN YAN 17 RASENIYS 5757 AATNLAD 6267 QHFWGTPWT 6777 NLA 157 IT- 5758 EGNTLRP 6268 LQSDNMPFT 6778 STDIDDD MN 27 MTSIDIDD 5759 EGNTLRP 6269 LQSDNMPFT 6779 DMN 158 RSST- 5760 GTSNRAP 6270 ALWYSTHYF 6780 GAVTTSN YAN 159 KASQSVD 5761 AASNLES 6271 QQSNEDPLT 6781 YD- GDSYMN 83 ASQSVDY 5762 AASNLES 6272 QQSNEDPLT 6782 DGDSYMN 160

KSS- 5763 WASTRES 6273 QQYYSYPPW 6783 QSLLYSTN QKNYLA 28 RSST- 5764 GTSNRAP 6274 ALWYSTHWV 6784 GAVTTSN YAN 84 SST- 5765 GTSNRAP 6275 ALWYSTHWV 6785 GAVTTSN YAN 29 RSST- 5766 GTNNRAP 6276 ALWYSNHLF 6786 GAVTTSN YAN 161 KASQSVD 5767 AASNLES 6277 QQSNEDPPT 6787 YD- GDSYMN 30 RSST- 5768 GTNNRAP 6278 ALWYSNHWV 6788 GAVTTSN YAN 162 RASENIYY 5769 NANSLED 6279 KQAYDVPYT 6789 SLA 18 KSS- 5770 FASTRES 6280 QQHYSTPLT 6790 QSLLNSSN QKNYLA 85 SSQSLLNS 5771 FASTRES 6281 QQHYSTPLT 6791 SNQKNYL A 163 RASQDISN 5772 YTSRLHS 6282 QQDNTLPRT 6792 YLN 86 ASQDISNY 5773 YTSRLHS 6283 QQDNTLPRT 6793 LN 164 RASQDISN 5774 YTSRLHS 6284 QQGNTLPWT 6794 YLN 165 RASENIYS 5775 AATNLAD 6285 QHFWGTPWT 6795 NLA 31 QATQDIV- 5776 YATELAE 6286 LQFYEFPYT 6796 KNLN 87 ATQDIV- 5777 YATELAE 6287 LQFYEFPYT 6797 KNLN 32 SASSSVSY 5778 DTSNLAS 6288 QQWSSYPLT 6798 MY 166 KASQSVD 5779 AASNLES 6289 QQSNEDPWT 6799 YD- GDSYMN 88 ASQSVDY 5780 AASNLES 6290 QQSNEDPWT 6800 DGDSYMN 167 RSST- 5781 GTNNRAP 6291 ALWYSNHWV 6801 GAVTTSN YAN 168 QATQDIV- 5782 YATELAE 6292 LQFYEFPLT 6802 KNLN 89 ATQDIV- 5783 YATELAE 6293 LQFYEFPLT 6803 KNLN 169 TLSSQHST 5784 GSHSTGD 6294 GVGDTIKEQ 6804 YTIEWYQ Q 170 KASQSVD 5785 AASNLES 6295 QQSNEDPPT 6805 YD- GDSYMN 171 RASQDIGI 5786 ATSSLDS 6296 LQYASSPYT 6806 SLN 172 RASENIYS 5787 YNAKNLA 6297 QHFWGTPYT 6807 YL 19 SASSSVSY 5788 STSNLAS 6298 QQRSSFPYT 6808 MH 90 ASSSVSYM 5789 STSNLAS 6299 QQRSSFPYT 6809 H 173 KASQDIN- 5790 RANRLVD 6300 LQYDEFPLT 6810 SYLS 91 ASQDIN- 5791 RANRLVD 6301 LQYDEFPLT 6811 SYLS 174 RASQDIH 5792 ETSNLDS 6302 LQYASSPLT 6812 GYLN 92 ASQDIHGY 5793 ETSNLDS 6303 LQYASSPLT 6813 LN 175 KASQDVG 5794 WASTRHT 6304 QQYSSYPFT 6814 TAVA 93 ASQDVG- 5795 WASTRHT 6305 QQYSSYPFT 6815 TAVA 20 KASQSVS 5796 YASNRYT 6306 QQDYTSLPT 6816 NDVA 94 ASQSVSN- 5797 YASNRYT 6307 QQDYTSLPT 6817 DVA 176 RSST- 5798 GTNNRAP 6308 ALWYSNHLV 6818 GAVTTSN YAN 177 KASQSVD 5799 VASNLES 6309 QQSHEDPRT 6819 YD- GDSYMN 178 SASSSVSY 5800 DTSKLAS 6310 FQGSGYPLT 6820 MH 33 IT- 5801 EGNTLRP 6311 LQSDNMPLM 6821 STDIDDD MN 179 RASENIYS 5802 AATNLAD 6312 QHFWGTPYT 6822 NLA 34 SASSSVSY 5803 STSNLAS 6313 QQRSTYPTF 6823 MH 180 IT- 5804 EGNTLRP 6314 LQSDNMPLT 6824 STDIDDD MN 95 TSTDIDDD 5805 EGNTLRP 6315 LQSDNMPLT 6825 MN 181 RASQDISN 5806 YTSRLHS 6316 QQGNTLPFT 6826 YLN 96 ASQDISNY 5807 YTSRLHS 6317 QQGNTLPFT 6827 LN 182 SAS- 5808 YTSSLHS 6318 QQYSKLPYT 6828 QGISNYLN 97 ASQGISNY 5809 YTSSLHS 6319 QQYSKLPYT 6829 LN 183 RASQSISD 5810 YASQSIS 6320 QNGHSFPWT 6830 YLH 98 ASQSISDY 5811 YASQSIS 6321 QNGHSFPWT 6831 LH 35 QATQDIV- 5812 YATELAE 6322 LQFYEFPYT 6832 KNLN 99 ATQDIV- 5813 YATELAE 6323 LQFYEFPYT 6833 KNLN 100 ATQDIV- 5814 YATELAE 6324 LQFYEFPYT 6834 KNLN 36 RASGNIH 5815 NAKTLAD 6325 QHFWSTPWT 6835 NYLA 101 AS- 5816 NAKTLAD 6326 QHFWSTPWT 6836 GNIHNYLA 184 RSST- 5817 GTSYRAP 6327 ALWYSTHYV 6837 GAVTTSN YAN 102 SST- 5818 GTSYRAP 6328 ALWYSTHYV 6838 GAVTTSN YAN 185 RASQEIS- 5819 AASTLDS 6329 LQYASYPWT 6839 GYLS 103 ASQEIS- 5820 AASTLDS 6330 LQYASYPWT 6840 GYLS 186 KASQSVD 5821 AASNLES 6331 QQSNEDPLP 6841 YD- GDSYMN 21 HASQNIN 5822 KASNLHT 6332 QQGQSYPLT 6842 VWLS 187 SASSSVSY 5823 DTSKLAS 6333 QQWSSNPLT 6843 MH 188 RASGNIH 5824 NAKTLAD 6334

QHFWSTPWT 6844 NYLA 189 RASENIYS 5825 NAKTLAE 6335 QHHYGTPFT 6845 YLA 190 RAS- 5826 YTSNLAP 6336 QQFTSSLTF 6846 SSVNYMY 191 RSST- 5827 STNNRAP 6337 TLWYSNHWV 6847 GAVTTSN YAN 192 KASQDVG 5828 WASTRHT 6338 QQYSSYPFT 6848 TAVA 193 KASQDIN- 5829 RANRLVD 6339 LQYDEFPPT 6849 SYLS 194 RASENIYS 5830 AATNLAD 6340 QHFWGTPFT 6850 NLA 195 RASQSISD 5831 YASQSIS 6341 QNGHSFPYT 6851 YLH 104 ASQSISDY 5832 YASQSIS 6342 QNGHSFPYT 6852 LH 37 RASENIYS 5833 AATNLAD 6343 QHFWGTPYT 6853 NLA 196 RSST- 5834 GTNNRAP 6344 ALWYSNHWV 6854 GAVTTSN YAN 197 IT- 5835 EGNTLRP 6345 LQSDNLPLT 6855 STDIDDD MN 198 SASSSVSY 5836 STSNLAS 6346 QQRSSYPPT 6856 MH 199 SASSSVSY 5837 DTSKLAS 6347 QQWSSNPLT 6857 MH 22 IT- 5838 EGNSLRP 6348 LQSDNLPLT 6858 STDIDDD MN 200 SASSSVSS 5839 STSNLAS 6349 HQWSSYPPT 6859 SYLY 201 TLSSQHST 5840 GSHSTGD 6350 GVGDTIKEQ 6860 YTIEWYQ Q 202 RSSQSIVH 5841 KVSNRFS 6351 FQGSHVPWT 6861 SNGNTYLE 203 KASQDIN- 5842 RANRLVD 6352 LQYDEFPPT 6862 SYLS 204 KSS- 5843 LVSKLDS 6353 WQGTHFPWT 6863 QSLLDSDG KTYLN 205 SASSSVSS 5844 STSNLAS 6354 HQWSSYPPT 6864 SYLY 206 RSSQSIVY 5845 KVSNRFS 6355 FQGSHVPPT 6865 SNGNTYLE 207 RASENI- 5846 AATNLAD 6356 QHFWGTPWT 6866 YNNLA 208 SASSSVSS 5847 STSNLAS 6357 HQWSSYPPT 6867 SYLY 209 SASSSVSY 5848 DTSNLAS 6358 QQWSTYPPI 6868 MY 210 SASSSVSY 5849 DTSNLAS 6359 QQWSSYPFT 6869 MY 211 RAS- 5850 ATSNLAS 6360 QQWSSNPYT 6870 SSVSYMH 212 RASENIYS 5851 AATNLAD 6361 QHFWGTPWT 6871 NLA 213 KASQDVS 5852 WASTRHT 6362 QQHYSTPWT 6872 TAVA 214 KSS- 5853 LVSKLDS 6363 WQGTHFPWT 6873 QSLLDSDG KTYLN 215 RSSQSIVH 5854 KVSNRFS 6364 FQGSHVPPT 6874 SNGNTYLE 216 IT- 5855 EGNTLRP 6365 LQSDNMPLT 6875 STDIDDD MN 217 KASQDINK 5856 YTSTLQP 6366 LQYDNLYMY 6876 YIA 218 HASQNIN 5857 KASNLHT 6367 QQGQSYPYT 6877 VWLS 219 SASSSVSY 5858 LTSNLAS 6368 QQWSSNPPT 6878 MY 220 KSS- 5859 LVSKLDS 6369 WQGTHFPWT 6879 QSLLDSDG KTYLN 221 SASLSVSD 5860 DTSNLAS 6370 QQWSSYPFT 6880 MY 222 SASSSVSS 5861 STSNLAS 6371 HQWSSYPPT 6881 SYLY 223 RAS- 5862 YTSNLAP 6372 QQFTSSPST 6882 SSVNYMY 105 ASSSVNY 5863 YTSNLAP 6373 QQFTSSPST 6883 MY 224 IT- 5864 EGNTLRP 6374 LQSDNLPLT 6884 NTDIDDD MN 106 TNTDIDDD 5865 EGNTLRP 6375 LQSDNLPLT 6885 MN 225 KASQSVD 5866 AASNLES 6376 QQSNEDPWT 6886 YD- GDSYMN 226 RASQSISN 5867 YASQSIS 6377 QQSNSWPFT 6887 NLH 227 SASSSVSS 5868 STSNLAS 6378 HQWSSYPPT 6888 SYLY 228 IT- 5869 EGNTLRP 6379 LQSDNMPLT 6889 STDIDDD MN 229 KSS- 5870 LVSKLDS 6380 WQGTHFPWT 6890 QSLLDSDG KTYLH 230 KASQDINK 5871 YTSTLQP 6381 LQYDNLRTF 6891 YIA 231 RSST- 5872 GTNNRAP 6382 VLWYSNHLV 6892 GAVTTSN YAN 232 KASEN- 5873 GASNRYT 6383 GQSYSYPPT 6893 VGTYVS 233 RASET- 5874 RASNLES 6384 QQSNEDPRT 6894 VDSYGYSF MH 234 KSS- 5875 WASTRES 6385 QQYYSYRTF 6895 QSLLYSTN QKNYLA 235 KSS- 5876 LVSKLDS 6386 WQGTHFPFT 6896 QSLLDSDG KTYLN 236 KASQSVD 5877 AASNLES 6387 QQSNEDPFT 6897 YD- GDSYMN 237 TLSSQHST 5878 GSHSTGD 6388 GVGDTIKEQ 6898 YTIEWYQ Q 107 LSSQHSTY 5879 GSHSTGD 6389 GVGDTIKEQFVYV 6899 TIEWYQQ 238 SASSSVSY 5880 DTSKLAS 6390 QQWSSNPLT 6900 MH 108 ASSSVSYM 5881 DTSKLAS 6391 QQWSSNPLT 6901 H 239 QATQDIV- 5882 YATELAE 6392 LQFYEFPLT 6902 KNLN 240 RAS- 5883 AASNQGS 6393 QQSKEVPPT 6903 ESVDNY- GISFMN 241 RASQSISD 5884 YASQSIS 6394 QNGHSFPLT 6904 YLH 242 RSSQSIVH 5885 KVSNRFS 6395

FQGSHVPLT 6905 SNGDTYLE 243 SASSSVSY 5886 DTSKLAS 6396 QQWSSNPLT 6906 MH 244 TASESLYS- 5887 GASNRYI 6397 AQFYSYPYT 6907 SKHKVHYL A 245 TLSSQHST 5888 GSHSTGD 6398 GVGDTIKEQ 6908 YTIEWYQ Q 246 RASQSVST 5889 YASNLES 6399 QHSWEIPLT 6909 SSYSYMH 247 SASSSVSY 5890 DTSKLAS 6400 QQWSSNPLT 6910 MH 248 KASDHIN 5891 GATSLET 6401 QQYWSTPLT 6911 NWLA 249 RASENIYS 5892 NAKTLAE 6402 QHHYGTPYT 6912 YLA 250 RASENIYS 5893 NAKTLAE 6403 QHHYGTPPT 6913 YLA 251 SASSSVSY 5894 DTSKLAS 6404 QQWSSNPPT 6914 MH 252 RAS- 5895 AASNQGS 6405 QQSKEVPPT 6915 ESVDNY- GISFMN 253 TASESLYS- 5896 GASNRYI 6406 AQFYSYPYT 6916 SKHKVHYL A 254 RSST- 5897 GTSNRAP 6407 ALWYSTHYV 6917 GAVTTSN YAN 255 SASSSI- 5898 DTSKLAS 6408 HQRSSYPTF 6918 SYMH HCDR1 HCDR2 HCDR3 Binder SEQ ID SEQ ID SEQ ID Name Sequence NO: Sequence NO: Sequence NO: 265 SYALS 9968 IINPSGGTNYAQKFQG 10230 DLGDPGMDV 10492 266 NYAIS 9969 IIDPSGGSTTYAQKFQG 10231 DLGDMGMDV 10493 267 NYAFS 9970 IINPSGGSTSYAQKFQG 10232 DVGDRGMDV 10494 263 SYALS 9971 IINPSGGTNYAQKFQG 10233 DLGDPGMDV 10495 268 GYYMH 9972 WIDPNGGGTQYAQKFQG 10234 DIVHDGTEYFQH 10496 269 SYYMH 9973 IINPSGGSTSYAQKFQG 10235 DIVHDGTEYFQH 10497 270 SYAIS 9974 IINPSGGSTNYAQKFQG 10236 EGRDHDAFDI 10498 271 DYGIS 9975 IINPSGGSTSYAQKFQG 10237 EGRSHDAFDI 10499 272 GYYMH 9976 WMNPHSGDTGYAQKFQG 10238 WVGTTEYYYYYMDV 10500 273 DYYLH 9977 IIDPSGGSTSIAQKFQG 10239 TAYYDFWSGYSMDV 10501 274 SHYMH 9978 IIDPSGGSTSYAQEFQG 10240 DMDNWNTGYYYYMD 10502 V 275 SYAIN 9979 WVNPNSGDTAYAQKFQG 10241 DQRGGDAWDV 10503 276 NYAIS 9980 IITPSGGSTTYAHKFQG 10242 DTAGHFDI 10504 277 NDVIN 9981 WMNPNSGNTGYAQKFQG 10243 DNPDLDGMDV 10505 278 SYAIN 9982 WISAYNGNTNYAQKFQG 10244 DLVGHFDY 10506 279 SYAIS 9983 WINPNSGGTNYAQKFQG 10245 DGYSGSYSD 10507 264 SYAIS 9984 WINPNSGGTNYAQKFQG 10246 DGYSGSYSD 10508 257 SYAIS 9985 WINPNSGGTNYAQKFQG 10247 DGYSGSYSD 10509 280 SHAIS 9986 IINPSGGSTSYAQKFQG 10248 DQGSSGTFDY 10510 281 SYAIS 9987 WINPNSGGTNYAQKFQG 10249 DSTDVIDY 10511 282 SYDIN 9988 WINPNSGDTKYAQNFQG 10250 DGGTVTPTEEYYYYG- 10512 MDV 283 SYAIS 9989 WISVYNGNTNYAQNLQG 10251 LDDLDY 10513 284 SYYIH 9990 WINPNNGGTHYAQKFQG 10252 DMVRDSAEYFQH 10514 285 TSYIH 9991 MINPSGGTTTYAQKFQG 10253 DSSGYPIDY 10515 286 SYDIN 9992 GIIPLSGAPNYAHKFQG 10254 GALYNWNDGWFDP 10516 258 SYDIN 9993 GIIPLSGAPNYAHKFQG 10255 GALYNWNDGWFDP 10517 287 SWYMS 9994 GIWYEGSNKYYADSVKG 10256 LGTASLPYFDY 10518 288 GYYMH 9995 WINPNRGDTKYAQKFQG 10257 ESGDGFDP 10519 289 NYYIH 9996 WMNPNSGNTGYAQKFQG 10258 DWPNWFDP 10520 290 DNYIH 9997 WIRSDNGETSYAQKFQG 10259 EVQLVGFDY 10521 291 DHHVH 9998 GIIPIFGTANYAQKFQG 10260 GSSWYLHFQH 10522 260 DHHVH 9999 GIIPIFGTANYAQKFQG 10261 GSSWYLHFQH 10523 292 SYAIY 10000 GIIPIFGTTNYAQKFQG 10262 GVDRYNWNDAFDY 10524 293 DYYMH 10001 WIHSNSGGTHSAQKFQG 10263 ESSGYDSSLDY 10525 294 SYGIS 10002 WINPNSGDTDYAQKFQG 10264 DPRLDSSDPGY 10526 295 NYGIN 10003 WISAYNGNTNYAQKFQG 10265 GGMDV 10527 296 RYGIA 10004 ISYPSDGSTSSAQKLQG 10266 DRLGDLDY 10528 261 RYGIA 10005 ISYPSDGSTSSAQKLQG 10267 DRLGDLDY 10529 297 SYAIS 10006 WMNPNSGNTGYAQKFQG 10268 DSIVGGYPFDY 10530 298

SYDIN 10007 TITPIFGTTDYAQKFQG 10269 EGYSSSWHDDAFDI 10531 299 NYAIS 10008
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WMNPKIGNTGYAQKFQG 10293 DSSGYDAFDI 10555 323 SYDIN 10032
RINPGTGGTDYAHKFQG 10294 ETPSDYYDSSGYYYN- 10556 DAFDI 324 SYAIS 10033
IIIPSGGTNYAQTFQG 10295 DLGTTFDI 10557 325 AYYLH 10034
WINPDNDNAYYAQKFQG 10296 DIAVAALAYGMDV 10558 326 SYAMS 10035
VISYDGSDQYYADSVKG 10297 QSLYYYYGMDV 10559 327 DYYVH 10036
WISTFTGNTDYAQNFQG 10298 DAPLAAAGTDYYYG- 10560 MDV 262 DYYVH 10037
WISTFTGNTDYAQNFQG 10299 DAPLAAAGTDYYYG- 10561 MDV 328 SYAMS 10038
FISDDGITKYYADSVKG 10300 DDSSGYGGMDV 10562 329 SYAMH 10039
VISYDGGDKYYADSVKG 10301 GSLVLGYYYMDV 10563 330 NYYIH 10040
WINPNTGGTDYAQKFQG 10302 GGGGSYYDAFDV 10564 331 SYAIS 10041
RINPNSGNTGYAQKFQG 10303 DIGEGYSMDV 10565 332 NHYTS 10042
VISYDGSNKYYADSVKG 10304 EEKYSSSWYVGVDAFDI 10566 333 SSAMH 10043
SISGSGDNAYYADSVKG 10305 DQEDYYYDSSGYGMDV 10567 334 SHAIS 10044
GIIPIFGTANYAQKFQG 10306 GDWGIVVVPAAIGAFDI 10568 335 AYYMH 10045
RISPVFGSTTYAQRFQG 10307 DLGYYDSSGYRYDAFDI 10569 336 SYDIN 10046
GISPMFGTANYAQKFQG 10308 DGWYYGMDV 10570 337 SYGIS 10047
WINPNSGGTKYAQKFQG 10309 GEAGNLDWYFDL 10571 338 NYGIS 10048
WINPNNGDTKYAQKFQG 10310 EDVWYFDL 10572 339 TYGIS 10049
WISTYDGKTNYAQKLQG 10311 HLGGDWYFDL 10573 340 GYYMH 10050
WINPNTGATYYAQKFQG 10312 QHGDYDWYFDL 10574 341 TYYVH 10051
WINPNSGNTGYAQKFQG 10313 DSGRH 10575 342 SYGIS 10052 RIIPMLGIANYAQKFQG
10314 EEVAGANWFDP 10576 343 SYAMN 10053 IINPSGGSTSYARKFQG 10315
EGDYGSGEFDY 10577 344 SSYMH 10054 WMNPRSGNTGYAQKFQG 10316
ERDDYGDYGWLDY 10578 345 GYYMH 10055 IINPSGGSTSYAQKFQG 10317 DLYDSSGY-
10579 WHYYYYMDV 346 SYAFS 10056 WINPNSGGTNYAQKFQG 10318 FSGYDYVDY
10580 347 SYAIS 10057 IINPNGGNTSYAQKFQG 10319 DVGEDFDL 10581 348 SYYIH

10058 VINPADGDTTYAQMFQG 10320 DFDWLFAMDV 10582 349 NYALN 10059 RINPNGGTTYYAKNFQG 10321 HGDHGFYV 10583 350 SYAIS 10060 MINPNVGSATYAQRFQG 10322 EDSGTSWFDP 10584 351 SYYMH 10061 IINPSDGSTSYAQRFQG 10323 DDRGSNYYYGMDV 10585 352 AYYVH 10062 WMNPNSGTTGYAQKFQG 10324 DSSDYYGDYRADAFDI 10586 353 SYDIN 10063 VISPSGDATLYAQSFQG 10325 GLDH 10587 354 DYGMH 10064 AIGGIGDSTYYADSVKG 10326 MNYGDSNYYYYYG- 10588 MDV 355 SYDIS 10065 MISPSDGSTTYAPKFQG 10327 GAVGFDY 10589 356 SYGIS 10066 WINTYSGYTDYAHKFQG 10328 DDFLSFGY 10590 357 DYYIH 10067 GIIPYFGTANYAQKFQG 10329 SISGSYVLDAFDI 10591 358 SYGIS 10068 GIIPIFGTANYAQKFQG 10330 DWGYGDYADDAFDI 10592 359 NNDIN 10069 WINPIYGSANYAQNFQG 10331 DWRGFDY 10593 360 EYAIH 10070 RMNPHNGDTGYAQKFQG 10332 EGDYLGYPIDC 10594 361 DYSMS 10071 AIWQDGNVKFYADSVKG 10333 DGNSGYVF 10595 362 TYYMH 10072 WINPNTGDTAYAQKIQG 10334 TAEAVAGLPAFDY 10596 363 NYAID 10073 GIIPLFGTTTYAQKFQG 10335 VTLYGDYDY 10597 364 THWMH 10074 MINPSDGVTYYAQTFQG 10336 EYYGEGFDY 10598 259 THWMH 10075 MINPSDGVTYYAQTFQG 10337 EYYGEGFDY 10599 365 SYAIS 10076 IINPSGGSTSNAQKFQG 10338 DLGDTAMDG 10600 366 SYYLH 10077 IITPSGGSTTYAHKFQG 10339 DGGLASFDY 10601 367 SYAIS 10078 WMNPNSGNTGYAQKFQG 10340 GGGWAMTDAFDI 10602 368 DYGMS 10079 LIYSGGDTYYADSVKG 10341 KEYYYDSSGYLRLFDY 10603 369 DYYMH 10080 GINPIFGTSNYAQKFQG 10342 DISGYDYYYYGMDV 10604 370 NYAFS 10081 MIDPSDGTIAYAQKFQG 10343 SDYDFWSGLGGYFDY 10605 371 SYAIS 10082 TIDPNSGGTMFAQKFQG 10344 DSAEWELGGSFDY 10606 372 NHYTS 10083 SIGVNGDTYYLDSVKG 10345 EGLVFSGRGHWYFDL 10607 373 NYAIS 10084 RINPNGGNTSNAQKFQG 10346 DYEDADFDG 10608 374 DHHVH 10085 WMNPDSGNTGYAQRFQG 10347 DSTSGVDY 10609 375 SYAMS 10086 VISYDGHDQFYADSVKG 10348 GEQQLEGFYYYYGMDV 10610 376 SYWMH 10087 VISYDGSKEYYADSVKG 10349 DYGDYGTYDY 10611 377 SYWMH 10088 GISGGGDDTYYADSVKG 10350 EPLAYCGGDCPGGFDY 10612 378 DHYMD 10089 AIGTGGDTYYADSVKG 10351 HEDTAIFLDY 10613 379 SYYMH 10090 MISPSDGSTTYAPKFQG 10352 DGYDAWSYGMDV 10614 380 GYYMH 10091 WMNPNSGNTGYAQKFQG 10353 DGVTGTDY 10615 381 SYVLH 10092 AISGAGDSTYYADSVKG 10354 EPTTVTDDWYFDL 10616 382 SHWMH 10093 AISGNGDNSYYADSVKG 10355 DRAPEYFDL 10617 383 SYAIS 10094 WINPNSGGTNYAQKFQG 10356 DDYGDYGGGMDV 10618 384 DYYMH 10095 WMNPNSGHTGYAEKFQG 10357 DTSPRYGDGFFDY 10619 385 SYWMH 10096 VTSYDGSNKYYADSVKG 10358 ESGFSAEYFQH 10620 386 SYAIS 10097 IINPSGGSTSYAQKFQG 10359 ATGLYCSGSCFDY 10621 LCDR1 LCDR2 LCDR3 Binder SEQ ID SEQ ID SEQ ID Name Sequence NO: Sequence NO: Sequence NO: 265 RASQDISN 12194 DASNLET 12454 QQSYSTPLT 12714 YLN 266 RASQSIS- 12195 AASSLQS 12455 QQSYSTPLT 12715 SYLN 267 QASQDIS 12196 KASSLET 12456 QQSFSSPLT 12716 NYLN 263 RASQDISN 12197 DASNLET 12457 QQSYSTPLT 12717 YLN 268 RASQNVN 12198 EASSLQS 12458 QQANSFPFT 12718 TWLA 269 RASQSISD 12199 AASSLQS 12459 AQHNHYPYT 12719 WLA 270 KSS- 12200 WASTRES 12460 QQYYTTPFT 12720 QSVLSSSY NKNYLA 271 KSS- 12201 WASTRAS 12461 QQYYSTPFT 12721 QSVLSSSY NKNYLA 272 RAS- 12202 DASHLEA 12462 QQANSFPIT 12722 QAIRDDLG 273 RASQGVG 12203 AASTLQT 12463 QQASSFPLT 12723 NDLA 274 RASQI- 12204 AASSLQS 12464 QQSYTFPVT 12724 IGTNLA 275 RASQSIST- 12205 DASSLES 12465 QQSYSTPFT 12725 WLA 276 KSS- 12206 WASTRES 12466 QQYYGSPLT 12726 QSVLSSS- NNKNYLA

277 KSS- 12207 WASTRES 12467 QQYYSSPPT 12727 QSVLSSSY NKNYLA 278 KSS- 12208 WASTRES 12468 QQYYSSPPT 12728 QSVLSSSY NKNYLA 279 KSS- 12209 WASTRES 12469 QQYYSTPWT 12729 QSVLSSSY NKNYLA 264 KSS- 12210 WASTRES 12470 QQYYSTPWT 12730 QSVLSSSY NKNYLA 257 KSS- 12211 WASTRES 12471 QQYYSTPWT 12731 QSVLSSSY NKNYLA 280 KSS- 12212 WASTRAS 12472 QQYYGSPPT 12732 QSVLSSSY NKNYLA 281 QASQDIR 12213 DASTLQS 12473 QQAYSFPWT 12733 NYLN 282 QASQDIS 12214 NASNLET 12474 QQLNSYPFT 12734 NYLN 283 QASQSIST 12215 AASTLRS 12475 LQHYTYPLT 12735 WLA 284 RASEDI- 12216 AASTLQS 12476 QQSHTIPWT 12736 STYLA 285 RASH- 12217 AASTLQS 12477 QQSYSSPYT 12737 HISDFLN 286 RASQDIG 12218 DASSLQS 12478 QQANSFPLT 12738 DYLA 258 RASQDIG 12219 DASSLQS 12479 QQANSFPLT 12739 DYLA 287 RASQDIRS 12220 AASSLQS 12480 QQSYTAPPT 12740 YLA 288 RASQDIS- 12221 AASTLQS 12481 LQHNTYPLT 12741 NNLN 289 RASQDIS- 12222 DASSLQS 12482 QQAISFPLT 12742 NWLA 290 RASQGIA 12223 AASSLQS 12483 QQADSFPLT 12743 NYLA 291 RASQGIAS 12224 AASTLQP 12484 QQFDSYPIT 12744 YLA 260 RASQGIAS 12225 AASTLQP 12485 QQFDSYPIT 12745 YLA 292 RASQGISN 12226 AASRLQS 12486 QQSSIIPFT 12746 YLA 293 RASQGISN 12227 AASTLQS 12487 QQAYSFPYT 12747 YLA 294 RASQSIGR 12228 DASNLET 12488 QQSYSTPRT 12748 WLA 295 RASQSINS 12229 DTSSLQS 12489 QQTYSTPYT 12749 WLA 296 RASQSISS 12230 AASTLQS 12490 QQGYSTPYI 12750 WLA 261 RASQSISS 12231 AASTLQS 12491 QQGYSTPYI 12751 WLA 297 RASQSIS- 12232 AASSLQS 12492 QQTDSIPIT 12752 SYLN 298 RASQSIS- 12233 AASTLQS 12493 QQSYSIPYT 12753 SYLN 299 RASQTIRS 12234 KASSLES 12494 QQTYTIPIT 12754 YLN 300 RASQTISN 12235 AASTLQS 12495 QQANSFPPT 12755 WLA 301 RASQYIGS 12236 DASNLET 12496 QQVDSYPLT 12756 YLN 302 RSSQSLLH 12237 LGSNRAS 12497 MQGTHWPPT 12757 SNGYNYL D 303 RSSQSLLH 12238 FGSNRAS 12498 MQALQAPVS 12758 SNGYNYL D 304 KSS- 12239 WASSRQS 12499 QQYYSTPLT 12759 QSVLSSSY NKNYLA 305 KSS- 12240 WASVRES 12500 QQYYSTPIT 12760 QSVSSSSY NKNYLA 306 KSTQNVLS 12241 WASTRES 12501 QQYYSTPFT 12761 SSNNN- SYLA 307 QASQDIG 12242 AASSLQS 12502 QQTYNTPLT 12762 NYLN 308 QASQDIS 12243 EASTLQS 12503 QQSYSTPFT 12763 NYLN 309 QASQDIST 12244 RASTLES 12504 QQSYSIPLT 12764 WLA 310 RASQNIN- 12245 AASRLQS 12505 QQSYSAPVT 12765 NYLN 311 RASQNINT 12246 AASSLQS 12506 QQAYSFPFT 12766 WLA 312 RASQRIGN 12247 AASSLQS 12507 QQSYSTPLT 12767 YLN 313 RASQSIST 12248 AASTLQS 12508 QQSYRTVT 12768 YLN 314 RASQSVG 12249 GASTRAT 12509 QQYDSSSQT 12769 SYLA 315 RASRSVST 12250 GASTRAT 12510 QQYDGSPYT 12770 YLA 316 RSSQSLLH 12251 DASNLET 12511 MQALQTPPA 12771 SNGYNYL D 317 KSS- 12252 WASTRES 12512 QQYYSAPPT 12772 QSVLSSSY NKNFLA 318 KSS- 12253 WASTRES 12513 QQYYSDPIT 12773 QSVLSSSY NKNFLA 319 KSS- 12254 WASARES 12514 QQYYSIPIA 12774 QSVLSSSY NKNYLA 320 KSS- 12255 WASTRDS 12515 QQYYSIPYT 12775 QSVLSSSY NKNYLA 321 KSS- 12256 WASTRAS 12516 QQYYTTPPT 12776 QSVLSTSY NKNYLA 322 KSS- 12257 WASTRQS 12517 QQYYSTPYT 12777 QSVLSTSY NRNFLA 323 KSS- 12258 WASTRES 12518 QQYYSTPLT 12778 QSVLYSSN NKNYLA 324 QASQDIS 12259 AASSLQS 12519 QQSYSTPT 12779 NYLN 325 QASQDIS 12260 GASTLQS 12520 QEADSFPLT 12780 NYLN 326 RASQGIRN 12261 DASSLHS 12521 QQAYSFPWT 12781 DLG 327 RASQGISN 12262 KASSLES 12522 QQSYNTPFT 12782 YLA 262 RASQGISN 12263 KASSLES 12523 QQSYNTPFT 12783 YLA 328 RASQSINR 12264 SASNLQS 12524 QQSYNTPLT 12784 WLA 329 RASQSINT 12265 AASSLQS 12525 QQANSFPFT 12785 WLA 330 RASQSIRT- 12266 DASSLET 12526 QQLNSYPLT 12786 WLA 331 RASQSIRT 12267 AASTLQS 12527 QQSYSAPLT 12787 YLN 332 RASQSIST 12268 AASSLHS 12528 QQSYSTPLT 12788 YLN 333 RASQSIT- 12269 AASTLQS 12529 QQSYSTPLT 12789 TYLN 334 RSSQSLLH 12270 AASSLQS 12530

MQARQTPLT 12790 SNGYNYL D 335 RSSQSLLH 12271 GASSLQS 12531 MQTLQTPFT 12791 SNGYNYL D 336 RSSQSLLH 12272 LGSDRAS 12532 MQALQTPLT 12792 SNGYNYL D 337 KSSQTVF- 12273 WASTRES 12533 QQYYSTPLT 12793 STSYN- KNYLA 338 KTSQSVF- 12274 WASTRES 12534 QQYYSSPPT 12794 STSYN- RDYLA 339 RASQSISS 12275 DASTLQS 12535 QQSYSTPFT 12795 WLA 340 RASQSIS- 12276 DASNLKT 12536 QQSYSFPT 12796 SYLN 341 RASQSVSS 12277 DTSSRAT 12537 QQYYDTPYT 12797 YLA 342 KSS- 12278 LASTREP 12538 QQYYSTPPT 12798 QSVLYSSN NKNYLA 343 KSS- 12279 WASTRES 12539 QQYYSTPLT 12799 QSVLSSSY NKNYVA 344 QASQDIS 12280 AAASLQS 12540 QQTYSTPWT 12800 NYLN 345 RASQDIN- 12281 AASSLQS 12541 QQSSSFPLT 12801 TYLA 346 QASQDIS 12282 AASSLQS 12542 QQLYNFPYT 12802 NYLN 347 RASQSISR 12283 GASTRES 12543 QQSYNTPLT 12803 YLA 348 RASQTLSG 12284 GASTLQG 12544 QQYYSYPPT 12804 WLA 349 FASQDI- 12285 EASNLET 12545 QQSYSTPLT 12805 INYLN 350 RASQSIS- 12286 DVFNLGT 12546 QQSYSSPFT 12806 SYLN 351 QASQDIS 12287 MASNLES 12547 QQTNSFPLT 12807 NYLN 352 RASQSIS- 12288 DASNLET 12548 QQSYSTPLT 12808 SYLN 353 RSSQSLLH 12289 LGSNRAS 12549 MQALQSPWT 12809 SNGYNYL D 354 KSS- 12290 WASTRES 12550 QQYYSSPLT 12810 QSVLYSSN NKNYLA 355 RSSQSLLH 12291 LGSNRAS 12551 MQALQTPPS 12811 SNGYNYL D 356 RAS- 12292 KASRLES 12552 QQSYKTPYT 12812 ESVST- WLA 357 KSS- 12293 WASTRES 12553 QQYFTTPLT 12813 QSVLYSSN NKNYLA 358 RASQSIS- 12294 AASSLQS 12554 QQSYSTPYT 12814 SYLN 359 KSS- 12295 WASTRAS 12555 QQYYDTPLT 12815 QSVLSSSY NKNYLA 360 RASQSIS- 12296 KASTLES 12556 QQNDSIPIT 12816 SYLN 361 RASQSISR 12297 DASNLET 12557 LQDYSYPLT 12817 WLA 362 KTSQSVF- 12298 WASTRAA 12558 QQYYYTST 12818 STSYN- RDYLA 363 RASQSIN- 12299 AASSLQS 12559 QQANSFPPT 12819 RYLN 364 RASQGISN 12300 SASNLQS 12560 QQSYSTPLT 12820 YLA 259 RASQGISN 12301 SASNLQS 12561 QQSYSTPLT 12821 YLA 365 RASQSIDS 12302 KASTLES 12562 QQSYSAPLT 12822 YLN 366 RASQDIST 12303 DASNLET 12563 QQVNSDPYT 12823 WLA 367 QASQDIS 12304 AASTLES 12564 QQGDSLPLT 12824 NYLN 368 RASQGISN 12305 AASSLQS 12565 QQSDSFPYT 12825 YLA 369 RASQSVST 12306 GASTRAT 12566 QQHDSYPLT 12826 YLA 370 RASQGIRN 12307 AASSLQS 12567 QQANSFPPT 12827 DLG 371 RAS- 12308 KASNLES 12568 QQTDSTFIT 12828 ESISTYLN 372 RASRNIHD 12309 AASTLQT 12569 QQTYSTPPT 12829 YLN 373 RASQSND 12310 KASTLES 12570 QQSYSSPLT 12830 SYLN 374 RASQSISD 12311 AASTLQS 12571 QQSYSSPYT 12831 FLN 375 QASQDIS 12312 AASSLQS 12572 QQANRFPLT 12832 NYLN 376 QASQDIS 12313 KASNLQS 12573 QQSYNFPAT 12833 NYLN 377 RSSQSLLH 12314 LGSNRAS 12574 MQGTHWPET 12834 SNGYNYL D 378 RASQSIS- 12315 DASNLET 12575 QQSYSTPLT 12835 SYLN 379 RASQGISD 12316 DASNLET 12576 QQSYILPLT 12836 YLA 380 RASQDIN 12317 AASSLQS 12577 QQSYSAPYT 12837 DFLA 381 RASQSISN 12318 AASKLES 12578 QQSYSSPWT 12838 WLA 382 RASQGIDS 12319 AASTLES 12579 QQAYSFPLT 12839 WLA 383 RASQNIGT 12320 RASSLES 12580 QQAYSFPWT 12840 WLA 384 RASQNIN 12321 KASTLQS 12581 QQADSFPPT 12841 NWLA 385 RASQDIS- 12322 AASTLQS 12582 QQLNRYPIT 12842 SYLA 386 RASQDISN 12323 AASILHS 12583 QQYDSSFIT 12843 YLA

TABLE-US-00024 TABLE 23 CDRs using the Chothia Numbering Scheme Table 5-Chothia CDR Sequences Binder Name HCDR1 HCDR2 HCDR3 SEQ ID SEQ ID SEQ ID Sequence NO: Sequence NO: Sequence NO: 109 GYTFSSY 3077 LPGSGS 3591 ARRAYGYDGGFDY 4105 110 GYTFSSY 3078 LPGSGS 3592 ARRAYGYDEGFDY 4106 111 GYTFSSY 3079 LPGSDS 3593 ARRAYGYDEGFDY 4107 112 AYTFSIY 3080 LPGSGS 3594 ARRAYGYDGGFDY 4108 1 GYTFSSY 3081 FPGSGH 3595 ARRGYGYDEGFDY 4109 113 GYTFSSY 3082 LPGSGS 3596 ARRGYGYDEGFDY 4110 114 GYTFSSY 3083 LPGSGS 3597 ARRGYGYDEGFDY 4111 115 GYTFSNY

3084 LPGSGS 3598 ARRGYGYDEGFDY 4112 23 GYTFSSY 3085 LPGSGS 3599 ARRGYGYDEGFDY 4113 116 GYTFSSY 3086 LPGSGY 3600 ARRGYGYDEGFDY 4114 117 GYTFSSY 3087 LPGSGS 3601 ARRAYGYDEGFDY 4115 2 GYTLSSY 3088 LPGSGS 3602 ARRGYGYDEGFDY 4116 118 GYTFSSY 3089 LPGSGS 3603 ARRAYGYDEGFDY 4117 119 GYTFSSY 3090 SPGSGS 3604 ARRGYGYDEGFDY 4118 120 GYTFGTY 3091 LPGSGT 3605 ARRAYGYDAGFDY 4119 121 GYTFSSY 3092 LPGSGS 3606 ARRGYGYDEGFDY 4120 122 GYTFSSY 3093 LPGSGR 3607 ARRGYGYDEGFDY 4121 123 GYTFSSY 3094 LPGSGR 3608 ARRGYGYDEGFDY 4122 38 GYTFSSY 3095 LPGSGS 3609 ARRAYGYDEGFDY 4123 39 GYTFSSY 3096 LPGSGS 3610 ARRAYGYDEGFDY 4124 40 GYTFSSY 3097 LPGSGR 3611 ARRGYGYDEGFDY 4125 41 GYTFSSY 3098 LPGSGS 3612 ARRAYGYDEGFDY 4126 42 GYTFSSY 3099 LPGSGR 3613 ARRGYGYDEGFDY 4127 43 GYTFSSY 3100 LPGSGS 3614 ARRAYGYDGGFDY 4128 44 GYTFSSY 3101 LPGSDS 3615 ARRAYGYDEGFDY 4129 45 AYTFSIY 3102 LPGSGS 3616 ARRAYGYDGGFDY 4130 46 GYTFSSY 3103 FPGSGH 3617 ARRGYGYDEGFDY 4131 47 GYTFSNY 3104 LPGSGS 3618 ARRGYGYDEGFDY 4132 47 GYTFSNY 3104 LPGSGS 3618 ARRGYGYDEGFDY 4132 48 GYTFSSY 3105 LPGSGS 3619 ARRGYGYDEGFDY 4133 48 GYTFSSY 3105 LPGSGS 3619 ARRGYGYDEGFDY 4133 49 GYTFSSY 3106 LPGSGY 3620 ARRGYGYDEGFDY 4134 50 GYTLSSY 3107 LPGSGS 3621 ARRGYGYDEGFDY 4135 51 GYTFSSY 3108 SPGSGS 3622 ARRGYGYDEGFDY 4136 52 GYTFGTY 3109 LPGSGT 3623 ARRAYGYDAGFDY 4137 53 GYTFSSY 3110 LPGSGS 3624 ARRGYGYDEGFDY 4138 54 GYTFSSY 3111 LPGSGR 3625 ARRGYGYDEGFDY 4139 55 GYTFSNY 3112 LPGSGS 3626 ARRGYGYDEGFDY 4140 56 GYTFSSY 3113 LPGSGS 3627 ARRGYGYDEGFDY 4141 3 GYSFTGY 3114 SSYNGA 3628 ARGRYGEYFDY 4142 4 GYSFTGY 3115 SSYNGV 3629 ARGRYGDYFDY 4143 5 GYSFTGY 3116 SSYNGV 3630 ARGRYGDYFDY 4144 6 GYSFTGY 3117 SSYNGV 3631 ARGRYGDYFDY 4145 7 GYSFTGY 3118 SSYNGA 3632 ARGRYGDYFDY 4146 8 GYSFTGY 3119 SSYNGV 3633 ARGRYGDYFDY 4147 9 GYSFTGY 3120 SSYNGV 3634 ARGRYGDYFDY 4148 10 GYSFTGY 3121 SSYNGV 3635 ARGRYGDYFDY 4149 11 GYSFTGY 3122 SSYNGV 3636 ARGRYGDYFDY 4150 12 GYSFTGF 3123 SSYNGA 3637 ARGRYGDYFDY 4151 13 GYSFTGY 3124 SSYNGA 3638 ARGRYGDYFDY 4152 57 GYSFTGY 3125 SSYNGV 3639 ARGRYGDYFDY 4153 58 GYSFTGY 3126 SSYNGA 3640 ARGRYGEYFDY 4154 124 GFSLSSY 3127 WRGGS 3641 AKNLYGHYVMDY 4155 125 GFSVTSY 3128 WRGGS 3642 AKNLYGHYVMDY 4156 126 GFSLTSY 3129 WRGGS 3643 AKNLYGHYVMDY 4157 127 GFSLTRY 3130 WRGGS 3644 AKNLYGHYVMDY 4158 128 GFSVTTY 3131 WRGGS 3645 AKNLYGHYVMDY 4159 129 GFSVTSY 3132 WRGGS 3646 AKNLYGHYVMDY 4160 130 GFSLTRY 3133 WRGGS 3647 AKNLYGHYVMDY 4161 59 GFSLSSY 3134 WRGGS 3648 AKNLYGHYVMDY 4162 60 GFSVTSY 3135 WRGGS 3649 AKNLYGHYVMDY 4163 61 GFSLTSY 3136 WRGGS 3650 AKNLYGHYVMDY 4164 62 GFSLTRY 3137 WRGGS 3651 AKNLYGHYVMDY 4165 63 GFSVTTY 3138 WRGGS 3652 AKNLYGHYVMDY 4166 131 GYTFTSY 3139 HPNSGS 3653 ARWGDGYSFAY 4167 132 GYTFTSY 3140 HPNSGS 3654 ARWGDGYSFAY 4168 133 GYTFTTY 3141 HPNSDN 3655 ARWGDGYSFAY 4169 14 GYTFTSY 3142 HPNSGT 3656 ARWGDGYSFAY 4170 134 GYTFTSY 3143 HPNSGN 3657 ARWGDGYSFAY 4171 64 GYTFTSY 3144 HPNSGS 3658 ARWGDGYSFAY 4172 65 GYTFTSY 3145 HPNSGT 3659 ARWGDGYSFAY 4173 135 GYTFTDY 3146 YPGSGS 3660 ARRGERGPWFAY 4174 136 GYTFTDY 3147 YPGSGS 3661 ARRGERGPWFAY

4175 137 GYTFTDY 3148 YPGSGS 3662 ARRGERGPWFAY 4176 138 GYTFTDY
3149 YPGSGS 3663 ARRGERGPWFAY 4177 15 GYTFTDY 3150 YPGSGS 3664
ARRGERGPWFAY 4178 66 GYTFTDY 3151 YPGSGS 3665 ARRGERGPWFAY
4179 67 GYTFTDY 3152 YPGSGS 3666 ARRGERGPWFAY 4180 68 GYTFTDY
3153 YPGSGS 3667 ARRGERGPWFAY 4181 69 GYTFTDY 3154 YPGSGS 3668
ARRGERGPWFAY 4182 24 GYTFTNY 3155 DPSDSE 3669 ATYDVYYRFAY 4183
139 GYTFTNY 3156 DPSDSE 3670 ATYDGYYRFAY 4184 140 GYTFTNY 3157
DPSDSE 3671 ATYDIYYRFAY 4185 16 GYTFTSY 3158 HPNSGS 3672
ARPGGYGFVY 4186 141 GYTFTSY 3159 HPNSDS 3673 ARPGGYGFAD 4187 142
GYTFTTY 3160 HPNSGS 3674 ARPGGYGFTY 4188 143 GYTFTSY 3161 HPNSGS
3675 ARPGGYGFAY 4189 70 GYTFTSY 3162 HPNSGS 3676 ARPGGYGFAY
4190 25 GYTFTSY 3163 YPSDSY 3677 TRGNYIDY 4191 144 GYTFTSY 3164
YPSDSY 3678 TRGNYIDY 4192 145 GYTFTDY 3165 YPSDSY 3679 TRGNYIDY
4193 146 GYTFTDY 3166 YPGSGS 3680 ARPGDLGFAY 4194 147 GYTFTDY 3167
YPGSGS 3681 ARPGDLGFAY 4195 148 GYTFTDY 3168 YPGSGS 3682
ARPGDLGFAY 4196 71 GYTFTDY 3169 YPGSGS 3683 ARPGDLGFAY 4197 72
GYTFTDY 3170 YPGSGS 3684 ARPGDLGFAY 4198 73 GYTFTDY 3171 YPGSGS
3685 ARPGDLGFAY 4199 74 GYTFTDY 3172 YPGSGS 3686 ARPGDLGFAY
4200 74 GYTFTDY 3172 YPGSGS 3686 ARPGDLGFAY 4200 75 GYTFTDY
3173 YPGSGS 3687 ARPGDLGFAY 4201 75 GYTFTDY 3173 YPGSGS 3687
ARPGDLGFAY 4201 76 GYTFTDY 3174 YPGSGS 3688 ARPGDLGFAY 4202 149
GFSLTNY 3175 WAGGI 3689 ARGDGYDDGYAMDY 4203 150 GFSLTSY 3176
WAGGI 3690 ARGDGYDDGYAMDY 4204 26 GFSLTSY 3177 WAGGT 3691
ARGDGYDDGYAMDY 4205 77 GFSLTNY 3178 WAGGI 3692
ARGDGYDDGYAMDY 4206 78 GFSLTSY 3179 WAGGI 3693
ARGDGYDDGYAMDY 4207 79 GFSLTSY 3180 WAGGT 3694
ARGDGYDDGYAMDY 4208 151 GYSFTSY 3181 DPSDSE 3695 ARTRNY 4209 152
GYSFTSY 3182 DPSDSE 3696 ARTRNY 4210 153 GYSFTSY 3183 DPSDSE 3697
ARTRNY 4211 154 GFNIKDY 3184 DPENGD 3698 NAPLLRYSSAMDY 4212 155
GFNIKDY 3185 DPENGD 3699 NAPLLRYSSSMDY 4213 156 GFNIKDY 3186
DPENGD 3700 NVALLRYSSAMDY 4214 80 GFNIKDY 3187 DPENGD 3701
NAPLLRYSSAMDY 4215 81 GFNIKDY 3188 DPENGD 3702 NAPLLRYSSSMDY
4216 82 GFNIKDY 3189 DPENGD 3703 NVALLRYSSAMDY 4217 17 GFNIKDT
3190 DPANGN 3704 ARGPDDGYFYYYSMDY 4218 157 GYTFSNY 3191 NPSNGD
3705 TSYYTHEAYYYAMDC 4219 27 GSTFTTY 3192 NPSNGG 3706
TSYYTHETYYYAMDY 4220 158 GFNIKDY 3193 DPEDGD 3707 TPYSIYDAMDY
4221 159 GYTFTDY 3194 YPGSGS 3708 ARRGERGPWFAY 4222 83 GYTFTDY
3195 YPGSGS 3709 ARRGERGPWFAY 4223 160 GYSFTDY 3196 STYYGD 3710
ARQMDYDYTYYYAMDY 4224 28 GYTFTSY 3197 DPSDSY 3711
ARAEYGYGNYPWFAY 4225 84 GYTFTSY 3198 DPSDSY 3712
ARAEYGYGNYPWFAY 4226 29 GYTFTSY 3199 HPSDSD 3713
AIPYYYGGWYFDV 4227 161 GYTFTDY 3200 YPGSGS 3714 ARMDGPWFAY 4228
30 GFTFSSY 3201 SSGGSY 3715 ARLYDAHWDYFDY 4229 162 GISLSTSGM
3202 WNND 3716 AWRPYYRYDSFAY 4230 18 GYTFTNY 3203 NTYTGE 3717
ARKYYDYEFAY 4231 85 GYTFTNY 3204 NTYTGE 3718 ARKYYDYEFAY 4232
163 GYTFTDY 3205 DPETGG 3719 TRLGDYDVMDY 4233 86 GYTFTDY 3206
DPETGG 3720 TRLGDYDVMDY 4234 164 GYTFTSY 3207 DPSDSY 3721
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4236 31 GYTFTSS 3209 HPNSGN 3723 AIYYDYDAYYFDY 4237 87 GYTFTSS
3210 HPNSGN 3724 AIYYDYDAYYFDY 4238 32 GYTFTSY 3211 HPNSGS

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KLLIYSTSRLHS 7611 QQGNTLPWT 6770 YLN 154 RSST- 5751 TGLIGGTSNRAP 7612 ALWYSTHYV 6771 GAVTTSN YAN 155 RSST- 5752 TGLIGGTSNRAP 7613

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KLWIYSTSNLAS 7650 QQRSSFPYT 6809 MH 173 KASQDIN- 5790 KTLIYRANRLVD
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6812 GYLN 92 RASQDIH 7230 KHLIYETSNLDS 7654 LQYASSPLT 6813 GYLN
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STDIDDD MN 179 RASENIYS 5802 QLLVYAATNLAD 7663 QHFWGTPYT 6822 NLA
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KLLIYYTSRLHS 7667 QQGNTLPFT 6826 YLN 96 RASQDISN 7234

KLLIYYTSRLHS 7668 QQGNTLPFT 6827 YLN 182 SAS- 5808 KLLIYYTSSLHS 7669 QQYSKLPYT 6828 QGISNYLN 97 SAS- 7235 KLLIYYTSSLHS 7670 QQYSKLPYT 6829 QGISNYLN 183 RASQSISD 5810 RLLIKYASQSIS 7671 QNGHSFPWT 6830 YLH 98 RASQSISD 7236 RLLIKYASQSIS 7672 QNGHSFPWT 6831 YLH 35 QATQDIV- 5812 SFLIYYATELAE 7673 LQFYEFPYT 6832 KNLN 99 QATQDIV- 7237 SFLIYYATELAE 7674 LQFYEFPYT 6833 KNLN 100 QATQDIV- 7238 SFLIYYATELAE 7675 LQFYEFPYT 6834 KNLN 36 RASGNIH 5815 QLLVYNAKTLAD 7676 QHFWSTPWT 6835 NYLA 101 RASGNIH 7239 QLLVYNAKTLAD 7677 QHFWSTPWT 6836 NYLA 184 RSST- 5817 TGLIGGTSYRAP 7678 ALWYSTHYV 6837 GAVTTSN YAN 102 RSST- 7240 TGLIGGTSYRAP 7679 ALWYSTHYV 6838 GAVTTSN YAN 185 RASQEIS- 5819 KRLIYAASTLDS 7680 LQYASYPWT 6839 GYLS 103 RASQEIS- 7241 KRLIYAASTLDS 7681 LQYASYPWT 6840 GYLS 186 KASQSVD 5821 KLLIYAASNLES 7682 QQSNEDPLPT 7776 YD-GDSYMN 21 HASQNIN 5822 KLLIYKASNLHT 7683 QQGQSYPLT 6842 VWLS 187 SASSSVSY 5823 KRWIYDTSKLAS 7684 QQWSSNPLT 6843 MH 188 RASGNIH 5824 QLLVYNAKTLAD 7685 QHFWSTPWT 6844 NYLA 189 RASENIYS 5825 QLLVYNAKTLAE 7686 QHHYGTPFT 6845 YLA 190 RAS- 5826 KLWIYYTSNLAP 7687 QQFTSSLT 7777 SSVNYMY 191 RSST- 5827 TGLIGSTNNRAP 7688 TLWYSNHWV 6847 GAVTTSN YAN 192 KASQDVG 5828 KLLIYWASTRHT 7689 QQYSSYPFT 6848 TAVA 193 KASQDIN- 5829 KTLIYRANRLVD 7690 LQYDEFPPT 6849 SYLS 194 RASENIYS 5830 QLLVYAATNLAD 7691 QHFWGTPFT 6850 NLA 195 RASQSISD 5831 RLLIKYASQSIS 7692 QNGHSFPYT 6851 YLH 104 RASQSISD 7242 RLLIKYASQSIS 7693 QNGHSFPYT 6852 YLH 37 RASENIYS 5833 QLLVYAATNLAD 7694 QHFWGTPYT 6853 NLA 196 RSST- 5834 TGLIGGTNNRAP 7695 ALWYSNHWV 6854 GAVTTSN YAN 197 IT- 5835 KLLISEGNTLRP 7696 LQSDNLPLT 6855 STDIDDD MN 198 SASSSVSY 5836 KLWIYSTSNLAS 7697 QQRSSYPPT 6856 MH 199 SASSSVSY 5837 KRWIYDTSKLAS 7698 QQWSSNPLT 6857 MH 22 IT- 5838 KLLISEGNSLRP 7699 LQSDNLPLT 6858 STDIDDD MN 200 SASSSVSS 5839 KLWIYSTSNLAS 7700 HQWSSYPPT 6859 SYLY 201 TLSSQHST 7243 KYVMELKKDGSH 7701 GVGDTIKEQFVYV 7778 YTIE STGD 202 RSSQSIVH 5841 KLLIYKVSNRFS 7702 FQGSHVPWT 6861 SNGNTYLE 203 KASQDIN- 5842 KTLIYRANRLVD 7703 LQYDEFPPT 6862 SYLS 204 KSS- 5843 KRLIYLVSKLDS 7704 WQGTHFPWT 6863 QSLLDSDG KTYLN 205 SASSSVSS 5844 KLWIYSTSNLAS 7705 HQWSSYPPT 6864 SYLY 206 RSSQSIVY 5845 KLLIYKVSNRFS 7706 FQGSHVPPT 6865 SNGNTYLE 207 RASENI- 5846 QLLVYAATNLAD 7707 QHFWGTPWT 6866 YNNLA 208 SASSSVSS 5847 KLWIYSTSNLAS 7708 HQWSSYPPT 6867 SYLY 209 SASSSVSY 5848 RLLIYDTSNLAS 7709 QQWSTYPPIT 7779 MY 210 SASSSVSY 5849 RLLIYDTSNLAS 7710 QQWSSYPFT 6869 MY 211 RAS- 5850 KPWIYATSNLAS 7711 QQWSSNPYT 6870 SSVSYMH 212 RASENIYS 5851 QLLVYAATNLAD 7712 QHFWGTPWT 6871 NLA 213 KASQDVS 5852 KLLIYWASTRHT 7713 QQHYSTPWT 6872 TAVA 214 KSS- 5853 KRLIYLVSKLDS 7714 WQGTHFPWT 6873 QSLLDSDG KTYLN 215 RSSQSIVH 5854 KFLIYKVSNRFS 7715 FQGSHVPPT 6874 SNGNTYLE 216 IT- 5855 KLLISEGNTLRP 7716 LQSDNMPLT 6875 STDIDDD MN 217 KASQDINK 5856 RLLIHYTSTLQP 7717 LQYDNLYMYT 7780 YIA 218 HASQNIN 5857 KLLIYKASNLHT 7718 QQGQSYPYT 6877 VWLS 219 SASSSVSY 5858 KPWIYLTSNLAS 7719 QQWSSNPPT 6878 MY 220 KSS- 5859 KRLIYLVSKLDS 7720 WQGTHFPWT 6879 QSLLDSDG KTYLN 221 SASLSVSD 5860 RLLIYDTSNLAS 7721 QQWSSYPFT 6880 MY 222 SASSSVSS 5861 KLWIYSTSNLAS 7722 HQWSSYPPT 6881 SYLY 223 RAS- 5862 KLWIYYTSNLAP 7723 QQFTSSPST 6882 SSVNYMY 105 RAS- 7244

KLWIYYTSNLAP 7724 QQFTSSPST 6883 SSVNYMY 224 IT- 5864 KLLISEGNTLRP 7725 LQSDNLPLT 6884 NTDIDDD MN 106 IT- 7245 KLLISEGNTLRP 7726 LQSDNLPLT 6885 NTDIDDD MN 225 KASQSVD 5866 KLLIYAASNLES 7727 QQSNEDPWT 6886 YD- GDSYMN 226 RASQSISN 5867 RLLIKYASQSIS 7728 QQSNSWPFT 6887 NLH 227 SASSSVSS 5868 KLWIYSTSNLAS 7729 HQWSSYPPT 6888 SYLY 228 IT- 5869 KLLISEGNTLRP 7730 LQSDNMPLT 6889 STDIDDD MN 229 KSS- 5870 KRLIYLVSKLDS 7731 WQGTHFPWT 6890 QSLLDSDG KTYLH 230 KASQDINK 5871 RLLIHYTSTLQP 7732 LQYDNLRT 7781 YIA 231 RSST- 5872 TGLIVGTNNRAP 7733 VLWYSNHLV 6892 GAVTTSN YAN 232 KASEN- 5873 KLLIYGASNRYT 7734 GQSYSYPPT 6893 VGTYVS 233 RASET- 5874 KLLIYRASNLES 7735 QQSNEDPRT 6894 VDSYGYSF MH 234 KSS- 5875 KLLLYWASTRES 7736 QQYYSYRT 7782 QSLLYSTN QKNYLA 235 KSS- 5876 KRLIYLVSKLDS 7737 WQGTHFPFT 6896 QSLLDSDG KTYLN 236 KASQSVD 5877 KLLIYAASNLES 7738 QQSNEDPFT 6897 YD- GDSYMN 237 TLSSQHST 7246 KYVMELKKDGSH 7739 GVGDTIKEQFVYV 7783 YTIE STGD 107 TLSSQHST 7247 KYVMELKKDGSH 7740 GVGDTIKEQFVYV 6899 YTIE STGD 238 SASSSVSY 5880 KRWIYDTSKLAS 7741 QQWSSNPLT 6900 MH 108 SASSSVSY 7248 KRWIYDTSKLAS 7742 QQWSSNPLT 6901 MH 239 QATQDIV- 5882 SFLIYYATELAE 7743 LQFYEFPLT 6902 KNLN 240 RAS- 5883 KLLIYAASNQGS 7744 QQSKEVPPT 6903 ESVDNY- GISFMN 241 RASQSISD 5884 RLLIKYASQSIS 7745 QNGHSFPLT 6904 YLH 242 RSSQSIVH 5885 KLLIYKVSNRFS 7746 FQGSHVPLT 6905 SNGDTYLE 243 SASSSVSY 5886 KRWIYDTSKLAS 7747 QQWSSNPLT 6906 MH 244 TASESLYS- 5887 KLLIYGASNRYI 7748 AQFYSYPYT 6907 SKHKVHYL A 245 TLSSQHST 7249 KYVMELKKDGSH 7749 GVGDTIKEQFVYV 7784 YTIE STGD 246 RASQSVST 5889 KLLIKYASNLES 7750 QHSWEIPLT 6909 SSYSYMH 247 SASSSVSY 5890 KRWIYDTSKLAS 7751 QQWSSNPLT 6910 MH 248 KASDHIN 5891 RLLISGATSLET 7752 QQYWSTPLT 6911 NWLA 249 RASENIYS 5892 QLLVYNAKTLAE 7753 QHHYGTPYT 6912 YLA 250 RASENIYS 5893 QLLVYNAKTLAE 7754 QHHYGTPPT 6913 YLA 251 SASSSVSY 5894 KRWIYDTSKLAS 7755 QQWSSNPPT 6914 MH 252 RAS- 5895 KLLIYAASNQGS 7756 QQSKEVPPT 6915 ESVDNY- GISFMN 253 TASESLYS- 5896 KLLIYGASNRYI 7757 AQFYSYPYT 6916 SKHKVHYL A 254 RSST- 5897 TGLIGGTSNRAP 7758 ALWYSTHYV 6917 GAVTTSN YAN 255 SASSSI- 5898 KRWIYDTSKLAS 7759 HQRSSYPT 7785 SYMH HCDR1 HCDR2 HCDR3 SEQ ID SEQ ID SEQ ID Sequence NO: Sequence NO: Sequence NO: 265 GGTFSSY 10885 NPSGG 11147 ARDLGDPGMDV 11409 266 GGTFSNY 10886 DPSGGS 11148 ARDLGDMGMDV 11410 267 GGTFSNY 10887 NPSGGS 11149 ARDVGDRGMDV 11411 263 GGTFSSY 10888 NPSGG 11150 ARDLGDPGMDV 11412 268 GSTFSGY 10889 DPNGGG 11151 AKDIVHDGTEYFQH 11413 269 GYTFTSY 10890 NPSGGS 11152 AKDIVHDGTEYFQH 11414 270 GGTFSSY 10891 NPSGGS 11153 AREGRDHDAFDI 11415 271 GFTFTDY 10892 NPSGGS 11154 AREGRSHDAFDI 11416 272 GYTFTGY 10893 NPHSGD 11155 ARWVGTTEYYYYYMDV 11417 273 GYTFTDY 10894 DPSGGS 11156 ATTAYYDFWSGYSMDV 11418 274 GYTFTSH 10895 DPSGGS 11157 ARDMDNWNTGYYYYMDV 11419 275 GGTFSSY 10896 NPNSGD 11158 ARDQRGGDAWDV 11420 276 GGTFSNY 10897 TPSGGS 11159 ARDTAGHFDI 11421 277 GGTFRND 10898 NPNSGN 11160 ARDNPDLDGMDV 11422 278 GGTFSSY 10899 SAYNGN 11161 ARDLVGHFDY 11423 279 GGTFSSY 10900 NPNSGG 11162 ARDGYSGSYSD 11424 264 GGTFSSY 10901 NPNSGG 11163 ARDGYSGSYSD 11425 257 GGTFSSY 10902 NPNSGG 11164 ARDGYSGSYSD 11426 280 GNTLSSH 10903 NPSGGS 11165 ARDQGSSGTFDY 11427 281 GGTLSSY 10904 NPNSGG 11166 ARDSTDVIDY 11428 282 GYIFTSY 10905 NPNSGD 11167 ARDGGTVTPTEEYYYYG- 11429 MDV 283 GGTFSSY

10906 SVYNGN 11168 ASLDDLDY 11430 284 GHTFTSY 10907 NPNNGG 11169 ARDMVRDSAEYFQH 11431 285 GYTFITS 10908 NPSGGT 11170 ARDSSGYPIDY 11432 286 GYTFTSY 10909 IPLSGA 11171 ARGALYNWNDGWFDP 11433 258 GYTFTSY 10910 IPLSGA 11172 ARGALYNWNDGWFDP 11434 287 GFTVGSW 10911 WYEGSN 11173 ARLGTASLPYFDY 11435 288 GYTFTGY 10912 NPNRGD 11174 ARESGDGFDP 11436 289 GYTFTNY 10913 NPNSGN 11175 ARDWPNWFDP 11437 290 GYSFTDN 10914 RSDNGE 11176 AREVQLVGFDY 11438 291 GYTFSDH 10915 IPIFGT 11177 ARGSSWYLHFQH 11439 260 GYTFSDH 10916 IPIFGT 11178 ARGSSWYLHFQH 11440 292 GGTFSSY 10917 IPIFGT 11179 AKGVDRYNWNDAFDY 11441 293 GYTFTDY 10918 HSNSGG 11180 ARESSGYDSSLDY 11442 294 GGTFSSY 10919 NPNSGD 11181 TTDPRLDSSDPGY 11443 295 GGTFGNY 10920 SAYNGN 11182 ARGGMDV 11444 296 GGTFSRY 10921 YPSDGS 11183 ARDRLGDLDY 11445 261 GGTFSRY 10922 YPSDGS 11184 ARDRLGDLDY 11446 297 GGTFSSY 10923 NPNSGN 11185 ARDSIVGGYPFDY 11447 298 GYTFTSY 10924 TPIFGT 11186 AREGYSSSWHDDAFDI 11448 299 GGTFSNY 10925 DPSGGS 11187 ARDLGDYGLDS 11449 300 GYTFTGY 10926 NPNSGD 11188 ATGGSDSSGYYYEGYFQH 11450 301 GGTFSSY 10927 SPNSGN 11189 ARDKGGYYDSSGYYWY 11451 302 GFSLSSY 10928 SSNGGS 11190 ARVGDGDGYNPDFDY 11452 303 GYTFTSY 10929 DPTSGA 11191 AKDPIVATEVDY 11453 304 GGTFSSY 10930 SPNSGN 11192 ARDSGAFDI 11454 305 GVTISNY 10931 NPNSGN 11193 AREGLLDAFDI 11455 306 GGTFSRY 10932 NPYDGN 11194 ARGGRHHDAFDI 11456 307 GGTFSSY 10933 NPSGDG 11195 ARDISNDAFDI 11457 308 GYILTGH 10934 SAYNGD 11196 ARGSSWDDAFDI 11458 309 GFTFSNH 10935 GAGGG 11197 AREGWNDDVFDI 11459 310 GGTFSSY 10936 NPSAGT 11198 ARDGNFGAFDI 11460 311 GYSFTTY 10937 IPIFGT 11199 ARDKSGWNYGSGSYN- 11461 DAFDI 312 GYAFTGY 10938 NPNSGK 11200 ARDGGLDFDY 11462 313 GYTFTTY 10939 NPNTGD 11201 AKDPAVTPDAFDI 11463 314 GGTLSSY 10940 DPSGGG 11202 AGSLYYYGMDV 11464 315 GGTFGSS 10941 IPIFGT 11203 AKEDDILPPRAFDI 11465 316 GFTFDDY 10942 SGGGGV 11204 ARVYSSGWLDAFDI 11466 317 GGTFSSY 10943 SGYNGN 11205 ASSDVSPDAFDI 11467 318 GGTQNIY 10944 NPNSGN 11206 ATPTSSSDDAFDI 11468 319 GGTFSSY 10945 NPNSGG 11207 ARASRGDDAFDI 11469 320 GIPFTSD 10946 NPSGGS 11208 ARERYEGGY- 11470 SSGPGNYYYGMDV 321 GGTFSNY 10947 NPNSGN 11209 ARDDDYGDYPV 11471 322 GDTFSDH 10948 NPKIGN 11210 VYDSSGYDAFDI 11472 323 GYTFTSY 10949 NPGTGG 11211 ARETPSDYYDSSGYYYN- 11473 DAFDI 324 GGTFSSY 10950 IPSGG 11212 ARDLGTTFDI 11474 325 GYTFTAY 10951 NPDNDN 11213 AKDIAVAALAYGMDV 11475 326 GFTFSSY 10952 SYDGSD 11214 ARQSLYYYYGMDV 11476 327 GYTFTDY 10953 STFTGN 11215 ARDAPLAAAGTDYYYG- 11477 MDV 262 GYTFTDY 10954 STFTGN 11216 ARDAPLAAAGTDYYYG- 11478 MDV 328 GFTFSSY 10955 SDDGIT 11217 ARDDSSGYGGMDV 11479 329 GFTFSSY 10956 SYDGGD 11218 ASGSLVLGYYYMDV 11480 330 GYTFTNY 10957 NPNTGG 11219 ATGGGGSYYDAFDV 11481 331 GGTFSSY 10958 NPNSGN 11220 ARDIGEGYSMDV 11482 332 GFTFSNH 10959 SYDGSN 11221 AREEKYSSSWY- 11483 VGVDAFDI 333 GFTFSSS 10960 SGSGDN 11222 ARDQEDYYYDSSGYG- 11484 MDV 334 GGTFSSH 10961 IPIFGT 11223 AKGDW- 11485 GIVVVPAAIGAFDI 335 GYTFTAY 10962 SPVFGS 11224 ARDLGYYDSSGYRYDAF 11486 DI 336 GYTFTSY 10963 SPMFGT 11225 AKDGWYYGMDV 11487 337 GGTFSSY 10964 NPNSGG 11226 ARGEAGNLDWYFDL 11488 338 GGTFSNY 10965 NPNNGD 11227 AREDVWYFDL 11489 339 GYTFTTY 10966 STYDGK 11228 ALHLGGDWYFDL 11490 340 GYTFTGY 10967 NPNTGA 11229 ARQHGDYDWYFDL 11491 341 GDTFTTY 10968 NPNSGN 11230 ARDSGRH 11492 342 GGTFSSY 10969 IPMLGI 11231 VREEVAGANWFDP 11493 343 GYTFTSY 10970 NPSGGS 11232 AREGDYGSGEFDY 11494 344 GYTFTSS 10971 NPRSGN 11233 ARERDDYGDYGWLDY 11495 345 GYTFTGY 10972 NPSGGS 11234 ARDLYDSSGY- 11496 WHYYYYMDV 346 GGTFSSY 10973 NPNSGG 11235

ARFSGYDYVDY 11497 347 GGTFSSY 10974 NPNGGN 11236 ARDVGEDFDL 11498 348 GYTFTSY 10975 NPADGD 11237 ARDFDWLFAMDV 11499 349 GGTFSNY 10976 NPNGGT 11238 AKHGDHGFYV 11500 350 GGTFSSY 10977 NPNVGS 11239 AREDSGTSWFDP 11501 351 GYTFTSY 10978 NPSDGS 11240 ARDDRGSNYYYGMDV 11502 352 GYTFTAY 10979 NPNSGT 11241 ARDSSDYYGDYRADAFDI 11503 353 GYTFTSY 10980 SPSGDA 11242 VKGLDH 11504 354 GFSFSDY 10981 GGIGDS 11243 ARMNYGDSNYYYYYG- 11505 MDV 355 GYTFTSY 10982 SPSDGS 11244 ARGAVGFDY 11506 356 GYTFTSY 10983 NTYSGY 11245 TTDDFLSFGY 11507 357 GYMFTDY 10984 IPYFGT 11246 ARSISGSYVLDAFDI 11508 358 GYTFNSY 10985 IPIFGT 11247 ARDWGYGDYADDAFDI 11509 359 GGTFSNN 10986 NPIYGS 11248 AADWRGFDY 11510 360 GYTFTEY 10987 NPHNGD 11249 AREGDYLGYPIDC 11511 361 GFTFSDY 10988 WQDGNV 11250 ARDGNSGYVF 11512 362 GYTFTTY 10989 NPNTGD 11251 ARTAEAVAGLPAFDY 11513 363 GGTSNNY 10990 IPLFGT 11252 ARVTLYGDYDY 11514 364 GYSLITH 10991 NPSDGV 11253 AREYYGEGFDY 11515 259 GYSLITH 10992 NPSDGV 11254 AREYYGEGFDY 11516 365 GGTFSSY 10993 NPSGGS 11255 ARDLGDTAMDG 11517 366 GYTFTSY 10994 TPSGGS 11256 ARDGGLASFDY 11518 367 GGTFSSY 10995 NPNSGN 11257 ARGGGWAMTDAFDI 11519 368 GFTFDDY 10996 YSGGD 11258 TRKEYYYDSSGYLRLFDY 11520 369 GYTFTDY 10997 NPIFGT 11259 ARDISGYDYYYYGMDV 11521 370 GGTLNNY 10998 DPSDGT 11260 ARSDYDFWSGLGGYFDY 11522 371 GGTFSSY 10999 DPNSGG 11261 ARDSAEWELGGSFDY 11523 372 GFTFSNH 11000 GVNGD 11262 AREGLVFSGRGHWYFDL 11524 373 GGTFSNY 11001 NPNGGN 11263 ARDYEDADFDG 11525 374 GYTFSDH 11002 NPDSGN 11264 ARDSTSGVDY 11526 375 GFTFSSY 11003 SYDGHD 11265 ARGEQQLEGFYYYYG- 11527 MDV 376 GFTFSSY 11004 SYDGSK 11266 ASDYGDYGTYDY 11528 377 GFTFSSY 11005 SGGGDD 11267 AREPLAYCGG- 11529 DCPGGFDY 378 GFTFSDH 11006 GTGGD 11268 ARHEDTAIFLDY 11530 379 GYTFTSY 11007 SPSDGS 11269 ARDGYDAWSYGMDV 11531 380 GYTFTGY 11008 NPNSGN 11270 ARDGVTGTDY 11532 381 GFAFSSY 11009 SGAGDS 11271 AREPTTVTDDWYFDL 11533 382 GFAFSSH 11010 SGNGDN 11272 ARDRAPEYFDL 11534 383 GGTFSSY 11011 NPNSGG 11273 ARDDYGDYGGGMDV 11535 384 GYTFTDY 11012 NPNSGH 11274 AKDTSPRYGDGFFDY 11536 385 GFTFSSY 11013 SYDGSN 11275 ARESGFSAEYFQH 11537 386 GGTFSSY 11014 NPSGGS 11276 ARATGLYCSGSCFDY 11538 388 RSILDFN 14119 ARAGA 14120 RVFDLPNDY 14121 LCDR1 LCDR2 LCDR3 SEQ ID SEQ ID SEQ ID Sequence NO: Sequence NO: Sequence NO: 265 RASQDISN 12194 KLLIYDASNLET 13104 QQSYSTPLT 12714 YLN 266 RASQSIS- 12195 KLLIYAASSLQS 13105 QQSYSTPLT 12715 SYLN 267 QASQDIS 12196 KLLIYKASSLET 13106 QQSFSSPLT 12716 NYLN 263 RASQDISN 12197 KLLIYDASNLET 13107 QQSYSTPLT 12717 YLN 268 RASQNVN 12198 KLLIYEASSLQS 13108 QQANSFPFT 12718 TWLA 269 RASQSISD 12199 KLLIYAASSLQS 13109 AQHNHYPYT 12719 WLA 270 KSS- 12200 KLLIYWASTRES 13110 QQYYTTPFT 12720 QSVLSSSY NKNYLA 271 KSS- 12201 KLLIYWASTRAS 13111 QQYYSTPFT 12721 QSVLSSSY NKNYLA 272 RAS- 12202 KLLIYDASHLEA 13112 QQANSFPIT 12722 QAIRDDLG 273 RASQGVG 12203 KLLIYAASTLQT 13113 QQASSFPLT 12723 NDLA 274 RASQI- 12204 KLLIYAASSLQS 13114 QQSYTFPVT 12724 IGTNLA 275 RASQSIST- 12205 KLLIYDASSLES 13115 QQSYSTPFT 12725 WLA 276 KSS- 12206 KLLIYWASTRES 13116 QQYYGSPLT 12726 QSVLSSS- NNKNYLA 277 KSS- 12207 KLLIYWASTRES 13117 QQYYSSPPT 12727 QSVLSSSY NKNYLA 278 KSS- 12208 KLLIYWASTRES 13118 QQYYSSPPT 12728 QSVLSSSY NKNYLA 279 KSS- 12209 KLLIYWASTRES 13119 QQYYSTPWT 12729 QSVLSSSY NKNYLA 264 KSS- 12210 KLLIYWASTRES 13120 QQYYSTPWT 12730 QSVLSSSY NKNYLA 257 KSS- 12211 KLLIYWASTRES 13121 QQYYSTPWT 12731 QSVLSSSY NKNYLA 280 KSS- 12212 KLLIYWASTRAS 13122 QQYYGSPPT 12732 QSVLSSSY NKNYLA 281 QASQDIR 12213 KLLIYDASTLQS 13123

QQAYSFPWT 12733 NYLN 282 QASQDIS 12214 KLLIYNASNLET 13124 QQLNSYPFT 12734 NYLN 283 QASQSIST 12215 KLLIYAASTLRS 13125 LQHYTYPLT 12735 WLA 284 RASEDI- 12216 KLLIYAASTLQS 13126 QQSHTIPWT 12736 STYLA 285 RASH- 12217 KLLIYAASTLQS 13127 QQSYSSPYT 12737 HISDFLN 286 RASQDIG 12218 KLLIYDASSLQS 13128 QQANSFPLT 12738 DYLA 258 RASQDIG 12219 KLLIYDASSLQS 13129 QQANSFPLT 12739 DYLA 287 RASQDIRS 12220 KLLIYAASSLQS 13130 QQSYTAPPT 12740 YLA 288 RASQDIS- 12221 KLLIYAASTLQS 13131 LQHNTYPLT 12741 NNLN 289 RASQDIS- 12222 KLLIYDASSLQS 13132 QQAISFPLT 12742 NWLA 290 RASQGIA 12223 KLLIYAASSLQS 13133 QQADSFPLT 12743 NYLA 291 RASQGIAS 12224 KLLIYAASTLQP 13134 QQFDSYPIT 12744 YLA 260 RASQGIAS 12225 KLLIYAASTLQP 13135 QQFDSYPIT 12745 YLA 292 RASQGISN 12226 KLLIYAASRLQS 13136 QQSSIIPFT 12746 YLA 293 RASQGISN 12227 KLLIYAASTLQS 13137 QQAYSFPYT 12747 YLA 294 RASQSIGR 12228 KLLIYDASNLET 13138 QQSYSTPRT 12748 WLA 295 RASQSINS 12229 KLLIYDTSSLQS 13139 QQTYSTPYT 12749 WLA 296 RASQSISS 12230 KLLIYAASTLQS 13140 QQGYSTPYI 12750 WLA 261 RASQSISS 12231 KLLIYAASTLQS 13141 QQGYSTPYI 12751 WLA 297 RASQSIS- 12232 KLLIYAASSLQS 13142 QQTDSIPIT 12752 SYLN 298 RASQSIS- 12233 KLLIYAASTLQS 13143 QQSYSIPYT 12753 SYLN 299 RASQTIRS 12234 KLLIYKASSLES 13144 QQTYTIPIT 12754 YLN 300 RASQTISN 12235 KLLIYAASTLQS 13145 QQANSFPPT 12755 WLA 301 RASQYIGS 12236 KLLIYDASNLET 13146 QQVDSYPLT 12756 YLN 302 RSSQSLLH 12237 QLLIYLGSNRAS 13147 MQGTHWPPT 12757 SNGYNYL D 303 RSSQSLLH 12238 QLLIYFGSNRAS 13148 MQALQAPVS 12758 SNGYNYL D 304 KSS- 12239 KLLIYWASSRQS 13149 QQYYSTPLT 12759 QSVLSSSY NKNYLA 305 KSS- 12240 KLLIYWASVRES 13150 QQYYSTPIT 12760 QSVSSSSY NKNYLA 306 KSTQNVLS 12241 KLLIYWASTRES 13151 QQYYSTPFT 12761 SSNNN- SYLA 307 QASQDIG 12242 KLLIYAASSLQS 13152 QQTYNTPLT 12762 NYLN 308 QASQDIS 12243 KLLIYEASTLQS 13153 QQSYSTPFT 12763 NYLN 309 QASQDIST 12244 KLLIYRASTLES 13154 QQSYSIPLT 12764 WLA 310 RASQNIN- 12245 KLLIYAASRLQS 13155 QQSYSAPVT 12765 NYLN 311 RASQNINT 12246 KLLIYAASSLQS 13156 QQAYSFPFT 12766 WLA 312 RASQRIGN 12247 KLLIYAASSLQS 13157 QQSYSTPLT 12767 YLN 313 RASQSIST 12248 KLLIYAASTLQS 13158 QQSYRTVT 12768 YLN 314 RASQSVG 12249 RLLIYGASTRAT 13159 QQYDSSSQT 12769 SYLA 315 RASRSVST 12250 RLLIYGASTRAT 13160 QQYDGSPYT 12770 YLA 316 RSSQSLLH 12251 QLLIYDASNLET 13161 MQALQTPPA 12771 SNGYNYL D 317 KSS- 12252 KLLIYWASTRES 13162 QQYYSAPPT 12772 QSVLSSSY NKNFLA 318 KSS- 12253 KLLIYWASTRES 13163 QQYYSDPIT 12773 QSVLSSSY NKNFLA 319 KSS- 12254 KLLIYWASARES 13164 QQYYSIPIA 12774 QSVLSSSY NKNYLA 320 KSS- 12255 KLLIYWASTRDS 13165 QQYYSIPYT 12775 QSVLSSSY NKNYLA 321 KSS- 12256 KLLIYWASTRAS 13166 QQYYTTPPT 12776 QSVLSTSY NKNYLA 322 KSS- 12257 KLLIYWASTRQS 13167 QQYYSTPYT 12777 QSVLSTSY NRNFLA 323 KSS- 12258 KLLIYWASTRES 13168 QQYYSTPLT 12778 QSVLYSSN NKNYLA 324 QASQDIS 12259 KLLIYAASSLQS 13169 QQSYSTPT 12779 NYLN 325 QASQDIS 12260 KLLIYGASTLQS 13170 QEADSFPLT 12780 NYLN 326 RASQGIRN 12261 KLLIYDASSLHS 13171 QQAYSFPWT 12781 DLG 327 RASQGISN 12262 KLLIYKASSLES 13172 QQSYNTPFT 12782 YLA 262 RASQGISN 12263 KLLIYKASSLES 13173 QQSYNTPFT 12783 YLA 328 RASQSINR 12264 KLLIYSASNLQS 13174 QQSYNTPLT 12784 WLA 329 RASQSINT 12265 KLLIYAASSLQS 13175 QQANSFPFT 12785 WLA 330 RASQSIRT- 12266 KLLIYDASSLET 13176 QQLNSYPLT 12786 WLA 331 RASQSIRT 12267 KLLIYAASTLQS 13177 QQSYSAPLT 12787 YLN 332 RASQSIST 12268 KLLIYAASSLHS 13178 QQSYSTPLT 12788 YLN 333 RASQSIT- 12269 KLLIYAASTLQS 13179 QQSYSTPLT 12789 TYLN 334 RSSQSLLH 12270 QLLIYAASSLQS 13180 MQARQTPLT 12790 SNGYNYL D 335 RSSQSLLH 12271 QLLIYGASSLQS 13181 MQTLQTPFT 12791 SNGYNYL D 336

RSSQSLLH 12272 QLLIYLGSDRAS 13182 MQALQTPLT 12792 SNGYNYL D 337 KSSQTVF- 12273 KLLIYWASTRES 13183 QQYYSTPLT 12793 STSYN- KNYLA 338 KTSQSVF- 12274 KLLIYWASTRES 13184 QQYYSSPPT 12794 STSYN- RDYLA 339 RASQSISS 12275 KLLIYDASTLQS 13185 QQSYSTPFT 12795 WLA 340 RASQSIS- 12276 KLLIYDASNLKT 13186 QQSYSFPT 12796 SYLN 341 RASQSVSS 12277 RLLIYDTSSRAT 13187 QQYYDTPYT 12797 YLA 342 KSS- 12278 KLLIYLASTREP 13188 QQYYSTPPT 12798 QSVLYSSN NKNYLA 343 KSS- 12279 KLLIYWASTRES 13189 QQYYSTPLT 12799 QSVLSSSY NKNYVA 344 QASQDIS 12280 KLLIYAAASLQS 13190 QQTYSTPWT 12800 NYLN 345 RASQDIN- 12281 KLLIYAASSLQS 13191 QQSSSFPLT 12801 TYLA 346 QASQDIS 12282 KLLIYAASSLQS 13192 QQLYNFPYT 12802 NYLN 347 RASQSISR 12283 KLLIYGASTRES 13193 QQSYNTPLT 12803 YLA 348 RASQTLSG 12284 KLLIYGASTLQG 13194 QQYYSYPPT 12804 WLA 349 FASQDI- 12285 KLLIYEASNLET 13195 QQSYSTPLT 12805 INYLN 350 RASQSIS- 12286 KLLIYDVFNLGT 13196 QQSYSSPFT 12806 SYLN 351 QASQDIS 12287 KLLIYMASNLES 13197 QQTNSFPLT 12807 NYLN 352 RASQSIS- 12288 KLLIYDASNLET 13198 QQSYSTPLT 12808 SYLN 353 RSSQSLLH 12289 QLLIYLGSNRAS 13199 MQALQSPWT 12809 SNGYNYL D 354 KSS- 12290 KLLIYWASTRES 13200 QQYYSSPLT 12810 QSVLYSSN NKNYLA 355 RSSQSLLH 12291 QLLIYLGSNRAS 13201 MQALQTPPS 12811 SNGYNYL D 356 RAS- 12292 KLLIYKASRLES 13202 QQSYKTPYT 12812 ESVST- WLA 357 KSS- 12293 KLLIYWASTRES 13203 QQYFTTPLT 12813 QSVLYSSN NKNYLA 358 RASQSIS- 12294 KLLIYAASSLQS 13204 QQSYSTPYT 12814 SYLN 359 KSS- 12295 KLLIYWASTRAS 13205 QQYYDTPLT 12815 QSVLSSSY NKNYLA 360 RASQSIS- 12296 KLLIYKASTLES 13206 QQNDSIPIT 12816 SYLN 361 RASQSISR 12297 KLLIYDASNLET 13207 LQDYSYPLT 12817 WLA 362 KTSQSVF- 12298 KLLIYWASTRAA 13208 QQYYYTST 12818 STSYN- RDYLA 363 RASQSIN- 12299 KLLIYAASSLQS 13209 QQANSFPPT 12819 RYLN 364 RASQGISN 12300 KLLIYSASNLQS 13210 QQSYSTPLT 12820 YLA 259 RASQGISN 12301 KLLIYSASNLQS 13211 QQSYSTPLT 12821 YLA 365 RASQSIDS 12302 KLLIYKASTLES 13212 QQSYSAPLT 12822 YLN 366 RASQDIST 12303 KLLIYDASNLET 13213 QQVNSDPYT 12823 WLA 367 QASQDIS 12304 KLLIYAASTLES 13214 QQGDSLPLT 12824 NYLN 368 RASQGISN 12305 KLLIYAASSLQS 13215 QQSDSFPYT 12825 YLA 369 RASQSVST 12306 RLLIYGASTRAT 13216 QQHDSYPLT 12826 YLA 370 RASQGIRN 12307 KLLIYAASSLQS 13217 QQANSFPPT 12827 DLG 371 RAS- 12308 KLLIYKASNLES 13218 QQTDSTFIT 12828 ESISTYLN 372 RASRNIHD 12309 KLLIYAASTLQT 13219 QQTYSTPPT 12829 YLN 373 RASQSND 12310 KLLIYKASTLES 13220 QQSYSSPLT 12830 SYLN 374 RASQSISD 12311 KLLIYAASTLQS 13221 QQSYSSPYT 12831 FLN 375 QASQDIS 12312 KLLIYAASSLQS 13222 QQANRFPLT 12832 NYLN 376 QASQDIS 12313 KLLIYKASNLQS 13223 QQSYNFPAT 12833 NYLN 377 RSSQSLLH 12314 QLLIYLGSNRAS 13224 MQGTHWPET 12834 SNGYNYL D 378 RASQSIS- 12315 KLLIYDASNLET 13225 QQSYSTPLT 12835 SYLN 379 RASQGISD 12316 KLLIYDASNLET 13226 QQSYILPLT 12836 YLA 380 RASQDIN 12317 KLLIYAASSLQS 13227 QQSYSAPYT 12837 DFLA 381 RASQSISN 12318 KLLIYAASKLES 13228 QQSYSSPWT 12838 WLA 382 RASQGIDS 12319 KLLIYAASTLES 13229 QQAYSFPLT 12839 WLA 383 RASQNIGT 12320 KLLIYRASSLES 13230 QQAYSFPWT 12840 WLA 384 RASQNIN 12321 KLLIYKASTLQS 13231 QQADSFPPT 12841 NWLA 385 RASQDIS- 12322 KLLIYAASTLQS 13232 QQLNRYPIT 12842 SYLA 386 RASQDISN 12323 KLLIYAASILHS 13233 QQYDSSFIT 12843 YLA

TABLE-US-00025 TABLE 24 CDRs using the IMGT Numbering Scheme Table 6- IMGT CDR Sequences Binder SEQ ID SEQ ID SEQ Name Sequence NO: Sequence NO: Sequence ID NO: HCDR1 HCDR2 HCDR3 109 GYTFSSYW 4361 ILPGSGST 4875 ARRAYGYDGGFDY 4105 110 GYTFSSYW 4362 ILPGSGST 4876 ARRAYGYDEGFDY 4106 111 GYTFSSYW 4363 ILPGSDST 4877 ARRAYGYDEGFDY 4107 112 AYTFSIYW 4364

ILPGSGST 4878 ARRAYGYDGGFDY 4108 1 GYTFSSYW 4365 IFPGSGHT 4879 ARRGYGYDEGFDY 4109 113 GYTFSSYW 4366 ILPGSGST 4880 ARRGYGYDEGFDY 4110 114 GYTFSSYW 4367 ILPGSGST 4881 ARRGYGYDEGFDY 4111 115 GYTFSNYW 4368 ILPGSGST 4882 ARRGYGYDEGFDY 4112 23 GYTFSSYW 4369 ILPGSGST 4883 ARRGYGYDEGFDY 4113 116 GYTFSSYW 4370 ILPGSGYT 4884 ARRGYGYDEGFDY 4114 117 GYTFSSYW 4371 ILPGSGST 4885 ARRAYGYDEGFDY 4115 2 GYTLSSYW 4372 ILPGSGST 4886 ARRGYGYDEGFDY 4116 118 GYTFSSYW 4373 VLPGSGST 4887 ARRAYGYDEGFDY 4117 119 GYTFSSYW 4374 ISPGSGST 4888 ARRGYGYDEGFDY 4118 120 GYTFGTYW 4375 ILPGSGTP 4889 ARRAYGYDAGFDY 4119 121 GYTFSSYW 4376 ILPGSGST 4890 ARRGYGYDEGFDY 4120 122 GYTFSSYW 4377 ILPGSGRT 4891 ARRGYGYDEGFDY 4121 123 GYTFSSYW 4378 ILPGSGRT 4892 ARRGYGYDEGFDY 4122 38 GYTFSSYW 4379 ILPGSGST 4893 ARRAYGYDEGFDY 4123 39 GYTFSSYW 4380 VLPGSGST 4894 ARRAYGYDEGFDY 4124 40 GYTFSSYW 4381 ILPGSGRT 4895 ARRGYGYDEGFDY 4125 41 GYTFSSYW 4382 VLPGSGST 4896 ARRAYGYDEGFDY 4126 42 GYTFSSYW 4383 ILPGSGRT 4897 ARRGYGYDEGFDY 4127 43 GYTFSSYW 4384 ILPGSGST 4898 ARRAYGYDGGFDY 4128 44 GYTFSSYW 4385 ILPGSDST 4899 ARRAYGYDEGFDY 4129 45 AYTFSIYW 4386 ILPGSGST 4900 ARRAYGYDGGFDY 4130 46 GYTFSSYW 4387 IFPGSGHT 4901 ARRGYGYDEGFDY 4131 47 GYTFSNYW 4388 ILPGSGST 4902 ARRGYGYDEGFDY 4132 47 GYTFSNYW 4388 ILPGSGST 4902 ARRGYGYDEGFDY 4132 48 GYTFSSYW 4389 ILPGSGST 4903 ARRGYGYDEGFDY 4133 48 GYTFSSYW 4389 ILPGSGST 4903 ARRGYGYDEGFDY 4133 49 GYTFSSYW 4390 ILPGSGYT 4904 ARRGYGYDEGFDY 4134 50 GYTLSSYW 4391 ILPGSGST 4905 ARRGYGYDEGFDY 4135 51 GYTFSSYW 4392 ISPGSGST 4906 ARRGYGYDEGFDY 4136 52 GYTFGTYW 4393 ILPGSGTP 4907 ARRAYGYDAGFDY 4137 53 GYTFSSYW 4394 ILPGSGST 4908 ARRGYGYDEGFDY 4138 54 GYTFSSYW 4395 ILPGSGRT 4909 ARRGYGYDEGFDY 4139 55 GYTFSNYW 4396 ILPGSGST 4910 ARRGYGYDEGFDY 4140 56 GYTFSSYW 4397 ILPGSGST 4911 ARRGYGYDEGFDY 4141 3 GYSFTGYY 4398 ISSYNGAT 4912 ARGRYGEYFDY 4142 4 GYSFTGYY 4399 ISSYNGVT 4913 ARGRYGDYFDY 4143 5 GYSFTGYY 4400 ISSYNGVT 4914 ARGRYGDYFDY 4144 6 GYSFTGYY 4401 ISSYNGVT 4915 ARGRYGDYFDY 4145 7 GYSFTGYY 4402 ISSYNGAN 4916 ARGRYGDYFDY 4146 8 GYSFTGYY 4403 ISSYNGVT 4917 ARGRYGDYFDY 4147 9 GYSFTGYY 4404 ISSYNGVT 4918 ARGRYGDYFDY 4148 10 GYSFTGYY 4405 ISSYNGVT 4919 ARGRYGDYFDY 4149 11 GYSFTGYY 4406 ISSYNGVT 4920 ARGRYGDYFDY 4150 12 GYSFTGFY 4407 ISSYNGAT 4921 ARGRYGDYFDY 4151 13 GYSFTGYY 4408 ISSYNGAT 4922 ARGRYGDYFDY 4152 57 GYSFTGYY 4409 ISSYNGVT 4923 ARGRYGDYFDY 4153 58 GYSFTGYY 4410 ISSYNGAT 4924 ARGRYGEYFDY 4154 124 GFSLSSYG 4411 IWRGGST 4925 AKNLYGHYVMDY 4155 125 GFSVTSYG 4412 IWRGGST 4926 AKNLYGHYVMDY 4156 126 GFSLTSYG 4413 IWRGGST 4927 AKNLYGHYVMDY 4157 127 GFSLTRYG 4414 IWRGGST 4928 AKNLYGHYVMDY 4158 128 GFSVTTYG 4415 IWRGGST 4929 AKNLYGHYVMDY 4159 129 GFSVTSYG 4416 IWRGGST 4930 AKNLYGHYVMDY 4160 130 GFSLTRYG 4417 IWRGGST 4931 AKNLYGHYVMDY 4161 59 GFSLSSYG 4418 IWRGGST 4932 AKNLYGHYVMDY 4162 60 GFSVTSYG 4419 IWRGGST 4933 AKNLYGHYVMDY 4163 61 GFSLTSYG 4420 IWRGGST 4934 AKNLYGHYVMDY 4164 62 GFSLTRYG 4421 IWRGGST 4935 AKNLYGHYVMDY 4165 63 GFSVTTYG 4422 IWRGGST 4936 AKNLYGHYVMDY 4166 131 GYTFTSYW 4423 IHPNSGST 4937 ARWGDGYSFAY 4167 132 GYTFTSYW 4424 IHPNSGST 4938 ARWGDGYSFAY 4168 133 GYTFTTYW 4425 IHPNSDNT 4939 ARWGDGYSFAY 4169 14 GYTFTSYW 4426 IHPNSGTT 4940 ARWGDGYSFAY 4170 134 GYTFTSYW 4427 IHPNSGNT 4941 ARWGDGYSFAY 4171 64 GYTFTSYW 4428 IHPNSGST 4942 ARWGDGYSFAY 4172 65 GYTFTSYW 4429 IHPNSGTT 4943 ARWGDGYSFAY 4173 135 GYTFTDYV 4430 IYPGSGST 4944 ARRGERGPWFAY 4174

136 GYTFTDYV 4431 IYPGSGSS 4945 ARRGERGPWFAY 4175 137 GYTFTDYV 4432 IYPGSGSS 4946 ARRGERGPWFAY 4176 138 GYTFTDYV 4433 IYPGSGSS 4947 ARRGERGPWFAY 4177 15 GYTFTDYV 4434 IYPGSGSS 4948 ARRGERGPWFAY 4178 66 GYTFTDYV 4435 IYPGSGST 4949 ARRGERGPWFAY 4179 67 GYTFTDYV 4436 IYPGSGSS 4950 ARRGERGPWFAY 4180 68 GYTFTDYV 4437 IYPGSGSS 4951 ARRGERGPWFAY 4181 69 GYTFTDYV 4438 IYPGSGSS 4952 ARRGERGPWFAY 4182 24 GYTFTNYW 4439 IDPSDSET 4953 ATYDVYYRFAY 4183 139 GYTFTNYW 4440 IDPSDSET 4954 ATYDGYYRFAY 4184 140 GYTFTNYW 4441 IDPSDSET 4955 ATYDIYYRFAY 4185 16 GYTFTSYW 4442 IHPNSGST 4956 ARPGGYGFVY 4186 141 GYTFTSYW 4443 IHPNSDST 4957 ARPGGYGFAD 4187 142 GYTFTTYW 4444 IHPNSGST 4958 ARPGGYGFTY 4188 143 GYTFTSYW 4445 IHPNSGSP 4959 ARPGGYGFAY 4189 70 GYTFTSYW 4446 IHPNSGSP 4960 ARPGGYGFAY 4190 25 GYTFTSYW 4447 IYPSDSYT 4961 TRGNYIDY 4191 144 GYTFTSYW 4448 IYPSDSYT 4962 TRGNYIDY 4192 145 GYTFTDYW 4449 IYPSDSYT 4963 TRGNYIDY 4193 146 GYTFTDYV 4450 IYPGSGSS 4964 ARPGDLGFAY 4194 147 GYTFTDYV 4451 IYPGSGSN 4965 ARPGDLGFAY 4195 148 GYTFTDYV 4452 IYPGSGSS 4966 ARPGDLGFAY 4196 71 GYTFTDYV 4453 IYPGSGSS 4967 ARPGDLGFAY 4197 72 GYTFTDYV 4454 IYPGSGSS 4968 ARPGDLGFAY 4198 73 GYTFTDYV 4455 IYPGSGSN 4969 ARPGDLGFAY 4199 74 GYTFTDYV 4456 IYPGSGSS 4970 ARPGDLGFAY 4200 74 GYTFTDYV 4456 IYPGSGSS 4970 ARPGDLGFAY 4200 75 GYTFTDYV 4457 IYPGSGSS 4971 ARPGDLGFAY 4201 75 GYTFTDYV 4457 IYPGSGSS 4971 ARPGDLGFAY 4201 76 GYTFTDYV 4458 IYPGSGSS 4972 ARPGDLGFAY 4202 149 GFSLTNYG 4459 VWAGGIT 4973 ARGDGYDDGYAMDY 4203 150 GFSLTSYG 4460 LWAGGIT 4974 ARGDGYDDGYAMDY 4204 26 GFSLTSYG 4461 IWAGGTT 4975 ARGDGYDDGYAMDY 4205 77 GFSLTNYG 4462 VWAGGIT 4976 ARGDGYDDGYAMDY 4206 78 GFSLTSYG 4463 LWAGGIT 4977 ARGDGYDDGYAMDY 4207 79 GFSLTSYG 4464 IWAGGTT 4978 ARGDGYDDGYAMDY 4208 151 GYSFTSYW 4465 IDPSDSET 4979 ARTRNY 4209 152 GYSFTSYW 4466 IDPSDSET 4980 ARTRNY 4210 153 GYSFTSYW 4467 IDPSDSET 4981 ARTRNY 4211 154 GFNIKDYY 4468 IDPENGDT 4982 NAPLLRYSSAMDY 4212 155 GFNIKDYY 4469 IDPENGDT 4983 NAPLLRYSSSMDY 4213 156 GFNIKDYY 4470 IDPENGDT 4984 NVALLRYSSAMDY 4214 80 GFNIKDYY 4471 IDPENGDT 4985 NAPLLRYSSAMDY 4215 81 GFNIKDYY 4472 IDPENGDT 4986 NAPLLRYSSSMDY 4216 82 GFNIKDYY 4473 IDPENGDT 4987 NVALLRYSSAMDY 4217 17 GFNIKDTS 4474 IDPANGNT 4988 ARGPDDGYFYYYSMDY 4218 157 GYTFSNYY 4475 INPSNGDT 4989 TSYYTHEAYYYAMDC 4219 27 GSTFTTYY 4476 INPSNGGT 4990 TSYYTHETYYYAMDY 4220 158 GFNIKDYY 4477 IDPEDGDT 4991 TPYSIYDAMDY 4221 159 GYTFTDYV 4478 IYPGSGST 4992 ARRGERGPWFAY 4222 83 GYTFTDYV 4479 IYPGSGST 4993 ARRGERGPWFAY 4223 160 GYSFTDYG 4480 ISTYYGDA 4994 ARQMDYDYTYYYAMDY 4224 28 GYTFTSYW 4481 IDPSDSYT 4995 ARAEYGYGNYPWFAY 4225 84 GYTFTSYW 4482 IDPSDSYT 4996 ARAEYGYGNYPWFAY 4226 29 GYTFTSYW 4483 IHPSDSDT 4997 AIPYYYGGWYFDV 4227 161 GYTFTDYV 4484 IYPGSGST 4998 ARMDGPWFAY 4228 30 GFTFSSYG 4485 ISSGGSYT 4999 ARLYDAHWDYFDY 4229 162 GISLSTSGMG 4486 IWNNDN 5000 AWRPYYRYDSFAY 4230 18 GYTFTNYG 4487 INTYTGEP 5001 ARKYYDYEFAY 4231 85 GYTFTNYG 4488 INTYTGEP 5002 ARKYYDYEFAY 4232 163 GYTFTDYE 4489 IDPETGGT 5003 TRLGDYDVMDY 4233 86 GYTFTDYE 4490 IDPETGGT 5004 TRLGDYDVMDY 4234 164 GYTFTSYW 4491 IDPSDSYT 5005 ARAGRYGSSFDY 4235 165 GFSLSTSGMG 4492 IYWDDDK 5006 AGRPDDYDGAWFPY 4236 31 GYTFTSSW 4493 IHPNSGNT 5007 AIYYDYDAYYFDY 4237 87 GYTFTSSW 4494 IHPNSGNT 5008 AIYYDYDAYYFDY 4238 32 GYTFTSYW 4495 IHPNSGST 5009 ANPYYGYDVGY 4239 166 GYTFTDYV 4496 IYPGSGSN 5010 AREEKIYFDY 4240 88 GYTFTDYV 4497 IYPGSGSN 5011 AREEKIYFDY 4241 167 GYTFTSYW 4498 IHPNSGST 5012 ARYDGYWFDY 4242 168

GYTFTSYW 4499 IYPGNSDT 5013 TSLITTAYYFDY 4243 89 GYTFTSYW 4500 IYPGNSDT 5014 TSLITTAYYFDY 4244 169 GYTFTSYW 4501 IHPNSGST 5015 APETGDYGSSYVWYFDV 4245 170 GYTFTDYV 4502 IYPGSGST 5016 ARGKVTRFAY 4246 171 GFTFSSYA 4503 ISDGGSYT 5017 ARDQDSNWEYFDY 4247 172 GYTFTDYS 4504 INTETGEP 5018 ARESWDRAMDY 4248 19 GFTFSSYA 4505 ISSGGSYT 5019 ARHEEANWAWFAY 4249 90 GFTFSSYA 4506 ISSGGSYT 5020 ARHEEANWAWFAY 4250 173 GYSFTNYW 4507 IDPSDSET 5021 AIPYYAMDY 4251 91 GYSFTNYW 4508 IDPSDSET 5022 AIPYYAMDY 4252 174 GYTFTSSW 4509 IHPNSGNT 5023 ATYYGNYVWYFDV 4253 92 GYTFTSSW 4510 IHPNSGNT 5024 ATYYGNYVWYFDV 4254 175 GYTFTSYW 4511 IHPNSGST 5025 ASYGSSYWYFDV 4255 93 GYTFTSYW 4512 IHPNSGST 5026 ASYGSSYWYFDV 4256 20 GFSLTSYG 4513 IWSGGST 5027 ASYYGSSRSYWYLDV 4257 94 GFSLTSYG 4514 IWSGGST 5028 ASYYGSSRSYWYLDV 4258 176 GYTFTSYN 4515 LYSGNGDT 5029 ARDYYGSSHLWYFDV 4259 177 GFSLSTSGMG 4516 IYWDDDK 5030 ARRAHYDYGWYFDV 4260 178 GYTFTSYW 4517 IHPNSGST 5031 AGYDYDWYFDV 4261 33 GFTFSSYG 4518 ISSGGSYT 5032 TRHDDSSYDWFAY 4262 179 GFTFSSYG 4519 ISSGGSYT 5033 ARHEDSNYHYFDY 4263 34 GYTFTNYW 4520 IHPNSGTT 5034 ARFGDGYHFDY 4264 180 GFTFSSYG 4521 ISSGGSYT 5035 ARQNDSSWAWFAY 4265 95 GFTFSSYG 4522 ISSGGSYT 5036 ARQNDSSWAWFAY 4266 181 GYTFTSYW 4523 IHPNSGST 5037 ALPYSNYGWYFDV 4267 96 GYTFTSYW 4524 IHPNSGST 5038 ALPYSNYGWYFDV 4268 182 GYTFTSYW 4525 IDPSDSET 5039 ARDYYGSYWYFDV 4269 97 GYTFTSYW 4526 IDPSDSET 5040 ARDYYGSYWYFDV 4270 183 GFNIKDYY 4527 IDPEDGET 5041 AAYGNSAWFAY 4271 98 GFNIKDYY 4528 IDPEDGET 5042 AAYGNSAWFAY 4272 35 GYTFTNYG 4529 INTNTGEP 5043 ARWYPYFDY 4273 99 GYTFTNYG 4530 INTNTGEP 5044 ARWYPYFDY 4274 100 GYTFTNYG 4531 INTNTGEP 5045 ARWYPYFDY 4275 36 GYTFTSYW 4532 INPSSGYT 5046 ARSDGSSGNWYFDV 4276 101 GYTFTSYW 4533 INPSSGYT 5047 ARSDGSSGNWYFDV 4277 184 GFSLTSYG 4534 IWAGGST 5048 AREGGYTGYFDV 4278 102 GFSLTSYG 4535 IWAGGST 5049 AREGGYTGYFDV 4279 185 GYTFTSYW 4536 IDPSDSET 5050 AYSNYVPYYAMDY 4280 103 GYTFTSYW 4537 IDPSDSET 5051 AYSNYVPYYAMDY 4281 186 GYTFTDYV 4538 IYPGSGSA 5052 ARRGFDY 4282 21 GFTFSSYG 4539 ISSGGSYT 5053 ARHNYSNWDWFAY 4283 187 GYTFTSYW 4540 IHPNSGST 5054 ARDYYGSGYGYYFDY 4284 188 GYTFTSYW 4541 IHPNSGST 5055 ARDYYGSSYGWYFDV 4285 189 GYTFTSYW 4542 IHPNSGST 5056 ARDYYGSSYGWYFDV 4286 190 GYTFTSYW 4543 IHPNSGST 5057 ASDYYGSSYGWYFDV 4287 191 GYTFTSYW 4544 IHPNSGST 5058 ARDYYGSSYGWYFDV 4288 192 GYTFTSYW 4545 IHPNSGST 5059 TRDYYGSGYGWYFDV 4289 193 GYTFTNYW 4546 IDPSDSET 5060 ATYDGYYRFAY 4290 194 GYTFTNYW 4547 IDPSDSET 5061 ATYDVYYRFAY 4291 195 GYTFTSYW 4548 IHPNSGST 5062 ARDYGNYDYAMDY 4292 104 GYTFTSYW 4549 IHPNSGST 5063 ARDYGNYDYAMDY 4293 37 GYTFTSYW 4550 IHPNSGST 5064 ARDYGNYDYAMDY 4294 196 GYTFTSYW 4551 IHPNSGST 5065 ARDYGNYDYAMDY 4295 197 GFTFSSYG 4552 ISSGGSYT 5066 ASQLTGTWYYFDY 4296 198 GFTFSSYG 4553 ISSGGSYT 5067 ASQLTGTWYYFDY 4297 199 GFTFSSYG 4554 ISSGGSYT 5068 ASQLTGTWYYFDY 4298 22 GFNIKDTS 4555 IDPANGNT 5069 ARGPDDGYFYYYSMDY 4299 200 GFTFSNYY 4556 INSNGGST 5070 ARQEGIGYAMDY 4300 201 GYTFTEYT 4557 IYPNNGGT 5071 ARGGWLLGY 4301 202 GFSLTSYG 4558 IWSGGST 5072 ARDGGIRGAMDY 4302 203 GYTFSSYW 4559 ILPGSGST 5073 ARRGYGYDEGFDY 4303 204 GYTFTDYE 4560 IDPETGGT 5074 TRNYDYAMDY 4304 205 GFTFSSYY 4561 INSNGGST 5075 ARQEGIGYALDY 4305 206 GFTFSSYA 4562 ISSGGST 5076 AREREWGVYYGSSLDY 4306 207 GFNIKDTY 4563 IDPANGNT 5077 ARSDGNYD 4307 208 GFTFSNYY 4564 INSNGGST 5078 ARQEGIGYGMDY 4308 209 GFTFNTYV 4565 IRSKSDNYAT 5079 VRHDGVVGFDV

4309 210 GYSITSGYY 4566 ISYDGSN 5080 ARGGGRG 4310 211 GYTFTDYS 4567 INTETGEP 5081 ARDYYDYYYAMDY 4311 212 GYTFTDYS 4568 INTETGEP 5082 ARESWDRAMDY 4312 213 GYTFTNYW 4569 IDPYDSET 5083 ARIYSDYDGAWFAY 4313 214 GYTFTDYY 4570 VNPYNGGT 5084 ARGTVGFAY 4314 215 GFTFSSYA 4571 ISSGGST 5085 AREREWGVFYGSSLDY 4315 216 GFTFSSYA 4572 ISSGGSYT 5086 ARHDDSSYGYFDY 4316 217 GFTFSNYA 4573 ISSGGTT 5087 ARTMPDV 4317 218 GFSLTSYG 4574 IWAGGST 5088 ARDTDGYYWAMDY 4318 219 GYSITSDHA 4575 ISYSGST 5089 ARKWGDY 4319 220 GYTFTDYE 4576 IDPETGGT 5090 TRNYDYALDY 4320 221 GYSITSGYY 4577 ISYDGSN 5091 ARGGGRG 4321 222 GFTFSNYY 4578 INSNGGST 5092 ARQEEIGYAMDY 4322 223 GFNIKDYF 4579 IDPETDNT 5093 ARSGNMGFTY 4323 105 GFNIKDYF 4580 IDPETDNT 5094 ARSGNMGFTY 4324 224 GFTFSSYA 4581 ISSGGSYT 5095 ASQGGSSWGAMDY 4325 106 GFTFSSYA 4582 ISSGGSYT 5096 ASQGGSSWGAMDY 4326 225 GFTFSSYA 4583 ISNGGSYT 5097 ARHEITTRFAY 4327 226 GYSITSGYY 4584 MSYDGSN 5098 AREAGYFDY 4328 227 GFSFNTYA 4585 IRSKSNNYAT 5099 VRQYGYDFDY 4329 228 GFTFSSYG 4586 ISSGGSYT 5100 ARHKGVNWDYFDY 4330 229 GYTFTDYE 4587 IDPETGGT 5101 TRGDGNYDSWYFDV 4331 230 GFTFSSYA 4588 ISSGGSYT 5102 ARLPVTTVVFDY 4332 231 GFTFSSYA 4589 ISSGGSYT 5103 ARRPVVVPFDY 4333 232 GFSLTSYG 4590 IWSGGST 5104 ARGWDADYFDY 4334 233 GYTFTNYW 4591 IHPNSGST 5105 TRYDYDDY 4335 234 GYTFTDYY 4592 INPNNGGT 5106 ARSELGLYAMDY 4336 235 GYTFTGYW 4593 ILPGSGST 5107 ARGRIHYFDY 4337 236 GYTFTGYW 4594 ILPGSGST 5108 ARGRIHYFDY 4338 237 GFSLTSYG 4595 IWSGGST 5109 ARKGYGYDWYFDV 4339 107 GFSLTSYG 4596 IWSGGST 5110 ARKGYGYDWYFDV 4340 238 GYTFTSYW 4597 IDPSDSYT 5111 ARSSYYYYAMDY 4341 108 GYTFTSYW 4598 IDPSDSYT 5112 ARSSYYYYAMDY 4342 239 GYSITSGYY 4599 ISYDGSN 5113 ARGGGRD 4343 240 GFSLTSYG 4600 IWSGGST 5114 ARGGDYDSYAMDY 4344 241 GYTFTSYW 4601 IYPGSGST 5115 ARESVYDGYSWYFDV 4345 242 GYSFTDYN 4602 INPNYGTT 5116 ASTYDYDDWYFDV 4346 243 GYTFTSYW 4603 IDPSDSYT 5117 ARSGNYLYAMDY 4347 244 GYSFTDYN 4604 INPNYGTT 5118 AREGTSWYFDV 4348 245 GFSLTSYG 4605 IWRGGST 5119 AKKGDGYDWYFDV 4349 246 GFSLTSYG 4606 IWSGGST 5120 AREGNYGSSYDAMDY 4350 247 GYTFTSYW 4607 IDPSDSYT 5121 ARSSNYPYAMDY 4351 248 GFNIKNTY 4608 IDPANGNT 5122 AYYSGLY 4352 249 GYTFTSYW 4609 IDPSDSET 5123 ARRGQIYYGYSWFAY 4353 250 GYTFTDYY 4610 INPNNGGT 5124 ARSTVVADWYFDV 4354 251 GYTFTSYG 4611 IYPRSGNT 5125 ARSGSSYGYFDV 4355 252 GFSLTSYG 4612 IWSGGST 5126 ARKGGYDAYAMDY 4356 253 GYSFTDYN 4613 INPNYGTT 5127 AREGFITTVVAVDY 4357 254 GYTFTDYE 4614 IDPETGGT 5128 TREGNYDAMDY 4358 255 GYTFTSYW 4615 IDPSDSYT 5129 ARWDYYGVDY 4359 256 GFTFSGYW 4616 ISPGGGST 5130 ASSLTATHTYEYDY 4360 LCDR1 LCDR2 LCDR3 97 QGISNY 8066 YTS QQYSKLPYT 6829 182 QGISNY 8067 YTS QQYSKLPYT 6828 86 QDISNY 8068 YTS QQDNTLPRT 6793 96 QDISNY 8069 YTS QQGNTLPFT 6827 151 QDISNY 8070 STS QQGNTLPWT 6768 152 QDISNY 8071 STS QQGNALPWT 6769 153 QDISNY 8072 STS QQGNTLPWT 6770 163 QDISNY 8073 YTS QQDNTLPRT 6792 164 QDISNY 8074 YTS QQGNTLPWT 6794 181 QDISNY 8075 YTS QQGNTLPFT 6826 202 QSIVHSNGNTY 8076 KVS FQGSHVPWT 6861 206 QSIVYSNGNTY 8077 KVS FQGSHVPPT 6865 215 QSIVHSNGNTY 8078 KVS FQGSHVPPT 6874 242 QSIVHSNGDTY 8079 KVS FQGSHVPLT 6905 204 QSLLDSDGKTY 8080 LVS WQGTHFPWT 6863 214 QSLLDSDGKTY 8081 LVS WQGTHFPWT 6873 220 QSLLDSDGKTY 8082 LVS WQGTHFPWT 6879 229 QSLLDSDGKTY 8083 LVS WQGTHFPWT 6890 235 QSLLDSDGKTY 8084 LVS WQGTHFPFT 6896 162 ENIYYS 8085 NAN KQAYDVPYT 6789 101 GNIHNY 8086 NAK QHFWSTPWT 6836 36 GNIHNY 8087 NAK QHFWSTPWT 6835 188 GNIHNY 8088 NAK QHFWSTPWT 6844 172 ENIYSY 8089 NAK QHFWGTPYT 6807 189

ENIYSY 8090 NAK QHHYGTPFT 6845 249 ENIYSY 8091 NAK QHHYGTPYT 6912 250 ENIYSY 8092 NAK QHHYGTPPT 6913 57 DNIYSN 8093 AAT QHFWGTPWT 6712 58 ENIYSN 8094 AAT QHFWGTPWT 6713 3 ENIYSN 8095 AAT QHFWGTPWT 6701 4 ENIYSN 8096 AAT QHFWGTPWT 6702 5 ENIYSN 8097 AAT QHFWGSPWT 6703 6 ENIYSN 8098 AAT QHFWGTPWT 6704 7 ENIYSN 8099 AAT QHFWGTPWT 6705 8 ENIYSN 8100 AAT QHFWGTPWT 6706 9 DNIYSN 8101 AAT QHFWGTPWT 6707 10 ENIYSN 8102 AAT QHFWGTPWT 6708 11 ENIYSN 8103 AAT QHFWGTPWT 6709 12 ENIYSN 8104 AAT QHFWGTPWT 6710 13 ENIYSN 8105 AAT QHFWGSPWT 6711 17 ENIYSN 8106 AAT QHFWGTPWT 6777 165 ENIYSN 8107 AAT QHFWGTPWT 6795 179 ENIYSN 8108 AAT QHFWGTPYT 6822 194 ENIYSN 8109 AAT QHFWGTPFT 6850 37 ENIYSN 8110 AAT QHFWGTPYT 6853 207 ENIYNN 8111 AAT QHFWGTPWT 6866 212 ENIYSN 8112 AAT QHFWGTPWT 6871 248 DHINNW 8113 GAT QQYWSTPLT 6911 38 QDINSY 8114 RAN LQYDEFPLT 6682 39 QDINGY 8115 RAN LQYDEFPPT 6683 40 QDINSY 8116 RAK LQYDEFPPT 6684 41 QDINGY 8117 RAN LQYDEFPPT 6685 42 QDINSY 8118 RAK LQYDEFPPT 6686 43 QDINSY 8119 RAN LHYDEFPPT 6687 44 QDINSY 8120 RAN LQYDEFPPT 6688 45 QDINSY 8121 RAN LQYDEFPPT 6689 46 QDINSY 8122 RAN LQYDEFPPT 6690 47 QDINSY 8123 RAN LQYDEFPPT 6691 48 QDINSY 8124 RAN LQYDEFPPT 6692 49 QDINSY 8125 RAN LQYDEFPPT 6693 50 QDINSY 8126 RAN LQYDEFPPT 6694 51 QDINSY 8127 RAN LQYDEFPPT 6695 52 QDINSY 8128 RAN LQYDEFPPT 6696 53 QDINSY 8129 RAN LQYDEFPPT 6697 54 QDINSY 8130 RAN LQYDEFPPT 6698 55 QDINSY 8131 RAN LQYDEFPPT 6699 56 QDINSY 8132 RAN LQYDEFPPT 6700 91 QDINSY 8133 RAN LQYDEFPLT 6811 109 QDINSY 8134 RAN LHYDEFPPT 6664 110 QDINSY 8135 RAN LQYDEFPPT 6665 111 QDINSY 8136 RAN LQYDEFPPT 6666 112 QDINSY 8137 RAN LQYDEFPPT 6667 1 QDINSY 8138 RAN LQYDEFPPT 6668 113 QDINSY 8139 RAN PQYVESPPT 6669 114 QDINSY 8140 RAN LQYDEFPPT 6670 115 QDINSY 8141 RAN LQYDEFPPT 6671 23 QDINSY 8142 RAN LQYDEFPPT 6672 116 QDINSY 8143 RAN LQYDEFPPT 6673 117 QDINSY 8144 RAN LQYDEFPLT 6674 2 QDINSY 8145 RAN LQYDEFPPT 6675 118 QDINGY 8146 RAN LQYDEFPPT 6676 119 QDINSY 8147 RAN LQYDEFPPT 6677 120 QDINSY 8148 RAN LQYDEFPPT 6678 121 QDINSY 8149 RAN LQYDEFPPT 6679 122 QDINSY 8150 RAN LQYDEFPPT 6680 123 QDINSY 8151 RAK LQYDEFPPT 6681 124 QDINSY 8152 RAN LQYDEFPPT 6714 173 QDINSY 8153 RAN LQYDEFPLT 6810 193 QDINSY 8154 RAN LQYDEFPPT 6849 203 QDINSY 8155 RAN LQYDEFPPT 6862 70 QDIVKN 8156 YAT LQFYEFPLT 6749 87 QDIVKN 8157 YAT LQFYEFPYT 6797 89 QDIVKN 8158 YAT LQFYEFPLT 6803 99 QDIVKN 8159 YAT LQFYEFPYT 6833 100 QDIVKN 8160 YAT LQFYEFPYT 6834 16 QDIVKN 8161 YAT LQFYEFPLT 6745 141 QDIVKN 8162 YAT LQFYEFPLT 6746 142 QDIVKN 8163 YAT LQFYEFPLT 6747 143 QDIVKN 8164 YAT LQFYEFPLT 6748 31 QDIVKN 8165 YAT LQFYEFPYT 6796 168 QDIVKN 8166 YAT LQFYEFPLT 6802 35 QDIVKN 8167 YAT LQFYEFPYT 6832 239 QDIVKN 8168 YAT LQFYEFPLT 6902 21 QNINVW 8169 KAS QQGQSYPLT 6842 218 QNINVW 8170 KAS QQGQSYPYT 6877 106 TDIDDD 8171 EGN LQSDNLPLT 6885 197 TDIDDD 8172 EGN LQSDNLPLT 6855 22 TDIDDD 8173 EGN LQSDNLPLT 6858 224 TDIDDD 8174 EGN LQSDNLPLT 6884 95 TDIDDD 8175 EGN LQSDNMPLT 6825 157 TDIDDD 8176 EGN LQSDNMPFT 6778 27 IDIDDD 8177 EGN LQSDNMPFT 6779 33 TDIDDD 8178 EGN LQSDNMPLM 6821 180 TDIDDD 8179 EGN LQSDNMPLT 6824 216 TDIDDD 8180 EGN LQSDNMPLT 6875 228 TDIDDD 8181 EGN LQSDNMPLT 6889 24 QDINKY 8182 YTS LQYDNLMYT 6742 139 QDINKY 8183 YTS LQYDILMYT 6743 140 QDINKY 8184 YTS LQYDILMYT 6744 217 QDINKY 8185 YTS LQYDNLYMYT 7780 230 QDINKY 8186 YTS LQYDNLRT 7781 240 ESVDNYGISF 8187 AAS QQSKEVPPT 6903 252 ESVDNYGISF 8188 AAS QQSKEVPPT 6915 66 QSVDYDGDSY 8189 AAS QQSNEDPLT 6738 67 QSVDYDGDSY

8190 AAS QQSNEDPLT 6739 68 QSVDYDGDSY 8191 AAS QQSNEDPLT 6740 69 QSVDYDGDSY 8192 AAS QQSNEDPLT 6741 71 QSVDYDGDSY 8193 AAS QQSNKDPLT 6756 72 QSVDYDGDSY 8194 AAS QQSNKDPLT 6757 73 QSVDYDGDSY 8195 AAS QQSNEDPLT 6758 74 QSVDYDGDSY 8196 AAS QQSNKDPFT 6759 75 QSVDYDGDSY 8197 AAS QQSNKDPFT 6760 76 QSVDYDGDSY 8198 AAS QQSNKDPFT 6761 83 QSVDYDGDSY 8199 AAS QQSNEDPLT 6782 88 QSVDYDGDSY 8200 AAS QQSNEDPWT 6800 135 QSVDYDGDSY 8201 AAS QQSNEDPLT 6733 136 QSVDYDGDSY 8202 AAS QQSNEDPLT 6734 137 QSVDYDGDSY 8203 AAS QQSNEDPLT 6735 138 QSVDYDGDSY 8204 AAS QQSNEDPLT 6736 15 QSVDYDGDSY 8205 AAS QQSNEDPLT 6737 146 QSVDYDGDSY 8206 AAS QQSNKDPLT 6753 147 QSVDYDGDSY 8207 AAS QQSNEDPLT 6754 148 QSVDYDGDSY 8208 AAS QQSNKDPFT 6755 159 QSVDYDGDSY 8209 AAS QQSNEDPLT 6781 161 QSVDYDGDSY 8210 AAS QQSNEDPPT 6787 166 QSVDYDGDSY 8211 AAS QQSNEDPWT 6799 170 QSVDYDGDSY 8212 AAS QQSNEDPPT 6805 177 QSVDYDGDSY 8213 VAS QQSHEDPRT 6819 186 QSVDYDGDSY 8214 AAS QQSNEDPLPT 7776 225 QSVDYDGDSY 8215 AAS QQSNEDPWT 6886 236 QSVDYDGDSY 8216 AAS QQSNEDPFT 6897 233 ETVDSYGYSF 8217 RAS QQSNEDPRT 6894 77 QSVSTSSYSY 8218 YAS QHSWEIPLT 6765 78 QSVSTSSYSY 8219 YAS QHSWEIPLT 6766 79 QSVSTSSYSY 8220 YAS QHSWEIPLT 6767 149 QSVSTSSYSY 8221 YAS QHSWEIPLT 6762 150 QSVSTSSYSY 8222 YAS QHSWEIPLT 6763 26 QSVSTSSYSY 8223 YAS QHSWEIPLT 6764 246 QSVSTSSYSY 8224 YAS QHSWEIPLT 6909 105 SSVNY 8225 YTS QQFTSSPST 6883 25 SSVNY 8226 YTS QQFTSSHT 7768 144 SSVNY 8227 YTS QQFTSSHT 7769 145 SSVNY 8228 YTS QQFTSSHT 7770 190 SSVNY 8229 YTS QQFTSSLT 7777 223 SSVNY 8230 YTS QQFTSSPST 6882 32 SSVSY 8231 DTS QQWSSYPLT 6798 209 SSVSY 8232 DTS QQWSTYPPIT 7779 210 SSVSY 8233 DTS QQWSSYPFT 6869 221 LSVSD 8234 DTS QQWSSYPFT 6880 90 SSVSY 8235 STS QQRSSFPYT 6809 19 SSVSY 8236 STS QQRSSFPYT 6808 34 SSVSY 8237 STS QQRSTYPT 7775 198 SSVSY 8238 STS QQRSSYPPT 6856 64 SSVSY 8239 DTS QQWSSNPLY 6731 65 SSVSY 8240 DTS QQWSSNPLY 6732 SSVSY 8241 DTS QQWSSNPLT 6901 131 SSVSY 8242 DTS QQWSSNPLYT 7763 132 SSVSY 8243 DTS QQWSSNPHVHV 7764 133 SSVSY 8244 DTS QQWSSNPLYT 7765 14 SSVSY 8245 DTS QQWSSNPLYT 7766 134 SSVSY 8246 DTS QQWSSNPLYT 7767 187 SSVSY 8247 DTS QQWSSNPLT 6843 199 SSVSY 8248 DTS QQWSSNPLT 6857 238 SSVSY 8249 DTS QQWSSNPLT 6900 243 SSVSY 8250 DTS QQWSSNPLT 6906 247 SSVSY 8251 DTS QQWSSNPLT 6910 251 SSVSY 8252 DTS QQWSSNPPT 6914 178 SSVSY 8253 DTS FQGSGYPLT 6820 219 SSVSY 8254 LTS QQWSSNPPT 6878 255 SSISY 8255 DTS HQRSSYPT 7785 211 SSVSY 8256 ATS QQWSSNPYT 6870 200 SSVSSSY 8257 STS HQWSSYPPT 6859 205 SSVSSSY 8258 STS HQWSSYPPT 6864 208 SSVSSSY 8259 STS HQWSSYPPT 6867 222 SSVSSSY 8260 STS HQWSSYPPT 6881 227 SSVSSSY 8261 STS HQWSSYPPT 6888 98 QSISDY 8262 YAS QNGHSFPWT 6831 104 QSISDY 8263 YAS QNGHSFPYT 6852 183 QSISDY 8264 YAS QNGHSFPWT 6830 195 QSISDY 8265 YAS QNGHSFPYT 6851 241 QSISDY 8266 YAS QNGHSFPLT 6904 226 QSISNN 8267 YAS QQSNSWPFT 6887 60 ENVVTY 8268 GAS GQSYSYPFT 6722 61 ENVVTY 8269 GAS GQSYSYPFT 6723 62 ENVVTY 8270 GAS GQSYSYPFT 6724 63 ENVVTY 8271 GAS GQSYSYPFT 6725 59 QDINSY 8272 RAN LQYDEFPPT 6721 125 ENVVTY 8273 GAS GQSYSYPFT 6715 126 ENVVTY 8274 GAS GQSYSYPFT 6716 127 ENVVTY 8275 GAS GQSYSYPFT 6717 128 ENVVTY 8276 GAS GQSYSYPFT 6718 129 ENVVTY 8277 GAS GQSYSYLIHVR 7761 130 ENVVTY 8278 GAS GQSYSYLIHVR 7762 232 ENVGTY 8279 GAS GQSYSYPPT 6893 93 QDVGTA 8280 WAS QQYSSYPFT 6815 175 QDVGTA 8281 WAS QQYSSYPFT 6814 192 QDVGTA 8282 WAS QQYSSYPFT 6848 213 QDVSTA 8283 WAS QQHYSTPWT 6872 94 QSVSND 8284 YAS QQDYTSLPT 6817 20 QSVSND 8285 YAS QQDYTSLPT 6816 244 ESLYSSKHKVHY 8286 GAS AQFYSYPYT 6907 253

ESLYSSKHKVHY 8287 GAS AQFYSYPYT 6916 85 QSLLNSSNQKNY 8288 FAS QQHYSTPLT 6791 18 QSLLNSSNQKNY 8289 FAS QQHYSTPLT 6790 160 QSLLYSTNQKNY 8290 WAS QQYYSYPPWT 7772 234 QSLLYSTNQKNY 8291 WAS QQYYSYRT 7782 171 QDIGIS 8292 ATS LQYASSPYT 6806 92 QDIHGY 8293 ETS LQYASSPLT 6813 103 QEISGY 8294 AAS LQYASYPWT 6840 174 QDIHGY 8295 ETS LQYASSPLT 6812 185 QEISGY 8296 AAS LQYASYPWT 6839 29 TGAVTTSNY 8297 GTN ALWYSNHL 7773 30 TGAVTTSNY 8298 GTN ALWYSNHWV 6788 167 TGAVTTSNY 8299 GTN ALWYSNHWV 6801 176 TGAVTTSNY 8300 GTN ALWYSNHLV 6818 191 TGAVTTSNY 8301 STN TLWYSNHWV 6847 196 TGAVTTSNY 8302 GTN ALWYSNHWV 6854 231 TGAVTTSNY 8303 GTN VLWYSNHLV 6892 80 TGAVTTSNY 8304 GTS ALWYSTHYV 6774 81 TGAVTTSNY 8305 GTS ALWYSTHYV 6775 82 TGAVTTSNY 8306 GTS ALWYSTHYV 6776 84 TGAVTTSNY 8307 GTS ALWYSTHWV 6785 102 TGAVTTSNY 8308 GTS ALWYSTHYV 6838 154 TGAVTTSNY 8309 GTS ALWYSTHYV 6771 155 TGAVTTSNY 8310 GTS ALWYSTHYV 6772 156 TGAVTTSNY 8311 GTS ALWYSTHYV 6773 158 TGAVTTSNY 8312 GTS ALWYSTHY 7771 28 TGAVTTSNY 8313 GTS ALWYSTHWV 6784 184 TGAVTTSNY 8314 GTS ALWYSTHYV 6837 254 TGAVTTSNY 8315 GTS ALWYSTHYV 6917 107 SQHSTYT 8316 LKKDGSH 8826 GVGDTIKEQFVYV 6899 169 SQHSTYT 8317 LKKDGSH 8827 GVGDTIKEQFVFV 7774 201 SQHSTYT 8318 LKKDGSH 8828 GVGDTIKEQFVYV 7778 237 SQHSTYT 8319 LKKDGSH 8829 GVGDTIKEQFVYV 7783 245 SQHSTYT 8320 LKKDGSH 8830 GVGDTIKEQFVYV 7784 HCDR1 HCDR2 HCDR3 265 GGTFSSYA 11540 INPSGGT 11802 ARDLGDPGMDV 11409 266 GGTFSNYA 11541 IDPSGGST 11803 ARDLGDMGMDV 11410 267 GGTFSNYA 11542 INPSGGST 11804 ARDVGDRGMDV 11411 263 GGTFSSYA 11543 INPSGGT 11805 ARDLGDPGMDV 11412 268 GSTFSGYY 11544 IDPNGGGT 11806 AKDIVHDGTEYFQH 11413 269 GYTFTSYY 11545 INPSGGST 11807 AKDIVHDGTEYFQH 11414 270 GGTFSSYA 11546 INPSGGST 11808 AREGRDHDAFDI 11415 271 GFTFTDYG 11547 INPSGGST 11809 AREGRSHDAFDI 11416 272 GYTFTGYY 11548 MNPHSGDT 11810 ARWVGTTEYYYYYYMDV 11417 273 GYTFTDYY 11549 IDPSGGST 11811 ATTAYYDFWSGYSMDV 11418 274 GYTFTSHY 11550 IDPSGGST 11812 ARDMDNWNTGYYYYMDV 11419 275 GGTFSSYA 11551 VNPNSGDT 11813 ARDQRGGDAWDV 11420 276 GGTFSNYA 11552 ITPSGGST 11814 ARDTAGHFDI 11421 277 GGTFRNDV 11553 MNPNSGNT 11815 ARDNPDLDGMDV 11422 278 GGTFSSYA 11554 ISAYNGNT 11816 ARDLVGHFDY 11423 279 GGTFSSYA 11555 INPNSGGT 11817 ARDGYSGSYSD 11424 264 GGTFSSYA 11556 INPNSGGT 11818 ARDGYSGSYSD 11425 257 GGTFSSYA 11557 INPNSGGT 11819 ARDGYSGSYSD 11426 280 GNTLSSHA 11558 INPSGGST 11820 ARDQGSSGTFDY 11427 281 GGTLSSYA 11559 INPNSGGT 11821 ARDSTDVIDY 11428 282 GYIFTSYD 11560 INPNSGDT 11822 ARDGGTVTPTEEYYYYG-11429 MDV 283 GGTFSSYA 11561 ISVYNGNT 11823 ASLDDLDY 11430 284 GHTFTSYY 11562 INPNNGGT 11824 ARDMVRDSAEYFQH 11431 285 GYTFITSY 11563 INPSGGTT 11825 ARDSSGYPIDY 11432 286 GYTFTSYD 11564 IIPLSGAP 11826 ARGALYNWNDGWFDP 11433 258 GYTFTSYD 11565 IIPLSGAP 11827 ARGALYNWNDGWFDP 11434 287 GFTVGSWY 11566 IWYEGSNK 11828 ARLGTASLPYFDY 11435 288 GYTFTGYY 11567 INPNRGDT 11829 ARESGDGFDP 11436 289 GYTFTNYY 11568 MNPNSGNT 11830 ARDWPNWFDP 11437 290 GYSFTDNY 11569 IRSDNGET 11831 AREVQLVGFDY 11438 291 GYTFSDHH 11570 IIPIFGTA 11832 ARGSSWYLHFQH 11439 260 GYTFSDHH 11571 IIPIFGTA 11833 ARGSSWYLHFQH 11440 292 GGTFSSYA 11572 IIPIFGTT 11834 AKGVDRYNWNDAFDY 11441 293 GYTFTDYY 11573 IHSNSGGT 11835 ARESSGYDSSLDY 11442 294 GGTFSSYG 11574 INPNSGDT 11836 TTDPRLDSSDPGY 11443 295 GGTFGNYG 11575 ISAYNGNT 11837 ARGGMDV 11444 296 GGTFSRYG 11576 SYPSDGST 11838 ARDRLGDLDY 11445 261 GGTFSRYG 11577 SYPSDGST 11839 ARDRLGDLDY 11446 297 GGTFSSYA 11578 MNPNSGNT 11840

ARDSIVGGYPFDY 11447 298 GYTFTSYD 11579 ITPIFGTT 11841 AREGYSSSWHDDAFDI 11448 299 GGTFSNYA 11580 IDPSGGST 11842 ARDLGDYGLDS 11449 300 GYTFTGYY 11581 MNPNSGDT 11843 ATGGSDSSGYYYEGYFQH 11450 301 GGTFSSYA 11582 MSPNSGNT 11844 ARDKGGYYDSSGYYWY 11451 302 GFSLSSYE 11583 ISSNGGST 11845 ARVGDGDGYNPDFDY 11452 303 GYTFTSYG 11584 IDPTSGAT 11846 AKDPIVATEVDY 11453 304 GGTFSSYA 11585 MSPNSGNT 11847 ARDSGAFDI 11454 305 GVTISNYA 11586 MNPNSGNT 11848 AREGLLDAFDI 11455 306 GGTFSRYG 11587 MNPYDGNT 11849 ARGGRHHDAFDI 11456 307 GGTFSSYA 11588 INPSGDGT 11850 ARDISNDAFDI 11457 308 GYILTGHY 11589 ISAYNGDT 11851 ARGSSWDDAFDI 11458 309 GFTFSNHY 11590 IGAGGGT 11852 AREGWNDDVFDI 11459 310 GGTFSSYA 11591 INPSAGTT 11853 ARDGNFGAFDI 11460 311 GYSFTTYA 11592 IIPIFGTA 11854 ARDKSGWNYGSGSYN- 11461 DAFDI 312 GYAFTGYY 11593 MNPNSGKT 11855 ARDGGLDFDY 11462 313 GYTFTTYY 11594 MNPNTGDT 11856 AKDPAVTPDAFDI 11463 314 GGTLSSYA 11595 IDPSGGGT 11857 AGSLYYYGMDV 11464 315 GGTFGSSA 11596 IIPIFGTA 11858 AKEDDILPPRAFDI 11465 316 GFTFDDYA 11597 ISGGGGVT 11859 ARVYSSGWLDAFDI 11466 317 GGTFSSYA 11598 ISGYNGNT 11860 ASSDVSPDAFDI 11467 318 GGTQNIYA 11599 VNPNSGNT 11861 ATPTSSSDDAFDI 11468 319 GGTFSSYA 11600 INPNSGGT 11862 ARASRGDDAFDI 11469 320 GIPFTSDD 11601 INPSGGST 11863 ARERYEGGY- 11470 SSGPGNYYYGMDV 321 GGTFSNYA 11602 MNPNSGNT 11864 ARDDDYGDYPV 11471 322 GDTFSDHA 11603 MNPKIGNT 11865 VYDSSGYDAFDI 11472 323 GYTFTSYD 11604 INPGTGGT 11866 ARETPSDYYDSSGYYYN- 11473 DAFDI 324 GGTFSSYA 11605 IPSGGT 11867 ARDLGTTFDI 11474 325 GYTFTAYY 11606 INPDNDNA 11868 AKDIAVAALAYGMDV 11475 326 GFTFSSYA 11607 ISYDGSDQ 11869 ARQSLYYYYGMDV 11476 327 GYTFTDYY 11608 ISTFTGNT 11870 ARDAPLAAAGTDYYYG- 11477 MDV 262 GYTFTDYY 11609 ISTFTGNT 11871 ARDAPLAAAGTDYYYG- 11478 MDV 328 GFTFSSYA 11610 ISDDGITK 11872 ARDDSSGYGGMDV 11479 329 GFTFSSYA 11611 ISYDGGDK 11873 ASGSLVLGYYYMDV 11480 330 GYTFTNYY 11612 INPNTGGT 11874 ATGGGGSYYDAFDV 11481 331 GGTFSSYA 11613 INPNSGNT 11875 ARDIGEGYSMDV 11482 332 GFTFSNHY 11614 ISYDGSNK 11876 AREEKYSSSWYVGVDAFDI 11483 333 GFTFSSSA 11615 ISGSGDNA 11877 ARDQEDYYYDSSGYGMDV 11484 334 GGTFSSHA 11616 IIPIFGTA 11878 AKGDWGIVVVPAAIGAFDI 11485 335 GYTFTAYY 11617 ISPVFGST 11879 ARDLGYYDSSGYRYDAFDI 11486 336 GYTFTSYD 11618 ISPMFGTA 11880 AKDGWYYGMDV 11487 337 GGTFSSYG 11619 INPNSGGT 11881 ARGEAGNLDWYFDL 11488 338 GGTFSNYG 11620 INPNNGDT 11882 AREDVWYFDL 11489 339 GYTFTTYG 11621 ISTYDGKT 11883 ALHLGGDWYFDL 11490 340 GYTFTGYY 11622 INPNTGAT 11884 ARQHGDYDWYFDL 11491 341 GDTFTTYY 11623 INPNSGNT 11885 ARDSGRH 11492 342 GGTFSSYG 11624 IIPMLGIA 11886 VREEVAGANWFDP 11493 343 GYTFTSYA 11625 INPSGGST 11887 AREGDYGSGEFDY 11494 344 GYTFTSSY 11626 MNPRSGNT 11888 ARERDDYGDYGWLDY 11495 345 GYTFTGYY 11627 INPSGGST 11889 ARDLYDSSGY- 11496 WHYYYYMDV 346 GGTFSSYA 11628 INPNSGGT 11890 ARFSGYDYVDY 11497 347 GGTFSSYA 11629 INPNGGNT 11891 ARDVGEDFDL 11498 348 GYTFTSYY 11630 INPADGDT 11892 ARDFDWLFAMDV 11499 349 GGTFSNYA 11631 INPNGGTT 11893 AKHGDHGFYV 11500 350 GGTFSSYA 11632 INPNVGSA 11894 AREDSGTSWFDP 11501 351 GYTFTSYY 11633 INPSDGST 11895 ARDDRGSNYYYGMDV 11502 352 GYTFTAYY 11634 MNPNSGTT 11896 ARDSSDYYGDYRADAFDI 11503 353 GYTFTSYD 11635 ISPSGDAT 11897 VKGLDH 11504 354 GFSFSDYG 11636 IGGIGDST 11898 ARMNYGDSNYYYYYG- 11505 MDV 355 GYTFTSYD 11637 ISPSDGST 11899 ARGAVGFDY 11506 356 GYTFTSYG 11638 INTYSGYT 11900 TTDDFLSFGY 11507 357 GYMFTDYY 11639 IIPYFGTA 11901 ARSISGSYVLDAFDI 11508 358 GYTFNSYG 11640 IIPIFGTA 11902 ARDWGYGDYADDAFDI 11509 359 GGTFSNND 11641 INPIYGSA 11903

AADWRGFDY 11510 360 GYTFTEYA 11642 MNPHNGDT 11904 AREGDYLGYPIDC 11511 361 GFTFSDYS 11643 IWQDGNVK 11905 ARDGNSGYVF 11512 362 GYTFTTYY 11644 INPNTGDT 11906 ARTAEAVAGLPAFDY 11513 363 GGTSNNYA 11645 IIPLFGTT 11907 ARVTLYGDYDY 11514 364 GYSLITHW 11646 INPSDGVT 11908 AREYYGEGFDY 11515 259 GYSLITHW 11647 INPSDGVT 11909 AREYYGEGFDY 11516 365 GGTFSSYA 11648 INPSGGST 11910 ARDLGDTAMDG 11517 366 GYTFTSYY 11649 ITPSGGST 11911 ARDGGLASFDY 11518 367 GGTFSSYA 11650 MNPNSGNT 11912 ARGGGWAMTDAFDI 11519 368 GFTFDDYG 11651 IYSGGDT 11913 TRKEYYYDSSGYLRLFDY 11520 369 GYTFTDYY 11652 INPIFGTS 11914 ARDISGYDYYYYGMDV 11521 370 GGTLNNYA 11653 IDPSDGTI 11915 ARSDYDFWSGLGGYFDY 11522 371 GGTFSSYA 11654 IDPNSGGT 11916 ARDSAEWELGGSFDY 11523 372 GFTFSNHY 11655 IGVNGDT 11917 AREGLVFSGRGHWYFDL 11524 373 GGTFSNYA 11656 INPNGGNT 11918 ARDYEDADFDG 11525 374 GYTFSDHH 11657 MNPDSGNT 11919 ARDSTSGVDY 11526 375 GFTFSSYA 11658 ISYDGHDQ 11920 ARGEQQLEGFYYYYGMDV 11527 376 GFTFSSYW 11659 ISYDGSKE 11921 ASDYGDYGTYDY 11528 377 GFTFSSYW 11660 ISGGGDDT 11922 AREPLAYCGGDCPGGFDY 11529 378 GFTFSDHY 11661 IGTGGDT 11923 ARHEDTAIFLDY 11530 379 GYTFTSYY 11662 ISPSDGST 11924 ARDGYDAWSYGMDV 11531 380 GYTFTGYY 11663 MNPNSGNT 11925 ARDGVTGTDY 11532 381 GFAFSSYV 11664 ISGAGDST 11926 AREPTTVTDDWYFDL 11533 382 GFAFSSHW 11665 ISGNGDNS 11927 ARDRAPEYFDL 11534 383 GGTFSSYA 11666 INPNSGGT 11928 ARDDYGDYGGGMDV 11535 384 GYTFTDYY 11667 MNPNSGHT 11929 AKDTSPRYGDGFFDY 11536 385 GFTFSSYW 11668 TSYDGSNK 11930 ARESGFSAEYFQH 11537 386 GGTFSSYA 11669 INPSGGST 11931 ARATGLYCSGSCFDY 11538 388 RSILDFNA 14122 IARAGAT 14123 NARVFDLPNDY 14124 LCDR1 LCDR2 LCDR3 265 QDISNY 13364 DAS QQSYSTPLT 12714 266 QSISSY 13365 AAS QQSYSTPLT 12715 267 QDISNY 13366 KAS QQSFSSPLT 12716 263 QDISNY 13367 DAS QQSYSTPLT 12717 268 QNVNTW 13368 EAS QQANSFPFT 12718 269 QSISDW 13369 AAS AQHNHYPYT 12719 272 QAIRDD 13370 DAS QQANSFPIT 12722 273 QGVGND 13371 AAS QQASSFPLT 12723 274 QIIGTN 13372 AAS QQSYTFPVT 12724 275 QSISTW 13373 DAS QQSYSTPFT 12725 281 QDIRNY 13374 DAS QQAYSFPWT 12733 282 QDISNY 13375 NAS QQLNSYPFT 12734 283 QSISTW 13376 AAS LOHYTYPLT 12735 284 EDISTY 13377 AAS QQSHTIPWT 12736 285 HHISDF 13378 AAS QQSYSSPYT 12737 286 QDIGDY 13379 DAS QQANSFPLT 12738 258 QDIGDY 13380 DAS QQANSFPLT 12739 287 QDIRSY 13381 AAS QQSYTAPPT 12740 288 QDISNN 13382 AAS LOHNTYPLT 12741 289 QDISNW 13383 DAS QQAISFPLT 12742 290 QGIANY 13384 AAS QQADSFPLT 12743 291 QGIASY 13385 AAS QQFDSYPIT 12744 260 QGIASY 13386 AAS QQFDSYPIT 12745 292 QGISNY 13387 AAS QQSSIIPFT 12746 293 QGISNY 13388 AAS QQAYSFPYT 12747 294 QSIGRW 13389 DAS QQSYSTPRT 12748 295 QSINSW 13390 DTS QQTYSTPYT 12749 296 QSISSW 13391 AAS QQGYSTPYI 12750 261 QSISSW 13392 AAS QQGYSTPYI 12751 297 QSISSY 13393 AAS QQTDSIPIT 12752 298 QSISSY 13394 AAS QQSYSIPYT 12753 299 QTIRSY 13395 KAS QQTYTIPIT 12754 300 QTISNW 13396 AAS QQANSFPPT 12755 301 QYIGSY 13397 DAS QQVDSYPLT 12756 307 QDIGNY 13398 AAS QQTYNTPLT 12762 308 QDISNY 13399 EAS QQSYSTPFT 12763 309 QDISTW 13400 RAS QQSYSIPLT 12764 310 QNINNY 13401 AAS QQSYSAPVT 12765 311 QNINTW 13402 AAS QQAYSFPFT 12766 312 QRIGNY 13403 AAS QQSYSTPLT 12767 313 QSISTY 13404 AAS QQSYRTVT 12768 324 QDISNY 13405 AAS QQSYSTPT 12779 325 QDISNY 13406 GAS QEADSFPLT 12780 326 QGIRND 13407 DAS QQAYSFPWT 12781 327 QGISNY 13408 KAS QQSYNTPFT 12782 262 QGISNY 13409 KAS QQSYNTPFT 12783 328 QSINRW 13410 SAS QQSYNTPLT 12784 329 QSINTW 13411 AAS QQANSFPFT 12785 330 QSIRTW 13412 DAS QQLNSYPLT 12786 331 QSIRTY 13413 AAS QQSYSAPLT 12787 332 QSISTY 13414 AAS QQSYSTPLT 12788 333 QSITTY 13415 AAS QQSYSTPLT 12789 339 QSISSW 13416 DAS QQSYSTPFT 12795 340

QSISSY 13417 DAS QQSYSFPT 12796 344 QDISNY 13418 AAA QQTYSTPWT 12800 345 QDINTY 13419 AAS QQSSSFPLT 12801 346 QDISNY 13420 AAS QQLYNFPYT 12802 347 QSISRY 13421 GAS QQSYNTPLT 12803 348 QTLSGW 13422 GAS QQYYSYPPT 12804 349 QDIINY 13423 EAS QQSYSTPLT 12805 350 QSISSY 13424 DVF QQSYSSPFT 12806 351 QDISNY 13425 MAS QQTNSFPLT 12807 352 QSISSY 13426 DAS QQSYSTPLT 12808 356 ESVSTW 13427 KAS QQSYKTPYT 12812 358 QSISSY 13428 AAS QQSYSTPYT 12814 360 QSISSY 13429 KAS QQNDSIPIT 12816 361 QSISRW 13430 DAS LODYSYPLT 12817 363 QSINRY 13431 AAS QQANSFPPT 12819 364 QGISNY 13432 SAS QQSYSTPLT 12820 259 QGISNY 13433 SAS QQSYSTPLT 12821 365 QSIDSY 13434 KAS QQSYSAPLT 12822 366 QDISTW 13435 DAS QQVNSDPYT 12823 367 QDISNY 13436 AAS QQGDSLPLT 12824 368 QGISNY 13437 AAS QQSDSFPYT 12825 370 QGIRND 13438 AAS QQANSFPPT 12827 371 ESISTY 13439 KAS QQTDSTFIT 12828 372 RNIHDY 13440 AAS QQTYSTPPT 12829 373 QSNDSY 13441 KAS QQSYSSPLT 12830 374 QSISDF 13442 AAS QQSYSSPYT 12831 375 QDISNY 13443 AAS QQANRFPLT 12832 376 QDISNY 13444 KAS QQSYNFPAT 12833 378 QSISSY 13445 DAS QQSYSTPLT 12835 379 QGISDY 13446 DAS QQSYILPLT 12836 380 QDINDF 13447 AAS QQSYSAPYT 12837 381 QSISNW 13448 AAS QQSYSSPWT 12838 382 QGIDSW 13449 AAS QQAYSFPLT 12839 383 QNIGTW 13450 RAS QQAYSFPWT 12840 384 QNINNW 13451 KAS QQADSFPPT 12841 385 QDISSY 13452 AAS QQLNRYPIT 12842 386 QDISNY 13453 AAS QQYDSSFIT 12843 302 QSLLHSNGYNY 13454 LGS MQGTHWPPT 12757 303 QSLLHSNGYNY 13455 FGS MQALQAPVS 12758 316 QSLLHSNGYNY 13456 DAS MQALQTPPA 12771 334 QSLLHSNGYNY 13457 AAS MQARQTPLT 12790 335 QSLLHSNGYNY 13458 GAS MQTLQTPFT 12791 336 QSLLHSNGYNY 13459 LGS MQALQTPLT 12792 353 QSLLHSNGYNY 13460 LGS MQALQSPWT 12809 355 QSLLHSNGYNY 13461 LGS MQALQTPPS 12811 377 QSLLHSNGYNY 13462 LGS MQGTHWPET 12834 314 QSVGSY 13463 GAS QQYDSSSQT 12769 315 RSVSTY 13464 GAS QQYDGSPYT 12770 341 QSVSSY 13465 DTS QQYYDTPYT 12797 369 QSVSTY 13466 GAS QQHDSYPLT 12826 270 QSVLSSSYNKNY 13467 WAS QQYYTTPFT 12720 271 QSVLSSSYNKNY 13468 WAS QQYYSTPFT 12721 276 QSVLSSSNNKNY 13469 WAS QQYYGSPLT 12726 277 QSVLSSSYNKNY 13470 WAS QQYYSSPPT 12727 278 QSVLSSSYNKNY 13471 WAS QQYYSSPPT 12728 279 QSVLSSSYNKNY 13472 WAS QQYYSTPWT 12729 264 QSVLSSSYNKNY 13473 WAS QQYYSTPWT 12730 257 QSVLSSSYNKNY 13474 WAS QQYYSTPWT 12731 280 QSVLSSSYNKNY 13475 WAS QQYYGSPPT 12732 304 QSVLSSSYNKNY 13476 WAS QQYYSTPLT 12759 305 QSVSSSSYNKNY 13477 WAS QQYYSTPIT 12760 306 QNVLSSSNNNSY 13478 WAS QQYYSTPFT 12761 317 QSVLSSSYNKNF 13479 WAS QQYYSAPPT 12772 318 QSVLSSSYNKNF 13480 WAS QQYYSDPIT 12773 319 QSVLSSSYNKNY 13481 WAS QQYYSIPIA 12774 320 QSVLSSSYNKNY 13482 WAS QQYYSIPYT 12775 321 QSVLSTSYNKNY 13483 WAS QQYYTTPPT 12776 322 QSVLSTSYNRNF 13484 WAS QQYYSTPYT 12777 323 QSVLYSSNNKNY 13485 WAS QQYYSTPLT 12778 337 QTVFSTSYNKNY 13486 WAS QQYYSTPLT 12793 338 QSVFSTSYNRDY 13487 WAS QQYYSSPPT 12794 342 QSVLYSSNNKNY 13488 LAS QQYYSTPPT 12798 343 QSVLSSSYNKNY 13489 WAS QQYYSTPLT 12799 354 QSVLYSSNNKNY 13490 WAS QQYYSSPLT 12810 357 QSVLYSSNNKNY 13491 WAS QQYFTTPLT 12813 359 QSVLSSSYNKNY 13492 WAS QQYYDTPLT 12815 362 QSVFSTSYNRDY 13493 WAS QQYYYTST 12818

TABLE-US-00026 TABLE 25 G and F Proteins SEQ ID NO: SEQUENCE ANNOTATION 9258 MVVILDKRCY CNLLILILMI SECSVGILHY Nipah virus NiV-F EKLSKIGLVK GVTRKYKIKS NPLIKDIVIK with signal sequence MIPNVSNMSQ CTGSVMENYK TRLNGILTPI (aa 1-546) KGALEIYKNN THDLVGDVRL AGVIMAGVAI Uniprot Q9IH63 GIATAAQITA GVALYEAMKN ADNINKLKSS

IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD
LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT
EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS
FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMIN
NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA
ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF
TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL
SIASLCIGLI TFISFIIVEK KRNTYSRLED RRVRPTSSGD LYYIGT 9259 ILHY
EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK Nipah virus NiV-F F0 MIPNVSNMSQ
CTGSVMENYK TRLNGILTPI (aa 27-546) KGALEIYKNN THDLVGDVRL
AGVIMAGVAI GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK
LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL
LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDEDDLLESD
SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS
IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMIN NMRECLTGST
EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL
MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI
SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI
TFISFIIVEK KRNTYSRLED RRVRPTSSGD LYYIGT 9260
ILHYEKLSKIGLVKGVTRKYKIKSNPLTKDIVIKMIPNVS Nipah virus NiV-F F2
NMSQCTGSVMENYKTRLNGILTPIKGALEIYKNNTHDLVG (aa 27-109) DVR 9261
LAGVIMAGVAIGIATAAQITAGVALYEAMKNADNINKLKS Nipah virus NiV F F1
SIESTNEAVVKLQETAEKTVYVLTALQDYINTNLVPTIDK (aa 110-546)
ISCKQTELSLDLALSKYLSDLLFVFGPNLQDPVSNSMTIQ
AISQAFGGNYETLLRTLGYATEDFDDLLESDSITGQIIYV
DLSSYYIIVRVYFPILTEIQQAYIQELLPVSFNNDNSEWI
SIVPNFILVRNTLISNIEIGFCLITKRSVICNQDYATPMT
NNMRECLIGSTEKCPRELVVSSHVPRFALSNGVLFANCIS
VTCQCQTTGRAISQSGEQTLLMIDNTTCPTAVLGNVIISL
GKYLGSVNYNSEGIAIGPPVFTDKVDISSQISSMNQSLQQ
SKDYIKEAQRLLDTVNPSLISMLSMIILYVLSIASLCIGL
ITFISFIIVEKKRNTYSRLEDRRVRPTSSGDLYYIGT 9262 ILHY EKLSKIGLVK
GVTRKYKIKS NPLTKDIVIK Nipah virus NiV-F F0 MIPNVSNMSQ
CTGSVMENYK TRLNGILTPI (aa T234 truncation KGALEIYKNN THDLVGDVRL
AGVIMAGVAI 525-544) GIATAAQITA GVALYEAMKN ADNINKLKSS IESTNEAVVK
LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL
LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD
SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS
IVPNFILVRN TLISNIEIGE CLITKRSVIC NQDYATPMIN NMRECLTGST
EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL
MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI
SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI
TFISFIIVEK KRNTGT 9263 LAGVIMAGVAIGIATAAQITAGVALYEAMKNADNINKLKS
Nipah virus NiV F F1 SIESTNEAVVKLQETAEKTVYVLTALQDYINTNLVPTIDK (aa
110-546) ISCKQTELSLDLALSKYLSDLLFVFGPNLQDPVSNSMTIQ truncation
AISQAFGGNYETLLRTLGYATEDFDDLLESDSITGQIIYV (aa 525-544)
DLSSYYIIVRVYFPILTEIQQAYIQELLPVSFNNDNSEWI
SIVPNFILVRNTLISNIEIGFCLITKRSVICNQDYATPMT
NNMRECLTGSTEKCPRELVVSSHVPRFALSNGVLFANCIS
VTCQCQTTGRAISQSGEQTLLMIDNTTCPTAVLGNVIISL

GKYLGSVNYNSEGIAIGPPVFTDKVDISSQISSMNQSLQQ
SKDYIKEAQRLLDTVNPSLISMLSMIILYVLSIASLCIGL ITFISFIIVEKKRNTGT 9264
ILHY EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK Nipah virus NiV-F F0
MIPNVSNMSQ CTGSVMENYK TRINGILIPI T234 truncation (aa KGALEIYKNQ
THDLVGDVRL AGVIMAGVAI 525-544) AND GIATAAQITA GVALYEAMKN
ADNINKLKSS mutation on N-linked IESTNEAVVK LQETAEKTVY VLTALQDYIN
glycosylation site TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD
PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD
LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGE
CLITKRSVIC NQDYATPMIN NMRECLIGST EKCPRELVVS SHVPRFALSN
GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG
KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL
LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTGT 9265
MVVILDKRCY CNLLILILMI SECSVGILHY Truncated NiV fusion EKLSKIGLVK
GVTRKYKIKS NPLTKDIVIK glycoprotein MIPNVSNMSQ CTGSVMENYK
TRLNGILTPI (FcDelta22) at KGALEIYKNN THDLVGDVRL AGVIMAGVAI
cytoplasmic tail GIATAAQITA GVALYEAMKN ADNINKLKSS (with signal sequence)
IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI SCKQTELSLD
LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT
EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS
FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMTN
NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA
ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF
TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL
SIASLCIGLI TFISFIIVEK KRNT 9266 MGPAENKKVR FENTTSDKGK IPSKVIKSYY
NiVG protein GTMDIKKINE GLLDSKILSA FNTVIALLGS attachment IVIIVMNIMI
IQNYTRSTDN QAVIKDALQG glycoprotein (602 aa) IQQQIKGLAD KIGTEIGPKV
SLIDTSSTIT IPANIGLLGS KISQSTASIN ENVNEKCKFT LPPLKIHECN
ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV
VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG
DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY
WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG
DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL
RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS
WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN
DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ
KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9267 MGKVR
FENTTSDKGK IPSKVIKSYY GTMDIKKINE NiVG protein GLLDSKILSA
FNTVIALLGS IVIIVMNIMI attachment IQNYTRSTDN QAVIKDALQG IQQQIKGLAD
glycoprotein KIGTEIGPKV SLIDTSSTIT IPANIGLLGS Truncated Δ5 KISQSTASIN
ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC
LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS
CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE
FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS
IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY
SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG
SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG
QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV
FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR
PKLFAVKIPE QC 9268 MGNTTSDKGK IPSKVIKSYY GTMDIKKINE NiVG protein

GLLDSKILSA FNTVIALLGS IVIIVMNIMI attachment IQNYTRSTDN QAVIKDALQG IQQQIKGLAD glycoprotein KIGTEIGPKV SLIDTSSTIT IPANIGLLGS Truncated Δ10 KISQSTASIN ENVNEKCKFT LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9269 MGKGK IPSKVIKSYY GIMDIKKINE GLLDSKILSA NiVG protein FNTVIALLGS IVIIVMNIMI IQNYTRSTDN attachment QAVIKDALQG IQQQIKGLAD KIGTEIGPKV glycoprotein SLIDTSSTIT IPANIGLLGS KISQSTASIN Truncated Δ15 ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGICITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRIEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9270 MGSKVIKSYY GTMDIKKINE GLLDSKILSA NiVG protein FNTVIALLGS IVIIVMNIMI IQNYTRSTDN attachment QAVIKDALQG IQQQIKGLAD KIGTEIGPKV glycoprotein SLIDTSSTIT IPANIGLLGS KISQSTASIN Truncated Δ20 ENVNEKCKFT LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRIEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9271 MGSYY GTMDIKKINE GLLDSKILSA FNTVIALLGS NiVG protein IVIIVMNIMI IQNYTRSTDN QAVIKDALQG attachment IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT glycoprotein IPANIGLLGS KISQSTASIN ENVNEKCKFT Truncated Δ25 LPPLKIHECN ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9272 MGTMDIKKINE GLLDSKILSA FNTVIALLGS NiVG protein IVIIVMNIMI IQNYTRSTDN QAVIKDALQG attachment IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT glycoprotein

IPANIGLLGS KISQSTASIN ENVNEKCKFT Truncated Δ30 LPPLKIHECN
ISCPNPLPER EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV
VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG
DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY
WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG
DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL
RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS
WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN
DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ
KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9273
MKKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI NiVG protein IQNYTRSTDN
QAVIKDALQG IQQQIKGLAD attachment KIGTEIGPKV SLIDTSSTIT IPANIGLLGS
glycoprotein KISQSTASIN ENVNEKCKFT LPPLKIHECN Truncated and ISCPNPLPFR
EYRPQTEGVS NLVGLPNNIC mutated LQKTSNQILK PKLISYTLPV VGQSGTCITD
(E501A, W504A, PLLAMDEGYF AYSHLERIGS CSRGVSKQRI Q530A, E533A)
NIV G IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV protein (Gc Δ 34)
YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG
YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND
SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI
EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW
RNNTVISRPG QSQCPRENTC PAICAEGVYN DAFLIDRINW ISAGVFLDSN
ATAANPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE
IYDTGDNVIR PKLFAVKIPE QCT 9274 MATQEVRLKC LLCGIIVLVL SLEGLGILHY
Hendra virus F protein EKLSKIGLVK GITRKYKIKS Uniprot O89342 (with
NPLTKDIVIK MIPNVSNVSK CTGTVMENYK signal sequence) SRLIGILSPI
KGAIELYNNN THDLVGDVKL AGVVMAGIAI GIATAAQITA GVALYEAMKN
ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDQI
SCKQTELALD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE
TLLRTLGYAT EDFDDLLESD SIAGQIVYVD LSSYYIIVRV YFPILTEIQQ
AYVQELLPVS FNNDNSEWIS IVPNFVLIRN TLISNIEVKY CLITKKSVIC
NQDYATPMTA SVRECLIGST DKCPRELVVS SHVPRFALSG GVLFANCISV
TCQCQTTGRA ISQSGEQTLL MIDNTTCTTV VLGNIIISLG KYLGSINYNS
ESIAVGPPVY TDKVDISSQI SSMNQSLQQS KDYIKEAQKI LDTVNPSLIS
MLSMIILYVL SIAALCIGLI TFISFVIVEK KRGNYSRLDD RQVRPVSNGD
LYYIGT 9275 MMADSKLVSL NNNLSGKIKD QGKVIKNYYG Hendra virus G
protein TMDIKKINDG LLDSKILGAF Uniprot O89343 NTVIALLGSI IIIVMNIMII
QNYTRTTDNQ ALIKESLQSV QQQIKALTDK IGTEIGPKVS LIDTSSTITI
PANIGLLGSK ISQSTSSINE NVNDKCKFTL PPLKIHECNI SCPNPLPFRE
YRPISQGVSD LVGLPNQICL QKTTSTILKP RLISYTLPIN TREGVCITDP
LLAVDNGFFA YSHLEKIGSC TRGIAKQRII GVGEVLDRGD KVPSMFMTNV
WTPPNPSTIH HCSSTYHEDF YYTLCAVSHV GDPILNSTSW TESLSLIRLA
VRPKSDSGDY NQKYIAITKV ERGKYDKVMP YGPSGIKQGD TLYFPAVGFL
PRTEFQYNDS NCPIIHCKYS KAENCRLSMG VNSKSHYILR SGLLKYNLSL
GGDIILQFIE IADNRLTIGS PSKIYNSLGQ PVFYQASYSW DTMIKLGDVD
TVDPLRVQWR NNSVISRPGQ SQCPRFNVCP EVCWEGTYND AFLIDRLNWV
SAGVYLNSNQ TAENPVFAVF KDNEILYQVP LAEDDTNAQK TITDCFLLEN
VIWCISLVEI YDTGDSVIRP KLFAVKIPAQ CSES 9276 MVVILDKRCY CNLLILILMI
SECSVGILHY Nipah virus NiV-F FO EKLSKIGLVK GVTRKYKIKS NPLIKDIVIK
T234 truncation (aa MIPNVSNMSQ CTGSVMENYK TRINGILTPI 525-544) (with
signal KGALEIYKNN THDLVGDVRL AGVIMAGVAI sequence) GIATAAQITA

GVALYEAMKN ADNINKLKSS IESTNEAVVK LQETAEKTVY VLTALQDYIN
TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA
ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV
YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGE CLITKRSVIC
NQDYATPMIN NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV
TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS
EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS
MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNTGT 9277 MVVILDKRCY
CNLLILILMI SECSVGILHY Nipah virus NiV-F F0 EKLSKIGLVK GVTRKYKIKS
NPLIKDIVIK T234 truncation (aa MIPNVSNMSQ CTGSVMENYK TRLNGILTPI 525-
544) AND KGALEIYKNQ THDLVGDVRL AGVIMAGVAI mutation on N-linked
GIATAAQITA GVALYEAMKN ADNINKLKSS glycosylation site IESTNEAVVK
LQETAEKTVY VLTALQDYIN (with signal sequence) TNLVPTIDKI SCKQTELSLD
LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT
EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ AYIQELLPVS
FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC NQDYATPMIN
NMRECLTGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA
ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF
TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL
SIASLCIGLI TFISFIIVEK KRNTGT 9278 MVVILDKRCY CNLLILILMI
SECSVGILHY Truncated NiV fusion EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK
glycoprotein MIPNVSNMSQ CTGSVMENYK TRLNGILTPI (FcDelta22) at
KGALEIYKNN THDLVGDVRL AGVIMAGVAI cytoplasmic tail (with GIATAAQITA
GVALYEAMKN ADNINKLKSS signal sequence) IESTNEAVVK LQETAEKTVY
VLTALQDYIN TNLVPTIDKI SCKQTELSLD LALSKYLSDL LFVFGPNLQD
PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD SITGQIIYVD
LSSYYIIVRV YFPILTEIQQ AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGE
CLITKRSVIC NQDYATPMIN NMRECLIGST EKCPRELVVS SHVPRFALSN
GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG
KYLGSVNYNS EGIAIGPPVE TDKVDISSQI SSMNQSLQQS KDYIKEAQRL
LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNT 9279
MKKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI NiVG protein IQNYTRSTDN
QAVIKDALQG IQQQIKGLAD attachment KIGTEIGPKV SLIDTSSTIT IPANIGLLGS
glycoprotein KISQSTASIN ENVNEKCKFT LPPLKIHECN Truncated (Gc Δ 34)
ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV
VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG
DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY
WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG
DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL
RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS
WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN
DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ
KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QCT 9280 ILHY
EKLSKIGLVK GVTRKYKIKS NPLTKDIVIK Truncated mature NiV MIPNVSNMSQ
CTGSVMENYK TRLNGILTPI fusion glycoprotein KGALEIYKNN THDLVGDVRL
AGVIMAGVAI (FcDelta22) at GIATAAQITA GVALYEAMKN ADNINKLKSS
cytoplasmic tail IESTNEAVVK LQETAEKTVY VLTALQDYIN TNLVPTIDKI
SCKQTELSLD LALSKYLSDL LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE
TLLRTLGYAT EDFDDLLESD SITGQIIYVD LSSYYIIVRV YFPILTEIQQ
AYIQELLPVS FNNDNSEWIS IVPNFILVRN TLISNIEIGF CLITKRSVIC

NQDYATPMIN NMRECLIGST EKCPRELVVS SHVPRFALSN GVLFANCISV TCQCQTTGRA ISQSGEQTLL MIDNTTCPTA VLGNVIISLG KYLGSVNYNS EGIAIGPPVF TDKVDISSQI SSMNQSLQQS KDYIKEAQRL LDTVNPSLIS MLSMIILYVL SIASLCIGLI TFISFIIVEK KRNT 9281
MSNKRITVLIIISYTLFYLNNAAIVGFDFDKLNKIGVVQG gb: JQ001776:6129-
RVLNYKIKGDPMTKDLVLKFIPNIVNITECVREPLSRYNE 8166|Organism: Cedar
TVRRLLLPIHNMLGLYLNNTNAKMTGLMIAGVIMGGIAIG virus|Strain
IATAAQITAGFALYEAKKNTENIQKLIDSIMKTQDSIDKL Name: CG1a|Protein
TDSVGTSILILNKLQTYINNQLVPNLELLSCRQNKIEFDL Name: fusion
MLTKYLVDLMTVIGPNINNPVNKDMTIQSLSLLFDGNYDI glycoprotein|Gene
MMSELGYTPQDFLDLIESKSITGQIIYVDMENLYVVIRTY Symbol: F (with signal
LPTLIEVPDAQIYEFNKITMSSNGGEYLSTIPNFILIRGN sequence)
YMSNIDVATCYMTKASVICNQDYSLPMSQNLRSCYQGETE
YCPVEAVIASHSPRFALINGVIFANCINTICRCQDNGKTI
TQNINQFVSMIDNSTCNDVMVDKFTIKVGKYMGRKDINNI
NIQIGPQIIIDKVDLSNEINKMNQSLKDSIFYLREAKRIL
DSVNISLISPSVQLFLIIISVLSFIILLIIIVYLYCKSKH
SYKYNKFIDDPDYYNDYKRERINGKASKSNNIYYVGD 9282
MALNKNMFSSLFLGYLLVYATTVQSSIHYDSLSKVGVIKG gb: NC_025352:5950-
LTYNYKIKGSPSTKLMVVKLIPNIDSVKNCTQKQYDEYKN 8712|Organism: Mojiang
LVRKALEPVKMAIDTMLNNVKSGNNKYRFAGAIMAGVALG virus|Strain
VATAATVTAGIALHRSNENAQAIANMKSAIQNTNEAVKQL Name: Tongguan1|Protein
QLANKQTLAVIDTIRGEINNNIIPVINQLSCDTIGLSVGI Name: fusion
RLTQYYSEIITAFGPALQNPVNTRITIQAISSVENGNFDE protein|Gene
LLKIMGYTSGDLYEILHSELIRGNIIDVDVDAGYIALEIE Symbol: F (with signal
FPNLTLVPNAVVQELMPISYNIDGDEWVTLVPRFVLTRIT sequence)
LLSNIDTSRCTITDSSVICDNDYALPMSHELIGCLQGDTS
KCAREKVVSSYVPKFALSDGLVYANCLNTICRCMDTDTPI
SQSLGATVSLLDNKRCSVYQVGDVLISVGSYLGDGEYNAD
NVELGPPIVIDKIDIGNQLAGINQTLQEAEDYIEKSEEFL
KGVNPSIITLGSMVVLYIFMILIAIVSVIALVLSIKLTVK
GNVVRQQFTYTQHVPSMENINYVSH 9283
MKKKTDNPTISKRGHNHSRGIKSRALLRETDNYSNGLIVE gb: NC_025256:6865-
NLVRNCHHPSKNNLNYTKTQKRDSTIPYRVEERKGHYPKI 8853|Organism: Bat
KHLIDKSYKHIKRGKRRNGHNGNIITIILLLILILKTQMS Paramyxovirus
EGAIHYETLSKIGLIKGITREYKVKGTPSSKDIVIKLIPN Eid_hel/GH-
VTGLNKCTNISMENYKEQLDKILIPINNIIELYANSTKSA M74a/GHA/2009|Strain
PGNARFAGVIIAGVALGVAAAAQITAGIALHEARQNAERI Name: BatPV/Eid_hel/
NLLKDSISATNNAVAELQEATGGIVNVITGMQDYININLV GH-M74a/GHA/2009|
PQIDKLQCSQIKTALDISLSQYYSEILTVFGPNLQNPVTT Protein
SMSIQAISQSFGGNIDLLLNLLGYTANDLLDLLESKSITG Name: fusion
QITYINLEHYFMVIRVYYPIMTTISNAYVQELIKISFNVD protein|Gene
GSEWVSLVPSYILIRNSYLSNIDISECLITKNSVICRHDF Symbol: F (with signal
AMPMSYTLKECLTGDTEKCPREAVVTSYVPRFAISGGVIY sequence)
ANCLSTTCQCYQTGKVIAQDGSQTLMMIDNQTCSIVRIEE
ILISTGKYLGSQEYNTMHVSVGNPVFTDKLDITSQISNIN
QSIEQSKFYLDKSKAILDKINLNLIGSVPISILFIIAILS
LILSIITFVIVMIIVRRYNKYTPLINSDPSSRRSTIQDVY IIPNPGEHSIRSAARSIDRDRD 9284
(GGGGGS)n wherein n is 1 to 6 Peptide Linker 9285
MPAENKKVRFENTTSDKGKIPSKVIKSYYGTMDIKKINEG gb: AF212302|Organism:

LLDSKILSAFNTVIALLGSIVIIVMNIMIIQNYTRSTDNQ Nipah virus|Strain
AVIKDALQGIQQQIKGLADKIGTEIGPKVSLIDTSSTITI Name: UNKNOWN-
PANIGLLGSKISQSTASINENVNEKCKFTLPPLKIHECNI AF212302|Protein
SCPNPLPFREYRPQTEGVSNLVGLPNNICLQKTSNQILKP Name: attachment
KLISYTLPVVGQSGTCITDPLLAMDEGYFAYSHLERIGSC glycoprotein|Gene
SRGVSKQRIIGVGEVLDRGDEVPSLFMTNVWTPPNPNTVY Symbol: G
HCSAVYNNEFYYVLCAVSTVGDPILNSTYWSGSLMMTRLA (Uniprot Q9IH62)
VKPKSNGGGYNQHQLALRSIEKGRYDKVMPYGPSGIKQGD
TLYFPAVGFLVRTEFKYNDSNCPITKCQYSKPENCRLSMG
IRPNSHYILRSGLLKYNLSDGENPKVVFIEISDQRLSIGS
PSKIYDSLGQPVFYQASFSWDTMIKFGDVLTVNPLVVNWR
NNTVISRPGQSQCPRFNTCPEICWEGVYNDAFLIDRINWI
SAGVFLDSNQTAENPVFTVFKDNEILYRAQLASEDTNAQK
TITNCFLLKNKIWCISLVEIYDTGDNVIRPKLFAVKIPEQ CT 9286
MLSQLQKNYLDNSNQQGDKMNNPDKKLSVNFNPLELDKGQ gb: JQ001776:8170-
KDLNKSYYVKNKNYNVSNLLNESLHDIKFCIYCIFSLLII 10275|Organism: Cedar
ITIINIITISIVITRLKVHEENNGMESPNLQSIQDSLSSL virus|Strain
TNMINTEITPRIGILVTATSVTLSSSINYVGTKTNQLVNE Name: CG1a|Protein
LKDYITKSCGFKVPELKLHECNISCADPKISKSAMYSTNA Name: attachment
YAELAGPPKIFCKSVSKDPDFRLKQIDYVIPVQQDRSICM glycoprotein|Gene
NNPLLDISDGFFTYIHYEGINSCKKSDSFKVLLSHGEIVD Symbol: G
RGDYRPSLYLLSSHYHPYSMQVINCVPVTCNQSSFVFCHI
SNNTKTLDNSDYSSDEYYITYFNGIDRPKTKKIPINNMTA
DNRYIHFTFSGGGGVCLGEEFIIPVTTVINTDVFTHDYCE
SFNCSVQTGKSLKEICSESLRSPINSSRYNLNGIMIISQN
NMTDFKIQLNGITYNKLSFGSPGRLSKTLGQVLYYQSSMS
WDTYLKAGFVEKWKPFTPNWMNNTVISRPNQGNCPRYHKC
PEICYGGTYNDIAPLDLGKDMYVSVILDSDQLAENPEITV
FNSTTILYKERVSKDELNTRSTTTSCFLFLDEPWCISVLE TNRFNGKSIRPEIYSYKIPKYC
9287 MPQKTVEFINMNSPLERGVSTLSDKKTLNQSKITKQGYFG gb: NC_025256:9117-
LGSHSERNWKKQKNQNDHYMTVSIMILEILVVLGIMENLI 11015|Organism: Bat
VLTMVYYQNDNINQRMAELTSNITVLNLNLNQLINKIQRE Paramyxovirus
IIPRITLIDTATTITIPSAITYILATLITRISELLPSINQ Eid_hel/GH-
KCEFKTPTLVLNDCRINCTPPLNPSDGVKMSSLATNLVAH M74a/GHA/2009|Strain
GPSPCRNFSSVPTIYYYRIPGLYNRTALDERCILNPRLTI Name: BatPV/Eid_hel/
SSTKFAYVHSEYDKNCTRGFKYYELMTFGEILEGPEKEPR GH-
MFSRSFYSPTNAVNYHSCTPIVTVNEGYFLCLECTSSDPL M74a/GHA/2009|Protein
YKANLSNSTFHLVILRHNKDEKIVSMPSFNLSTDQEYVQI Name: glycoprotein|
IPAEGGGTAESGNLYFPCIGRLLHKRVTHPLCKKSNCSRT Gene Symbol: G
DDESCLKSYYNQGSPQHQVVNCLIRIRNAQRDNPTWDVIT
VDLTNTYPGSRSRIFGSFSKPMLYQSSVSWHTLLQVAEIT
DLDKYQLDWLDTPYISRPGGSECPFGNYCPTVCWEGTYND
VYSLTPNNDLFVTVYLKSEQVAENPYFAIFSRDQILKEFP
LDAWISSARTTTISCFMFNNEIWCIAALEITRLNDDIIRP
IYYSFWLPTDCRTPYPHTGKMTRVPLRSTYNY 9288
MATNRDNTITSAEVSQEDKVKKYYGVETAEKVADSISGNK gb: NC_025352:8716-
VFILMNTLLILTGAIITITLNITNLTAAKSQQNMLKIIQD 11257|Organism: Mojiang
DVNAKLEMFVNLDQLVKGEIKPKVSLINTAVSVSIPGQIS virus|Strain
NLQTKFLQKYVYLEESITKQCTCNPLSGIFPTSGPTYPPT Name: Tongguan1|
DKPDDDTTDDDKVDTTIKPIEYPKPDGCNRTGDHFTMEPG Protein Name:

ANFYTVPNLGPASSNSDECYTNPSFSIGSSIYMFSQEIRK attachment
TDCTAGEILSIQIVLGRIVDKGQQGPQASPLLVWAVPNPK glycoprotein|Gene
IINSCAVAAGDEMGWVLCSVTLTAASGEPIPHMEDGEWLY Symbol: G
KLEPDTEVVSYRITGYAYLLDKQYDSVFIGKGGGIQKGND
LYFQMYGLSRNRQSFKALCEHGSCLGTGGGGYQVLCDRAV
MSFGSEESLITNAYLKVNDLASGKPVIIGQTFPPSDSYKG
SNGRMYTIGDKYGLYLAPSSWNRYLRFGITPDISVRSTTW
LKSQDPIMKILSTCTNTDRDMCPEICNTRGYQDIFPLSED
SEYYTYIGITPNNGGTKNFVAVRDSDGHIASIDILQNYYS
ITSATISCFMYKDEIWCIAITEGKKQKDNPQRIYAHSYKI
RQMCYNMKSATVTVGNAKNITIRRY 9289 FNTVIALLGS IVIIVMNIMI IQNYTRSTDN
NivG protein QAVIKDALQG IQQQIKGLAD KIGTEIGPKV attachment SLIDTSSTIT
IPANIGLLGS KISQSTASIN glycoprotein ENVNEKCKFT LPPLKIHECN ISCPNPLPER
Without cytoplasmic EYRPQTEGVS NLVGLPNNIC LQKTSNQILK tail PKLISYTLPV
VGQSGTCITD PLLAMDEGYF Uniprot Q9IH62 AYSHLERIGS CSRGVSKQRI
IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST
VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM
PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM
GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG
QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG PEICWEGVYN
DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ
KTITNCFLLK NKIWCISLVE IYDIGDNVIR PKLFAVKIPE QC 9290 FNTVIALLGSI
IIIVMNIMII QNYTRTTDNQ Hendra virus G protein ALIKESLQSV QQQIKALTDK
Uniprot O89343 IGTEIGPKVS LIDTSSTITI PANIGLLGSK Without cytoplasmic
ISQSTSSINE NVNDKCKFTL tail PPLKIHECNI SCPNPLPFRE YRPISQGVSD
LVGLPNQICL QKITSTILKP RLISYTLPIN TREGVCITDP LLAVDNGFFA
YSHLEKIGSC TRGIAKQRII GVGEVLDRGD KVPSMFMTNV WTPPNPSTIH
HCSSTYHEDF YYTLCAVSHV GDPILNSTSW TESLSLIRLA VRPKSDSGDY
NQKYIAITKV ERGKYDKVMP YGPSGIKQGD TLYFPAVGFL PRTEFQYNDS
NCPIIHCKYS KAENCRLSMG VNSKSHYILR SGLLKYNLSL GGDIILQFIE
IADNRLTIGS PSKIYNSLGQ PVFYQASYSW DTMIKLGDVD TVDPLRVQWR
NNSVISRPGQ SQCPRFNVCP EVCWEGTYND AFLIDRLNWV SAGVYLNSNQ
TAENPVFAVF KDNEILYQVP LAEDDTNAQK TITDCFLLEN VIWCISLVEI
YDIGDSVIRP KLFAVKIPAQ CSES 9291 MVVILDKRCY CNLLILILMI SECSVG
Signal sequence 9292 GGGGGS Peptide linker 9293 (GGGGS)n wherein n is 1 to
10 Peptide linker 9294 GGGGS Peptide linker 9295 PAENKKVR FENTTSDKGK
IPSKVIKSYY NiVG protein GTMDIKKINE GLLDSKILSA FNTVIALLGS attachment
IVIIVMNIMI IQNYTRSTDN QAVIKDALQG glycoprotein (602 aa) IQQQIKGLAD
KIGTEIGPKV SLIDTSSTIT Without N-terminal IPANIGLLGS KISQSTASIN
ENVNEKCKFT methionine LPPLKIHECN ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC
LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS
CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE
FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS
IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY
SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG
SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG
QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV
FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR
PKLFAVKIPE QC 9296 KVR FENTTSDKGK IPSKVIKSYY GTMDIKKINE NiVG
protein GLLDSKILSA FNTVIALLGS IVIIVMNIMI attachment IQNYTRSTDN

QAVIKDALQG IQQQIKGLAD glycoprotein KIGTEIGPKV SLIDTSSTIT IPANIGLLGS Truncated Δ5 Without KISQSTASIN ENVNEKCKFT LPPLKIHECN N-terminal methionine ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9297 NTTSDKGK IPSKVIKSYY GTMDIKKINE NiVG protein GLLDSKILSA FNTVIALLGS IVIIVMNIMI attachment IQNYTRSTDN QAVIKDALQG IQQQIKGLAD glycoprotein KIGTEIGPKV SLIDTSSTIT IPANIGLLGS Truncated Δ10 KISQSTASIN ENVNEKCKFT LPPLKIHECN Without N-terminal ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC methionine LQKTSNQILK PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9298 KGK IPSKVIKSYY GTMDIKKINE GLLDSKILSA NiVG protein FNTVIALLGS IVIIVMNIMI IQNYTRSTDN attachment QAVIKDALQG IQQQIKGLAD KIGTEIGPKV glycoprotein SLIDTSSTIT IPANIGLLGS KISQSTASIN Truncated Δ15 ENVNEKCKFT LPPLKIHECN ISCPNPLPER Without N-terminal EYRPQTEGVS NLVGLPNNIC LQKTSNQILK methionine PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9299 SKVIKSYY GTMDIKKINE GLLDSKILSA NiVG protein FNTVIALLGS IVIIVMNIMI IQNYTRSTDN attachment QAVIKDALQG IQQQIKGLAD KIGTEIGPKV glycoprotein SLIDTSSTIT IPANIGLLGS KISQSTASIN Truncated Δ20 ENVNEKCKFT LPPLKIHECN ISCPNPLPER Without N-terminal EYRPQTEGVS NLVGLPNNIC LQKTSNQILK methionine PKLISYTLPV VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRFNTC PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9300 SYY GTMDIKKINE GLLDSKILSA FNTVIALLGS NiVG protein IVIIVMNIMI

IQNYTRSTDN QAVIKDALQG attachment IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT glycoprotein IPANIGLLGS KISQSTASIN ENVNEKCKFT Truncated Δ25 LPPLKIHECN ISCPNPLPFR EYRPQTEGVS Without N-terminal NLVGLPNNIC LQKTSNQILK PKLISYTLPV methionine VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDINAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9301 TMDIKKINE GLLDSKILSA FNTVIALLGS NiVG protein IVIIVMNIMI IQNYTRSTDN QAVIKDALQG attachment IQQQIKGLAD KIGTEIGPKV SLIDTSSTIT glycoprotein IPANIGLLGS KISQSTASIN ENVNEKCKFT Truncated Δ30 LPPLKIHECN ISCPNPLPER EYRPQTEGVS Without N-terminal NLVGLPNNIC LQKTSNQILK PKLISYTLPV methionine VGQSGTCITD PLLAMDEGYF AYSHLERIGS CSRGVSKQRI IGVGEVLDRG DEVPSLEMTN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PEICWEGVYN DAFLIDRINW ISAGVELDSN QTAENPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QC 9302 KKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI NiVG protein IQNYTRSTDN QAVIKDALQG IQQQIKGLAD attachment KIGTEIGPKV SLIDTSSTIT IPANIGLLGS glycoprotein KISQSTASIN ENVNEKCKFT LPPLKIHECN Truncated and ISCPNPLPFR EYRPQTEGVS NLVGLPNNIC mutated LQKTSNQILK PKLISYTLPV VGQSGTCITD (E501 A, W504A, PLLAMDEGYF AYSHLERIGS CSRGVSKQRI Q530A, E533A) NIV G IGVGEVLDRG DEVPSLFMTN VWTPPNPNTV protein (Gc Δ 34) YHCSAVYNNE FYYVLCAVST VGDPILNSTY Without N-terminal WSGSLMMTRL AVKPKSNGGG YNQHQLALRS methionine IEKGRYDKVM PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC PAICAEGVYN DAFLIDRINW ISAGVFLDSN ATAANPVFTV FKDNEILYRA QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QCT 9303 MADSKLVSL NNNLSGKIKD QGKVIKNYYG Hendra virus G protein TMDIKKINDG LLDSKILGAF Uniprot O89343 NTVIALLGSI IIIVMNIMII QNYTRTTDNQ Without N-terminal ALIKESLQSV QQQIKALTDK IGTEIGPKVS methionine LIDTSSTITI PANIGLLGSK ISQSTSSINE NVNDKCKFTL PPLKIHECNI SCPNPLPFRE YRPISQGVSD LVGLPNQICL QKTTSTILKP RLISYTLPIN TREGVCITDP LLAVDNGFFA YSHLEKIGSC TRGIAKQRII GVGEVLDRGD KVP SMFMTNV WTPPNPSTIH HCSSTYHEDF YYTLCAVSHV GDPILNSTSW TESLSLIRLA VRPKSDSGDY NQKYIAITKV ERGKYDKVMP YGPSGIKQGD TLYFPAVGFL PRTEFQYNDS NCPIIHCKYS KAENCRLSMG VNSKSHYILR SGLLKYNLSL GGDIILQFIE IADNRLTIGS PSKIYNSLGQ PVFYQASYSW DTMIKLGDVD TVDPLRVQWR NNSVISRPGQ SQCPRFNVCP EVCWEGTYND AFLIDRLNWV SAGVYLNSNQ TAENPVFAVF KDNEILYQVP LAEDDINAQK TITDCFLLEN VIWCISLVEI YDTGDSVIRP KLFAVKIPAQ CSES 9304 KKINEGLLDSKILSA FNTVIALLGS IVIIVMNIMI NiVG protein IQNYTRSTDN QAVIKDALQG

IQQQIKGLAD attachment KIGTEIGPKV SLIDTSSTIT IPANIGLLGS glycoprotein
KISQSTASIN ENVNEKCKFT LPPLKIHECN Truncated (Gc Δ 34) ISCPNPLPER
EYRPQTEGVS NLVGLPNNIC Without N-terminal LQKTSNQILK PKLISYTLPV
VGQSGICITD methionine PLLAMDEGYF AYSHLERIGS CSRGVSKQRI
IGVGEVLDRG DEVPSLFMIN VWTPPNPNTV YHCSAVYNNE FYYVLCAVST
VGDPILNSTY WSGSLMMTRL AVKPKSNGGG YNQHQLALRS IEKGRYDKVM
PYGPSGIKQG DTLYFPAVGF LVRTEFKYND SNCPITKCQY SKPENCRLSM
GIRPNSHYIL RSGLLKYNLS DGENPKVVFI EISDQRLSIG SPSKIYDSLG
QPVFYQASFS WDTMIKFGDV LTVNPLVVNW RNNTVISRPG QSQCPRENTC
PEICWEGVYN DAFLIDRINW ISAGVFLDSN QTAENPVFTV FKDNEILYRA
QLASEDTNAQ KTITNCFLLK NKIWCISLVE IYDTGDNVIR PKLFAVKIPE QCT 9305
LSQLQKNYLDNSNQQGDKMNNPDKKLSVNFNPLELDKGQK gb: JQ001776:8170-
DLNKSYYVKNKNYNVSNLLNESLHDIKFCIYCIFSLLIII 10275|Organism: Cedar
TIINIITISIVITRLKVHEENNGMESPNLQSIQDSLSSLT virus|Strain
NMINTEITPRIGILVTATSVTLSSSINYVGTKTNQLVNEL Name: CG1a|Protein
KDYITKSCGFKVPELKLHECNISCADPKISKSAMYSTNAY Name: attachment
AELAGPPKIFCKSVSKDPDFRLKQIDYVIPVQQDRSICMN glycoprotein|Gene
NPLLDISDGFFTYIHYEGINSCKKSDSFKVLLSHGEIVDR Symbol: G Without N-
GDYRPSLYLLSSHYHPYSMQVINCVPVTCNQSSFVFCHIS terminal methionine
NNTKTLDNSDYSSDEYYITYFNGIDRPKTKKIPINNMTAD
NRYIHFTFSGGGGVCLGEEFIIPVTTVINTDVFTHDYCES
FNCSVQTGKSLKEICSESLRSPINSSRYNLNGIMIISQNN
MTDFKIQLNGITYNKLSFGSPGRLSKTLGQVLYYQSSMSW
DTYLKAGFVEKWKPFTPNWMNNTVISRPNQGNCPRYHKCP
EICYGGTYNDIAPLDLGKDMYVSVILDSDQLAENPEITVF
NSTTILYKERVSKDELNTRSTTTSCFLFLDEPWCISVLET NRFNGKSIRPEIYSYKIPKYC
9306 PQKTVEFINMNSPLERGVSTLSDKKTLNQSKITKQGYFGL gb: NC_025256:9117-
GSHSERNWKKQKNQNDHYMTVSTMILEILVVLGIMENLIV 11015|Organism: Bat
LTMVYYQNDNINQRMAELISNITVLNLNLNQLINKIQREI Paramyxovirus
IPRITLIDTATTITIPSAITYILATLTTRISELLPSINQK Eid_hel/GH-
CEFKTPTLVLNDCRINCTPPLNPSDGVKMSSLATNLVAHG M74a/GHA/2009|Strain
PSPCRNFSSVPTIYYYRIPGLYNRTALDERCILNPRLTIS Name: BatPV/Eid_hel/
STKFAYVHSEYDKNCTRGFKYYELMTFGEILEGPEKEPRM GH-
FSRSFYSPTNAVNYHSCTPIVTVNEGYFLCLECTSSDPLY M74a/GHA/2009|Protein
KANLSNSTFHLVILRHNKDEKIVSMPSFNLSTDQEYVQII Name: glycoprotein|Gene
PAEGGGTAESGNLYFPCIGRLLHKRVTHPLCKKSNCSRID Symbol: G Without
DESCLKSYYNQGSPQHQVVNCLIRIRNAQRDNPTWDVITV N-terminal methionine
DLTNTYPGSRSRIFGSFSKPMLYQSSVSWHILLQVAEITD
LDKYQLDWLDTPYISRPGGSECPFGNYCPTVCWEGTYNDV
YSLTPNNDLFVTVYLKSEQVAENPYFAIFSRDQILKEFPL
DAWISSARTTTISCFMFNNEIWCIAALEITRLNDDIIRPI
YYSFWLPTDCRTPYPHTGKMTRVPLRSTYNY 9307
ATNRDNTITSAEVSQEDKVKKYYGVETAEKVADSISGNKV gb: NC_025352:8716-
FILMNTLLILTGAIITITLNITNLTAAKSQQNMLKIIQDD 11257|Organism:
VNAKLEMFVNLDQLVKGEIKPKVSLINTAVSVSIPGQISN Mojiang virus|Strain
LQTKFLQKYVYLEESITKQCTCNPLSGIFPTSGPTYPPTD Name: Tongguan1|
KPDDDTTDDDKVDTTIKPIEYPKPDGCNRTGDHFTMEPGA Protein Name:
NFYTVPNLGPASSNSDECYINPSFSIGSSIYMFSQEIRKT attachment
DCTAGEILSIQIVLGRIVDKGQQGPQASPLLVWAVPNPKI glycoprotein|Gene
INSCAVAAGDEMGWVLCSVTLTAASGEPIPHMEDGFWLYK Symbol: G Without N-

LEPDTEVVSYRITGYAYLLDKQYDSVFIGKGGGIQKGNDL terminal methionine
YFQMYGLSRNRQSFKALCEHGSCLGTGGGGYQVLCDRAVM
SFGSEESLITNAYLKVNDLASGKPVIIGQTFPPSDSYKGS
NGRMYTIGDKYGLYLAPSSWNRYLRFGITPDISVRSTTWL
KSQDPIMKILSTCTNTDRDMCPEICNTRGYQDIFPLSEDS
EYYTYIGITPNNGGTKNFVAVRDSDGHIASIDILQNYYSI
TSATISCFMYKDEIWCIAITEGKKQKDNPQRIYAHSYKIR
QMCYNMKSATVTVGNAKNITIRRY 9308
DFDKLNKIGVVQGRVLNYKIKGDPMTKDLVLKFIPNIVNI gb: JQ001776:6129-
TECVREPLSRYNETVRRLLLPIHNMLGLYLNNTNAKMTGL 8166|Organism: Cedar
MIAGVIMGGIAIGIATAAQITAGFALYEAKKNTENIQKLT virus|Strain
DSIMKTQDSIDKLTDSVGTSILILNKLQTYINNQLVPNLE Name: CG1a|Protein
LLSCRQNKIEFDLMLTKYLVDLMTVIGPNINNPVNKDMTI Name: fusion
QSLSLLFDGNYDIMMSELGYTPQDFLDLIESKSITGQIIY glycoprotein|Gene
VDMENLYVVIRTYLPTLIEVPDAQIYEFNKITMSSNGGEY Symbol: F (without
LSTIPNFILIRGNYMSNIDVATCYMTKASVICNQDYSLPM signal sequence)
SQNLRSCYQGETEYCPVEAVIASHSPRFALINGVIFANCI
NTICRCQDNGKTITQNINQFVSMIDNSTCNDVMVDKFTIK
VGKYMGRKDINNINIQIGPQIIIDKVDLSNEINKMNQSLK
DSIFYLREAKRILDSVNISLISPSVQLFLIIISVLSFIIL
LIIIVYLYCKSKHSYKYNKFIDDPDYYNDYKRERINGKAS KSNNIYYVGD 9309
SRALLRETDNYSNGLIVENLVRNCHHPSKNNLNYTKTQKR gb: NC_025256:6865-
DSTIPYRVEERKGHYPKIKHLIDKSYKHIKRGKRRNGHNG 8853|Organism: Bat
NIITIILLLILILKTQMSEGAIHYETLSKIGLIKGITREY Paramyxovirus
KVKGTPSSKDIVIKLIPNVTGLNKCTNISMENYKEQLDKI Eid_hel/GH-
LIPINNIIELYANSTKSAPGNARFAGVIIAGVALGVAAAA M74a/GHA/2009|Strain
QITAGIALHEARQNAERINLLKDSISATNNAVAELQEATG Name: BatPV/Eid_hel/
GIVNVITGMQDYINTNLVPQIDKLQCSQIKTALDISLSQY GH-
YSEILTVFGPNLQNPVTTSMSIQAISQSFGGNIDLLLNLL M74a/GHA/2009|Protein
GYTANDLLDLLESKSITGQITYINLEHYFMVIRVYYPIMT Name: fusion
TISNAYVQELIKISFNVDGSEWVSLVPSYILIRNSYLSNI protein|Gene
DISECLITKNSVICRHDFAMPMSYTLKECLTGDTEKCPRE Symbol: F (without
AVVTSYVPRFAISGGVIYANCLSTTCQCYQTGKVIAQDGS signal sequence)
QTLMMIDNQTCSIVRIEEILISTGKYLGSQEYNTMHVSVG
NPVFTDKLDITSQISNINQSIEQSKFYLDKSKAILDKINL
NLIGSVPISILFIIAILSLILSIITFVIVMIIVRRYNKYT
PLINSDPSSRRSTIQDVYIIPNPGEHSIRSAARSIDRDRD 9310 ILHY EKLSKIGLVK
GITRKYKIKS Hendra virus F protein NPLTKDIVIK MIPNVSNVSK CTGTVMENYK
Uniprot O89342 SRLIGILSPI KGAIELYNNN (without signal THDLVGDVKL
AGVVMAGIAI GIATAAQITA sequence) GVALYEAMKN ADNINKLKSS IESTNEAVVK
LQETAEKTVY VLTALQDYIN TNLVPTIDQI SCKQTELALD LALSKYLSDL
LFVFGPNLQD PVSNSMTIQA ISQAFGGNYE TLLRTLGYAT EDFDDLLESD
SIAGQIVYVD LSSYYIIVRV YFPILTEIQQ AYVQELLPVS FNNDNSEWIS
IVPNFVLIRN TLISNIEVKY CLITKKSVIC NQDYATPMTA SVRECLIGST
DKCPRELVVS SHVPRFALSG GVLFANCISV TCQCQTTGRA ISQSGEQTLL
MIDNTTCTTV VLGNIIISLG KYLGSINYNS ESIAVGPPVY TDKVDISSQI
SSMNQSLQQS KDYIKEAQKI LDTVNPSLIS MLSMIILYVL SIAALCIGLI
TFISFVIVEK KRGNYSRLDD RQVRPVSNGD LYYIGT 9311
IHYDSLSKVGVIKGLTYNYKIKGSPSTKLMVVKLIPNIDS gb: NC_025352:5950-
VKNCTQKQYDEYKNLVRKALEPVKMAIDTMLNNVKSGNNK 8712|Organism: Mojiang

YRFAGAIMAGVALGVATAATVTAGIALHRSNENAQAIANM virus|Strain KSAIQNTNEAVKQLQLANKQTLAVIDTIRGEINNNIIPVI Name: Tongguan1| NQLSCDTIGLSVGIRLTQYYSEIITAFGPALQNPVNTRIT Protein Name: fusion IQAISSVENGNFDELLKIMGYTSGDLYEILHSELIRGNII protein|Gene DVDVDAGYIALEIEFPNLILVPNAVVQELMPISYNIDGDE Symbol: F (without WVTLVPRFVLTRTILLSNIDTSRCTITDSSVICDNDYALP signal sequence) MSHELIGCLQGDTSKCAREKVVSSYVPKFALSDGLVYANC LNTICRCMDTDTPISQSLGATVSLLDNKRCSVYQVGDVLI SVGSYLGDGEYNADNVELGPPIVIDKIDIGNQLAGINQTL QEAEDYIEKSEEFLKGVNPSIITLGSMVVLYIFMILIAIV SVIALVLSIKLTVKGNVVRQQFTYTQHVP SMENINYVSH 9312 (GGGS)n wherein n is 1 to 10 Peptide linker 9313 GGGGSGGGGSGGGGS Peptide linker 9314 TTAASGSSGGSSSGA Peptide linker 9315 GSTSGSGKPGSGEGSTKG Peptide linker

Claims

1. An antibody or antigen binding fragment thereof that specifically binds CD4, comprising a heavy chain variable region (VH) and/or a light chain variable region (VL), wherein the heavy chain variable region comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3), and the light chain variable region comprises three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3, respectively, comprise: a) SEQ ID NOs: 1280, 1794, 2308, 5644, 6154, 6664, respectively; b) SEQ ID NOs: 1281, 1795, 2309, 5645, 6155, 6665, respectively; c) SEQ ID NOs: 1282, 1796, 2310, 5646, 6156, 6666, respectively; d) SEQ ID NOs: 1283, 1797, 2311, 5647, 6157, 6667, respectively; e) SEQ ID NOs: 1284, 1798, 2312, 5648, 6158, 6668, respectively; f) SEQ ID NOs: 1285, 1799, 2313, 5649, 6159, 6669, respectively; g) SEQ ID NOs: 1286, 1800, 2314, 5650, 6160, 6670, respectively; h) SEQ ID NOs: 1287, 1801, 2315, 5651, 6161, 6671, respectively; i) SEQ ID NOs: 1288, 1802, 2316, 5652, 6162, 6672, respectively; j) SEQ ID NOs: 1289, 1803, 2317, 5653, 6163, 6673, respectively; k) SEQ ID NOs: 1290, 1804, 2318, 5654, 6164, 6674, respectively; l) SEQ ID NOs: 1291, 1805, 2319, 5655, 6165, 6675, respectively; m) SEQ ID NOs: 1292, 1806, 2320, 5656, 6166, 6676, respectively; n) SEQ ID NOs: 1293, 1807, 2321, 5657, 6167, 6677, respectively; o) SEQ ID NOs: 1294, 1808, 2322, 5658, 6168, 6678, respectively; p) SEQ ID NOs: 1295, 1809, 2323, 5659, 6169, 6679, respectively; q) SEQ ID NOs: 1296, 1810, 2324, 5660, 6170, 6680, respectively; r) SEQ ID NOs: 1297, 1811, 2325, 5661, 6171, 6681, respectively; s) SEQ ID NOs: 1298, 1812, 2326, 5662, 6172, 6682, respectively; t) SEQ ID NOs: 1299, 1813, 2327, 5663, 6173, 6683, respectively; u) SEQ ID NOs: 1300, 1814, 2328, 5664, 6174, 6684, respectively; v) SEQ ID NOs: 1301, 1815, 2329, 5665, 6175, 6685, respectively; w) SEQ ID NOs: 1302, 1816, 2330, 5666, 6176, 6686, respectively; x) SEQ ID NOs: 1303, 1817, 2331, 5667, 6177, 6687, respectively; y) SEQ ID NOs: 1304, 1818, 2332, 5668, 6178, 6688, respectively; z) SEQ ID NOs: 1305, 1819, 2333, 5669, 6179, 6689, respectively; aa) SEQ ID NOs: 1306, 1820, 2334, 5670, 6180, 6690, respectively; bb) SEQ ID NOs: 1307, 1821, 2335, 5671, 6181, 6691, respectively; cc) SEQ ID NOs: 1308, 1822, 2336, 5672, 6182, 6692, respectively; dd) SEQ ID NOs: 1309, 1823, 2337, 5673, 6183, 6693, respectively; ee) SEQ ID NOs: 1310, 1824, 2338, 5674, 6184, 6694, respectively; ff) SEQ ID NOs: 1311, 1825, 2339, 5675, 6185, 6695, respectively; gg) SEQ ID NOs: 1312, 1826, 2340, 5676, 6186, 6696, respectively; hh) SEQ ID NOs: 1313, 1827, 2341, 5677, 6187, 6697, respectively; ii) SEQ ID NOs: 1314, 1828, 2342, 5678, 6188, 6698, respectively; jj) SEQ ID NOs: 1315, 1829, 2343, 5679, 6189, 6699, respectively; kk) SEQ ID NOs: 1316, 1830, 2344, 5680, 6190, 6700, respectively; ll) SEQ ID NOs: 1317, 1831, 2345, 5681, 6191, 6701, respectively; mm) SEQ ID NOs: 1318, 1832, 2346, 5682, 6192, 6702, respectively; nn) SEQ ID NOs: 1319, 1833, 2347, 5683, 6193, 6703, respectively; oo) SEQ ID

NOs: 1320, 1834, 2348, 5684, 6194, 6704, respectively; pp) SEQ ID NOs: 1321, 1835, 2349, 5685, 6195, 6705, respectively; qq) SEQ ID NOs: 1322, 1836, 2350, 5686, 6196, 6706, respectively; rr) SEQ ID NOs: 1323, 1837, 2351, 5687, 6197, 6707, respectively; ss) SEQ ID NOs: 1324, 1838, 2352, 5688, 6198, 6708, respectively; tt) SEQ ID NOs: 1325, 1839, 2353, 5689, 6199, 6709, respectively; uu) SEQ ID NOs: 1326, 1840, 2354, 5690, 6200, 6710, respectively; vv) SEQ ID NOs: 1327, 1841, 2355, 5691, 6201, 6711, respectively; ww) SEQ ID NOs: 1328, 1842, 2356, 5692, 6202, 6712, respectively; xx) SEQ ID NOs: 1329, 1843, 2357, 5693, 6203, 6713, respectively; yy) SEQ ID NOs: 1330, 1844, 2358, 5694, 6204, 6714, respectively; zz) SEQ ID NOs: 1331, 1845, 2359, 5695, 6205, 6715, respectively; aaa) SEQ ID NOs: 1332, 1846, 2360, 5696, 6206, 6716, respectively; bbb) SEQ ID NOs: 1333, 1847, 2361, 5697, 6207, 6717, respectively; ccc) SEQ ID NOs: 1334, 1848, 2362, 5698, 6208, 6718, respectively; ddd) SEQ ID NOs: 1335, 1849, 2363, 5699, 6209, 6719, respectively; eee) SEQ ID NOs: 1336, 1850, 2364, 5700, 6210, 6720, respectively; fff) SEQ ID NOs: 1337, 1851, 2365, 5701, 6211, 6721, respectively; ggg) SEQ ID NOs: 1338, 1852, 2366, 5702, 6212, 6722, respectively; hhh) SEQ ID NOs: 1339, 1853, 2367, 5703, 6213, 6723, respectively; iii) SEQ ID NOs: 1340, 1854, 2368, 5704, 6214, 6724, respectively; jjj) SEQ ID NOs: 1341, 1855, 2369, 5705, 6215, 6725, respectively; kkk) SEQ ID NOs: 1342, 1856, 2370, 5706, 6216, 6726, respectively; lll) SEQ ID NOs: 1343, 1857, 2371, 5707, 6217, 6727, respectively; mmm) SEQ ID NOs: 1344, 1858, 2372, 5708, 6218, 6728, respectively; nnn) SEQ ID NOs: 1345, 1859, 2373, 5709, 6219, 6729, respectively; ooo) SEQ ID NOs: 1346, 1860, 2374, 5710, 6220, 6730, respectively; ppp) SEQ ID NOs: 1347, 1861, 2375, 5711, 6221, 6731, respectively; qqq) SEQ ID NOs: 1348, 1862, 2376, 5712, 6222, 6732, respectively; rrr) SEQ ID NOs: 1349, 1863, 2377, 5713, 6223, 6733, respectively; sss) SEQ ID NOs: 1350, 1864, 2378, 5714, 6224, 6734, respectively; ttt) SEQ ID NOs: 1351, 1865, 2379, 5715, 6225, 6735, respectively; uuu) SEQ ID NOs: 1352, 1866, 2380, 5716, 6226, 6736, respectively; vvv) SEQ ID NOs: 1353, 1867, 2381, 5717, 6227, 6737, respectively; www) SEQ ID NOs: 1354, 1868, 2382, 5718, 6228, 6738, respectively; xxx) SEQ ID NOs: 1355, 1869, 2383, 5719, 6229, 6739, respectively; yyy) SEQ ID NOs: 1356, 1870, 2384, 5720, 6230, 6740, respectively; zzz) SEQ ID NOs: 1357, 1871, 2385, 5721, 6231, 6741, respectively; aaaa) SEQ ID NOs: 1358, 1872, 2386, 5722, 6232, 6742, respectively; bbbb) SEQ ID NOs: 1359, 1873, 2387, 5723, 6233, 6743, respectively; cccc) SEQ ID NOs: 1360, 1874, 2388, 5724, 6234, 6744, respectively; dddd) SEQ ID NOs: 1361, 1875, 2389, 5725, 6235, 6745, respectively; eeee) SEQ ID NOs: 1362, 1876, 2390, 5726, 6236, 6746, respectively; ffff) SEQ ID NOs: 1363, 1877, 2391, 5727, 6237, 6747, respectively; gggg) SEQ ID NOs: 1364, 1878, 2392, 5728, 6238, 6748, respectively; hhhh) SEQ ID NOs: 1365, 1879, 2393, 5729, 6239, 6749, respectively; iiii) SEQ ID NOs: 1366, 1880, 2394, 5730, 6240, 6750, respectively; jjjj) SEQ ID NOs: 1367, 1881, 2395, 5731, 6241, 6751, respectively; kkkk) SEQ ID NOs: 1368, 1882, 2396, 5732, 6242, 6752, respectively; llll) SEQ ID NOs: 1369, 1883, 2397, 5733, 6243, 6753, respectively; mmmm) SEQ ID NOs: 1370, 1884, 2398, 5734, 6244, 6754, respectively; nnnn) SEQ ID NOs: 1371, 1885, 2399, 5735, 6245, 6755, respectively; oooo) SEQ ID NOs: 1372, 1886, 2400, 5736, 6246, 6756, respectively; pppp) SEQ ID NOs: 1373, 1887, 2401, 5737, 6247, 6757, respectively; qqqq) SEQ ID NOs: 1374, 1888, 2402, 5738, 6248, 6758, respectively; rrrr) SEQ ID NOs: 1375, 1889, 2403, 5739, 6249, 6759, respectively; ssSS) SEQ ID NOs: 1376, 1890, 2404, 5740, 6250, 6760, respectively; tttt) SEQ ID NOs: 1377, 1891, 2405, 5741, 6251, 6761, respectively; uuuu) SEQ ID NOs: 1378, 1892, 2406, 5742, 6252, 6762, respectively; vvvv) SEQ ID NOs: 1379, 1893, 2407, 5743, 6253, 6763, respectively; wwww) SEQ ID NOs: 1380, 1894, 2408, 5744, 6254, 6764, respectively; xxxx) SEQ ID NOs: 1381, 1895, 2409, 5745, 6255, 6765, respectively; yyyy) SEQ ID NOs: 1382, 1896, 2410, 5746, 6256, 6766, respectively; zzzz) SEQ ID NOs: 1383, 1897, 2411, 5747, 6257, 6767, respectively; aaaaa) SEQ ID NOs: 1384, 1898, 2412, 5748, 6258, 6768, respectively; bbbbb) SEQ ID NOs: 1385, 1899, 2413, 5749, 6259, 6769, respectively; ccccc) SEQ ID NOs: 1386, 1900, 2414, 5750, 6260, 6770, respectively; ddddd) SEQ ID NOs: 1387, 1901, 2415, 5751, 6261, 6771, respectively; eeeee) SEQ ID NOs: 1388, 1902, 2416, 5752, 6262, 6772, respectively; fffff) SEQ ID NOs: 1389, 1903, 2417, 5753, 6263, 6773, respectively; ggggg) SEQ ID NOs: 1390, 1904, 2418, 5754, 6264, 6774, respectively; hhhhh) SEQ ID NOs: 1391, 1905, 2419, 5755, 6265, 6775, respectively; iiiii) SEQ ID NOs: 1392, 1906, 2420, 5756, 6266, 6776, respectively; jjjjj) SEQ ID NOs: 1393, 1907, 2421, 5757, 6267, 6777, respectively; kkkkk) SEQ ID NOs: 1394, 1908, 2422, 5758, 6268, 6778, respectively; lllll) SEQ ID NOs: 1395, 1909, 2423, 5759, 6269, 6779, respectively; mmmmm) SEQ ID NOs: 1396, 1910, 2424, 5760, 6270, 6780, respectively; nnnnn) SEQ ID NOs: 1397, 1911, 2425, 5761, 6271, 6781, respectively; ooooo) SEQ ID NOs: 1398, 1912, 2426, 5762, 6272, 6782, respectively; ppppp) SEQ ID NOs: 1399, 1913, 2427, 5763, 6273, 6783, respectively; qqqqq) SEQ ID NOs: 1400, 1914, 2428, 5764, 6274, 6784, respectively; rrrrr) SEQ ID NOs: 1401, 1915, 2429, 5765, 6275, 6785, respectively; sssss) SEQ ID NOs: 1402, 1916, 2430, 5766, 6276, 6786, respectively; ttttt) SEQ ID NOs: 1403, 1917, 2431, 5767, 6277, 6787, respectively; uuuuu) SEQ ID NOs: 1404, 1918, 2432, 5768, 6278, 6788, respectively; vvvvv) SEQ ID NOs: 1405, 1919, 2433, 5769, 6279, 6789, respectively; wwwww) SEQ ID NOs: 1406, 1920, 2434, 5770, 6280, 6790, respectively; xxxxx) SEQ ID NOs: 1407, 1921, 2435, 5771, 6281, 6791, respectively; yyyyy) SEQ ID NOs: 1408, 1922, 2436, 5772, 6282, 6792, respectively; zzzzz) SEQ ID NOs: 1409, 1923, 2437, 5773, 6283, 6793, respectively; aaaaaa) SEQ ID NOs: 1410, 1924, 2438, 5774, 6284, 6794, respectively; bbbbbb) SEQ ID NOs: 1411, 1925, 2439, 5775, 6285, 6795, respectively; cccccc) SEQ ID NOs: 1412, 1926, 2440, 5776, 6286, 6796, respectively; dddddd) SEQ ID NOs: 1413, 1927, 2441, 5777, 6287, 6797, respectively; eeeeee) SEQ ID NOs: 1414, 1928, 2442, 5778, 6288, 6798, respectively; ffffff) SEQ ID NOs: 1415, 1929, 2443, 5779, 6289, 6799, respectively; gggggg) SEQ ID NOs: 1416, 1930, 2444, 5780, 6290, 6800, respectively; hhhhhh) SEQ ID NOs: 1417, 1931, 2445, 5781, 6291, 6801, respectively; iiiiii) SEQ ID NOs: 1418, 1932, 2446, 5782, 6292, 6802, respectively; jjjjjj) SEQ ID NOs: 1419, 1933, 2447, 5783, 6293, 6803, respectively; kkkkkk) SEQ ID NOs: 1420, 1934, 2448, 5784, 6294, 6804, respectively; llllll) SEQ ID NOs: 1421, 1935, 2449, 5785, 6295, 6805, respectively; mmmmmm) SEQ ID NOs: 1422, 1936, 2450, 5786, 6296, 6806, respectively; nnnnnn) SEQ ID NOs: 1423, 1937, 2451, 5787, 6297, 6807, respectively; oooooo) SEQ ID NOs: 1424, 1938, 2452, 5788, 6298, 6808, respectively; pppppp) SEQ ID NOs: 1425, 1939, 2453, 5789, 6299, 6809, respectively; qqqqqq) SEQ ID NOs: 1426, 1940, 2454, 5790, 6300, 6810, respectively; rrrrrr) SEQ ID NOs: 1427, 1941, 2455, 5791, 6301, 6811, respectively; ssssss) SEQ ID NOs: 1428, 1942, 2456, 5792, 6302, 6812, respectively; tttttt) SEQ ID NOs: 1429, 1943, 2457, 5793, 6303, 6813, respectively; uuuuuu) SEQ ID NOs: 1430, 1944, 2458, 5794, 6304, 6814, respectively; vvvvvv) SEQ ID NOs: 1431, 1945, 2459, 5795, 6305, 6815, respectively; wwwwww) SEQ ID NOs: 1432, 1946, 2460, 5796, 6306, 6816, respectively; xxxxxx) SEQ ID NOs: 1433, 1947, 2461, 5797, 6307, 6817, respectively; yyyyyy) SEQ ID NOs: 1434, 1948, 2462, 5798, 6308, 6818, respectively; zzzzzz) SEQ ID NOs: 1435, 1949, 2463, 5799, 6309, 6819, respectively; aaaaaaa) SEQ ID NOs: 1436, 1950, 2464, 5800, 6310, 6820, respectively; bbbbbbb) SEQ ID NOs: 1437, 1951, 2465, 5801, 6311, 6821, respectively; ccccccc) SEQ ID NOs: 1438, 1952, 2466, 5802, 6312, 6822, respectively; ddddddd) SEQ ID NOs: 1439, 1953, 2467, 5803, 6313, 6823, respectively; eeeeeee) SEQ ID NOs: 1440, 1954, 2468, 5804, 6314, 6824, respectively; fffffff) SEQ ID NOs: 1441, 1955, 2469, 5805, 6315, 6825, respectively; ggggggg) SEQ ID NOs: 1442, 1956, 2470, 5806, 6316, 6826, respectively; hhhhhhh) SEQ ID NOs: 1443, 1957, 2471, 5807, 6317, 6827, respectively; iiiiiii) SEQ ID NOs: 1444, 1958, 2472, 5808, 6318, 6828, respectively; jjjjjjj) SEQ ID NOs: 1445, 1959, 2473, 5809, 6319, 6829, respectively; kkkkkkk) SEQ ID NOs: 1446, 1960, 2474, 5810, 6320, 6830, respectively; lllllll) SEQ ID NOs: 1447, 1961, 2475, 5811, 6321, 6831, respectively; mmmmmmm) SEQ ID NOs: 1448, 1962, 2476, 5812, 6322, 6832, respectively; nnnnnnn) SEQ ID NOs: 1449, 1963, 2477, 5813, 6323, 6833, respectively; ooooooo) SEQ ID NOs: 1450, 1964, 2478, 5814, 6324, 6834, respectively; ppppppp) SEQ ID NOs: 1451, 1965, 2479, 5815, 6325, 6835, respectively; qqqqqqq) SEQ ID NOs: 1452, 1966, 2480, 5816, 6326, 6836, respectively; rrrrrrr) SEQ ID NOs: 1453, 1967, 2481, 5817, 6327, 6837, respectively; sssssss) SEQ ID NOs: 1454, 1968, 2482, 5818, 6328, 6838, respectively; ttttttt) SEQ ID NOs: 1455, 1969, 2483, 5819, 6329, 6839, respectively; uuuuuuu) SEQ ID NOs: 1456, 1970, 2484, 5820, 6330, 6840, respectively; vvvvvvv) SEQ ID NOs: 1457, 1971, 2485, 5821, 6331, 6841, respectively; wwwwwww) SEQ ID NOs: 1458, 1972, 2486, 5822, 6332, 6842, respectively; xxxxxxx) SEQ ID NOs: 1459, 1973, 2487, 5823, 6333, 6843, respectively; yyyyyyy) SEQ ID NOs: 1460, 1974, 2488, 5824, 6334, 6844, respectively; zzzzzzz) SEQ ID NOs: 1461, 1975, 2489, 5825, 6335, 6845, respectively; aaaaaaaa) SEQ ID NOs: 1462, 1976, 2490, 5826, 6336, 6846, respectively; bbbbbbbb) SEQ ID NOs: 1463, 1977, 2491, 5827, 6337, 6847, respectively; cccccccc) SEQ ID NOs: 1464, 1978, 2492, 5828, 6338, 6848, respectively; dddddddd) SEQ ID NOs: 1465, 1979, 2493, 5829, 6339, 6849, respectively; eeeeeeee) SEQ ID NOs: 1466, 1980, 2494, 5830, 6340, 6850, respectively; ffffffff) SEQ ID NOs: 1467, 1981, 2495, 5831, 6341, 6851, respectively; gggggggg) SEQ ID NOs: 1468, 1982, 2496, 5832, 6342, 6852, respectively; hhhhhhhh) SEQ ID NOs: 1469, 1983, 2497, 5833, 6343, 6853, respectively; iiiiiiii) SEQ ID NOs: 1470, 1984, 2498, 5834, 6344, 6854, respectively; jjjjjjjj) SEQ ID NOs: 1471, 1985, 2499, 5835, 6345, 6855, respectively; kkkkkkkk) SEQ ID NOs: 1472, 1986, 2500, 5836, 6346, 6856, respectively; llllllll) SEQ ID NOs: 1473, 1987, 2501, 5837, 6347, 6857, respectively; mmmmmmmm) SEQ ID NOs: 1474, 1988, 2502, 5838, 6348, 6858, respectively; nnnnnnnn) SEQ ID NOs: 1475, 1989, 2503, 5839, 6349, 6859, respectively; oooooooo) SEQ ID NOs: 1476, 1990, 2504, 5840, 6350, 6860, respectively; pppppppp) SEQ ID NOs: 1477, 1991, 2505, 5841, 6351, 6861, respectively; qqqqqqqq) SEQ ID NOs: 1478, 1992, 2506, 5842, 6352, 6862, respectively; rrrrrrrr) SEQ ID NOs: 1479, 1993, 2507, 5843, 6353, 6863, respectively; ssssssss) SEQ ID NOs: 1480, 1994, 2508, 5844, 6354, 6864, respectively; tttttttt) SEQ ID NOs: 1481, 1995, 2509, 5845, 6355, 6865, respectively; uuuuuuuu) SEQ ID NOs: 1482, 1996, 2510, 5846, 6356, 6866, respectively; vvvvvvvv) SEQ ID NOs: 1483, 1997, 2511, 5847, 6357, 6867, respectively; wwwwwwww) SEQ ID NOs: 1484, 1998, 2512, 5848, 6358, 6868, respectively; xxxxxxxx) SEQ ID NOs: 1485, 1999, 2513, 5849, 6359, 6869, respectively; yyyyyyyy) SEQ ID NOs: 1486, 2000, 2514, 5850, 6360, 6870, respectively; zzzzzzzz) SEQ ID NOs: 1487, 2001, 2515, 5851, 6361, 6871, respectively; aaaaaaaaa) SEQ ID NOs: 1488, 2002, 2516, 5852, 6362, 6872, respectively; bbbbbbbbb) SEQ ID NOs: 1489, 2003, 2517, 5853, 6363, 6873, respectively; ccccccccc) SEQ ID NOs: 1490, 2004, 2518, 5854, 6364, 6874, respectively; ddddddddd) SEQ ID NOs: 1491, 2005, 2519, 5855, 6365, 6875, respectively; eeeeeeeee) SEQ ID NOs: 1492, 2006, 2520, 5856, 6366, 6876, respectively; fffffffff) SEQ ID NOs: 1493, 2007, 2521, 5857, 6367, 6877, respectively; ggggggggg) SEQ ID NOs: 1494, 2008, 2522, 5858, 6368, 6878, respectively; hhhhhhhhh) SEQ ID NOs: 1495, 2009, 2523, 5859, 6369, 6879, respectively; iiiiiiiii) SEQ ID NOs: 1496, 2010, 2524, 5860, 6370, 6880, respectively; jjjjjjjjj) SEQ ID NOs: 1497, 2011, 2525, 5861, 6371, 6881, respectively; kkkkkkkkk) SEQ ID NOs: 1498, 2012, 2526, 5862, 6372, 6882, respectively; lllllllll) SEQ ID NOs: 1499, 2013, 2527, 5863, 6373, 6883, respectively; mmmmmmmmm) SEQ ID NOs: 1500, 2014, 2528, 5864, 6374, 6884, respectively; nnnnnnnnn) SEQ ID NOs: 1501, 2015, 2529, 5865, 6375, 6885, respectively; ooooooooo) SEQ ID NOs: 1502, 2016, 2530, 5866, 6376, 6886, respectively; ppppppppp) SEQ ID NOs: 1503, 2017, 2531, 5867, 6377, 6887, respectively; qqqqqqqqq) SEQ ID NOs: 1504, 2018, 2532, 5868, 6378, 6888, respectively; rrrrrrrrr) SEQ ID NOs: 1505, 2019, 2533, 5869, 6379, 6889, respectively; sssssssss) SEQ ID NOs: 1506, 2020, 2534, 5870, 6380, 6890, respectively; ttttttttt) SEQ ID NOs: 1507, 2021, 2535, 5871, 6381, 6891, respectively; uuuuuuuuu) SEQ ID NOs: 1508, 2022, 2536, 5872, 6382, 6892, respectively; vvvvvvvvv) SEQ ID NOs: 1509, 2023, 2537, 5873, 6383, 6893, respectively; wwwwwwwww) SEQ ID NOs: 1510, 2024, 2538, 5874, 6384, 6894, respectively; xxxxxxxxx) SEQ ID NOs: 1511, 2025, 2539, 5875, 6385, 6895, respectively; yyyyyyyyy) SEQ ID NOs: 1512, 2026, 2540, 5876, 6386, 6896, respectively; zzzzzzzzz) SEQ ID NOs: 1513, 2027, 2541, 5877, 6387, 6897, respectively; aaaaaaaaaa) SEQ ID NOs: 1514, 2028, 2542, 5878, 6388, 6898, respectively; bbbbbbbbbb) SEQ ID NOs: 1515, 2029, 2543, 5879, 6389, 6899, respectively; cccccccccc) SEQ

ID NOs: 1516, 2030, 2544, 5880, 6390, 6900, respectively; dddddddddd) SEQ ID NOs: 1517, 2031, 2545, 5881, 6391, 6901, respectively; eeeeeeeeee) SEQ ID NOs: 1518, 2032, 2546, 5882, 6392, 6902, respectively; ffffffffff) SEQ ID NOs: 1519, 2033, 2547, 5883, 6393, 6903, respectively; gggggggggg) SEQ ID NOs: 1520, 2034, 2548, 5884, 6394, 6904, respectively; hhhhhhhhhh) SEQ ID NOs: 1521, 2035, 2549, 5885, 6395, 6905, respectively; iiiiiiiiii) SEQ ID NOs: 1522, 2036, 2550, 5886, 6396, 6906, respectively; jjjjjjjjjj) SEQ ID NOs: 1523, 2037, 2551, 5887, 6397, 6907, respectively; kkkkkkkkkk) SEQ ID NOs: 1524, 2038, 2552, 5888, 6398, 6908, respectively; llllllllll) SEQ ID NOs: 1525, 2039, 2553, 5889, 6399, 6909, respectively; mmmmmmmmmm) SEQ ID NOs: 1526, 2040, 2554, 5890, 6400, 6910, respectively; nnnnnnnnnn) SEQ ID NOs: 1527, 2041, 2555, 5891, 6401, 6911, respectively; oooooooooo) SEQ ID NOs: 1528, 2042, 2556, 5892, 6402, 6912, respectively; pppppppppp) SEQ ID NOs: 1529, 2043, 2557, 5893, 6403, 6913, respectively; qqqqqqqqqq) SEQ ID NOs: 1530, 2044, 2558, 5894, 6404, 6914, respectively; rrrrrrrrrr) SEQ ID NOs: 1531, 2045, 2559, 5895, 6405, 6915, respectively; ssssssssss) SEQ ID NOs: 1532, 2046, 2560, 5896, 6406, 6916, respectively; tttttttttt) SEQ ID NOs: 1533, 2047, 2561, 5897, 6407, 6917, respectively; uuuuuuuuuu) SEQ ID NOs: 1534, 2048, 2562, 5898, 6408, 6918, respectively; vvvvvvvvvv) SEQ ID NOs: 9968, 10230, 10492, 12194, 12454, 12714, respectively; wwwwwwwwww) SEQ ID NOs: 9969, 10231, 10493, 12195, 12455, 12715, respectively; xxxxxxxxxx) SEQ ID NOs: 9970, 10232, 10494, 12196, 12456, 12716, respectively; yyyyyyyyyy) SEQ ID NOs: 9971, 10233, 10495, 12197, 12457, 12717, respectively; zzzzzzzzzz) SEQ ID NOs: 9972, 10234, 10496, 12198, 12458, 12718, respectively; aaaaaaaaaaa) SEQ ID NOs: 9973, 10235, 10497, 12199, 12459, 12719, respectively; bbbbbbbbbbb) SEQ ID NOs: 9974, 10236, 10498, 12200, 12460, 12720, respectively; ccccccccccc) SEQ ID NOs: 9975, 10237, 10499, 12201, 12461, 12721, respectively; ddddddddddd) SEQ ID NOs: 9976, 10238, 10500, 12202, 12462, 12722, respectively; eeeeeeeeeee) SEQ ID NOs: 9977, 10239, 10501, 12203, 12463, 12723, respectively; fffffffffff) SEQ ID NOs: 9978, 10240, 10502, 12204, 12464, 12724, respectively; ggggggggggg) SEQ ID NOs: 9979, 10241, 10503, 12205, 12465, 12725, respectively; hhhhhhhhhhh) SEQ ID NOs: 9980, 10242, 10504, 12206, 12466, 12726, respectively; iiiiiiiiiii) SEQ ID NOs: 9981, 10243, 10505, 12207, 12467, 12727, respectively; jjjjjjjjjjj) SEQ ID NOs: 9982, 10244, 10506, 12208, 12468, 12728, respectively; kkkkkkkkkkk) SEQ ID NOs: 9983, 10245, 10507, 12209, 12469, 12729, respectively; lllllllllll) SEQ ID NOs: 9984, 10246, 10508, 12210, 12470, 12730, respectively; mmmmmmmmmmm) SEQ ID NOs: 9985, 10247, 10509, 12211, 12471, 12731, respectively; nnnnnnnnnnn) SEQ ID NOs: 9986, 10248, 10510, 12212, 12472, 12732, respectively; ooooooooooo) SEQ ID NOs: 9987, 10249, 10511, 12213, 12473, 12733, respectively; ppppppppppp) SEQ ID NOs: 9988, 10250, 10512, 12214, 12474, 12734, respectively; qqqqqqqqqqq) SEQ ID NOs: 9989, 10251, 10513, 12215, 12475, 12735, respectively; rrrrrrrrrrr) SEQ ID NOs: 9990, 10252, 10514, 12216, 12476, 12736, respectively; sssssssssss) SEQ ID NOs: 9991, 10253, 10515, 12217, 12477, 12737, respectively; ttttttttttt) SEQ ID NOs: 9992, 10254, 10516, 12218, 12478, 12738, respectively; uuuuuuuuuuu) SEQ ID NO: 9993, 10255, 10517, 12219, 12479, 12739, respectively; vvvvvvvvvvv) SEQ ID NOs: 9994, 10256, 10518, 12220, 12480, 12740, respectively; wwwwwwwwwww) SEQ ID NOs: 9995, 10257, 10519, 12221, 12481, 12741, respectively; xxxxxxxxxxx) SEQ ID NOs: 9996, 10258, 10520, 12222, 12482, 12742, respectively; yyyyyyyyyyy) SEQ ID NOs: 9997, 10259, 10521, 12223, 12483, 12743, respectively; zzzzzzzzzzz) SEQ ID NOs: 9998, 10260, 10522, 12224, 12484, 12744, respectively; aaaaaaaaaaaa) SEQ ID NOs: 9999, 10261, 10523, 12225, 12485, 12745, respectively; bbbbbbbbbbbb) SEQ ID NOs: 10000, 10262, 10524, 12226, 12486, 12746, respectively; cccccccccccc) SEQ ID NOs: 10001, 10263, 10525, 12227, 12487, 12747, respectively; dddddddddddd) SEQ ID NOs: 10002, 10264, 10526, 12228, 12488, 12748, respectively; eeeeeeeeeeee) SEQ ID NOs: 10003, 10265, 10527, 12229, 12489, 12749, respectively; ffffffffffff) SEQ ID NOs: 10004, 10266, 10528, 12230, 12490, 12750, respectively; gggggggggggg) SEQ ID NOs: 10005, 10267, 10529, 12231, 12491, 12751, respectively; hhhhhhhhhhhh) SEQ ID NOs:

10006, 10268, 10530, 12232, 12492, 12752, respectively; iiiiiiiiiii) SEQ ID NOs: 10007, 10269, 10531, 12233, 12493, 12753, respectively; jjjjjjjjjjj) SEQ ID NOs: 10008, 10270, 10532, 12234, 12494, 12754, respectively; kkkkkkkkkkk) SEQ ID NOs: 10009, 10271, 10533, 12235, 12495, 12755, respectively; lllllllllll) SEQ ID NOs: 10010, 10272, 10534, 12236, 12496, 12756, respectively; mmmmmmmmmmm) SEQ ID NOs: 10011, 10273, 10535, 12237, 12497, 12757, respectively; nnnnnnnnnnn) SEQ ID NOs: 10012, 10274, 10536, 12238, 12498, 12758, respectively; ooooooooooo) SEQ ID NOs: 10013, 10275, 10537, 12239, 12499, 12759, respectively; ppppppppppp) SEQ ID NOs: 10014, 10276, 10538, 12240, 12500, 12760, respectively; qqqqqqqqqqq) SEQ ID NOs: 10015, 10277, 10539, 12241, 12501, 12761, respectively; rrrrrrrrrrr) SEQ ID NOs: 10016, 10278, 10540, 12242, 12502, 12762, respectively; sssssssssss) SEQ ID NOs: 10017, 10279, 10541, 12243, 12503, 12763, respectively; ttttttttttt) SEQ ID NOs: 10018, 10280, 10542, 12244, 12504, 12764, respectively; uuuuuuuuuuu) SEQ ID NOs: 10019, 10281, 10543, 12245, 12505, 12765, respectively; vvvvvvvvvvv) SEQ ID NOs: 10020, 10282, 10544, 12246, 12506, 12766, respectively; wwwwwwwwwww) SEQ ID NOs: 10021, 10283, 10545, 12247, 12507, 12767, respectively; xxxxxxxxxxx) SEQ ID NOs: 10022, 10284, 10546, 12248, 12508, 12768, respectively; yyyyyyyyyyy) SEQ ID NOs: 10023, 10285, 10547, 12249, 12509, 12769, respectively; zzzzzzzzzzz) SEQ ID NOs: 10024, 10286, 10548, 12250, 12510, 12770, respectively; aaaaaaaaaaaa) SEQ ID NOs: 10025, 10287, 10549, 12251, 12511, 12771, respectively; bbbbbbbbbbbb) SEQ ID NOs: 10026, 10288, 10550, 12252, 12512, 12772, respectively; cccccccccccc) SEQ ID NOs: 10027, 10289, 10551, 12253, 12513, 12773, respectively; dddddddddddd) SEQ ID NOs: 10028, 10290, 10552, 12254, 12514, 12774, respectively; eeeeeeeeeeee) SEQ ID NOs: 10029, 10291, 10553, 12255, 12515, 12775, respectively; ffffffffffff) SEQ ID NOs: 10030, 10292, 10554, 12256, 12516, 12776, respectively; gggggggggggg) SEQ ID NOs: 10031, 10293, 10555, 12257, 12517, 12777, respectively; hhhhhhhhhhhh) SEQ ID NOs: 10032, 10294, 10556, 12258, 12518, 12778, respectively; iiiiiiiiiiii) SEQ ID NOs: 10033, 10295, 10557, 12259, 12519, 12779, respectively; jjjjjjjjjjjj) SEQ ID NOs: 10034, 10296, 10558, 12260, 12520, 12780, respectively; kkkkkkkkkkkk) SEQ ID NOs: 10035, 10297, 10559, 12261, 12521, 12781, respectively; llllllllllll) SEQ ID NOs: 10036, 10298, 10560, 12262, 12522, 12782, respectively; mmmmmmmmmmmm) SEQ ID NOs: 10037, 10299, 10561, 12263, 12523, 12783, respectively; nnnnnnnnnnnn) SEQ ID NOs: 10038, 10300, 10562, 12264, 12524, 12784, respectively; oooooooooooo) SEQ ID NOs: 10039, 10301, 10563, 12265, 12525, 12785, respectively; pppppppppppp SEQ ID NOs: 10040, 10302, 10564, 12266, 12526, 12786, respectively; qqqqqqqqqqqq) SEQ ID NOs: 10041, 10303, 10565, 12267, 12527, 12787, respectively; rrrrrrrrrrrr) SEQ ID NOs: 10042, 10304, 10566, 12268, 12528, 12788, respectively; ssssssssssss) SEQ ID NOs: 10043, 10305, 10567, 12269, 12529, 12789, respectively; tttttttttttt) SEQ ID NOs: 10044, 10306, 10568, 12270, 12530, 12790, respectively; uuuuuuuuuuuu) SEQ ID NOs: 10045, 10307, 10569, 12271, 12531, 12791, respectively; vvvvvvvvvvvv) SEQ ID NOs: 10046, 10308, 10570, 12272, 12532, 12792, respectively; wwwwwwwwwwww) SEQ ID NOs: 10047, 10309, 10571, 12273, 12533, 12793, respectively; xxxxxxxxxxxx) SEQ ID NOs: 10048, 10310, 10572, 12274, 12534, 12794, respectively; yyyyyyyyyyyy) SEQ ID NOs: 10049, 10311, 10573, 12275, 12535, 12795, respectively; zzzzzzzzzzzz) SEQ ID NOs: 10050, 10312, 10574, 12276, 12536, 12796, respectively; aaaaaaaaaaaaa) SEQ ID NOs: 10051, 10313, 10575, 12277, 12537, 12797, respectively; bbbbbbbbbbbbb) SEQ ID NOs: 10052, 10314, 10576, 12278, 12538, 12798, respectively; ccccccccccccc) SEQ ID NOs: 10053, 10315, 10577, 12279, 12539, 12799, respectively; ddddddddddddd) SEQ ID NOs: 10054, 10316, 10578, 12280, 12540, 12800, respectively; eeeeeeeeeeeee) SEQ ID NOs: 10055, 10317, 10579, 12281, 12541, 12801, respectively; fffffffffffff) SEQ ID NOs: 10056, 10318, 10580, 12282, 12542, 12802, respectively; ggggggggggggg) SEQ ID NOs: 10057, 10319, 10581, 12283, 12543, 12803, respectively; hhhhhhhhhhhhh) SEQ ID NOs: 10058, 10320, 10582, 12284, 12544, 12804, respectively; iiiiiiiiiiiii) SEQ ID NOs: 10059, 10321, 10583, 12285, 12545, 12805, respectively; jjjjjjjjjjjjj) SEQ

ID NOs: 10060, 10322, 10584, 12286, 12546, 12806, respectively; kkkkkkkkkkkkk) SEQ ID NOs: 10061, 10323, 10585, 12287, 12547, 12807, respectively; lllllllllllll) SEQ ID NOs: 10062, 10324, 10586, 12288, 12548, 12808, respectively; mmmmmmmmmmmmm) SEQ ID NOs: 10063, 10325, 10587, 12289, 12549, 12809, respectively; nnnnnnnnnnnnn) SEQ ID NOs: 10064, 10326, 10588, 12290, 12550, 12810, respectively; ooooooooooooo) SEQ ID NOs: 10065, 10327, 10589, 12291, 12551, 12811, respectively; ppppppppppppp SEQ ID NOs: 10066, 10328, 10590, 12292, 12552, 12812, respectively; qqqqqqqqqqqqq) SEQ ID NOs: 10067, 10329, 10591, 12293, 12553, 12813, respectively; rrrrrrrrrrrrr) SEQ ID NOs: 10068, 10330, 10592, 12294, 12554, 12814, respectively; sssssssssssss) SEQ ID NOs: 10069, 10331, 10593, 12295, 12555, 12815, respectively; ttttttttttttt) SEQ ID NOs: 10070, 10332, 10594, 12296, 12556, 12816, respectively; uuuuuuuuuuuuu) SEQ ID NOs: 10071, 10333, 10595, 12297, 12557, 12817, respectively; vvvvvvvvvvvvv) SEQ ID NOs: 10072, 10334, 10596, 12298, 12558, 12818, respectively; wwwwwwwwwwwww) SEQ ID NOs: 10073, 10335, 10597, 12299, 12559, 12819, respectively; xxxxxxxxxxxxx) SEQ ID NOs: 10074, 10336, 10598, 12300, 12560, 12820, respectively; yyyyyyyyyyyyy) SEQ ID NOs: 10075, 10337, 10599, 12301, 12561, 12821, respectively; zzzzzzzzzzzzz) SEQ ID NOs: 10076, 10338, 10600, 12302, 12562, 12822, respectively; aaaaaaaaaaaaaa) SEQ ID NOs: 10077, 10339, 10601, 12303, 12563, 12823, respectively; bbbbbbbbbbbbbb) SEQ ID NOs: 10078, 10340, 10602, 12304, 12564, 12824, respectively; cccccccccccccc) SEQ ID NOs: 10079, 10341, 10603, 12305, 12565, 12825, respectively; dddddddddddddd) SEQ ID NOs: 10080, 10342, 10604, 12306, 12566, 12826, respectively; eeeeeeeeeeeeee) SEQ ID NOs: 10081, 10343, 10605, 12307, 12567, 12827, respectively; ffffffffffffff) SEQ ID NOs: 10082, 10344, 10606, 12308, 12568, 12828, respectively; gggggggggggggg) SEQ ID NOs: 10083, 10345, 10607, 12309, 12569, 12829, respectively; hhhhhhhhhhhhhh) SEQ ID NOs: 10084, 10346, 10608, 12310, 12570, 12830, respectively; iiiiiiiiiiiiii) SEQ ID NOs: 10085, 10347, 10609, 12311, 12571, 12831, respectively; jjjjjjjjjjjjjj) SEQ ID NOs: 10086, 10348, 10610, 12312, 12572, 12832, respectively; kkkkkkkkkkkkkk) SEQ ID NOs: 10087, 10349, 10611, 12313, 12573, 12833, respectively; llllllllllllll) SEQ ID NOs: 10088, 10350, 10612, 12314, 12574, 12834, respectively; mmmmmmmmmmmmmm) SEQ ID NOs: 10089, 10351, 10613, 12315, 12575, 12835, respectively; nnnnnnnnnnnnnn) SEQ ID NOs: 10090, 10352, 10614, 12316, 12576, 12836, respectively; oooooooooooooo) SEQ ID NOs: 10091, 10353, 10615, 12317, 12577, 12837, respectively; pppppppppppppp) SEQ ID NOs: 10092, 10354, 10616, 12318, 12578, 12838, respectively; qqqqqqqqqqqqqq) SEQ ID NOs: 10093, 10355, 10617, 12319, 12579, 12839, respectively; rrrrrrrrrrrrrr) SEQ ID NOs: 10094, 10356, 10618, 12320, 12580, 12840, respectively; ssssssssssssss) SEQ ID NOs: 10095, 10357, 10619, 12321, 12581, 12841, respectively; tttttttttttttt) SEQ ID NOs: 10096, 10358, 10620, 12322, 12582, 12842, respectively; uuuuuuuuuuuuuu) SEQ ID NO: 10097, 10359, 10621, 12323, 12583, 12843, respectively; or wherein the HCDR1, HCDR2, and HCDR3, respectively, comprise: vvvvvvvvvvvvvv) SEQ ID NOs: 1535, 2049, 2563, respectively; wwwwwwwwwwwwww) SEQ ID NOs: 9249, 9252, 9255, respectively; xxxxxxxxxxxxxx) SEQ ID NOs: 9250, 9253, 9256, respectively; or yyyyyyyyyyyyyy) wherein the HCDR2 and HCDR3 comprise SEQ ID NOs: 9254 and 9257, respectively.

2-4. (canceled)

5. The antibody or antigen binding fragment thereof of claim 1, comprising a VH having an amino acid sequence with at least 90% identity to a sequence selected from SEQ ID NOs: 303, 304, 331, 9469, 9476, or 9554, and a VL having an amino acid sequence with at least 90% identity to a sequence selected from SEQ ID NOs: 558, 559, 586, 9599, 9606, or 9684.

6-7. (canceled)

8. The antibody or antigen binding fragment thereof of claim 1, wherein the antigen binding fragment is a Fab, Fab′, F(ab′).sub.2, Fd, scFv, (scFv).sub.2, scFv-Fc, sdAb, VHH, or Fv fragment.

9. The antibody or antigen binding fragment thereof of claim 8, wherein the antigen binding

fragment is a scFv comprising a linker connecting the VH and VL, wherein the linker comprises an amino acid sequence selected from the group SEQ ID NOs: 9312-9315.
10-15. (canceled)
16. An isolated polynucleotide encoding the antibody or antigen binding fragment thereof of claim 1.
17. An isolated vector comprising the polynucleotide of claim 16.
18. An isolated host cell comprising the polynucleotide of claim 16, and/or the vector of claim 17.
19. A fusion protein comprising a glycoprotein G (G protein), hemagglutinin (H protein), or hemagglutinin-neuraminidase (HN protein) of the Paramyxoviridae family, or a biologically active portion thereof and at least one antibody or antigen binding fragment thereof of claim 1, wherein the antibody or antigen binding fragment is fused to the C-terminus of the G protein or the biologically active portion thereof.
20. The fusion protein of claim 19, wherein the antibody or antigen binding fragment thereof is fused to the G protein via a peptide linker comprising a sequence selected from the group GS, GGS, GGGS (SEQ ID NO: 14125), GGGGS (SEQ ID NO: 9294), GGGGGS (SEQ ID NO: 9292), and combinations thereof.
21-33. (canceled)
34. The fusion protein of claim 19, wherein the G protein or a biologically active portion thereof is a Henipavirus G protein or a functionally active variant or a biologically active portion thereof comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295, or wherein the functionally active variant or a biologically active portion thereof lacks up to 40 contiguous amino acid residues at or near the N-terminus of a wild-type NiV-G protein SEQ ID NO: 9266, SEQ ID NO: 9285, or SEQ ID NO: 9295.
35-51. (canceled)
52. The fusion protein of claim 19, in which the protein is pseudotyped onto a lentiviral particle.
53. (canceled)
54. A fusosome comprising at least one antibody or antigen binding fragment thereof that specifically binds CD4 of claim 1 and at least one fusogen.
55-58. (canceled)
59. A viral vector comprising: i) a F protein molecule or biologically active portion thereof of the Paramyxoviridae family; ii) an envelope glycoprotein G (G protein), hemagglutinin (H protein), or hemagglutinin-neuraminidase (HN Protein) of the Paramyxoviridae family, or a biologically active portion thereof; and iii) at least one antibody or antigen binding fragment thereof of claim 1, wherein the antibody or antigen binding fragment thereof is attached to the C-terminus of the G protein or the biologically active portion thereof.
60-63. (canceled)
64. The viral vector of claim 59, wherein the F protein or the biologically active portion thereof is a Henipavirus F protein or a functionally active variant or biologically active portion thereof comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 9259, 9265, or 9278, or fragments thereof lacking about 20 contiguous amino acid residues from the C-terminus.
65-72. (canceled)
73. The viral vector of claim 59, wherein the F-protein or the biologically active portion thereof comprises an F1 subunit or a fusogenic portion thereof, wherein the F1 subunit comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO:9261.
74. The viral vector of claim 73, wherein the F1 subunit is a proteolytically cleaved portion of the F0 precursor.
75-101. (canceled)
102. A method for selectively modulating the activity of CD4+ T cells, comprising contacting a

viral vector according to claim 59 with cells comprising CD4+ T cells.
103-104. (canceled)
105. A method of delivering an exogenous agent to a subject, comprising administering to the subject a viral vector according to claim 59, wherein the viral vector further comprises an exogenous agent.
106. The method of claim 105, wherein the exogenous agent encodes a therapeutic agent or a diagnostic agent.
107-109. (canceled)
110. A method of treating cancer in a subject, comprising administering to the subject a viral vector according to claim 59, wherein the viral vector further comprises an exogenous agent.
111. The method of claim 110, wherein the exogenous agent encodes a therapeutic agent or a diagnostic agent.
112-114. (canceled)
115. A method of transducing a cell that expresses CD4, comprising contacting the cell with the viral vector of claim 59.
116-126. (canceled)